

AN ANTISYMMETRIC SYNTAX OF WORD ORDER IN FARSI: THE PREDICATE
DOMAIN

By

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ABSTRACT

This dissertation closely reexamines the structure of the predicate domain in Farsi/Persian and evaluates whether its surface OV order is best explained by the widely adopted head-final VP/vP analyses or by an alternative framework that universally derives all linear orders from the same base structure. In favor of the latter approach, I argue that the traditional head-final verb phrase accounts face significant theoretical issues highlighted by three empirical domains: (i) complex predicates and their interaction with auxiliary verbs and phenomena such as ellipsis and verb-preposing, (ii) postverbal clausal complements and their subextraction, binding, and ellipsis patterns, and (iii) the licensing of post-nominal arguments and modifiers, particularly within the *ezâfe* construction. Crucially, such analyses cannot straightforwardly account for the core data without relying on additional assumptions, such as inconsistent directionality in head-adjunction, reliance on head-excorporation out of complex heads, and head-directionality parametrization within the VP depending on the (non)-clausal status of the verb's argument. Alternatively, an analysis within the Antisymmetry framework of Kayne (1994) is developed that captures the full range of ordering facts with the single, consistent derivational mechanism of leftward movement, eliminates rightward adjunction and excorporation, and unifies the treatment of predicate-argument and predicate-modifier configurations.

A main claim of this dissertation is that Farsi complex predicates (combinations of Non-Verbal elements (NV) and Light-Verbs (LV)) and larger verb clusters in general form phrasal XP constituents derived from a head-initial structure and various leftward movement operations. In the current analysis, such movements are defined as the raising of predicate-internal elements from their base position to dedicated licensing functional projections to the left. Focusing on complex predicate constructions and their interaction with auxiliary verbs, I propose that not only the internal and external arguments, but also the Non-Verbal (NV) predicates of complex verbs undergo movement. This analysis systematically accounts for Auxiliary-Verb ordering facts and the leftmost position of NV elements in any verbal cluster. It also straightforwardly deals with verb cluster movement in the context of data from verb-stranding vP ellipsis and Topic/Focus fronting by treat-

ing such cases as the movement of derived verbal phrases to the relevant left-peripheral specifier positions.

Further evidence in favor of the developed antisymmetric analysis comes from clausal arguments of verbs, which always surface postverbally in contrast to non-clausal arguments, which precede the verb. Previous head-final vP analyses require either exceptional rightward selection or rightward extraposition of CPs, both of which are inadequate in the context of complex predicates and auxiliary constructions. I propose that in such constructions the same leftward movements licensing the NV predicate as well as the DPs and PPs still apply. However, the crucial claim is that CP arguments remain in their base position and are not subject to the same licensing requirements above the predicate. Since the base-generated position of CP arguments is always hierarchically lower than the projections of the verb and the auxiliary head, it follows that they are always in a final position in the clause independent of whether the verb is simplex, complex, or part of a larger verbal cluster. Consequently, this proposal correctly accounts for the subextraction, binding, and ellipsis facts that are indicative of the low interpretation site of the CP.

A final puzzle for the traditional accounts is raised by post-predicate constituents in the non-verbal domain of copular and complex predicate constructions, where internal arguments or modifiers appear to the right of the predicate, often linked by the *ezâfe* morpheme. Based on a close investigation of the relevant patterns, I adopt and extend Kahnemuyipour's (2014) roll-up phrasal movement analysis, originally proposed for postnominal dependents in the NP/DP domain, to also cover predicate–modifier and predicate–argument orders in the verbal domain. Under this proposal, both modifiers and arguments are base-generated in specifiers of functional projections above the predicate head (whether nominal, adjectival, or prepositional). Roll-up movement of the predicate (or a projection containing it) across these specifiers derives the observed [Predicate –Ezf–XP] orders, while the *ezâfe* morpheme spells out the agreement relation established in this process. Crucially, I claim that this derivation represents a licensing operation available to arguments and modifiers in certain constructions and is further evidence that elements must be functionally licensed within the antisymmetric framework pursued in the dissertation.

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This thesis is dedicated to my parents, Hamid Ahari and Maryam Hassanzadeh, who are my true role models in life.

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LIST OF ABBREVIATIONS

1	1st Person
2	2nd Person
3	3rd Person
AUX	Auxiliary
CL	Clitic
DOM	Differential Object Marking
Ezf	Ezafe
FUT	Future
GEN	Genitive
IPFV	Imperfective
LV	Light Verb
NEG	Negation
NV	Non-Verb(al)
PL	Plural
PRF	Perfect(ive)
PTCP	Participle
SBJV	Subjunctive
SG	Singular

CHAPTER 1

INTRODUCTION

1.1 About this dissertation

The discussion of mixed headedness in languages, and headedness in general, seeks to answer the central question of whether word orders are freely generated or whether there is an underlying order from which linear order is derived. Word order differences among languages, particularly the distinction between OV and VO orders, reveals fundamental issues at hand that require further explanation. Crucially, such variations are not adequately accounted for by simply adhering to the Head-Directionality Parameter (Baker 2003; Chomsky 1981; Stowell 1981; Travis 1984) and assuming language-specific choices for head placement. For one, Kayne (1994, 2011, 2013, 2022) observes that we do not find languages that form true mirror image pairs of each other, casting doubts on the OV/VO parameter.

For example, while both OV and VO languages overwhelmingly raise elements to the left, such as in *wh*-movement to the Spec of CP and to the left periphery, subject movement to the Spec of TP, scrambling of arguments to the Spec of vP, TopP, FocP, and so on, there is an absence of languages that overwhelmingly do so to the right. So, although not directly related to headedness, the fact that specifiers tend to uniformly precede heads crosslinguistically requires a systematic explanation. The Head-Directionality Parameter cannot provide such an explanation and is unable to predict/motivate the observed pattern (Kusmer 2020).

It is also observed that there are systematic verb second languages (such as the Germanic languages) and an absence of languages where verbs systematically surface in a penultimate position. In seemingly OV languages, Kayne (2011) points out that there are examples suggesting that the surface order does not reflect a configuration in which the object is occupying the complement/sister of V position to the left, such as the S-O-Neg-V order in Korean, where Whitman (2005) argues that the pre Neg position of the object is due to movement. Examples of other elements such as auxiliary verbs intervening between the object and the verb (resulting in the S-O-Aux-V order) are found in languages as well, such as the Future Tense auxiliary in Farsi/Persian.

Another inconsistency that surfaces in a number of languages that have been traditionally analyzed as OV languages is that clausal complements surface post-verbally while non-clausal complements precede the verb. This is observed in Bangali, Dutch, Hindi, German, Turkish, Farsi, among others.

Kusmer (2020) further highlights the core empirical issue of a Headedness Parameter from two perspectives. If the parameter is taken as a single, language-wide setting, it predicts that all phrases in a language share the same headedness. This is empirically false, since many languages, such as German, exhibit mixed headedness (head-initial in some domains and head-final in others). Data from the *World Atlas of Linguistic Structures* (Dryer 2013) also shows a number of languages where the ordering of verb-object contrasts with adposition-noun order. On the other hand, if the parameter can be set independently for each phrase category, then it predicts logically possible head-complement combinations that seem to be unattested. In particular, it would wrongly predict configurations of the Final-over-Final Condition (FOFC), which states that head-final phrases cannot dominate head-initial ones within the same extended projection (Sheehan et al. 2017). A sample case reported in Kusmer (2020) concerns the pattern observed with respect to Complementizers and their TP complements. Essentially, while the orders $[[VP\ T]\ C]$, $[C\ [T\ VP]]$, and $[C\ [VP\ T]]$ are all found crosslinguistically, there is a typological gap with respect to the $[[T\ VP]\ C]$ configuration.

The aim of this dissertation is to highlight that Farsi adds to the larger issues of mixed-headedness in several ways. Farsi is a canonical SOV language that has long been analyzed as underlyingly head-final in the verb phrase and predicate domain (Dabir-Moghaddam 1982; Darzi 1996; Ganjavi 2007; Ghomeshi 1996; Hashemipour 1989; Karimi 1989, 2001, 2003, 2005; Mahootian 1993; Megerdooomian 2012; Rasekhi 2018; Samiiian 1983; Shafiei 2015, 2016; Taleghani 2006; Vahedi-Langrudi 1996, among many others), and otherwise as head-initial in phrases of other categories, such as PPs, CP, TPs, and arguably NPs. The OV order is shown in (1) with respect to the simplex verb *dâdan* ‘give’ and the objects *pul* ‘money’ and *be Mohammad* ‘to Mohammad’, and in (2) with respect to the complex verb *gharz dâdan* ‘rent’ and the same objects.

- (1) *Ehsân be Mohammad pul dâd.*
 Ehsan to Mohammad money gave.3SG
 ‘Ehsan gave money to Mohammad.’
- (2) *Ehsân be Mohammad pul gharz dâd.*
 E. to M. money loan gave.3SG
 ‘Ehsan lent money to Mohammad.’

I argue however that the standard OV analyses cannot be maintained based on several lingering problems that systematically undermine a head-final verb phrase structure. One such problem concerns the non-uniform distribution of auxiliaries with respect to verbs. In (3) and (4), the Perfect auxiliary *budan* ‘be’ follows the predicate, whereas the Future Tense auxiliary *xâstan* ‘want’ precedes a simplex verb (5) or the light verbal component of a complex predicate (6).

- (3) *Ehsân be Mohammad pul dâd-e ast.*
 Ehsan to Mohammad money give-PTCP AUX.PRF.3SG
 ‘Ehsan has given money to Mohammad.’
- (4) *Ehsân be Mohammad pul gharz dâd-e ast.*
 E. to M. money loan give-PTCP AUX.PRF.3SG
 ‘Ehsan has lent money to Mohammad.’
- (5) *Ehsân be Mohammad pul xâhad dâd.*
 Ehsan to Mohammad money AUX.FUT.3SG give
 ‘Ehsan will give money to Mohammad.’
- (6) *Ehsân be Mohammad pul gharz xâhad dâd.*
 E. to M. money loan AUX.FUT.3SG give
 ‘Ehsan will lend money to Mohammad.’

In the broad sense, this disharmony poses an inconsistency question as to what systematically accounts for the V-Aux versus the Aux-V order under the head-final directionality parameter assumed for Farsi verb phrases. Crucially, if the auxiliaries in question simply select for the verb phrase to the left, we expect ‘will’ to surface in final position as well (as also pointed out by Darzi (1996)). Consequently, traditional accounts would have to assume adjunction of verbs to both the left and the right of licensing heads depending on the auxiliary verb, and it is unclear what this asymmetry stems from.

Verb-Aux variation is further crucial in revealing the nature of verb cluster formation in Farsi once complex predicates and ellipsis come into the picture. Data from complex predicates and their interaction with language phenomena that can target verbal elements further leaves a gap in explaining how verbal sequences of different sizes move as a complex head unit in the structure under the standard analyses. For example, verb-stranding ellipsis cases such as (7) have been taken to propose that the complex verb *hedye dâdan* ‘gift’ can survive deletion in (7c) because the nominal predicate *hedye* ‘gift’ and the light verb *dad* ‘give’ form a unit together and can escape the deleted constituent (Rasekhi 2018). Rasekhi proposes that such cases involve the deletion of the vP, out of which either the light verb alone (7b) or the entire complex verb (7c) escapes through successive V-v-Foc head movement.

- (7) a. *Aydâ be dokhtar-esh ye ketâb hedye na-dâd.*
 Ayda to daughter-CL.3SG a book gift NEG.give.3SG
 “Ayda didn’t gift a book to her daughter.”
- b. *Vali Aydâ be pesar-esh ye ketâb hedye dâd.*
 but Ayda to son-CL.3SG a book gift gave.3SG
 “But (she) gifted (a book) to her son.”
- c. *Vali Aydâ be pesar-esh ye ketâb hedye dâd.*
 but Ayda to son-CL.3SG a book gift gave.3SG

When auxiliaries are added to the mix, the components of a complex verb can remain together in ellipsis cases involving both the Perfect in (8b) and the Future in (9b).

- (8) a. *Aydâ be doxtar-esh ye ketâb hedye na-dâd-e ast.*
 Ayda to daughter-CL.3SG a book gift NEG-GIVE-PTCP AUX.PRF.3SG
 “Ayda has not gifted a book to her daughter.”
- b. *Vali Aydâ be pesar-esh ye ketâb hedye dâd-e ast.*
 but Ayda to son-CL.3SG a book gift GIVE-PTCP AUX.PRF.3SG
 “But (she) has gifted (a book) to her son.”
- (9) a. *Aydâ be dokhtar-esh ye ketâb hedye na-xâhad dâd.*
 Ayda to daughter-CL.3SG a book gift NEG-AUX.FUT.3SG give
 “Ayda will not gift a book to her daughter.”

- b. *Vali Ayda be pesar-esh ye-ketâb hedyê xâhad dâd.*
 but Ayda to son-CL.3SG a-book gift AUX.FUT.3SG give
 “But (she) will gift (a book) to her son.”

On the assumption that complex verbs can successively adjoin to higher auxiliaries by head movement, the separability of the nominal predicate and the light verb by the Future Aux and the possibility of the three elements escaping ellipsis as a unit in (9b) remains to be adequately explained.

This thesis further revisits the puzzle of linear ordering of clausal complements with respect to the predicate as an additional argument against the head-final verb phrase analyses. The postverbal position of CP arguments reveals that it cannot be maintained that verbs consistently take their complements to the left.

- (10) *Ehsân be man goft [ke Ali xune n-ist].*
 Ehsan to me told/said.3SG that Ali home NEG-is.3SG
 “Ehsan told me that Ali isn’t home.”

Such analyses are weakened if extra assumptions are made to account for the clausal complement data, such as verbs exceptionally merging with clauses to the right in their base position (Aghaei 2006; Darzi 1996; Karimi 2005). This parametrization is in itself insufficient in explaining the consistent final position of CPs in the context of complex predicates in (11) and auxiliary constructions in (12) and (13) given that the V-CP adjacency is lost in these cases.

- (11) *Ehsân be Mohammad ghol dâd [ke ketâb-e Ali-ro be-xune].*
 E. to M. promise gave.3SG that book-Ezf A.-DOM SBJV-read.3SG
 “Ehsan promised Mohammad to read Ali’s book.”
- (12) *Ehsân be man ghol dad-e ast [ke ketâb-e Ali-ro be-xune].*
 Ehsan to me promise give-PTCP AUX.PRF.3SG that book-Ezf Ali-DOM SBJV-read.3SG
 “Ehsan has promised me to read Ali’s book.”
- (13) *Ehsân be man ghol xâhad dâd [ke ketâb-e Ali-ro be-xune].*
 Ehsan to me promise AUX.FUT.3SG give that book-Ezf Ali-DOM SBJV-read.3SG
 “Ehsan will promise me to read Ali’s book.”

Lastly, I argue that sentences involving post-predicate arguments/modifiers in complex verb constructions challenge the standard view that the constituent ordering within the predicate domain

closely reflects base-generated order. The data concerns cases such as in (14) and (15), where the PP argument *be dars xundan* ‘to study’ and the DP argument *dâru* ‘medicine’ can either precede or follow their respective non-verbal predicate. Additionally, when the non-verbal component of a complex predicate is attributively modified as in (16c), the modifier must follow the predicate.

- (14) a. *moallem* [_{DP} *dâneshamuz-o*] [_{PP} *be dars xundan*] ***vâdâr kard.***
 teacher student-DOM to lesson read-INF forced did.3SG
 “The teacher forced the student to study.”
- b. *moallem* [_{DP} *dâneshamuz-o*] ***vâdâr*** (-e) [_{PP} *be dars xundan*] ***kard.***
 teacher student-DOM forced -Ezf to lesson read-INF did.3SG
- (15) a. *man* [_{PP} *be Ali*] [_{DP} *dâru*] ***tajviz kardam.***
 I to Ali medicine prescription did.1SG
 “I prescribed Ali medicines.”
- b. *man* [_{PP} *be Ali*] ***tajviz-e*** [_{DP} *dâru*] ***kardam.***
 I to Ali prescription-Ezf medicine did.1SG
- (16) a. *Ehsân* [_{PP} *be dust-esh*] ***komak kard.***
 Ehsan to friend-CL.3SG help did.3SG
 “Ehsan helped his friend.”
- b. *Ehsân* [_{PP} *be dust-esh*] [_z*iyâd*] ***komak kard.***
 Ehsan to friend-CL.3SG much help did.3SG
 “Ehsan helped his friend often.”
- c. *Ehsân* [_{PP} *be dust-esh*] ***komak-e*** [_z*iyâdi*] ***kard.***
 Ehsan to friend-CL.3SG help-Ezf much did.3SG
 “Ehsan helped his friend a lot.”

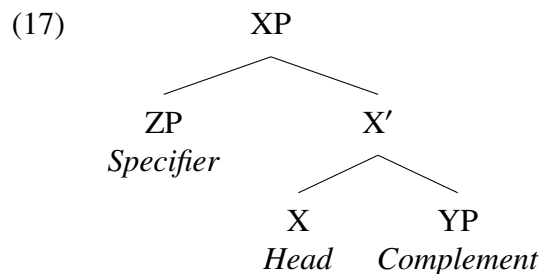
Given the consensus in the standard head-final VP/vP analyses that in Farsi complex predicates the non-verbal elements select their arguments to the left like a simplex V does (Folli, Harley, and Karimi 2005; Karimi 2005), we should perhaps expect nominal predicates to strictly follow their arguments in such constructions. The facts in (14) and (15) are at odds with this expectation and therefore require a more principled explanation within the traditional accounts.

To deal with the above issues, this dissertation adopts the approach to the analyses of headedness and word order variation that assumes there are no directionality parameters in line with the

Antisymmetry framework of Kayne (1994). Instead, the canonical Verb-Object (VO) vs. Object-Verb (OV) order distinction is attributed to a difference in movement possibilities in the derivation due to a difference in properties of functional heads. The following section provides a brief review of this framework, accompanied by a discussion of two common derivational approaches to the OV order under this theory and how Farsi fits in the picture.

1.2 Kayne's Antisymmetry framework

The core of Kayne's (1994) Antisymmetry theory is the proposal that hierarchical structure in syntax determines linear word order in a highly restricted way, formalized in the Linear Correspondence Axiom (LCA). Without going into the technical details, the LCA directly translates asymmetric c-command relations into linear precedence. Asymmetric c-command is simply defined here as non-mutual c-command; *X c-commands Y and Y does not c-command X*. As a result of this theory, all phrases are underlying head-initial, in the sense that Specifier-Head-Complement (S-H-C) is the only constituent ordering available, as suggested in (17). Following this configuration, heads asymmetrically c-command their complements, and specifiers (a type of adjunct) asymmetrically c-command heads.



This theory has several other consequences, some of which are given below.

- (18)
- a. All adjunction is to the left. Right-adjunction is prohibited as it's not LCA compatible.
 - b. Movements of heads and phrases are uniformly leftward.
 - c. No phrase can have more than one head.
 - d. No head can have more than one other head adjoined to it.
 - e. No phrase can have more one specifier

A crucial implication of the antisymmetry framework is that the Head-Directionality Parameter (in terms of head-initial vs. head-final) is eliminated from the grammar. Instead, different word orders are assumed to be the outcome of different combinations of movements that maintain the S-H-C configuration. Surface head-final orders, such the OV order observed in Farsi, are therefore derived from head-initial structures through the leftward movement of the object or other phrasal constituents to some specifier position(s). This theory further predicts mixed-headedness patterns to be possible and in fact common given that movements may apply in some domains but not others.

This framework has previously been adopted in Koster (1999, 2000), Struik (2022), and Zwart (1993, 1994, 1997) for Dutch, Koopman and Szabolcsi (2000) for Hungarian, Mahajan (2003) for Hindi, Farudi (2007), Kahnemuyipour (2004, 2009), and Moinszadeh (2001) for Farsi, Aboh (2004) for Gbe, Smith (2024a,b) for Mande, Pearson (2000) for Malagasy, Haegeman (2000, 2001, 2002) for West Flemish, Kim (2023) for Korean, Kandybowicz and Baker (2003) for Nupe, among others. Although the analyses proposed within such framework are all based on Kayne's (1994) proposal that phrase structure is head-initial and that head-final orders (such as OV) are derived by leftward movement, there are differing views with respect to how such movements proceed.

1.2.1 Individual licensing

One common approach to the antisymmetric analyses is to capture the OV word order through the movements of elements (VP-internal material) individually to various functional projections above the verb. A major representative of this approach is Zwart's (1993, 1994, 1997) and Koster's (1994) analysis of Dutch clausal structure. They argue that Dutch is underlying VO, and the OV word order observed in embedded clauses is the result of various movements. In (19), the DP object *Marie* has moved to the specifier of the functional projection AgrOP, as triggered by the strong N-features of AgrO head (Zwart 1993, 1997). Similarly, the subject DP has raised to the specifier of an AgrSP functional projection to satisfy the AgrS head's features.

- (19) ..*dat* [_{AgrSP} *Jan*_j [_{AgrOP} *Marie*_i [_{VP} ⟨*Jan*⟩_j *kust* ⟨*Marie*⟩_i]]].
 that John Mary kisses

”..that John kisses Mary.”

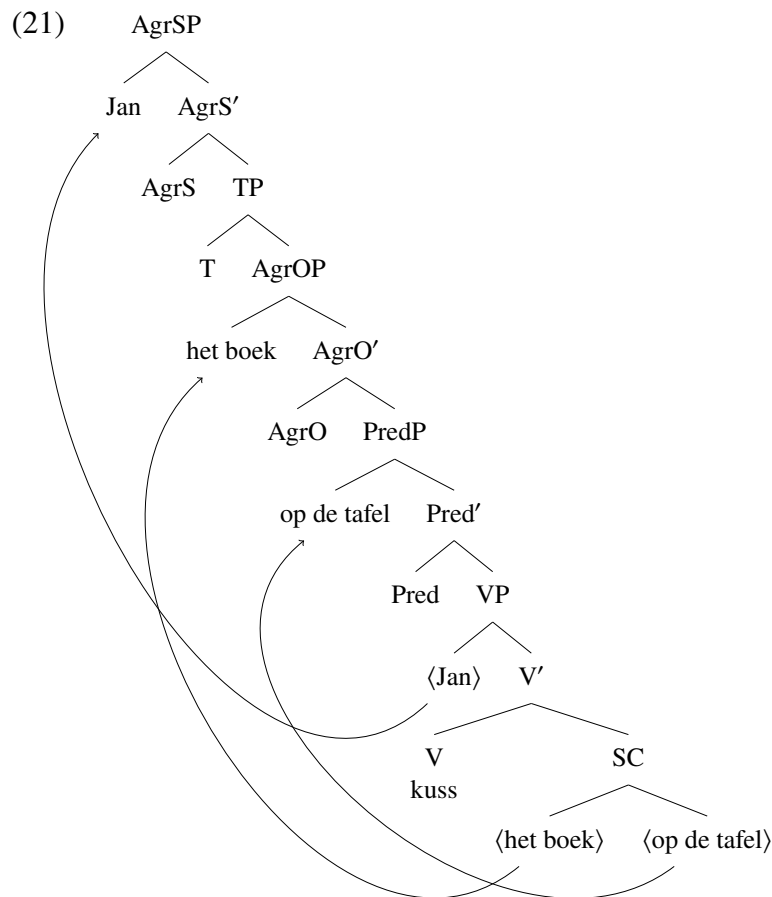
(Adapted from Zwart 1993, p. 90)

An additional assumption of Zwart's theory is that in Small Clause (SC) constructions such as (20), the SC predicate's position to the left of the verb is the result of the movement of the predicate to a functional projection *PredP* located in the selecting verb's functional domain. This is observed in the (21), where the SC predicate *op de tafel* 'on the table', the subject of the SC predicate *het boek* 'the book', and the external argument *Jan* have each moved to their respective licensing positions.

- (20) ..*dat* [_{AgrSP} *Jan*_k [_{AgrOP} *het boek*_j [_{PredP} *op de tafel*_i [_{VP} *heeft* *⟨Jan⟩*_k *gelegd* *⟨het boek⟩*_j *⟨op de tafel⟩*_i]]]].
that John the book on the table has put

"..that John has put the book on the table."

(Adapted from Zwart 1993, p. 322)



Similar to Zwart's and Koster's analysis for Dutch, Hinterhölzl (2000) proposes that phrase structure in West Germanic is uniformly head-initial, and that not only nominal arguments of the verb, but also non-verbal predicates including idioms and directional PPs, as well as clausal complements, undergo individual leftward movement to licensing positions before spellout. The obligatory

movement of the internal arguments is exemplified in (22), where the (in)definite direct object *den Plan / einen Plan* ‘the plan / a plan’ has moved to the left of the inflected verb *ausführte* ‘executed’.

- (22) ..weil Hans den Plan / einen Plan₁ genau ausführte <den Plan / einen Plan>₁.
 since Hans the plan / a plan exactly executed
 “..since Hans executed the/a plan exactly.” (Hinterhölzl 2000, ex. 19a)

As schematically shown in (23), the Small Clause projection containing the non-verbal predicate *intelligent* and the subject of the predicate *Maria* is assumed to start as the complement of the verb *findet* ‘finds’ and undergo leftward movement to the PredP above the VP.

- (23) ..weil Hans Maria₂ [_{PredP} [_{SC} <Maria>₂ intelligent]₁] [_{VP} findet [_{SC} <Maria intelligent>₁]].
 since Hans Maria intelligent finds
 “..since Hans finds Maria intelligent.” (Hinterhölzl 2000, ex. 32c)

Hinterhölzl (2000) derives the postverbal position of the CP argument in (24a) through the licensing of the CP in the spec of a Functional Projection (SpecF3P) right above the VP, with the verb also having moved to the head of a F2 above the CP, as shown in the schematic representation in (24c).

- (24) a. ..ohne der Maria zu sagen daß Peter krank ist.
 without the Maria.DAT to say that Peter sick is
 “..without telling Maria that Peter is sick”
 b. ohne der Maria [_{F1} zu [_{VP} sagen daß Peter krank ist]]
 c. ohne der Maria [_{F1} zu [_{F2} sagen [_{F3} CP [_{VP} t_V t_{CP}]]]]

(Hinterhölzl 2000, ex. 38b-d)

1.2.2 (Remnant) VP movement

A different view on deriving the linear order from a base SVO structure is to assume that elements do not move individually but are instead displaced due to the movement of a larger constituent containing them. In line with this view, Pearson (2000), Koopman and Szabolcsi (2000) and Haegeman (2000, 2001, 2002) propose a remnant movement analysis to account for verb-final constructions in Malagasy, Hungarian, and West Flemish. respectively. Taking West Flemish for instance, Haegeman discusses for the OV order of embedded clauses in (25-27).

- (25) ..*da Valère nen boek leest.*
 that Valère a book reads
 “..that Valère reads a book”
- (26) ..*da Valère nen boek no Gent stiert.*
 that Valère a book to Ghent sends
 “..that Valère sends a book to Ghent.”
- (27) ..*da Valère Marie dienen boek geeft.*
 that Valère Marie that book gives
 “that Valère gives Mary that book.”

The basic idea of Haegeman’s analysis is that two movements take place; (i) the Verb moves to “a high inflectional head”, and (ii) an extended projection of the V remnant moves to a projection above the V’s landing site in the I/T system. She assumes that this accounts extends to the same OV order phenomena in other West Germanic languages such as Dutch and German. In the following schematic representations, Haegeman (2002) illustrates that the verbs *leest* ‘reads’ (28a), *stiert* ‘sends’ (28b), and *geeft* ‘gives’ (28c) have adjoined to the F head, and the extended verbal projections (XP) have moved to the specifier position of FP.

- (28) a. *da* [_{TP} *Valère*_{su} [_{FP} [_{XP}... [_{AgrOP} [_{VP} *t*_{su} *t*_v *nen boek*]]] [_F *leest*] *t*_{xp}]].
 that Valère a book reads
- b. *da* [_{TP} *Valère*_{su} [_{FP} [_{XP}... [_{AgrOP} [_{VP} *t*_{su} *t*_v [*nen boek no Gent*]]] [_F *stiert*_v] *t*_{xp}]].
 that Valère a book to Ghent sends
- c. *da* [_{TP} *Valère*_{su} [_{FP} [_{XP}... [_{AgrIOP} *Marie*_{io} [_{VP} *t*_{su} *t*_v *t*_{io} *nen boek*]]] [_F *geeft*_v] *t*_{xp}]].
 that Valère Marie a book gives
 (Haegeman 2002, ex. 8a-c)

Also within the Antisymmetry framework and under the VP movement umbrella, Mahajan (2003) proposes an analysis for deriving the SOV word order in Hindi by which head movement (such as Verb-to-I/T) is entirely eliminated from the system, and instead, the whole VP moves to the specifier of IP/TP to satisfy the V features of I/T. Also crucial to Mahajan’s (2003) analysis is the universal requirement that the object moves leftward to a position to the left of the verb as “driven for case reasons”. Putting the two assumptions together, the SOV word order is achieved

as suggested in (29), where the object is raised within the VP before the movement of the VP takes place.

(29) SUB [_{VP} OBJ V _{t_{OBJ}}] [_{PredP} _{t_{SUB}} _{t_{VP}}] (Mahajan 2003, ex. 10)

Mahajan discusses the sample Hindi data in (30), in which the direct object *saare phal* ‘all fruits’ precedes the participle verb *khaaye* ‘eaten’, which in turn precedes the auxiliary *the* ‘be’. As the derivation in (31) illustrates, the S-O-V-Aux order is the result of (i) object raising within the VP and (ii) the subsequent leftward movement of the VP from its post-Aux position to the specifier of the IP.

(30) *raam -ne saare phal khaaye the.*
 Ram -ERG all fruits.MASC eat-PERF.MASC.PL be.MASC.PL.PAST
 “Ram had eaten all the fruits.” (Mahajan 2003, ex. 13)

(31) [*raam -ne* [_{IP} [_{VP} *saare phal khaaye t_{obj}* *the* [_{PredP}
 Ram -ERG all fruits.MASC eat-PERF.MASC.PL be.MASC.PL.PAST
t_{VP}]]].

(Mahajan 2003, ex. 14)

Under Mahajan’s (2003) analysis, the derivation of SVO word order also involves object raising and VP-movement, as well as an additional movement step targeting VP-internal arguments prior to VP-movement. This means that VP-movement in VO cases is an instance of remnant movement since the VP contains the verb and the object’s trace, as schematically illustrated in (32).

(32) SUB [_{VP} _{t’_{OBJ}} V _{t_{OBJ}}] OBJ [_{PredP} _{t_{SUB}} _{t_{VP}}] (Mahajan 2003, ex. 9)

This proposal is in part based on the analysis of Kayne (1998) for English sentences such as *Mary likes no one*, by which *no one* first moves the specifier of a NEG head above the VP and the remnant VP subsequently moves to a position to the left of *no one* in (33). It also builds on the account of Sportiche (1997) for sentences such as *Mary likes the book*, where it is assumed that the determiner *the* originates external to the VP constituent (34a), the noun *book* moves to its right (34b), and the VP remnant moves to its left (34c).¹

¹See Mahajan (2003) for the independent motivation behind generalizing this idea to SVO languages.

(33) Mary [_{VP} likes t_{no one}] no one t_{VP} (Mahajan 2003, ex. 11)

(34) a. Mary the [_{VP} likes book] →

b. Mary the book [_{VP} likes t_{book}] →

c. Mary [_{VP} likes t_{book}] the book t_{VP} (Mahajan 2003, ex. 12)

1.2.3 The approach to Farsi OV order

The antisymmetric analysis developed in this dissertation, to be laid in detail in Chapter 2, is in the spirit of Zwart's (1993, 1994, 1997) individual checking approach for Dutch. It builds on aspects of previous accounts of Farsi phrase structure within the same framework (Farudi 2007; Kahnemuyipour 2004, 2009; Moinszadeh 2001) and expands upon them by focusing on the word order facts in larger verbal clusters involving auxiliaries, which those accounts have generally left unexplored.

1.3 Outline of the dissertation

The rest of the dissertation is structured as follows. Chapter 2 provides a brief background on word order in Farsi in simplex and complex verb constructions. Focusing on complex verbs, in particular, it discusses some of the existing head-final verb phrase analyses and shows that a head-final structure for the verbal domain fails to consistently account for the clustering behavior of the predicates through rightward head-adjunction, and without the additional stipulation that heads can excorporate out of complex heads. This is highlighted in sentences involving the Future Tense Auxiliary *xâstan* 'want/will'. The Future Aux (unlike the Passive and Perfect) never surfaces in the final position in any verbal cluster and always appears in a fixed penultimate position, resulting in the strict S-O-Aux-V order with simplex predicates and the S-O-NV-Aux-LV order with complex verbs. More specifically, data from verb-stranding verb phrase ellipsis and Focus/Topic fronting reveals that auxiliaries in Farsi (including the Future Aux) tend to form constituents with other verbal elements and undergo syntactic movements that are best systematically captured as instances of leftward XP movements. This chapter develops the foundational antisymmetric framework that derives the verb-final order with different auxiliaries via uniform leftward head and phrasal move-

ments and without resorting to excorporation. It also offers an analysis for the verb-stranding ellipsis and verb fronting in line with this framework.

Chapter 3 highlights the shortcomings of the head-final verb phrase analyses when it comes to accounting for the post-verbal position of clausal arguments in Farsi (as opposed to the pre-verbal position of non-clausal arguments). I discuss previous proposals that attempt to account for the postverbal position of CPs under the head-final VP umbrella by parametrizing the directionality of selection based on the argument type; clausal arguments are selected as righthand complements while nonclausal ones are selected to the left. I show that this parametrization can only explain the distribution of CP with respect simplex verbs, but it's inadequate as far as constructions involving complex verbs and larger verbal complexes are concerned. Reconciling these facts with the head-final verb phrase structure requires implementing some movement operation such as rightward extraposition or remnant movement, which is argued to be unattainable given that data from subextraction, binding, and ellipsis suggest a low structural position for the CPs. The antisymmetric analysis is shown to derive the consistent final position of the CPs independent of the complexity of the verbal clusters while also accounting for the subextraction and binding facts.

Chapter 4 concerns a deeper investigation of the non-verbal predicate domain in Farsi and reveals cases where complements and modifiers follow the predicate, introducing additional inconsistencies to the directionality parameter and further evidence that all arguments are licensed in their surface positions throughout the derivation. Relevant to the post-predicate constituents, the *ezâfe* construction is examined and its implications for argument/modifier licensing are laid out. The antisymmetric analysis is extended to predicate-modifier and predicate-argument configurations.

Chapter 5 concludes.

CHAPTER 2

THE HEAD-ADJUNCTION ISSUE: EVIDENCE FROM ELLIPSIS AND FRONTING

This chapter addresses the verb phrase domain in Farsi and the Complex Verb domain, in particular, and highlights empirical data from verb-stranding vP ellipsis and from optional verb fronting as pieces of evidence against the traditional SOV analyses and in favor of adopting Kayne (1994) and invariable leftward movements. I argue that head-final verb phrase analyses such as Karimi (2005) and Rasekhi (2018) face inconsistencies with respect to the ordering of auxiliaries and main verbs and will have to rely on ad hoc head movement operations to both the left and right of licensing heads requiring both excorporation and incorporation. The focus of this discussion will be on how such analyses would deal with ellipsis and verb cluster movement in sentences containing a complex predicate and auxiliary verbs, paying special attention to the Future Tense Aux *xâstan* ‘want/will’. By proposing an antisymmetric approach to deriving Farsi complex predicates, I argue that verb clusters form phrasal constituents (as opposed to head constituents) in the derivation, and that both ellipsis and verb fronting involve the movement of maximal projections to left-peripheral XP projections. This eliminates the need for excorporation while keeping movement consistently leftward.

2.1 Brief background on the Farsi word order

2.1.1 Canonical word order with Simplex Verbs

Persian/Farsi is canonically a Subject-Object-Verb (SOV) language where the internal arguments of a verb linearly precede the main verb/predicate in a sentence. This is shown in the following ditransitive constructions in (1a)-(1b), where the simplex verb *dâdan* ‘give’ follows both the direct object DP *pul* ‘money’ and the indirect object PP *be Mohammad* ‘to Mohammad’ (main predicates are boldfaced throughout the examples).

- (1) a. *Ehsân be Mohammad pul **dâd.***
Ehsan to Mohammad money gave.3SG
“Ehsan gave money to Mohammad.”
- b. *Ehsân pul-ra be Mohammad **dâd.***
Ehsan money-DOM to Mohammad gave.3SG

“Ehsan gave the money to Mohammad.”

It should be noted that there is a difference in the two sentences with respect to the distribution of the direct object ‘money’. When the direct object is non-specific as in (1a), it immediately precedes the main predicate and follows the indirect object. On its specific reading, however, the DP in (1b) bears the differential marker *râ* and linearly precedes the indirect object (as well as the verb).

- (2) a. (S) (PP) (O) V
- b. (S) (O-ra) (PP) V

Unlike the nominal (non-clausal) arguments, clausal arguments of simplex verbs in Farsi surface postverbally. As (3a) shows, the CP *ke Ali xune n-ist* ‘that Ali isn’t home’ must follow the predicate *goft* ‘said’ and cannot surface preverbally as suggested in (3b).

- (3) a. *Ehsân be Mohammad goft* [*ke Ali xune n-ist*].
Ehsan to Mohammad told/said.3SG that Ali home NEG-is.3SG
“Ehsan told Mohammad that Ali isn’t home.”
- b. * *Ehsân be Mohammad* [*ke Ali xune n-ist*] *goft*.
Ehsan to Mohammad that Ali home NEG-is.3SG told/said.3SG

2.1.2 Canonical word order with Complex Verbs

In addition to simplex verbs, Farsi primarily consists of a wide range of complex verbs, which are formed by a Non-Verbal (NV) predicate (Adj, N, Adv, P,...) and a Light Verb (LV). When it comes to the linear order, complex verbs (like simplex verbs) also follow their non-clausal arguments. This is shown in (4a) and (4b) where the predicate *gharz dâdan* ‘lend’ surfaces to the right of the direct object ‘money’ and the indirect object ‘to Mohammad’.

- (4) a. *Ehsân be Mohammad pul gharz dâd*.
E. to M. money loan gave.3SG
“Ehsan lent money to Mohammad.”
- b. *Ehsân pul-o be Mohammad gharz dâd*.
E. money-DOM to M. loan gave.3SG
“Ehsan lent the money to Mohammad.”

Clausal arguments follow complex verbs as they do simplex verbs. In (5a) and (6a), the bracketed clauses follow the predicates *ghol dâdan* ‘promise’ and *ghâneh kardan* ‘convince’, respectively. As the (b) and (c) examples show, these clauses can neither precede the predicate nor intervene between the NV and the LV components of the predicate.

- (5) a. *Ehsân be Mohammad ghol dâd* [*ke ketâb-e Ali-ro be-xune*].
 E. to M. promise gave.3SG that book-Ezf A.-DOM SBJV-read.3SG
 “Ehsan promised Mohammad to read Ali’s book.”
- b. **Ehsân be Mohammad* [*ke ketâb-e Ali-ro be-xune*] *ghol dâd*.
 E. to M. that book-Ezf A.-DOM SBJV-read.3SG promise gave.3SG
- c. **Ehsân be Mohammad ghol* [*ke ketâb-e Ali-ro be-xune*] *dâd*.
 E. to M. promise that book-Ezf A.-DOM SBJV-read.3SG gave.3SG
- (6) a. *Ehsân Mohammad-o ghâneh kard* [*ke ketâb-e Ali jâleb-e*].
 E. M.-DOM convinced did.3SG that book-Ezf A. interesting-is.3SG
 “Ehsan convinced Mohammad that Ali’s book is interesting.”
- b. **Ehsân Mohammad-o* [*ke ketâb-e Ali jâleb-e*] *ghâneh kard*.
 E. M.-DOM that book-Ezf A. interesting-is.3SG convinced did.3SG
- c. **Ehsân Mohammad-o ghâneh* [*ke ketâb-e Ali jâleb-e*] *kard*.
 E. M.-DOM convinced that book-Ezf A. interesting-is.3SG did.3SG

On these observations, Farsi displays mixed word order properties for the predicate and its arguments depending on whether the argument is clausal or not, suggesting an absence of uniform verb-finality despite its surface SOV structure. Additionally, Preposition Phrases such as the indirect PP object *to Mohammad* ‘to Mohammad’ in the above examples reveal that prepositions precede their arguments. Similarly, Complementizer Phrases such as *ke Ali xune nist* ‘that Ali isn’t home’ and *ke ketâb-e Ali-ro bexune* ‘that read Ali’s book/to read Ali’s book’ highlight the same fact for the position of complementizer *ke* ‘that’ with respect to its propositional argument. In the next section, I will discuss some of the more recent existing accounts in the literature for the Farsi phrase structure and how they extend to the data presented here. This discussion will not be an extensive review of all the previous work however; rather, it will focus on some of the prominent approaches to Farsi word order that assume the head parameter approach for cross-linguistic variation. I will lay out aspects

of such analyses, including what they assume the underlying structure to be for the verbal domain and the clauses with either simplex or complex predicates, as well as how the derivation proceeds to account for the movements that result in the direct object-indirect object ordering variation observed in (1) and (4). The difference in the distribution of the non-clausal arguments vs. the clausal ones with respect to the predicate and its comprehensive and in-depth characterization will however be the postponed to the following chapter.

2.2 A brief review of prominent head-final verb phrase analyses

The standard and generally adopted approaches to Farsi phrase structure and word order in the recent decades analyze the lexical VP/vP or PredP domain to be underlyingly head-final where verbs select their complements to the left, while all other categories generally show head-initiality and selection of complements to the right, therefore explaining the mixed-headedness properties of Farsi. This head-parameter symmetric approach is adopted in a vast body of literature, such as Dabir-Moghaddam (1982), Darzi (1996), Ganjavi (2007), Ghomeshi (1996), Hashemipour (1989), Karimi (1989, 2001, 2003, 2005), Mahootian (1993), Megerdooomian (2012), Rasekhi (2018), Samiian (1983), Shafiei (2015, 2016), Taleghani (2006), and Vahedi-Langrudi (1996), among many others. My overview of these analyses will focus more extensively on Farsi complex predicates (those containing of a non-verbal unit and light verb) and on two primary theoretical approaches that arise from Karimi (1989, 2001, 2005), Shafiei (2016), and Rasekhi (2018).

2.2.1 Karimi's account

Considering (1), repeated below as (7), the general approach taken in Karimi (1989, 2001, 2003, 2005) is that the main verb V head selects for its arguments to the left.

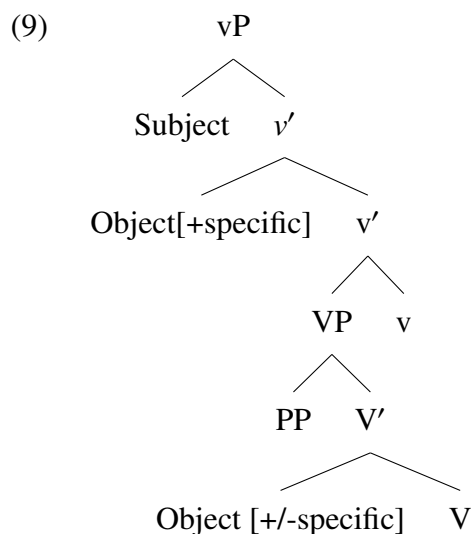
- (7) a. *Ehsân be Mohammad pul dâd.*
 Ehsan to Mohammad money gave.3sg
 ‘‘Ehsan gave money to Mohammad.’’
- b. *Ehsân pul-râ be Mohammad dâd.*
 Ehsan money-DOM to Mohammad gave.3sg
 ‘‘Ehsan gave the money to Mohammad.’’

With respect to the indefinite/non-specific versus the definite/specific direct object, Karimi (1989,

2003) originally proposed that the two originate in two different positions. In addition to the presence of the *râ* morpheme on the specific DO and its absence on the nonspecific one, there are semantic (such as nonspecific objects being part of the description of the verb) as well as syntactic properties (such as specific objects being able to bind indirect objects) for which Karimi (2003) proposes the Two Object Position Hypothesis in (8) to capture these asymmetries. Under this theory, the non-specific argument *pul* ‘money’ in (7a) originates as the complement of the verb and the indirect object ‘to Mohammad’ merges as the specifier of VP, while the specific *râ* marked object in (7b) originates as the specifier of the verb and the indirect object as its complement.

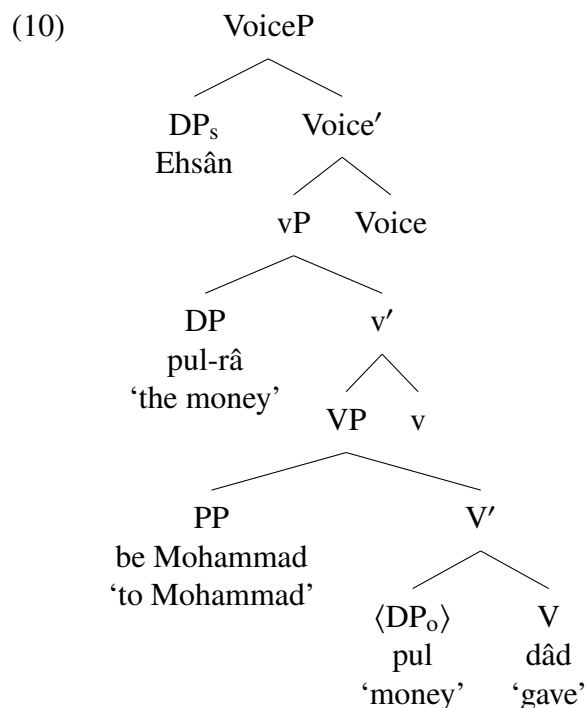
- (8) a. [VP DP_[+Specific] [V' PP V]]
 b. [VP [V' PP [V' DP_[-Specific] V]]

This analysis is further updated in Karimi (2005), however, proposing that both objects are base-generated in the same position. Karimi provides the schematic structure in (9) and argues that a [+specific] DP obligatorily scrambles to the vP edge. The internal arguments stay inside the VP, otherwise and are interpreted in their base position. The external argument in turn merges as an outer specifier of vP.



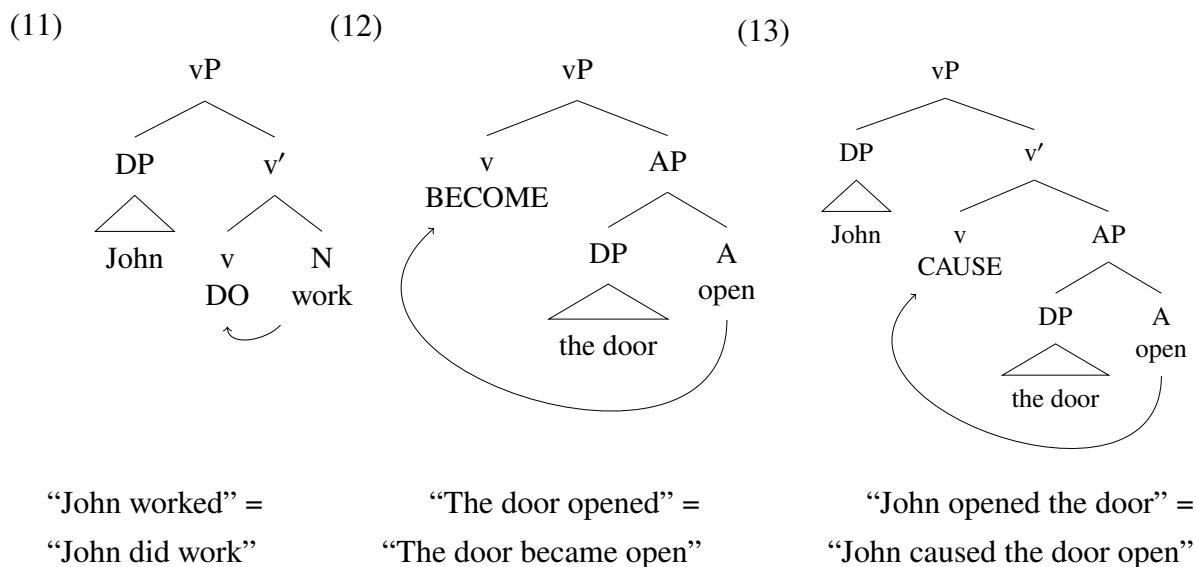
In Karimi and Smith (2020), it is assumed that the external argument is instead introduced in VoiceP (Kratzer 1996) and Specifier of vP is the position to which specific direct objects move. Following Karimi and Smith, (7a) is therefore updated to reflect this in (10). In this structure, the

verb *dâd* ‘buy’ selects for the nonspecific direct object *pul* ‘money’ as its complement and for the indirect object *be Mohammad* ‘to Mohammad’ as its specifier. The head-final vP and the VoiceP project higher, and the external argument ‘Ehsan’ is introduced in the specifier of VoiceP.

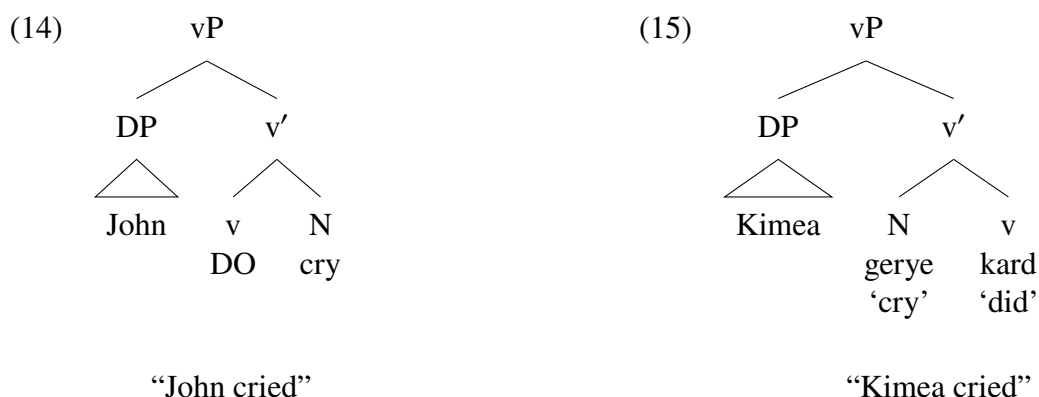


2.2.2 Complex Predicates: The syntactic independence approach

Folli, Harley, and Karimi (2005) adopt a constructionalist approach to representing Farsi complex predicates syntactically, in line with Hale and Keyser (1993 and subsequent work) and building on Karimi (2005). They argue for a correspondence between the following structures in (11)-(13) and Persian complex predicate constructions. In (11), the unergative ‘John worked’/‘John did work’ incorporates the object ‘work’ into the abstract v head. The unaccusative ‘The door opened’/‘The door became open’ incorporates the adjectival predicate into the abstract BECOME v head in (12), while its causative alternation ‘John opened the door’ is derived by the incorporation of the adjective into a CAUSE v head in (13).



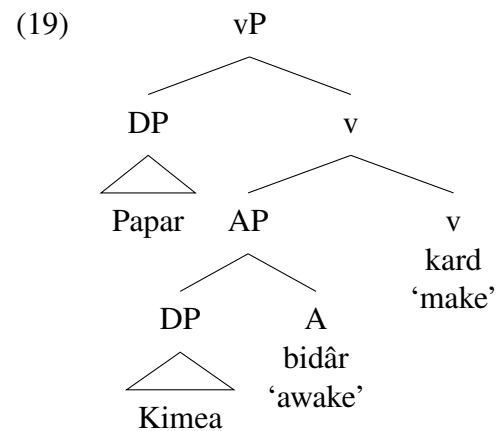
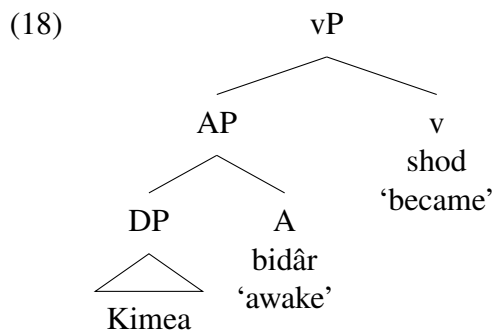
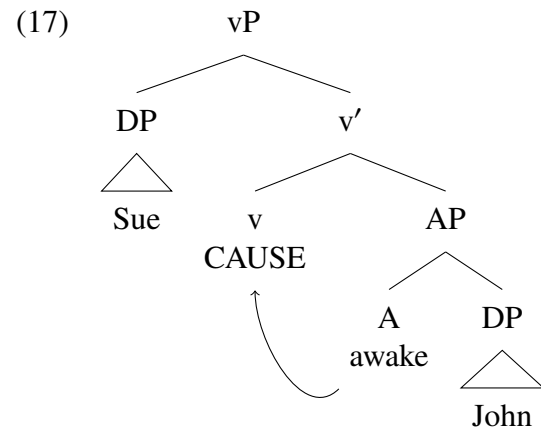
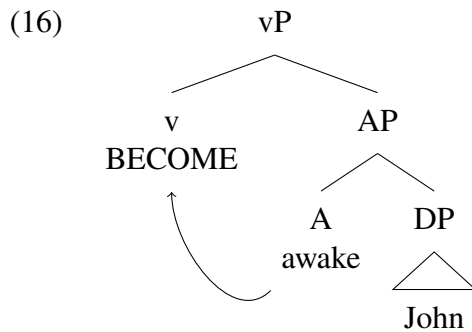
Folli, Harley, and Karimi (2005) and Karimi (2005) propose that in Farsi complex predicates, the Non-Verbal (NV) predicate does not incorporate into the LV. Taking the English-Farsi pair of unergative structures in (14) and (15) for instance, this means that the noun ‘cry’ in ‘John cried’ incorporates into v whereas the Farsi *gerye* ‘cry’ does not incorporate into the light verb.²



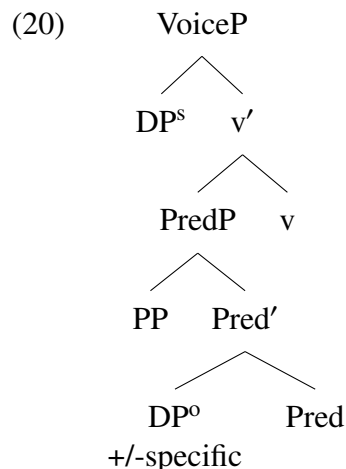
Extending this to an inchoative construction, Folli, Harley, and Karimi (2005) and Karimi (2005) assume the structure in (16) for the English ‘John awoke’ and the one in (18) for the Farsi *Kimeâ bidâr shod* ‘Kimea awoke’. On its causative alternation, the English ‘awake’ incorporates

²Folli, Harley, and Karimi (2005) propose that the verb finality shown in (15) is due to post-syntactic linearization at Spell Out, whereas Karimi (2005) adopts an underlying head-final verbal projection in agreement with Jazayery and Paper (1961), Bateni (1969,1977), Moyne (1970), Soheili Isfahani (1976), Farrokhphey (1979), Dabir-Moghaddam (1982), Samiian (1983), Darzi (1996), and Ghomeshi (1996).

into the CAUSE *v* in (17), and in Farsi, changing the light verb *shod* ‘became’ to *kard* ‘make’ does the job in (19).



On these assumptions, non-verbal predicates head their own small clause lexical projections (in this case AdjP) and select for the internal arguments similar to V in VPs. The *v* head, in turn, hosts the light verb in complex predicates. Consequently, Folli, Harley, and Karimi (2005) relabel the VP in Karimi’s underlying structure in (9) to PredP to capture the fact that the complement to *v* can also be the projection of the non-verbal element that carries the main predicate force of in complex predicate constructions. Incorporating both the relabeling of VP to PredP and Karimi and Smith’s (2020) VoiceP (as the projection that introduces the external argument) yields the structure in (20).



Folli, Harley, and Karimi (2005) and Megerdooomian (2001) discuss the role of each of the components of a complex predicate and agree that the non-verbal element is responsible for telicity and theta role assignment to internal arguments whereas the light verb introduces the external argument and is responsible for transitivity facts. A crucial point of the discussion here is that Folli et al. and Karimi argue for the treatment of the non-verbal and the light-verbal units as syntactically independent based on empirical data that shows the possibility of separating of the two components. For instance, the negative prefix *na-* in (21), the habitual/imperfective aspectual marker *mi-* in (22) and inflectional morphemes can intervene between the Non-Verb and the Light Verb of the complex predicate. Lexical units, such as the Future Tense auxiliary *xâstan* ‘want’, can also break up complex predicates as exemplified in (23).

- (21) *Ehsân be man pul gharz na-dâd.*
 Ehsan to me money loan NEG-gave.3SG
 “Ehsan didn’t lend me money.”

- (22) *Ehsân be man pul gharz ne-mi-dahad.*
 Ehsan to me money loan NEG-IPFV-give.3SG
 “Ehsan does not lend me money.”

- (23) *Ehsân be man pul gharz xâhad dâd.*
 Ehsan to me money loan AUX.FUT.3SG give
 “Ehsan will lend me money.”

Additionally, Folli, Harley, and Karimi (2005, p. 7) show in (24) and (25) that adjectives that modify the NV predicate surface inbetween the NV and the LV. They further argue that the pos-

sibility for gapping the NV in (26) and the limited scrambling of a quantified NV element in (27) supports their claim that the components of complex predicates are “separately generated and combined in syntax”.

- (24) *Kimeâ az ra'is-e edâre* [CV [NV **da'vat-e** rasmi] **kard**]].
 Kimea of boss-Ezf office invitation-Ezf formal did.3SG
 “Kimea extended a formal invitation to the boss of the office.”

- (25) *Kimeâ barâye in xune* [CV [NV **chune-ye** xubi] **zad**]].
 Kimea for this house chin-Ezf good hit.3SG
 “Kimea performed a good negotiation for this house.”

(Folli, Harley, and Karimi 2005, ex. 10a-b)

- (26) *Kimeâ faghat man-o da'vat kard-e, to-ro ke na-kard-e.*
 Kimea only me-DOM invitation do-PTCP.3SG, you-DOM emph NEG-do-PTCP.3SG
 “Kimea has only invited me, not you.”

(Folli, Harley, and Karimi 2005, ex. 11)

- (27) *Kimeâ [che zamin-e saxti]_i diruz [CV t_i **xord**]].*
 Kimea what earth-Ezf hard yesterday collided.3SG
 “What a hard fall Kimea had yesterday.” Lit. Kimea what a hard earth yesterday collided.

(Folli, Harley, and Karimi 2005, ex. 12a)

The exact details and analysis of the modification, gapping, and scrambling constructions in (24-27) will not be the focus of the current discussion. I will come back to the crucial cases of the Future tense auxiliary in (23) in Section 2.3.1.3 to argue that such cases pose serious questions about the head final assumptions for Farsi predicate phrases. The next order of business, however, is a discussion of an alternative view to Folli et al.’s (2005) and Karimi’s complex predicate approach. In the following section, I will briefly review proposals by Shafiei (2015, 2016) and Rasekhi (2018) for Farsi phrase structure and complex verbs, which aim to deal with both the separability facts and with the apparent ability for the entire complex predicate to move as a unit in data from verb stranding vP ellipsis.

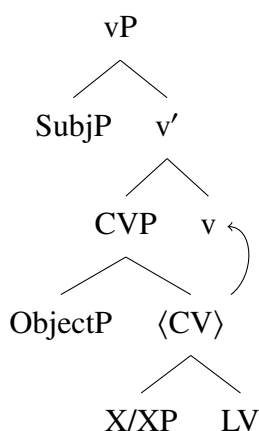
2.2.3 Complex Predicates: The constituent V head approach

A slightly different approach to Folli et al.'s (2005) and Karimi's (2005) analysis for the base structure of complex predicate treats the Non-Verbal and the Light Verbal components of the predicate as a syntactic unit (Rasekhi 2018; Shafiei 2015, 2016). This approach is primarily motivated by data from verb-stranding ellipsis that suggests that the NV and LV components of the complex predicate can raise together as a complex unit and escape the ellided constituent. Therefore, this section focuses on such elliptical constructions and introduces the base complex predicate structures proposed by Shafiei and Rasekhi.

2.2.3.1 Shafiei 2015, 2016

Shafiei (2015, 2016) proposes that the Non-Verbal (NV) element and the Light Verb (LV) constitute a Complex Predicate (CPr) that behaves like the head of a projection. This Complex Verb (CV) head takes internal arguments similar to a normal V and projects a Complex Verb Phrase (CVP). Shafiei proposes that in (28), the light verbal head (LV) can select for either a Non-verbal X head or a maximal projection XP (such as NP, AdjP, Adv, and PP), and the two m-merge (Matushansky 2006) to constitute the CV head.³

(28) Adapted from Shafiei (2016, p. 39)



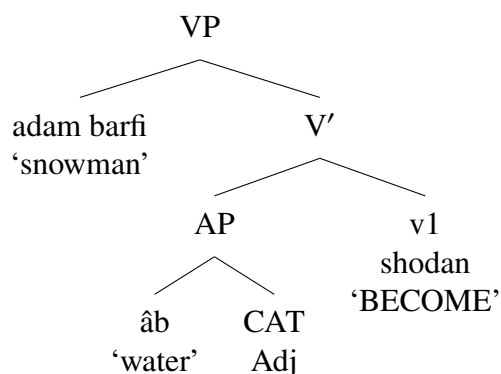
This proposal is based on Megerdooomian's (2001, 2012) analysis according to which in an intransitive construction such as (29a) has the structure in (29b), in which the lower v layer combines

³M-merger as adopted by Shafiei, is defined as the operation that takes place when a head selects for another head and results in projecting a complex head.

a light verb with the adjectival preverb prior to selecting for an internal argument.

- (29) a. *Âdam-barfi* [_{CPr} *âb* *shod*].
 man-snow [_{CPr} water became.PAST.3SG].
 “The snowman melted.” (Megerdooonian 2001, ex. 15a)

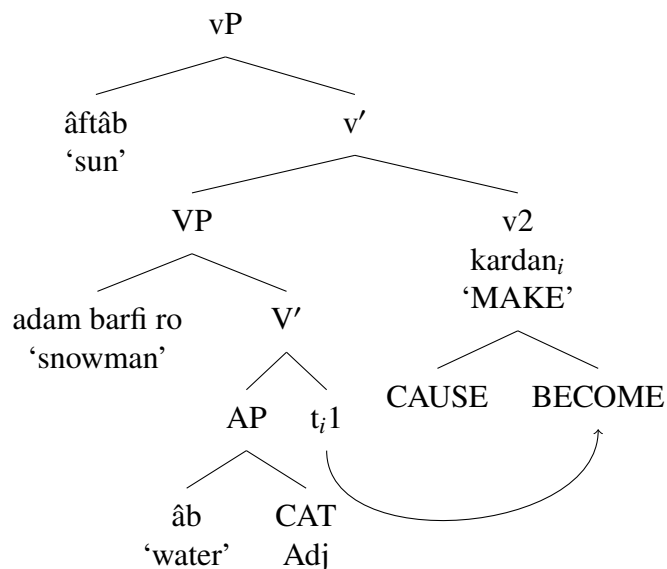
- b. The inner event (Megerdooonian 2001, ex. 37)



The CAUSE event in (30a) is built on top of the BECOME event through the outer v layer, where the lower v adjoins to the higher v and the external argument is introduced, as shown in (30b).

- (30) a. *Âftâb Âdam-barfi ro* [_{CPr} *âb* *kard*].
 sun man-snow DOM [_{CPr} water make.PAST.3SG].
 “The sun melted the snowman.” (Megerdooonian 2001, ex. 15b)

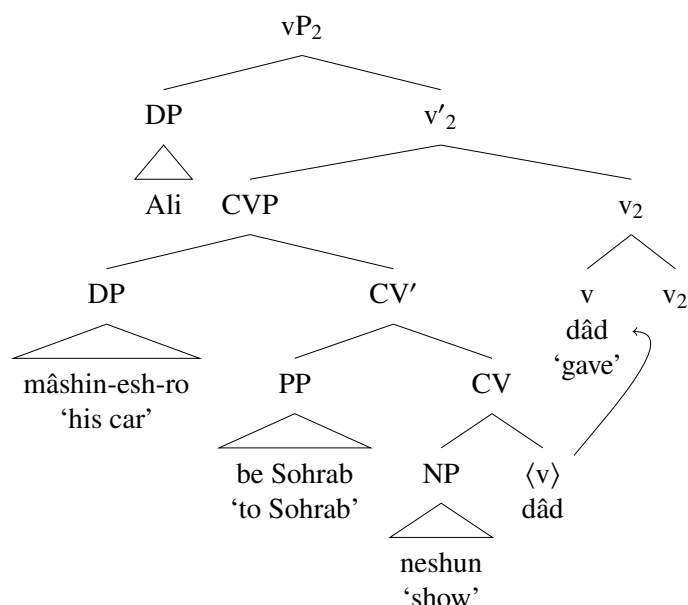
- b. The outer event (Megerdooonian 2001, ex. 38b)



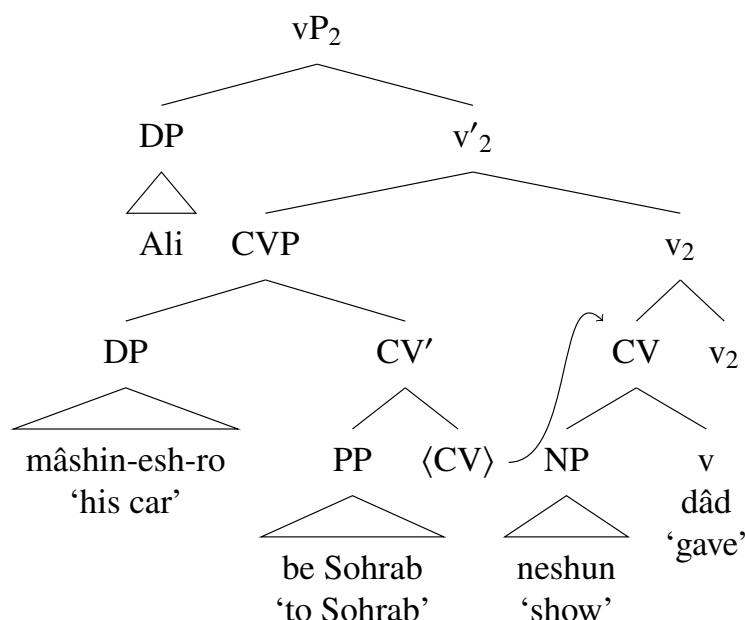
Shafiei proposes that in addition to raising for aspectual readings, verbs can also move to escape the ellipsis site in elliptical constructions. Crucially, cases such as (31c) show that the full predicate (containing the NV and the LV) can survive deletion, which previous verb phrase ellipsis literature had left unmentioned and unaccounted for (including Folli, Harley, and Karimi 2005). Building on the structure in (28), Shafiei (2016) accounts for instances where the NV is elided in (31b) through the movement of the LV by itself out of CV and stranding the NV head. The suggested derivation is given in (32). Whenever the NV preverb survives ellipsis as in (31c), however, the entire CV head is proposed to undergo movement, as shown in (33).

- (31) a. *Shahrâm mâshin-esh ro be Sohrâb* [_{CP} *neshun dâd*].
 Shahram car-CL.3SG DOM to Sohrab [_{CP} show gave.3SG]
 “Shahram showed his car to Sohrab.”
- b. *Ali ham* [*mâshin-esh-ro be Sohrâb neshun*] *dâd*.
 Ali also [car-CL.3SG DOM to Sohrab show] gave.3SG
 “Ali also did.”
- c. *Ali ham* [*mâshin-esh-ro be Sohrâb*] [_{NV} *neshun*] *dâd*.
 Ali also [car-CL.3SG DOM to Sohrab] [_{NV} show] gave.3SG
 “(lit.) Ali also showed.”

- (32) The Non-Verb is elided (Shafiei 2016, p. 42)



(33) The Non-Verb survives ellipsis (Shafiei 2016, p. 42)

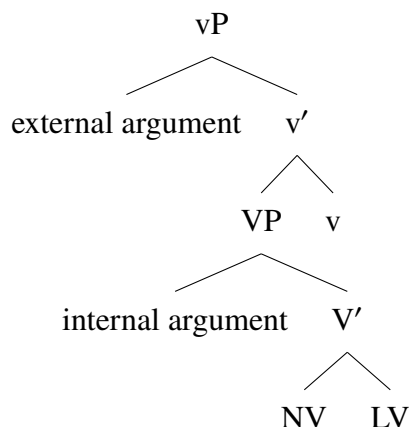


2.2.3.2 Rasekhi 2018

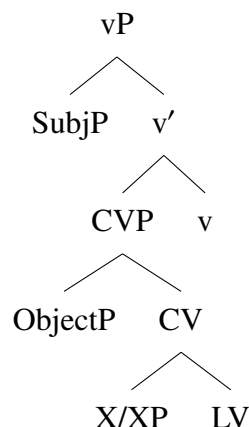
The Farsi verb-stranding ellipsis data of the sort discussed by Shafiei further leads Rasekhi (2018) to propose the following structure in (34) for complex predicates in Farsi. Similar to Shafiei, Rasekhi follows Megerdooian's (2001, 2012) treatment of the complex predicates and assumes that the non-verbal (NV) predicate and the light verb (LV) form a constituent. Comparing Shafiei's base structure in (35) with that of Rasekhi's, the only difference is the labeling of the complex constituent as V' with the internal arguments being generated in the specifier(s) of VP.⁴ Both proposals further claim that the light verb (LV), alone, can be targeted for movement to the v head, leaving behind the NV in the VP.

⁴Rasekhi does not go into the details of the V' but she follows Shafiei (2016) in proposing that the [NV+LV] can be targeted for head movement as one complex head. This suggests that the V' ends up being treated as a V node, perhaps following the V'-reanalysis of Larson (1988).

(34) (Rasekhi 2018, p. 86)



(35) (Shafiei 2016, p. 39)



Shafiei (2016) acknowledges the LV movement out of the derived CV head in (32) would be an instance of excorporation, against Matushansky's (2006) claim on the impossibility of such operation post m-merger. This is predicted following the assumption that m-merger takes place in the morphological component and renders the complex heads syntactically opaque to subextraction. Nonetheless, Shafiei's analysis relies on excorporation due to the LV's accessibility for movement. Rasekhi, on the other hand, doesn't make any claim for the need for excorporation, and it's not clear whether the first instance of movement of LV out of V' should be classified as such. I show in Section 2.3.1.3 that her account will also need to rely on excorporation, against which I will bring up issues and inconsistencies. At this point, I will focus on how Rasekhi deals with the verb-stranding vP ellipsis in (36), where not only the sole LV in (36b) but also the whole NV+LV predicate can avoid elision in (32b).

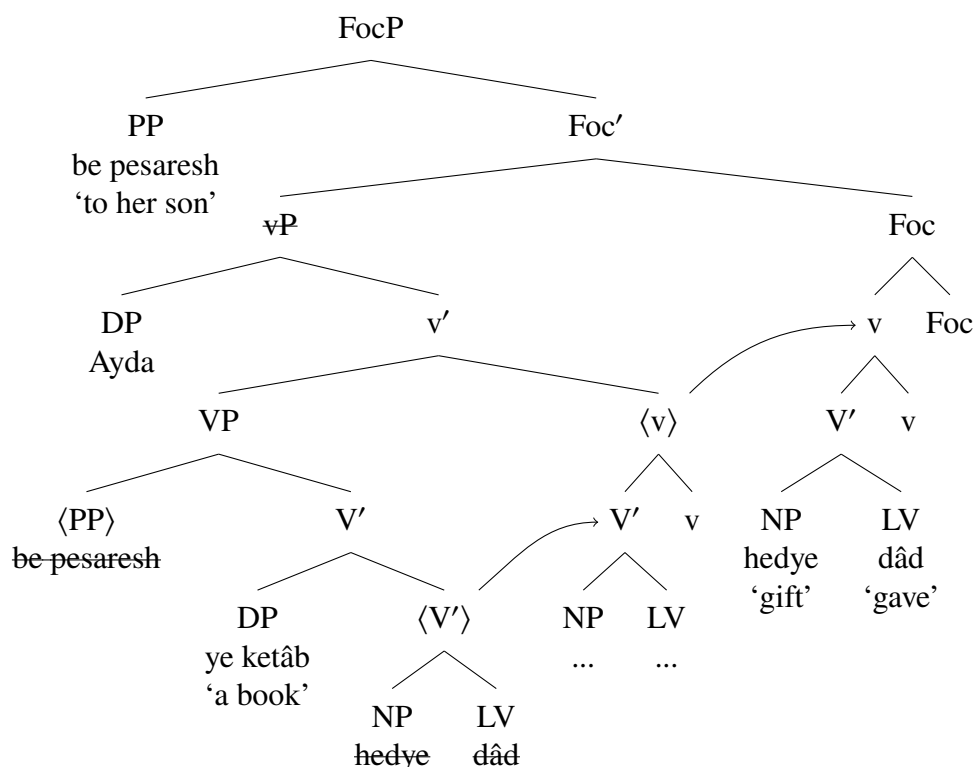
- (36) a. *Âyâ be dokhtar-esh ye ketâb **hedye** na-dâd.*
 Ayda to daughter-CL.3SG a book gift NEG.give.3SG
 "Ayda didn't gift a book to her daughter."
- b. *Vali Âyâ be pesar-esh ~~ye ketâb~~ **hedye** dâd.*
 but Ayda to son-CL.3SG a-book-gift gave.3SG
 "But (she) gifted (a book) to her son."
- c. *Vali Âyâ be pesar-esh ~~ye ketâb~~ **hedye** dâd.*
 but Aydaa to son-CL.3SG a-book gift gave.3SG

With (34) in place, Rasekhi proposes that in such cases the vP is the ellipsis site and gets deleted at

(37) Raising the LV alone



(38) Raising the NV-LV complex



The complex predicate analysis proposed in Folli, Harley, and Karimi (2005) and Karimi (2005) would therefore need to postulate that the non-verbal head can optionally adjoin to the light verb *v* head given that the NV and the LV start as the heads of their own projections. Since head movement is claimed to be key in data involving verb phrase ellipsis, I will turn to the discussion of head-adjunction next as it will provide the ground for two of my arguments against a traditional head-final VP analysis for Farsi: (1) an inconsistency in the nature of head-adjunction in terms of whether incorporation is to the left or right of the licensing heads, and (2) a non-uniform head-incorporation approach due to there being a need to resort to an odd excorporation operation out of complex heads of the sort that is inconsistent with the theoretical assumptions for excorporation and head-adjunction in the first place.

2.3 Some inconsistencies with head-final verb phrase analyses

Sections 2.2.2 and 2.2.3 provided an overview of two theoretical syntactic approaches to complex predicate constructions: (1) the projection independence approach for the Non-Verbal predicate and the Light Verb (Folli, Harley, and Karimi 2005; Karimi 2005), and (2) the syntactic

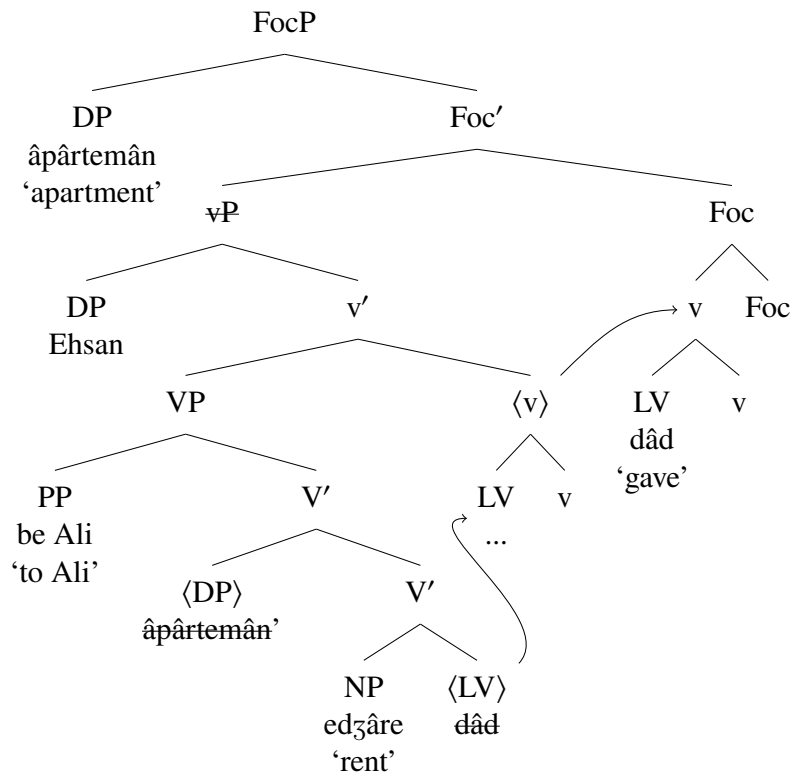
constituent treatment for the two components (Rasekhi 2018; Shafiei 2015, 2016). This section highlights some theoretical inconsistencies and assumptions that arise from adopting a head-final vP structure and rightward head-movement when accounting for more complex sentences. I argue that extending Rasekhi’s account for verb-stranding vP ellipsis to verbal clusters containing auxiliary verbs (such as the Passive, the Perfect, and the Future Tense) casts doubts on the traditional vP analyses. I further show that extending Karimi’s (2001, 2003, 2005) analysis of the Topic/Focus movement of verbal elements also falls short in accounting for larger clusters involving these auxiliaries.

2.3.1 The interaction between Ellipsis and verbal clusters containing auxiliary verbs

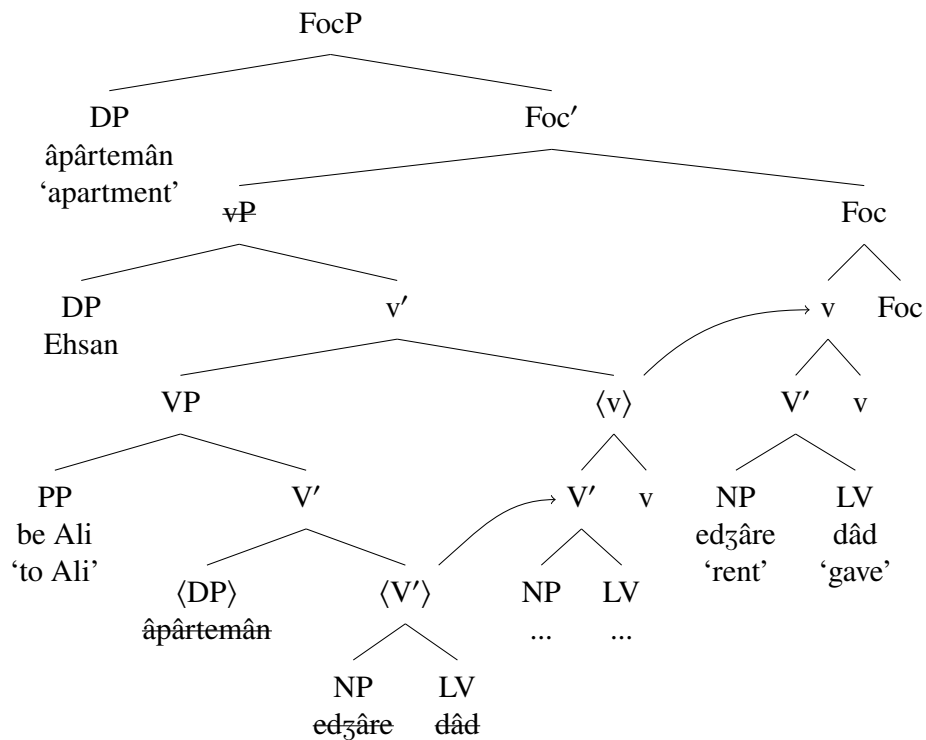
Rasekhi (2018) leaves out a discussion of verb phrase ellipsis data of complex predicates that involve the Passive, the Perfect, and the Future Tense auxiliary verbs. This section focuses on the ellipsis facts in these very contexts and brings to question the technicalities necessary if Rasekhi’s account is on the right track. This discussion will be based on the simple ellipsis case in (39). In (39), the negated active voice construction (39a) acts as the antecedent clause and contains the complex predicate *edžâre dâdan* ‘rent’, the direct object DP *xune* ‘house’, and the indirect object PP *be Ali* ‘to Ali’. Once again, the ellipsis clauses show that either the light verb ‘give’ alone can be stranded and escape deletion in (39b) or the entire complex predicate ‘rent giving’ can do so in (39c).

- (39) a. *Ehsân be Ali xune edžâre na-dâd.*
 Ehsan to Ali house rent NEG-gave.3SG
 “Ehsan did not rent house(s) to Ali.”
- b. *Vali Ehsân be Ali âpârtêmân edžâre dâd.*
 but Ehsan to Ali apartment rent gave.3SG
 “But (Ehsan) rented apartment(s) (to Ali).”
- c. *Vali Ehsân be Ali âpârtêmân edžâre dâd.*
 but Ehsan to Ali apartment rent gave.3SG

(40) Raising the LV alone



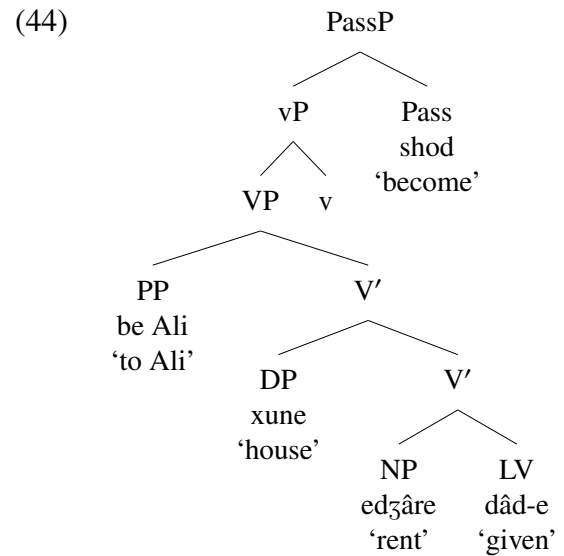
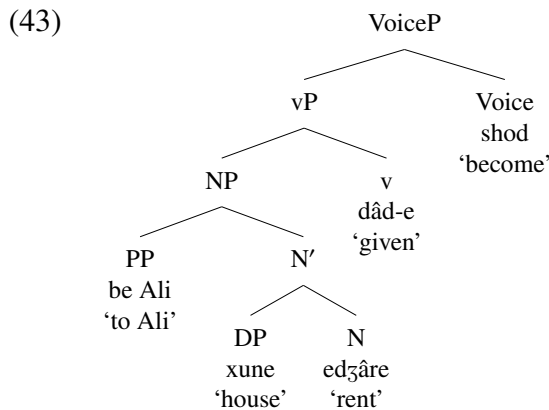
(41) Raising the NP-LV complex



2.3.1.1 Ellipsis and the Passive Auxiliary *shodan* ‘become’

Building on the data in (39), the addition of the Passive auxiliary *shodan* ‘become’ in (42) shows both the distribution of this auxiliary with respect to a complex predicate and the interaction between such constructions and the ellipsis cases discussed so far. In the non-elliptical sentence (42a), the Passive Aux *shodan* follows the main predicate *edžâre dâd-e* ‘rent’. Under the standard head-final verb phrase analyses, this auxiliary must be assumed to project a higher head-final verbal projection selecting for the lexical vP. Such projection can be assumed to be the VoiceP for Karimi and Smith (2020) as suggested in (43), and presumably some Passive Phrase (PassP) for Rasekhi (2018) as suggested in (44).

- (42) a. *be Ali xune edžâre dâd-e na-shod.*
to Ali house rent give-PTCP NEG-became.3SG
‘A house wasn’t rented to Ali./No House was rented to Ali.’
- b. *Vali be-Ali âpârtêmân edžâre dâd-e shod.*
but to-Ali apartment rent give-PTCP became.3SG
‘But an apartment was rented (to Ali).’
- c. *Vali be-Ali âpârtêmân edžâre dâd-e shod.*
but to-Ali apartment rent give-PTCP became.3SG

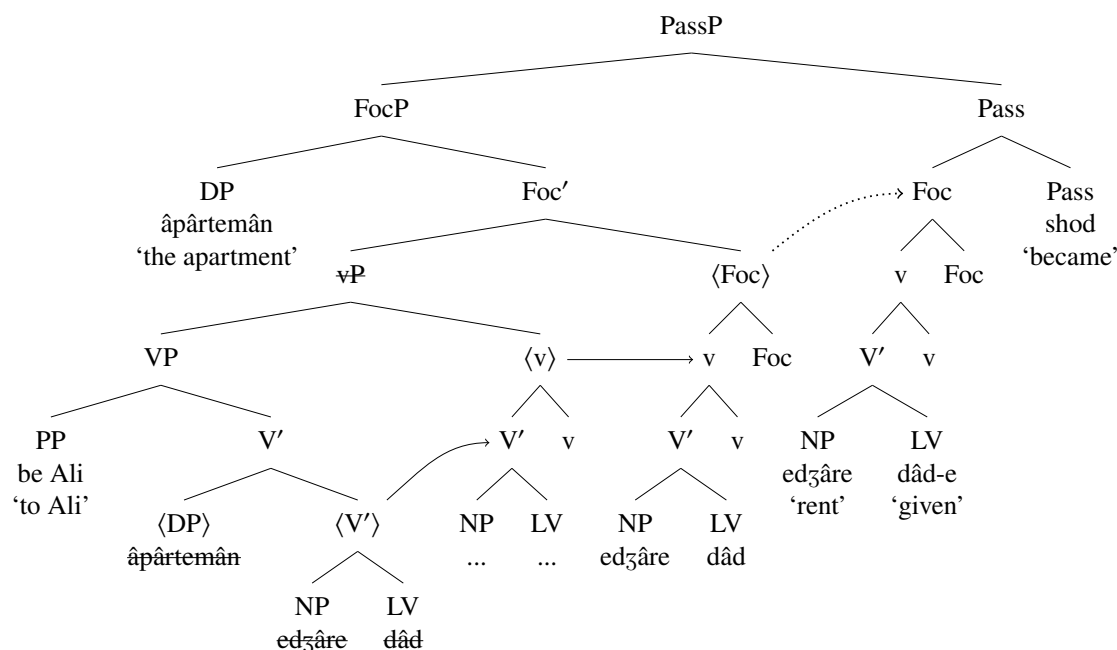


While is not obvious as to whether the predicate adjoins to the Passive auxiliary, such movement would be predicted to raise either the LV or the NV-LV complex to v and successively left-adjoin

(45) Raising the LV alone



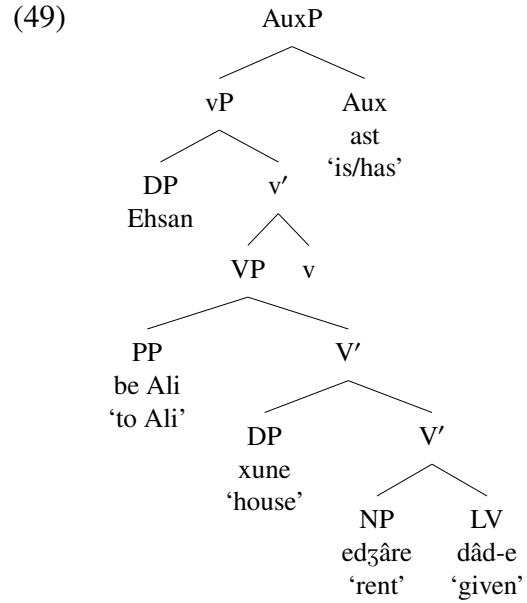
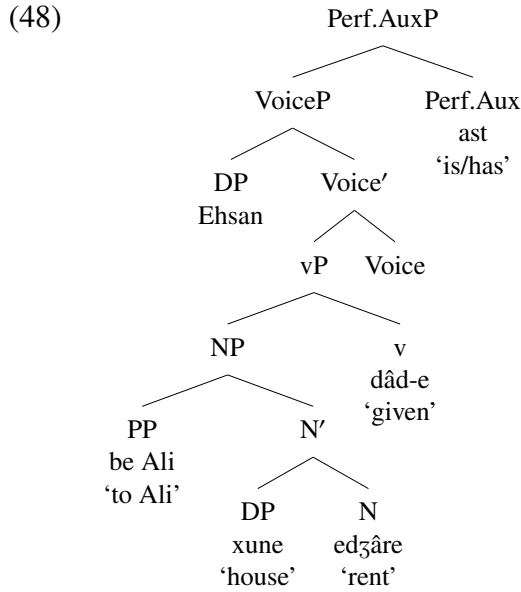
(46) Raising the NP-LV complex



2.3.1.2 Ellipsis and the Perfect Auxiliary *budan* ‘be’

Similar to the Passive Aux cases, the Perfect Auxiliary *budan* ‘be’ follows the main predicate *edžâre dâdan* in (47a). Again, the extension of either Karimi and Smith’s (2020) or Rasekhi’s (2018) proposed complex predicate structures to these cases should project a head-final verbal/aspectual projection above the VoiceP in (48) and vP in (49), respectively.

- (47) a. *Ehsân be Ali xune ed₃âre na-dâd-e ast.*
 Ehsan to Ali house rent NEG-give-PTCP NEG-AUX.PRF.3SG
 “Ehsan has not rented house(s) to Ali.”
- b. *Vali Ehsân be Ali âpârtêmân ed₃âre dâd-e ast.*
 but Ehsan to Ali apartment rent give-PTCP AUX.PRF.3SG
 “But (Ehsan) has rented apartment(s) (to Ali).”
- c. *Vali Ehsân be Ali âpârtêmân ed₃âre dâd-e ast.*
 but Ehsan to Ali apartment(s) rent give-PTCP AUX.PRF.3SG
- d. * *Vali Ehsân be Ali âpârtêmân ed₃âre dâd-e ast.*
 but Ehsan to Ali apartment(s) rent give-PTCP AUX.PRF.3SG

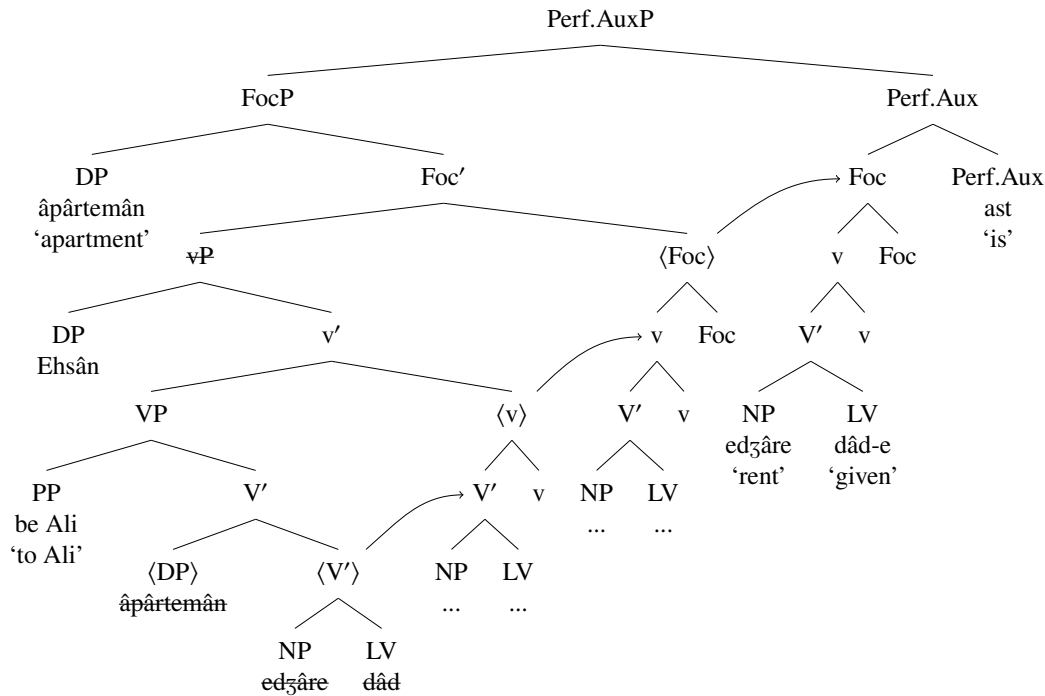


It can further be stipulated that (at least) the verbal component of the complex predicate adjoins to the Perfect auxiliary “overly” (Farudi 2013), given the attachment of the negative prefix *na* ‘not’ to the participle LV in (47a) and its inability to intervene between the LV and the Perfect auxiliary in (50).

- (50) * *Ehsân be Ali xune edzâre dâd-e na-ast.*
 Ehsan to Ali house rent give-PTCP NEG-AUX.PRF.3SG
 “Ehsan has not rented house(s) to Ali.”

Assuming Rasekhi’s complex predicate structure, adjunction to the Perfect auxiliary would therefore involve the movement of either the LV or the NV-LV to the Aux head since both options are proposed to be possible for adjunction to *v* and successively to higher heads. Crucially, incorporation into the Perfect Aux head appears to be possible only to the left of the auxiliary. When paired with ellipsis, (47b) and (47c) show that either the participle light verbal component *dâd-e* alone or the entire complex predicate *edzâre dâde* must be stranded and escape ellipsis. Cases such as (47d), where both components of the complex predicate are deleted and the auxiliary remains, are therefore ungrammatical. Assuming Rasekhi (2018), (47b) would be derived through the adjunction of the light verb to *v* and successively to the Foc head, leaving the non-verbal element in the ellipsis site. Presumably the Foc head to which the LV incorporates further left-adjoins to the Perfect Aux

(52) Raising the NP-LV complex



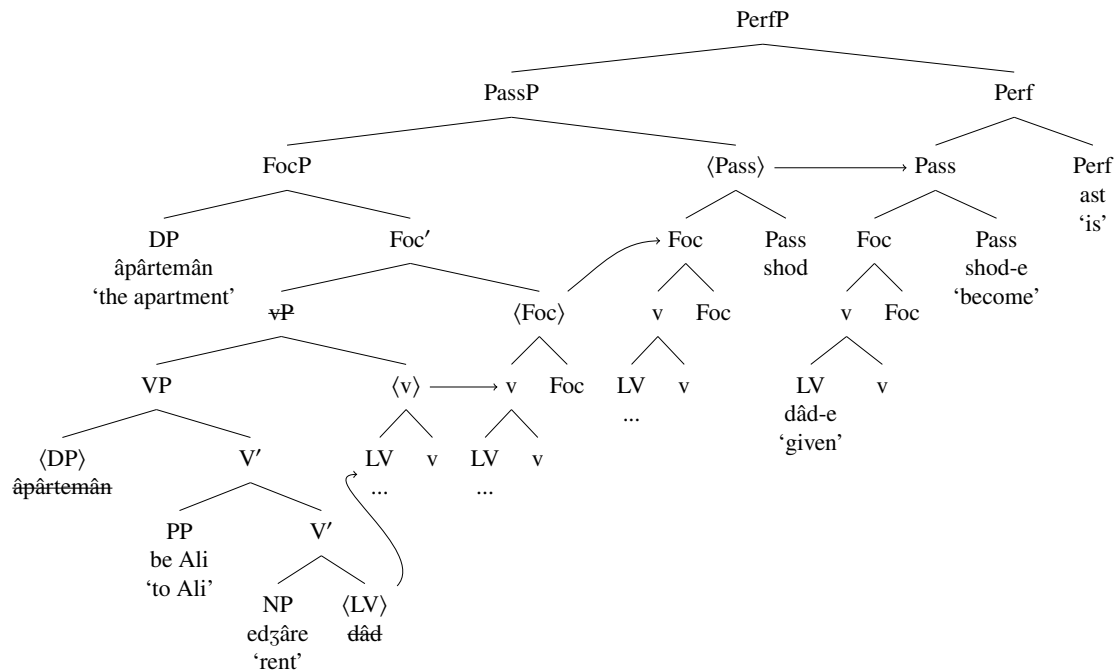
Rasekhi's analysis seems to structurally account for the Perfect Aux cases here in a similar fashion to what is predicted for the Passive in (45-46); left adjunction of either the LV or the [NV LV] constituent to the Foc head allows these items to be stranded and be further adjoined to the Aux (v-to-Foc-to-Aux movement) without posing any apparent inconsistencies.

The interaction between ellipsis and constructions that include the Perfect Aux, the Passive Aux, and a complex verb all within the same sentence, reveals that either the light verb *dâde* in (53b) or both the NV and the LV *edzâre dâde* in (53c) survive ellipsis alongside the Passive and the Perfect auxiliaries.

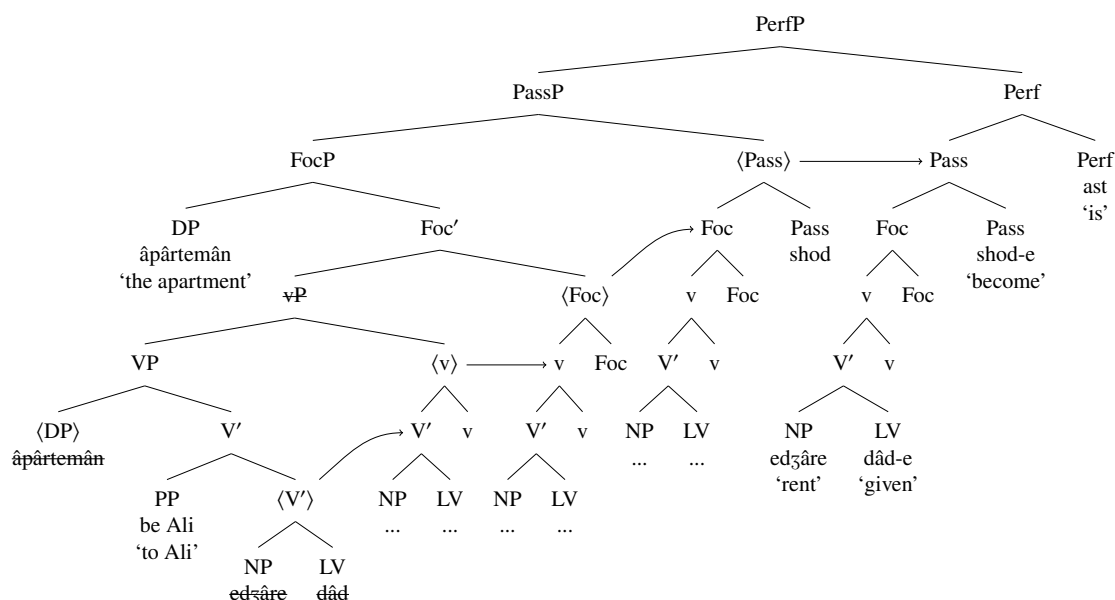
- (53) a. *be Ali xune ed₃âre dâd-e na-shod-e ast.*
to Ali house rent give-PTCP NEG-become-PTCP AUX.PRF.3SG
“A house hasn’t been rented to Ali./No house has been rented to Ali.”
- b. *Vali be Ali âpârtēmân ed₃âre dâd-e shod-e ast.*
but to Ali apartment rent give-PTCP become-PTCP AUX.PRF.3SG
“But an apartment has been rented (to Ali).”
- c. *Vali be Ali âpârtēmân ed₃âre dâd-e shod-e ast.*
but to Ali apartment rent give-PTCP become-PTCP AUX.PRF.3SG

To derive these cases, I assume that Rasekhi's analysis yields structures akin to (54) and (55). The tree in (54) corresponds to (53b), where only the LV has moved to Foc, whereas in (55) the non-verbal element is pied-piped and has moved along with the LV, resulting in both components escaping deletion.

(54) Raising the LV alone



(55) Raising the NP-LV complex



This consistency of left-adjunction in verb movement and cluster formation does not hold for all Farsi auxiliaries however. The Future Tense Auxiliary *xâstan* ‘want’/‘will’ breaks this consistency and will be the discussed next.

2.3.1.3 The ellipsis issue with the Future Auxiliary verb *xâstan* ‘want’

In addition to the Passive and the Perfect, Farsi also makes use of the Future Tense aux *xâstan* ‘want’, which reveals quite a bit about the language’s word order. As observed in the literature (Darzi 1996; Taleghani 2006; among others), the Future Tense auxiliary verb differs from the Past and the Perfect auxiliaries in its distribution with respect to the main predicate.⁵ When the Future Aux interacts with a simplex verb in (56a), it does not surface as the final element in the sentence (as evident in the ungrammaticality of (56b)) and must precede the main verb *dâd* ‘give’. In such con-

⁵The auxiliary *xâstan* ‘want’ as used in example (56) and throughout the dissertation does not function the same way as its control counterpart. Therefore the sentence in (56) cannot also mean ‘Ehsan wants/wanted to give Ali a house’. When used as a control verb in the following example, the argument of ‘want’ is a subjunctive clause as indicated by the subjunctive morpheme *be-* on the embedded verb. In such constructions, both the embedded verb and the matrix verb are inflected for tense and agreement.

- (i) *Ehsân mi-xâhad ke be Ali xune be-dâhad.*
 Ehsan 1PFV-wants.3SG that to Ali house SBJV-give.3SG
 “Ehsan wants to give Ali a house.”

As (56) shows, however, the Future Aux ‘want’ is the only verbal item inflected for subject agreement and the main verb is “in the form of bare infinitive” (Taleghani 2006).

structions, the auxiliary receives the negation and agreement inflection, and the main verb appears in its (short) infinitive form.

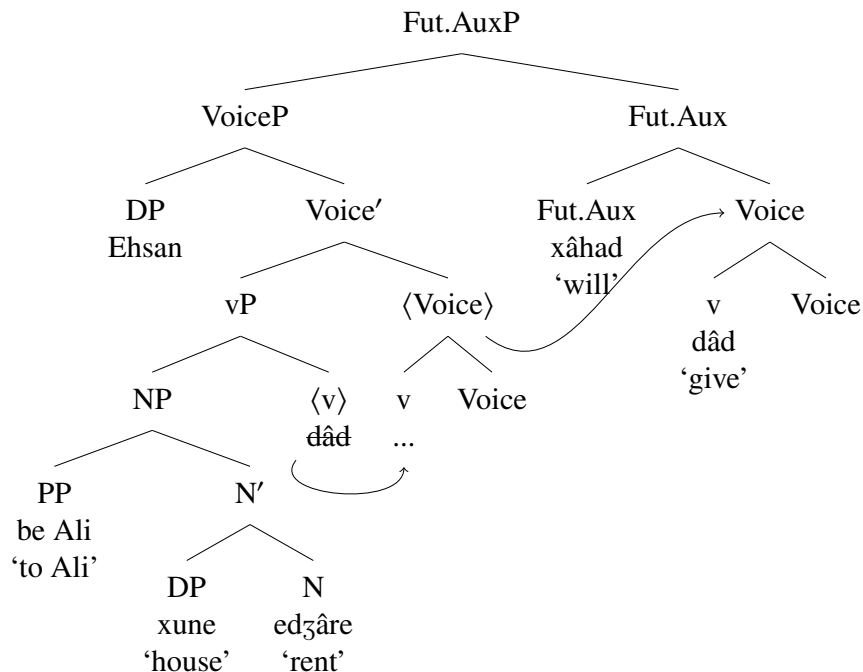
- (56) a. *Ehsân be Ali xune xâhad dâd.*
 Ehsan to Ali house AUX.FUT.3SG give
 “Ehsan will give Ali a house.”
- b. **Ehsân be Ali xune dâd xâhad.*
 Ehsan to Ali house give AUX.FUT.3SG

In complex predicate constructions, the Future Aux separates the components of the predicate and surfaces penultimately. As (2) shows, the inflected auxiliary *xâhad* intervenes between the NV *edžâre* ‘rent’ and the infinitive *dâd* ‘give’, and it can neither precede the entire complex verb in (57b) nor follow it in (57c).

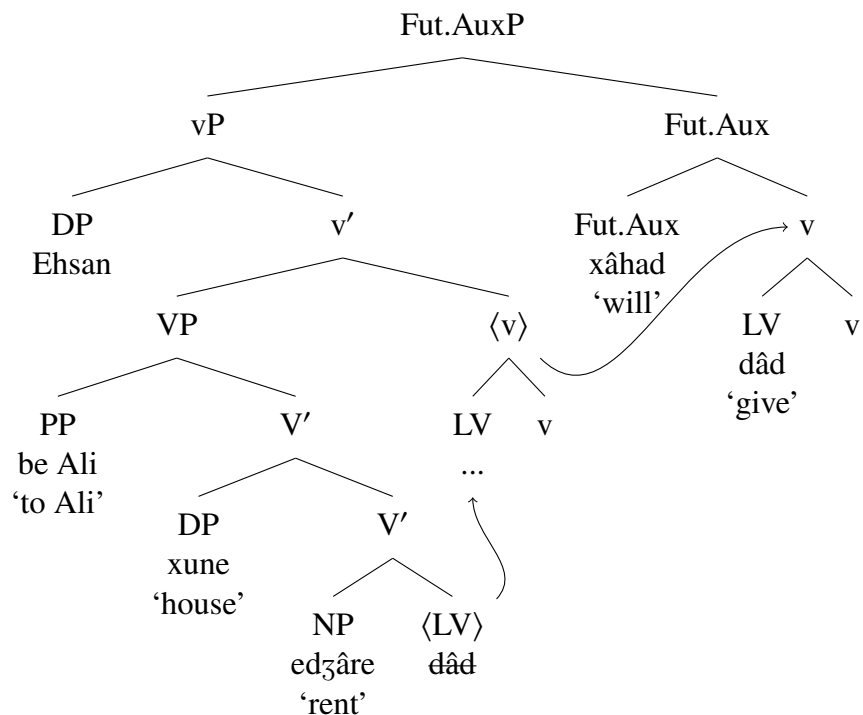
- (57) a. *Ehsân be Ali xune edžâre xâhad dâd.*
 Ehsan to Ali house rent AUX.FUT.3SG give
 “Ehsan will rent house(s) to Ali.”
- b. **Ehsân be Ali xune xâhad edžâre dâd.*
 Ehsan to Ali house AUX.FUT.3SG rent give
- c. **Ehsân be Ali xune edžâre dâd xâhad.*
 Ehsan to Ali house rent give AUX.FUT.3SG

Maintaining a head-final vP structure, Karimi (2005) and Rasekhi (2018) would need to assume that unlike the Perfect, the Future Aux attracts either the simplex verb in (56a) or light verbal item in (2) to its right and never to its left. Focusing on the complex predicate cases, the suggested derivations are given in (58) and (59) for Karimi and Rasekhi, respectively.

(58)

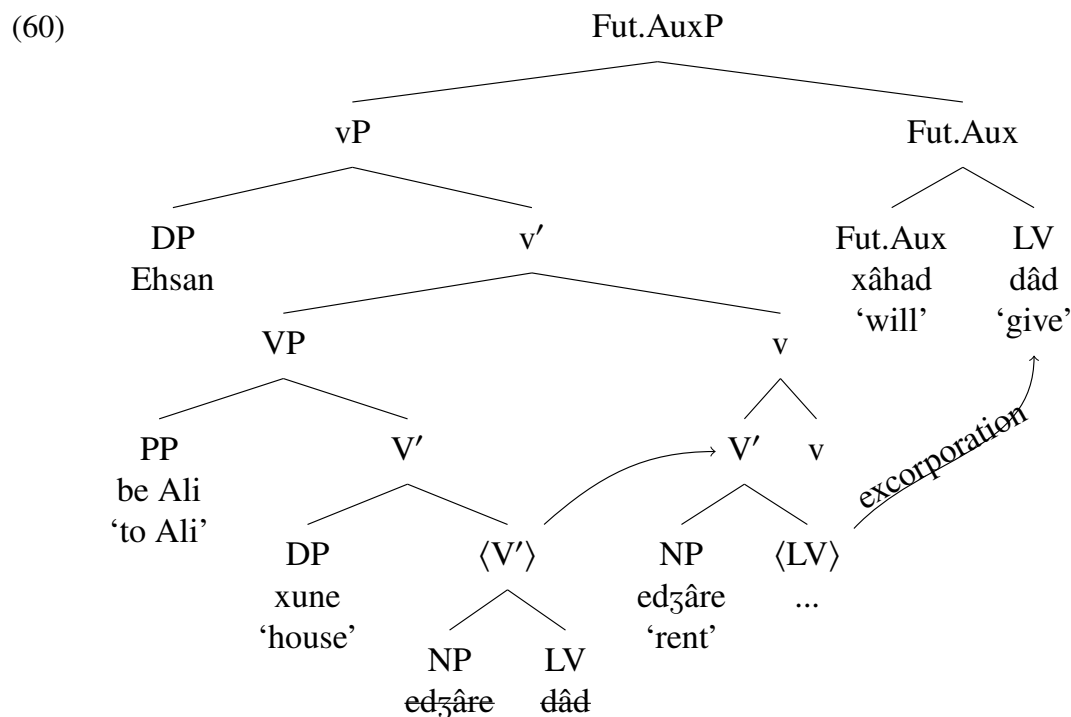


(59)



Therefore, either analysis faces the inconsistency issue of postulating either (a) a head directionality parameter variation for the Future vs. the Passive and the Perfect auxiliaries or (b) adjunction to both the left and the right of licensing heads depending on the auxiliary verb and it not being clear what this asymmetry stems from. Additionally, if Shafiei's (2015, 2016) and Rasekhi's (2018)

proposals that the NV and the LV can move as a verbal constituent are on the right track, it remains to be explained as to why movement of this sort should be unavailable for adjunction to the Future Aux and as to what prevents the infelicitous sequences *[Non-Verb Light-Verb Fut.Aux] in (57b) and *[Fut.Aux Non-Verb Light-Verb] in (57c) from surfacing. This shows that the NV and the LV components of a complex predicate cannot consistently incorporate into higher heads as a unit; they appear to be able to incorporate into the v head together, but the derivation can only proceed if the LV excorporates out of the complex v head and right-adjoins to the Future Aux, as suggested in (60).

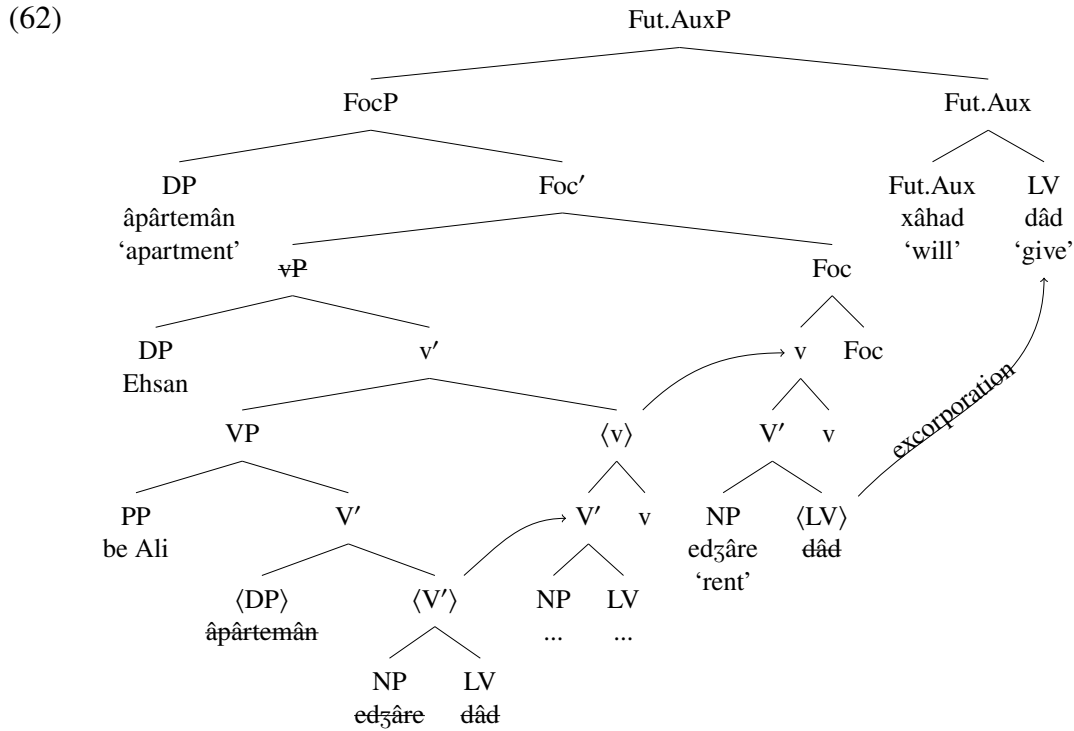


The inability for the entire complex predicate constituent to successively incorporate is further highlighted in ellipsis cases involving the Future Aux. In (61), the verb movement requirement in the ellipsis clause must be captured by either the Aux+LV *xâhad dad* ‘will give’ together in (61b) or the whole cluster NV+Aux+LV *edʒâre xâhad dad* ‘rent will give’ in (61c) escaping the ellipsis site.

- (61) a. *Ehsân be Ali xune edʒâre na-xâhad dâd.*
 Ehsan to Ali house rent NEG-AUX.FUT.3SG give
 “Ehsan will not rent house(s) to Ali.”

- b. *Vali Ehsân be-Ali apârtêmân edžâre xâhad dâd.*
 but Ehsan to-Ali apartment rent AUX.FUT.3SG give.
 “But (Ehsan) will rent apartment(s) (to Ali).”
- c. *Vali Ehsân be-Ali apârtêmân edžâre xâhad dâd.*
 but Ehsan to-Ali apartment rent AUX.FUT.3SG give.

To derive (61c), the [NV LV] head *edžâre dâdan* would have to first incorporate into the *v* head. The complex *v* head should be able to further raise to the Foc head. To account for the position of the Aux in the only possible NV-Aux-LV order, however, the LV would have to excorporate out of the complex Foc head to right-adjoin to the Aux as suggested in (62).

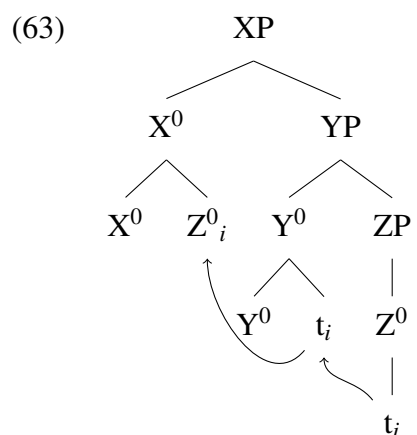


This is not desirable as the derivation would have to impose an ad hoc excorporation requirement inconsistent with all other head movements as incorporation. In addition to head excorporation generally being deemed unavailable, the excorporations needed to derive the Future auxiliary data, with or without ellipsis, do not follow from the head-adjunction theories of excorporation (Koopman 1995; Roberts 1991, and subsequent work) in a straightforward way. The following section illustrates how the suggested movement of the light verbs to the right of the Future Aux must proceed from within heads that are structurally too complex for excorporation.

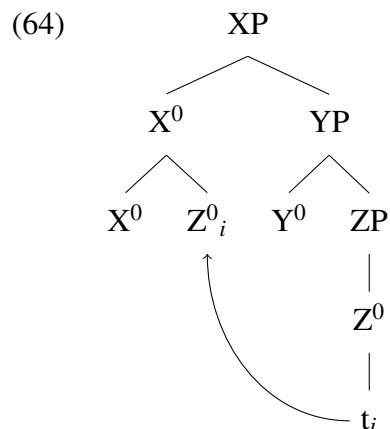
2.3.2 The Issue of Excorporation with respect to Farsi

The excorporation approach has not received a lot of attention in the Persian literature (beyond Shafiei's (2016) claim for the movement of the Light Verb out of a Complex Verb (CVP) in the structure in (32)), and it seems rather ad hoc and unmotivated for the Future Aux to impose such a requirement for deriving the constructions discussed in Section 2.3.1.3. Even under the assumption that verb movement to certain auxiliary verbs must involve excorporation, I argue that the implementation of such operation in Farsi would violate the earlier proposals of Roberts (1991) and Koopman (1995) as well as the subsequent accounts in Matushansky (2006) and Roberts (2010), and others. I will begin with presenting a brief background of the earlier accounts, focusing primarily on the technicalities of Roberts (1991) as the less restrictive theory compared to Koopman (1995).

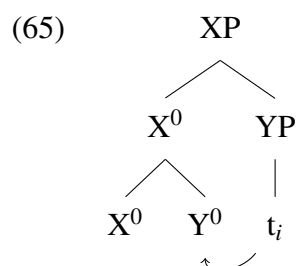
Roberts' (1991) definition of excorporation can be interpreted as *an instance of head-movement where a head travels through another head by first incorporating to that head and further moving on to a higher selecting head*. Robert includes the schematic structure in (63) to show this.



In (63), the Y^0 to which Z^0 adjoins would be termed the “host” and the target of this movement (Z^0) would be the “incorporee”. In turn, the X^0 head is the host that triggers the movement of incorporee Z^0 head. Roberts' excorporation assumes the Head Movement Constraint (Travis 1984) by which head movement cannot ignore intervening heads; the incorporation of Z^0 directly into the X^0 head in (64) (skipping over the governing Y^0 head) would be a violation of such constraint.

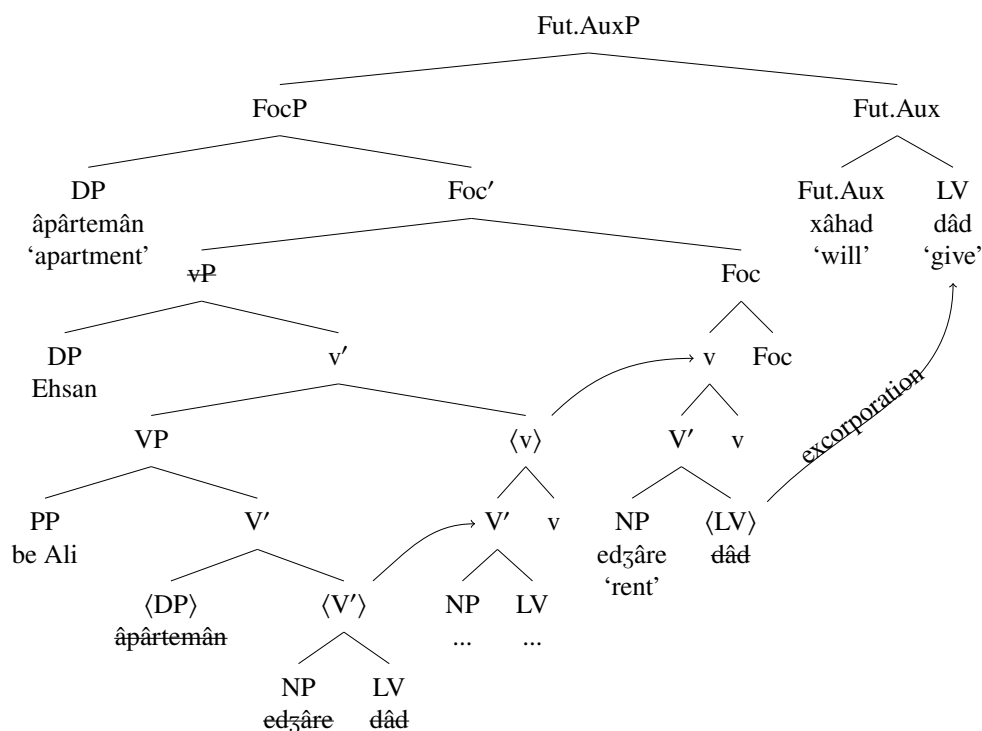


When it comes to head-adjunction, in particular, Roberts assumes the following schematics tree in line with May (1985) and Chomsky (1986):



An important assumption for each instance of movement is that the trace must be properly governed. Assuming that a higher ZP, headed by Z, could project above XP in (65), Y⁰ should be able to be targeted by Z⁰ for excorporation out of X⁰. We would expect that the X⁰ sister to Y⁰ should act as a minimal governor and further block the proper government of the trace Y⁰ from a higher adjoined position. Roberts points out, however, that the configuration (65) is one in which the two instances of X⁰ are segments of the same head and neither of the two sisters X⁰ or Y⁰ are “heads” of the complex X head. The incorporee Y⁰ should therefore be able to move on and its trace be properly governed. Roberts treats this as parallel to the movement of XP adjuncts, which avoids barrierhood. The prediction of this theory of head-adjunction is that three movement types should be allowed: (1) the movement of the host out the complex head; (2) the movement of the incorporee out of the complex head; and (3) the movement of the entire complex head. Given the three possibilities, I will now discuss how this has the immediate consequence that the adjunction of the Light Verb to the right of the Future Aux in (62), repeated below as (66), should be prohibited.

(66)



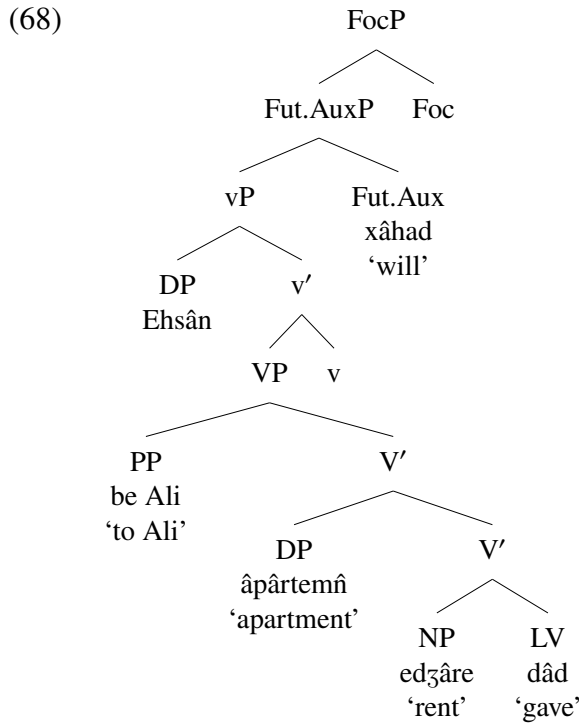
In (66), the complex predicate constituent [NV LV] *edžâre dâdan* ‘to rent’ that moves to the v head is the incorporee and the v head is the host. The complex v head further moves to the Foc head, and the newly formed complex Foc head [v Foc] will have the Foc as the host and the v head [[NV LV] v] as the incorporee. Assuming that the Future auxiliary merges next and further head movements can take place in the course of the derivation, Roberts (1991) would predict the following movement options to be available next.

- (67)
- a. The adjunction of the host, the Foc head, to the Aux head.
 - b. The adjunction of the incorporee, the v head that is sister to Foc, to the Aux head.
 - c. The adjunction of the entire complex Foc head to the Aux head.

We can rule out (67a) as the movement of Foc to Aux is string vacuous and cannot be evaluated one way or the other. Option (67b) must be out too since it will result in the unattested orders [*xâhad edžâre dâd*] if v adjoins to the left of the Future Aux head or [*edžâre dâd xâhad*] if v adjoins to its right. Lastly, (67c) is problematic for the same reason, as the movement of the entire complex Foc head derives the same unattested orders. Given the word order facts, it is only the Light Verb that ends up to the right of this auxiliary. The movement of the LV out of the Foc head would be instance

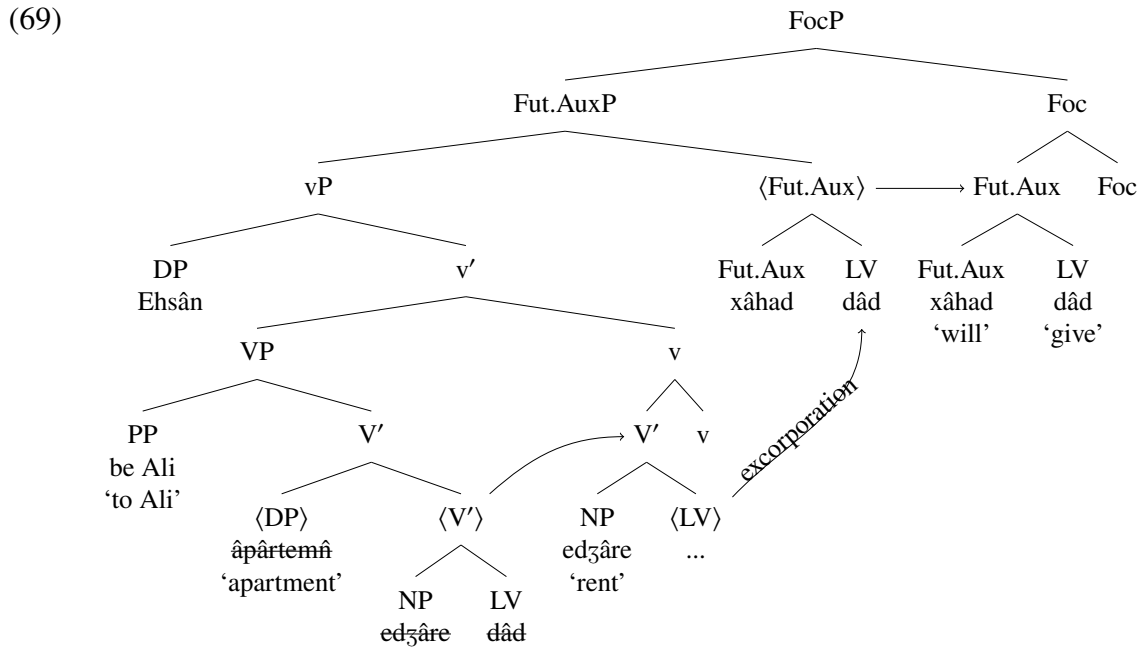
of raising from inside the incorporee *v* head. This Foc head would be considered too complex and such LV movement should therefore not be permitted without further assumptions under Roberts's (1996) excorporation story.

One may argue for an alternative analysis for the position of Rasekhi's TP-internal Foc head with respect to the auxiliaries. Under this alternative, the FocP merges above all the verbal heads as well as above the auxiliaries, as suggested in (68).



In this structure, the Fut.AuxP would be the ellipsis site if we maintain Rasekhi's proposal that the complement of the Foc head is the constituent that is deleted at PF. Considering the verb movement possibilities, the Light Verb is assumed to either move to *v* alone or pied-pipe the Non-Verbal predicate along with it. If the LV raises to *v* alone, the derived [*dâd v*] head should be able to further adjoin to the right of the Fut.Aux. The subsequent adjunction of the complex Aux head [*xâhad dâd*] to the Foc head would therefore move the Aux and the LV out of the ellided Fut.AuxP and account for the ellipsis data where only these two survive deletion. Alternatively, if the entire complex verb *edžâre dâd* raises to *v* first and forms [*edžâre dâd v*] as shown in (69), any subsequent head movement to the Aux would have to target the LV alone in order to capture the only permitted

order [NV Aux LV] *edžâre xâhad dâd*. A consequence of the movement of LV out of v in (69) is that it would not be targetting either the incorporee (the V head [*edžâre dâd*]) or the host (the v head) and is therefore unpredicted. Furthermore, once the derived Aux head adjoins to Foc, it is again only the auxiliary and LV that escape the ellipsis site, and the NV remains trapped inside it and is deleted. This would therefore leave unaccounted for the possible option where the NV, the Aux, and the LV *edžâre xâhad dâd* all escape the ellipsis site.



Robert's (2010) updated account of head movement and excorporation similarly requires additional stipulation to account for Future Aux cases in Farsi. In Roberts (2010), head movement is analyzed as an Agree relation established between a probing head and a goal head that is defective in features; a defective goal is defined by Roberts (2010) as follows:

(70) Defective goal (Roberts 2010: 62)

A goal G is defective iff G's formal features are a proper subset of those of G's Probe P.

An Agree operation results in the formal features of the goal valuing those on the probe. Consequently, the goal incorporates into the probe, and the goal's features are pronounced on the probe. Under Robert's Agree account, the Head Movement Constraint does not exist, the locality of head movement is attributed to phases and the Phase Impenetrability Condition. More specifically,

Roberts assumes that non-adjacent heads can establish an Agree relation as long as these heads are within the same phase at the point the probing head is introduced, therefore predicting cases of long head movement. This is schematically shown in (71), in which case H_1 can probe H_3 directly assuming that (i) H_3 as the head of the lower phrase is accessible to H_1 and (ii) “ H_2 ’s features are distinct from those of H_1 and H_3 ” (Roberts 2010, p. 194).

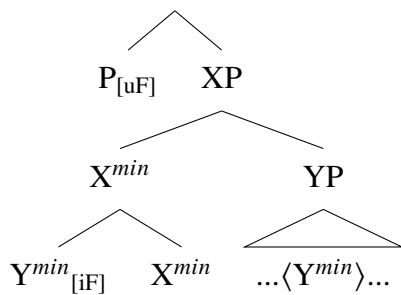
(71) ... $[PH\ H_1[uF]\ H_2[G]\ [PH\ H_3[iF]\ [PH\ ...]$ (Roberts 2010, p. 193)

Roberts (2010) proposes this to be the case in standard Germanic V2 cases where V/v raises to C without moving through T , as well as in English quotative inversion constructions as in (72) (citing Collins 1997, 2003) where the quotation occupies SpecCP and the verb directly moves to C .

(72) “Never!” cried/shouted/whispered/averred/scream/said John.

Furthermore, this Agree-based system does not impose a ban on excorporation, as it is assumed that a defective head that initially serves as the goal to a probing head can subsequently be probed by an even higher head. Given the schematic structure in (73), the Y^{min} can travel through the X^{min} host and be probed by P and incorporate into it, “as long as X lacks $[iF]$, and ... Y^{min} is a defective Goal in relation to P ” (Roberts 2010, p. 207).⁶ Consequently, the features of Y^{min} are ultimately pronounced on the higher head.

(73)



In Robert’s (2010) account, words and heads are treated as phases. Therefore, for a head to be able to excorporate it must be at the left edge of the larger head (left edge of the phase). The cases

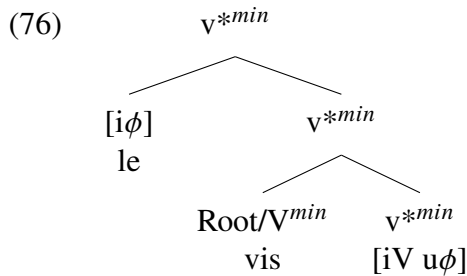
⁶Roberts (2010) proposes that the host of incorporation (in this case the lower X^{min}) however cannot excorporate, departing from the assumption made in Roberts (1991) that this movement is available.

provided by Roberts (2010) to argue in favor of the possibility of excorporation come primarily from Romance clitic climbing data in periphrastic tenses as exemplified in the Italian data in (74).

- (74) a. *Ho visto/*a la casa.*
 I-have seen-MASC/*FEM the house
 “I have seen the house.”
- b. *Ho visto/*a lei.*
 I-have seen-MASC/*FEM her
 “I have seen her.”
- c. *L’ho vista/*o.*
 her-I-have seen-FEM/*MASC
 “I have seen her.”

Roberts assumes the base structure in (75) for (74c), where the RootP contains the lexical root and its clitic complement, and the Participle and the Aux head project vPs above the transitive vP. Roberts proposes that initially the V adjoins to v and values its V-feature, while the clitic also attaches to v valuing its ϕ features, as suggested in (76).

- (75) ...[_{VP} [_v Aux] [_{VP} Part [_{VP} EA v RootP]]



As required by Part’s V feature, v* is probed and incorporates into the Part head (v*’s formal features being a subset of Part’s features), as shown in (77). It is further assumed that Aux’s ϕ features are valued by the clitic’s features, resulting in the incorporation of the clitic into the Aux head and them forming a complex unit. The excorporation of the clitic out of the v* head in (77) is proposed to be permitted given it is on the left edge of the Part^{min} phase.

(77)

```

graph TD
    Aux[Aux] --- Aux_min1[Auxmin]
    Aux --- Part_max[Partmax]
    Aux_min1 --- i_phi["[iφ]  
la"]
    Aux_min1 --- Aux_min2[Auxmin  
[uφ]  
ha]
    Part_max --- Part_min1[Partmin]
    Part_max --- v_star_min1[v*min]
    Part_min1 --- v_star_min2[v*min]
    Part_min1 --- Part_min2[Partmin  
-ta]
    v_star_min2 --- i_phi_in_brackets["⟨[iφ]⟩"]
    v_star_min2 --- la_in_brackets["⟨la⟩"]
    la_in_brackets --- vis[vis]
  
```

A challenge faced by the extension of this analysis to account for the apparent light verb excorporation case in Farsi in (66), repeated here as (78), is determining what counts as the left edge of the complex V' head that contains the NV and the LV.

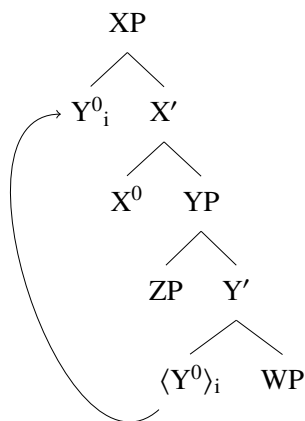
[illegible]

Since the initial movement of the V' constituent together to the v head is analyzed as the LV pied-piping the NV along, it follows that any subsequent head probing for the verb as its defective goal should be able to target the constituent containing both elements, which seems plausible for both the V' -to- v and the v -to-Foc movements. The seemingly obligatory extraction of the LV out of the

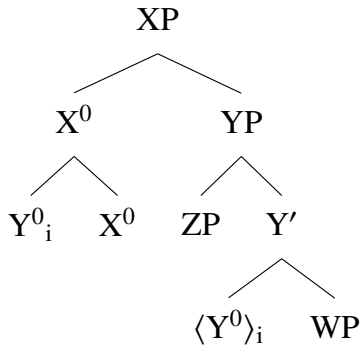
lower complex head in such cases therefore relies on proposing that the V-feature on the Fut.Aux must fundamentally be valued differently than that of other verbal heads (such as the transitive *v* and the Foc), which is an ad hoc stipulation that does not follow from any principled theory of head movement independently motivated for the Farsi verbal domain. This issue persists even if we assume that the *V'* directly excorporates out of the *v* head and incorporates to Foc. This shows that both excorporation accounts of Roberts (1991) and Roberts (2010) require additional assumptions to work for the Farsi cases.

A different view on head-adjunction unifies head movement and phrasal movement as both targeting specifier positions of probing heads (Harizanov 2019; Harizanov and Gribanova 2018; Matushansky 2006; Vicente 2007; among others). In Matushansky (2006), two operations are assumed to take place when forming a complex head: (i) the moving head lands in the specifier position of the probing head as illustrated in (79), resulting in two adjacent heads in a spec-head relation; and (ii) the two heads further undergo a Morphological Merger (m-merger) operation, creating the head adjunction structure in (80).

(79) Movement of Y^0 to Spec XP (Matushansky 2006, ex. 10b)



(80) M-merger yielding the head-adjunction structure (Matushansky 2006, ex. 10c)



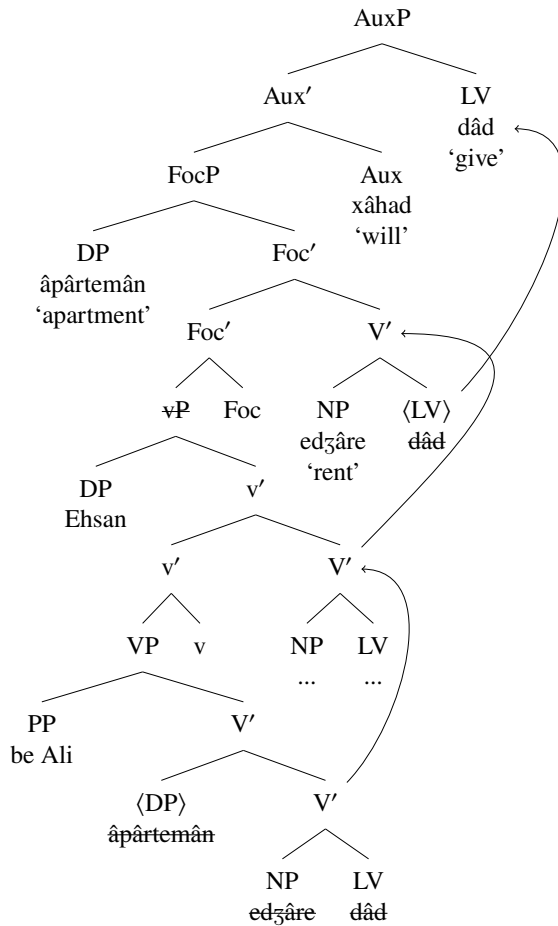
Matushansky addresses the locality of head movement and derives the Head Movement Constraint by proposing that the trigger of head movement is a c(ategorial)-selection feature on the higher head that is satisfied by raising the head of its complement (the lower head) to its specifier. As a consequence of m-merger in (80), the derived complex head is treated as a single unit that may undergo further movement operations, and this unit becomes a “syntactic phase” (Matushansky 2006, p. 95) in that it is inaccessible to any excorporation operation targetting the individual heads.⁷ Matushansky (2006) discusses apparent cases of long head movement and heads skipping over or travelling thorough intervening heads, which seem to represent limited cases of head excorporation. She suggests that if such cases do indeed involve excorporation, one way to implement this in the theory would be to assume that m-merger does not obligatorily apply and that the moved head can “excorporate” and raise further. What this theory should not permit however is the movement of elements internal to the moving heads. This means that the excorporation of the sort needed in Farsi, as suggested in (78), is problematic even if we adopt the head movement theory of Matushansky. Crucially, if we were to treat the movement of the complex predicate V' head *edžâre dâdan* as targeting specifier positions of the higher heads as in (81), any cyclic head movement up to the Fut.Aux head would be targeting the entire V' head and deriving the illicit orders.⁸ So, the required

⁷While Matushansky claims that each instance of m-merger is spelled out in the morphological component and is sent back to the syntax for further operations (multiple spell-out), Vicente (2007) proposes that m-merger happens in the narrow syntax, based on data suggesting that the internal structure of complex heads are accessible to operations other than movement.

⁸The head movements represented in (81) presumably target righthand specifiers of the probing heads, which raises the question as to why the directionality of head-to-spec movement should be parametrized in this way and as to how linearization is accounted for generally.

movement of the light-verbal element *dâdan* alone out of the V' constituent is not permitted at any step and must be exceptionally stipulated.

(81)



In Harizanov (2019) and Harizanov and Gribanova (2018), two types of head movements are assumed: (i) a word-formation operation that is post-syntactic, morphologically driven, obeys the Head Movement Constraint, and represents cases like affixation; and (ii) a narrow syntactic operation which assumes that both head and phrasal movement involve internal merge in specifier positions (in line with Matushansky 2006). In their analysis, syntactic head movements (akin to phrasal movements) are not subject to the HMC and can be long-distance. Relatedly, there is no ban on excorporation in the syntactic movement cases because no head-adjunction is involved. It should be clear that the suggested excorporation of the LV out of the V' should also be ruled out under the syntactic movement theory of Harizanov (2019) and Harizanov and Gribanova (2018). That is because although successive-cyclic movements of heads are allowed in such cases, subextraction

out of the moving V' head is still not predicted without further assumptions.

Given the observation that various head movement and excorporation theories would require additional stipulations in order to implement the type of excorporations needed in Farsi to explain the linear order facts with the Future Auxiliary, I argue that excorporation is absent altogether from the syntactic operations that derive the various Predicate + Aux constructions and can in fact be avoided (as will be shown in Section 2.5.3).

Based on the discussion in this section, two outstanding issues are highlighted so far:

- (82) a. There is no consistency to the nature of head-adjunction in terms of whether incorporation is to the left or the right of selecting heads. (Section 2.3.1)
- b. There is no uniform head-incorporation approach; excorporation out of complex heads + additional assumptions must also be resorted to. (this section)

The following section shows that a head-final vP analysis is also insufficient in explaining the extended data from the leftward movement of verbal clusters. This will add to the list of inconsistencies in (82).

2.3.3 Topic/Focus fronting and optional verb raising

This section addresses constructions that seem to involve verb movement for the purpose of information structure (topicalization/focalization), as it relates to Karimi's (2005) discussion of A-bar movement and scrambling. I will show that data involving the movement of larger verbal clusters containing auxiliary verbs challenge a head-adjunction approach to verb-fronting as there is no straightforward way to raise all the moved elements as a complex head. I will argue that the Syntactic Word Formation rule proposed by Karimi (2001, 2003) to deal with [V + object/modifier] movement cases also falls short in adequately explaining the auxiliary containing constructions.

2.3.3.1 Karimi's V-movement account

Karimi (1989, 1994) presents cases of verb preposing in Farsi where the verb may optionally move to the sentence initial position as motivated by discourse functions "contrast", "emphatic", and "interrogative" (Karimi 1989, p. 159). The emphatic interpretation is exemplified in (83) and

(84), where the verbs *xaridan* ‘buy’ and *dâdan* ‘give’ have fronted, respectively.

- (83) a. *man pirhan-o xarid-am.*
I shirt-DOM bought-I
“I bought the shirt.”

- b. *xarid-am man pirhan-o.* = V S O
bought-I I shirt-DOM
“I DID buy the shirt.” (Karimi 1989, p. 157)

- (84) a. *man be Sâsân pirhan-o dâd-am.*
I to Sasan shirt-DOM gave-I
“I gave the shirt to Sasan.”

- b. *dâd-am man be Sâsân pirhan-o.* = V S PP O
gave-I I to Sasan shirt-DOM
“I DID give the shirt to Sasan.” (Karimi 1989, p. 157)

Karimi (2005) observes that the verb may optionally move as allowed by the scrambling of a contrastive nominal element to the left. In (85), the simplex verb *didan* ‘see’ can optionally front following the scrambling of the direct object ‘Kimea’. In (86), the same possibility exists for the verb *dâdan* ‘give’ following the scrambling of the subject ‘Kimea’.

- (85) *KIMEA-ro_i didam_j <Kimeâ-ro>_i tu xiâbun <didam>_j*
Kimea-DOM saw.1SG in street
“It was Kimea that I saw in the street.” (Karimi 2005, p. 160)

- (86) *KIMEA_i dâd_j <Kimeâ>_i ketâb-ro be dust-esh <dâd>_j*
K gave.3SG book-DOM to friend-her
“It was Kimea who gave the book to her friend.” (Karimi 2005, p. 160)

Karimi proposes that such cases involve the head movement of the verb to a higher Focus position; in these examples, the subject DPs have moved to the Spec of FocP and the verb has adjoined to Foc. The same sort of verb movement is triggered in topicalization as well, such as in (87), for which Karimi assumes direct object movement to the Spec of TP and verb movement to T.

- (87) *un film-ro_i didan_j unâ <un film-o>_i <didan>_j.*
that movie-DOM saw.3PL they
“As for that movie, they saw it.” (Karimi 2005, p. 159)

It is shown in Karimi (2001) that verb preposing may move other elements together with the verb. The complex DP data in (88) containing a relative clause shows that the movement of the verb *dâdan* ‘give’ to the Foc head (whose specifier hosts the focalized direct object ‘that book’) brings along the indirect object PP *be Kimeâ* ‘to Kimea’.

- (88) a. *Man* [_{DP} [*un ketâb-i*]-ro] [_{CP} *ke Sepide diruz* ***xar-id***] *be Kimeâ dâd-am*.
 I that book-REL-DOM that Sepide yesterday bought.3SG to Kimea gave.1SG
 “I gave that book to Kimea that Sepide bought yesterday.”
- b. *Man* [_{FocP} [*un KETÂB-i*]-ro]_i [_{Foc} *be Kimeâ dâd-am*]_j [_{DP} <*un ketâb-i*-ro>_i [_{CP} *ke Sepide diruz* ***xar-id***]] <*be Kimeâ dâd-am*>_j.
 I that book-REL-DOM to Kimea gave.1SG that
 Sepide yesterday bought.3SG
 “I gave THAT BOOK (as opposed to a different book) to Kimea that Sepide bought yesterday.”

Additionally, (89) shows that when the verb *xaridan* ‘buy’ takes a non-specific direct object *xune* ‘house’, verb movement to Focus must target both elements in (89b).

- (89) a. *Man* [_{DP} [*PP az mard-i*] [_{CP} *ke to be man* ***mo’arefi kard-i***]] *xune xarid-am*.
 I of man-REL that you to me introduction did.2SG house bought.1SG
 “I bought a house from a man that you introduced to me.”
- b. *Man* [_{FocP} [*az MARD-i*]_i [_{Foc} *xune xarid-am*]_j] [_{DP} <*az mard-i*>_i [_{CP} *ke to be man* ***mo’arefi kard-i***]] <*xune xarid-am*>_j.
 I of man-REL house bought.1SG that you to me
 introduction did.2SG
 “I bought a house from a MAN that you introduced to me.” (not someone else.)

Not only arguments, but also modifiers may move with the verb. As (90) shows, raising the negated verb *xândan* ‘read’ to the Foc head moves both the verb and the modifier *âmadan* ‘intentionally’ in (90b).

- (90) a. *Man* [_{DP} [*un ketâb-i*]-ro] [_{CP} *ke Âlam* ***neveshte bud***] *amdan na-xund-am*.
 I that book-REL-DOM that Alam written was.3SG intentionally
 NEG-read.1SG
 “I intentionally did not read that book that Alam had written.”

- b. *Man* [_{FocP} [*un KETÂB-i-ro*]_i [_{Foc} *amdan na-xund-am*]_j][_{DP} *⟨un ketâb-i-ro⟩_i*
 I that book-REL-DOM intentionally NEG-read.1SG
 [_{CP} *ke Âlam neveshte bud*]] *⟨amdan na-xund-am⟩_j*.
 that Alam written was.3SG

“I intentionally did not read the book that Alam had written.” (As opposed to some other book.)

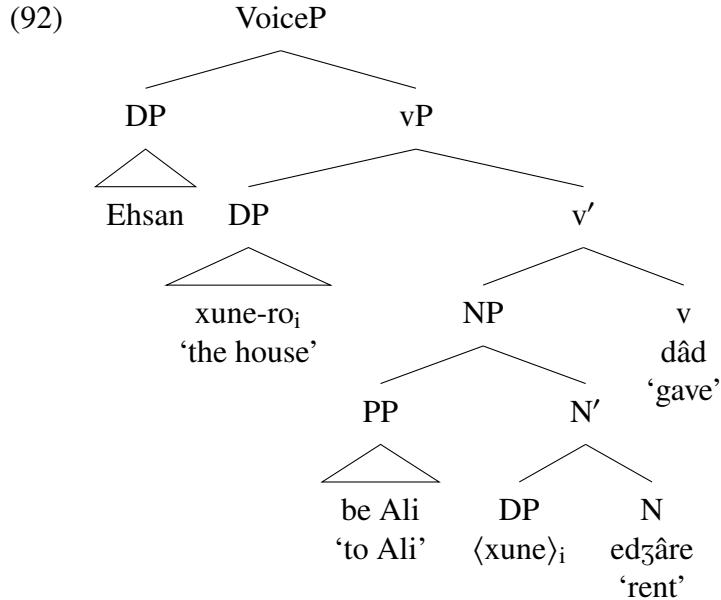
Maintaining a theory of verb-movement to account for the cases discussed so far, Karimi (2001, 2003) proposes the rule of Syntactic Word Formation (SWF), based on Marantz (1997) and Halle and Marantz (1993), to account for the movement of the indirect object and the verb in (88b), the non-specific direct object and the verb in (89b), and the modifier and the verb in (90b).⁹ This rule refers to a syntactic operation in which multiple components, such as a verb and its modifiers or arguments, form complex units that denote special meanings and can be targeted for further movement operations. Karimi (2001, 2003) suggests V' to be the domain on which Syntactic Word Formation operates in Farsi and “where the Encyclopedic knowledge consisting of the special meaning of the verb and its modifiers is obtained”. The V' domain can therefore adjoin to heads of discourse related Topic and Focus heads. From this point of view, the interaction between verb raising and a complex predicate in the following example suggests that what moves must be an X' level phrase that minimally contains both the NV and the LV elements *edžâre dâdan* ‘rent’ to raise, as shown in (91c).

- (91) a. *Ehsân in xune-ro be Ali edžâre dâd.*
 Ehsan this house-DOM to Ali rent gave.3SG
 “Ehsan rented this house to Ali.”
- b. **BE ALI_i dâd_j Ehsân in xune-ro ⟨be Ali⟩_i edžâre ⟨dâd⟩_j.*
 to Ali gave.3SG Ehsan this house-DOM rent
- c. *BE ALI_i edžâre dâd_j Ehsân in xune-ro ⟨be Ali⟩_i ⟨edžâre dâd⟩_j.*
 to Ali rent gave.3SG Ehsan this house-DOM

⁹Karimi (2001) argues against analyzing (88b), (89b) and (90b) as rightward CP extraposition due to such movement violating the minimalist principles of (a) being motivated by an uninterpretable feature and (b) applying leftward to a specifier position. The CP would further not be expected to escape the complex DP given the specifier of DP would already be occupied by the determiner and the head noun. Karimi further rules out V' reanalysis as it would only account for the combination of the verb and the non-specific object in (89) but not for the verb and its modifier in (90).

“It was to Ali that Ehsan rented this house.”

Since Folli, Harley, and Karimi (2005), Karimi (2005), and Karimi and Smith (2020) treat the non-verbal element of a complex verb parallel to a lexical V that introduces the internal arguments, and the light verb in turn occupies the v head that selects for the non-verbal phrase, the structure corresponding to (91a) would be as given in (92).



Based on this structure, Karimi’s (2001, 2003) SWF rule that operates on V’ as its domain would not extend to (92) since V’’s equivalent (N’) contains just the non-verbal element and the copy of the direct object. The SWF domain under the updated complex predicate structure would have to minimally be a v’ and it would need be stated as a requirement of verb movement that all verbal elements must prepose together. As the structure stands, however, the first v’ also contains the indirect object PP. So, presumably the PP would have to evacuate the NP as well if the v’ is what moves in (91c).

It is clear that a simple theory of head adjunction does not straightforwardly derive the verb movement data discussed so far. While Karimi utilizes the Syntactic Word Formation rule and specifies a domain for capturing special meanings between the verb and its arguments/modifiers and allowing them to adjoin to heads, this domain tends to expand as large as it gets to in order to contain everything that moves along with the verb. This is specially highlighted by the same

Verb-Auxiliary constructions discussed earlier, where the Past, the Perfect and the Future are added to the mix. The following section addresses these cases.

2.3.3.2 The verb movement issue with the auxiliaries

In data involving a complex verb and the auxiliary verbs (the Passive, the Perfect, and the Future), all three elements in the verbal cluster (the NV, the LV, and the Aux) must move to the left suggesting that they behave as a unit. With respect to the passive construction in (93), it is ungrammatical to front only the auxiliary *shodan* in (93b). The movement of either the LV in isolation in (93c) or the NV in isolation in (93d) is also ruled out. Raising only the Aux and the LV together in (93e) is ungrammatical as well. The only acceptable option is to front the NV, the LV, and the Aux together in (93f).

- (93) a. *In xune be Ali ed̤âre dâde shod.*
 this house to Ali rent give-PTCP became.3SG
 “This house was rented to Ali.”
- b. **BE ALI_i shod_j in xune <be Ali>_i ed̤âre dâde <shod>_j.*
 to Ali became.3SG this house rent give-PTCP
- c. **BE ALI_i dâde_j in xune <be Ali>_i ed̤âre <dâde>_j shod.*
 to Ali give-PTCP this house rent became.3SG
- d. **BE ALI_i ed̤âre_j in xune <be Ali>_i <ed̤âre>_j dâde shod.*
 to Ali rent this house give-PTCP became.3SG
- e. **BE ALI_i dâde shod_j in xune <be Ali>_i ed̤âre <dâde shod>_j.*
 to Ali give-PTCP became.3SG this house rent
- f. *BE ALI_i ed̤âre dâde shod_j in xune <be Ali>_i <ed̤âre dâde shod>_j.*
 to Ali rent give-PTCP became.3SG this house
 “It was to Ali that this house was rented.”

The same is observed in the Perfect auxiliary data in (94). It is ungrammatical for the Aux alone to raise to the left in (94b), for the NV alone to raise in (94c), for the LV alone to move in (94d), and for just the LV and the Aux to move together in (94e). Once again, all elements must raise in (94f).

- (94) a. *Ehsan in xune-ro be Ali ed̤âre dâde ast.*
 Ehsan this house-DOM to Ali rent given.PART AUX.PRF.3SG
 “Ehsan has rented this house to Ali.”
- b. **BE ALI_i ast_j Ehsan in xune-ro <be Ali>_i ed̤âre dâde <ast>_j.*
 to Ali AUX.PRF.3SG Ehsan this house-DOM rent give-PTCP
- c. **BE ALI_i dâde_j Ehsan in xune-ro <be Ali>_i ed̤âre <dâde>_j ast.*
 to Ali give-PTCP Ehsan this house-DOM rent AUX.PRF.3SG
- d. **BE ALI_i ed̤âre_j Ehsan in xune-ro <be Ali>_i <ed̤âre>_j dâde ast.*
 to Ali rent Ehsan this house-DOM give-PTCP AUX.PRF.3SG
- e. **BE ALI_i dâde ast_j Ehsan in xune-ro <be Ali>_i ed̤âre*
 to Ali give-PTCP AUX.PRF.3SG Ehsan this house-DOM rent
<dâde ast>_j.
- f. *BE ALI_i ed̤âre dâde_j ast Ehsan in xune-ro <be Ali>_i*
 to Ali rent give-PTCP AUX.PRF.3SG Ehsan this house-DOM
<ed̤âre dâde ast>_j.

“It is to Ali that Ehsan has rented this house.”

Moving on to constructions containing the Future auxiliary *xâstan* ‘want/will’, such as in (95a), the three elements must similarly move. It is ungrammatical to raise only the Aux in (95b), to raise only the LV in (95c), or to raise only the NV in (95d). Fronting just the Aux and the LV in (95e) doesn’t make it any better. The fronting of the entire cluster in (95f) in the strict order NV-Aux-LV *ed̤âre xâhad dâd* seems to be the only option for most speakers.

- (95) a. *Ehsan in xune-ro be Ali ed̤âre xâhad dâd.*
 Ehsan this house-DOM to Ali rent AUX.FUT.3SG give
 “Ehsan will rent this house to Ali.”
- b. **BE ALI_i xâhad_j Ehsan in xune-ro <be Ali>_i ed̤âre <xâhad>_j dâd.*
 to Ali AUX.FUT.3SG Ehsan this house-DOM rent give
- c. **BE ALI_i dâd_j Ehsan in xune-ro <be Ali>_i ed̤âre xâhad <dâd>_j.*
 to Ali give Ehsan this house-DOM rent AUX.FUT.3SG
- d. **BE ALI_i ed̤âre_j Ehsan in xune-ro <be Ali>_i <ed̤âre>_j xâhad dâd.*
 to Ali rent Ehsan this house-DOM AUX.FUT.3SG give

- e. * *BE ALI_i xâhad dâd_j Ehsan in xune-ro <be Ali>_i ed₃âre <xâhad dâd>_j.*
to Ali AUX.FUT.3SG give Ehsan this house-DOM rent
- f. *BE ALI_i ed₃âre xâhad dâd_j Ehsan in xune-ro <be Ali>_i*
to Ali rent AUX.FUT.3SG give Ehsan this house-DOM
<ed₃âre xâhad dâd>_j.

“It is to Ali that Ehsan will rent this house.”

This is further highlighted by even more complex constructions, such as (96), involving a complex predicate and both the Passive and the Future auxiliaries. As (96a) shows, the Passive aux *shodan* must surface to the right of the Future aux *xâstan*, and any leftward verb movement operation cannot strand any individual elements in the cluster. This again leaves us with the only grammatical option in (96d) where the entire sequence [NV-LV-Fut-Pass] *ed₃âre dâde xâhad shod* moves to the left.

- (96) a. *In xune be Ali ed₃âre dâde xâhad shod_i.*
this house to Ali rent given-PTCP AUX.FUT.3SG become
“This house will be rented to Ali.”
- b. * *BE ALI_i xâhad shod_j in xune <be Ali>_i ed₃âre dâde <xâhad shod>_j.*
to Ali AUX.FUT.3SG become this house rent give-PTCP
- c. * *BE ALI_i dâde xâhad shod_j in xune <be Ali>_i ed₃âre*
to Ali give-PTCP AUX.FUT.3SG become this house rent
<dâde xâhad shod>_j.
- d. *BE ALI_i ed₃âre dâde xâhad shod_j in xune <be Ali>_i*
to Ali rent give-PTCP AUX.FUT.3SG become this house
<ed₃âre dâde xâhad shod>_j.

“It is to Ali that this house will be rented.”

Karimi’s rule of Syntactic Word Formation is challenged by the auxiliary data because the phrasal domain for the encyclopedic knowledge and obtaining special meanings must also accommodate the auxiliaries. Auxiliary verbs however are not expected to be in that domain because they are not part of the special meanings obtained between the verb and its arguments/modifiers. Therefore,

widening this domain to be as large as the intermediate projection including the auxiliary heads is rather too arbitrary and unrestricted. It seems that some (remnant) phrasal movement of the XP containing all the moving elements is necessary to account for the data, an approach that I will discuss and adopt in Section 2.6. I take this observation as another argument against assuming a head-final vP and relying on head-adjunction as the source of movement. Whether one adopts a base structure in line with (a) Folli, Harley, and Karimi (2005) and Karimi (2005) or in line with (b) Shafiei (2015, 2016) and Rasekhi (2018), the extension of these accounts to larger verbal clusters highlights a number of inconsistencies (summarized in (97)) that I consider as grounds for pursuing an alternative approach.

- (97)
- a. There is no consistency to the nature of head-adjunction in terms of whether incorporation is to the left or to the right of licensing heads. (Section 2.3.1)
 - b. There is no uniform head-incorporation approach; excorporation out of complex heads + additional assumptions must also be resorted to. (Section 2.3.2)
 - c. There is no straightforward mechanism for deriving leftward verb movement through head-adjunction, without imposing rules for allowing non-head constituents to adjoin to heads. (this section)

In the section that follows, I propose an analysis for deriving Farsi complex predicates that assumes a base head-initial structure for all projections following Kayne (1994) and derives the canonical word order through consistent leftward movements.

2.4 An antisymmetric XP movement analysis for deriving complex predicates

In this section, I propose a syntactic analysis within the antisymmetry framework of Kayne (1994), by which all phrases are underlyingly head-initial and Specifier-Head-Complement (S-H-C) is the only constituent order possible; the head asymmetrically c-commands the complement, and the specifier asymmetrically c-commands both the head and the complement. Under this restrictive theory of phrase structure, which maps hierarchical structure to linear order, different word orders are due to different combinations of movements. This means that languages with an OV order

involve the movement of the object to the left of the verb to some specifier position, and that the derived structures also confirm to a S-H-C configuration. Following Ahari (to appear), I analyze elements that survive deletion in verb-stranding ellipsis data and those that raise in Topic/Focus fronting data as derived XP constituents, rather than complex heads. More specifically, two types of constituents are formed in the derivation of verbal clusters: (a) a VoiceP/AuxP that contains the light verb (and the auxiliary verb, when it is present), and (b) a PredP that contains the non-verbal element, the light verb, and the auxiliary and forms the complex predicate. This approach unifies ellipsis and fronting under XP movements to the left periphery; while ellipsis allows for either the VoiceP/AuxP or the PredP to escape the ellipsis site, Topic/Focus fronting requires the highest functional projection in the verbal cluster to move. I propose that the linear order facts in Farsi are captured through a series of feature driven movements that license the arguments as well as the Non-Verbal predicate in the Spec of designated Functional Projections, therefore deriving the preverbal distribution of the predicate's internal arguments (the direct and indirect objects). In Section 2.4.1, I show how this analysis is implemented in detail to derive a complex predicate construction without the auxiliary verbs and how to account for the verb-stranding ellipsis data as a result. Sections 2.5.2 and 2.5.3 will subsequently deal with how this implementation extends to complex predicate constructions involving the Perfect Aux and the Future Aux, and how ellipsis interacts with them.

2.4.1 Complex predicate derivation without auxiliaries

I assume a base complex predicate structure inspired by the small clause analysis adopted in Ghomeshi (1996). Ghomeshi proposes that in constructions such as (98) and (99), there is a predication relation between the adjectival non-verbal predicates *tamiz* 'clean' and *bidâr* 'awake' and the direct objects *miz* 'table' and *Bijan*, respectively. Ghomeshi shows that such relations hold in the copular construction counterparts in (100) and (100) as well.

- (98) *miz-o tamiz kardam.*
 table-DOM clean made.1SG

"I cleaned the table."

(Ghomeshi 1996, p. 266)

- (99) *Bijan-o bidâr kardam.*
 Bijan-DOM awake made.1SG
 “I woke Bijan up.”

(Ghameshi 1996, p. 266)

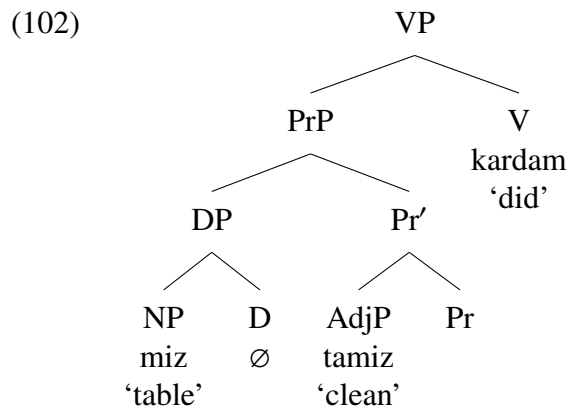
- (100) *miz tamiz-e.*
 table clean-is3SG
 “The table is clean.”

(Ghameshi 1996, p. 266)

- (101) *Bijan bidâr-e.*
 Bijan-DOM awake-is3SG
 “Bijan is awake.”

(Ghameshi 1996, p. 266)

To capture such predication relations, Ghomeshi (1996) adopts a small clause structure in line with Bowers (1993), in which predication is mediated by a functional head Pr^0 that takes the predicate as its complement and the object as its specifier. The proposed base structure for (98) is given in (102). In this structure, the Pr^0 selects for the maximal projection of the adjective ‘clean’ which assigns a theta role to the definite DP ‘table’ in the Spec of PrP. This PrP is in turn selected for by V (the light verbal component), the maximal projection of which is assumed to be further selected for by a higher Pr head that introduces the external argument.

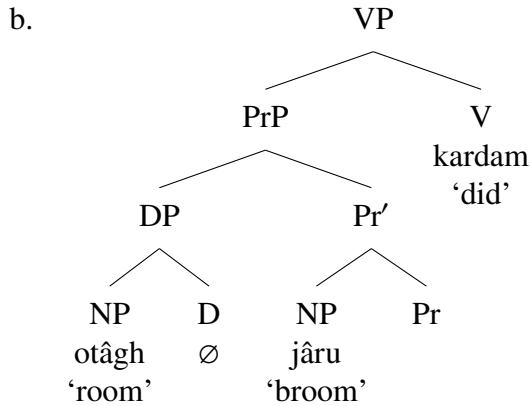


Ghameshi (1996) assumes a similar configuration for complex predicate constructions formed by a nominal non-verbal predicate as in (103a). In (103b) the projection of the non-verbal element *jâru* ‘broom’ is an NP category which assigns a theta role to the object *otâgh* ‘room’ as mediated by the Pr head.

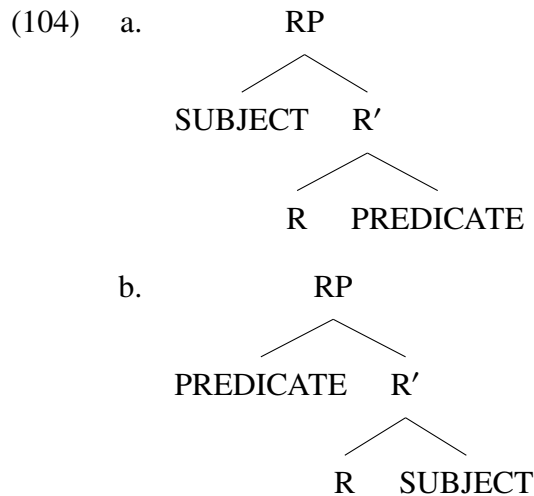
- (103) a. *otâgh-o jâru kardam.*
 room-DOM brood did.1SG

“I swept the room.”

(Ghameshi 1996, p. 275)



The Bowers (1993) style predication structures are fundamentally the same as that developed in Den Dikken (2006), in which the subject and the predicate are connected via the Relator (Rel) functional head. Den Dikken proposes that such relation is “always hierarchically asymmetrical” and argues that both cases in (104a) or (104b) exist.



I follow Ghameshi (1996) in adopting the Bowers and Den Dikken style SC configurations. However, in line with the antisymmetric framework and the assumption that Spec-Head-Comp is the only ordering possibility, I take the Pr or Relator heads to take their predicate complements to the right, as opposed to what’s suggested in the structures in (102) and (103b). In my analysis, not only a DP object but also a PP object is introduced in a functional projection above the NV head. This is in line with the the noun phrase domain analyses in Atlamaz (2016) and Kahnemuyipour (2014), which essentially follow the framework of Cinque (2005, 2010) and propose that modifiers and arguments

are hierarchically merged in Spec of FPs in the extended domain of the head noun. I extend this view to the verbal domain and propose that the Non-Verbal predicates and their arguments are introduced in the same way. This account will be further motivated in Chapters 3 and 4 where I discuss data involving clausal CP complements and predicate modification, respectively. To see how this all works out, I start with the derivation of (105) (the non-ellipsis counterpart to (39b)).

- (105) *Ehsân be Ali âpârtemân edžâre dâd.*
 Ehsan to Ali apartment rent gave.3SG
 “Ehsan rented houses to Ali.”

In this construction, the NV element *edžâre* ‘rent’ is related to its internal arguments *âpârtemân* ‘apartment’ and *be Ali* ‘to Ali’ in specifier positions of functional projections (labeled YP and XP) above the predicate. The light verb heads its vP projection above the XP. This directly follows Folli, Harley, and Karimi (2005) and Karimi (2005) in assuming that the components of the complex verb head separate projections. In line with Karimi and Smith (2020), the external argument ‘Ehsan’ is further introduced in a VoiceP, head of which probes for the light verb (the v head).¹⁰ These steps are shown in (106).

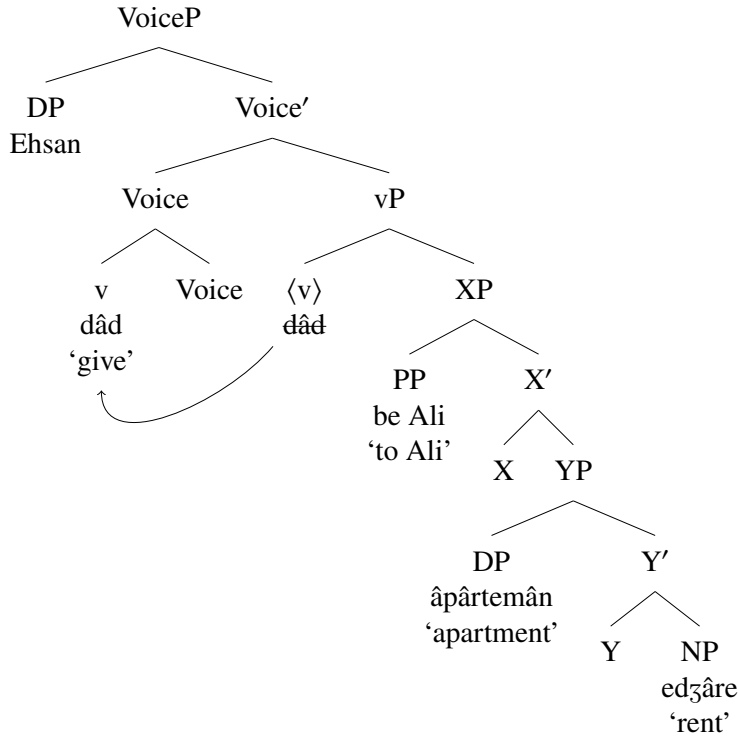
¹⁰There is evidence to argue that the subject merges external to the light verbal vP (and not in the specifier of vP). One such argument is based on the following example from Megerdooomian (2012) in which the causative is realized on the LV with an additional morpheme *ân*. This suggests that a separate head causativizes the light verb, and introduces the external argument.

- (i) a. *jang be pâyân resid.*
 war to end arrived.3SG
 “The war came to an end.”
 b. *soqut-e in rejim jang-râ be pâyân res-ân-d.*
 fall-Ezf this regime war-DOM to end arrived-CAUS.3SG
 “The fall of this regime brought the war to an end.” (Megerdooomian 2012, ex. 67)

Additionally, (ii) below exemplifies a construction in which the passive voice alternative to (iia) is derived with an additional verb *shodan* ‘become’ in (iib). In this case, the active-passive voice alternation is not due to an alternation in the light verb *dâd* but rather due to a separate head external to the vP.

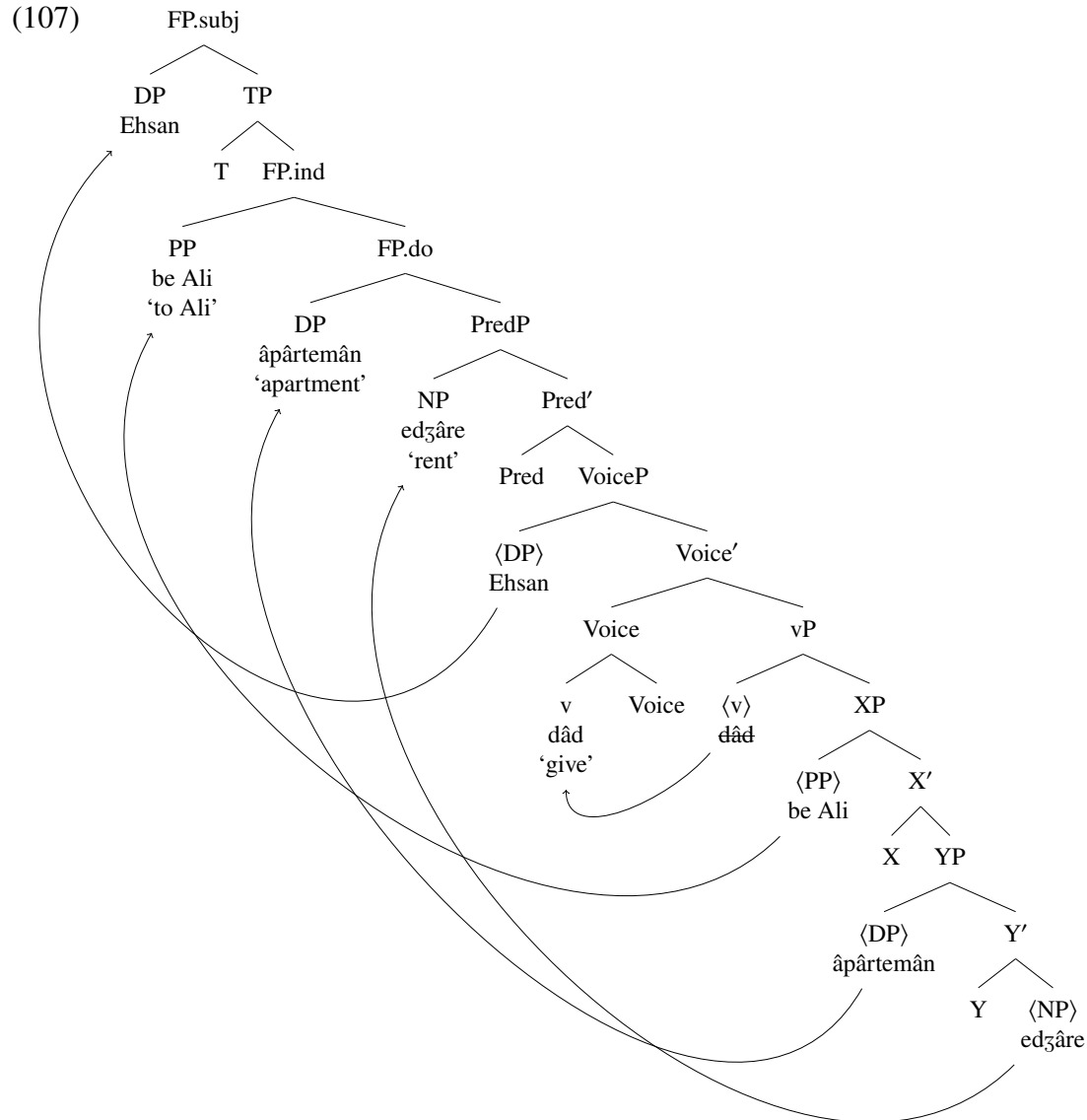
- (ii) a. *unâ âpârtemân-o be Ali edžâre dâdand.*
 they apartment-DOM to Ali rent gave.3PL
 “They rented the apartment to Ali.”
 b. *âpârtemân be Ali edžâre dâd-e shod.*
 apartment to Ali rent give-PTCP became.3SG
 “The apartment was rented to Ali.”

(106)



Taking inspiration from Zwart (1993, 1996, 1997), I assume that a PredP functional projection merges above all other verbal items and below the derived position of the object. Zwart (1993, 1996, 1997) proposes specifier of PredP as a licensing position for small clause predicates and particles in Dutch, and Koster (1999, 2000) assumes that XPs that are introduced by a verb but are not licensed as independent arguments throughout the derivation end up being part of the verb through licensing in spec of PredP. This projection as adopted for Farsi will merge below the licensing position of the object(s) and higher than the position of the (light verb), with its specifier as the landing site for the non-verbal predicate. One of the conditions for licensing this position is that this phrase can only project above the verbal projection that ends up hosting the non-verbal phrase's original licenser the (light) verb. This turns out to be the case in (106) since the LV adjoins to Voice. In fact, all complex predicate derivations prepose the non-verbal projection through movement to [Spec, PredP]. Zwart (1997) originally proposed that the Pred head has a strong N feature that must be satisfied locally with the non-verbal predicate in its specifier. In my analysis, I take this movement to be a reflex of an Edge feature, leaving out a detailed discussion of whether it must be the head (Chomsky 2001) or the moving element itself (Bošković 2007) hosting the feature. I further assume

that the internal arguments must move and be locally licensed in the specifiers of their Functional Projections (FP.do for the functional projection licensing the direct object and FP.ind for the one licensing the indirect object). The external argument must also raise to its designated licensing functional position FP.subj. These phrases can be alternatively thought of Agr projections for the direct object, the indirect object, and the subject. The full derivation is given in (107) below.¹¹



This derivation is in line with the feature checking approach of Zwart (1993, 1996, 1997), which assumes that a series of licensing operations apply, triggering the movement of elements from their

¹¹In this derivation and subsequent ones, only the specifiers of FP.do, FP.ind, and FP.subj are represented in the trees.

base merge positions to specifiers of functional licensing projections. In this case, such operations take place in the narrow syntax and the licensing is satisfied locally due to the functional heads' EPP features. The movements proposed for deriving the linear orders are therefore necessary given the antisymmetric framework of the analysis, and they ultimately follow from a set of universal conditions as well as language-specific choices. I also assume that the licensing movement of the direct object DP takes place independent of whether the object is non-specific as in (105) or specific (*râ*-marked) as in (108) below, in which case it typically precedes the indirect object PP.

- (108) *Ehsân un âpârtêmân-râ be Ali edžâre dâd.*
 Ehsan that apartment-DOM to Ali rent gave.3SG
 “Ehsan rented that apartment to Ali.”

I follow Farudi (2007) and Kahnemuyipour (2004, 2009) in proposing that specific objects such as in (108) undergo a subsequent movement to a higher position where they receive the *râ* marking.

2.5 Complex predicate derivation and Ellipsis

2.5.1 Ellipsis without the auxiliaries

My account of deriving the verb-stranding ellipsis data discussed in Rasekhi (2018) will essentially differ from Rasekhi in two aspects: (i) I assume that the verb-stranding operation involves maximal projection to specifier movement, rather than head-movement; and (ii) I consider the ellipsis site to be the full (FP.subj) clause rather than the vP. My analysis is still in line with Rasekhi's proposal that ellipsis is constrained by contrastive focus and that this comes in three possible interpretations in Farsi: contrastive, additive, and corrective. Rasekhi provides evidence that focus plays a role for licensing ellipsis by showing that the verbs across the antecedent and ellipsis clauses can be lexically contrastive and need not be identical. The contrastive interpretation across the two clauses can be shown in the pair in (109), where the verbs *xaridan* ‘buy’ and *furuxtan* ‘sell’ are lexically contrastive.

- (109) Modified from Rasekhi (2018, p. 71) to remove embedding.

- a. *Ali pârsâl yeho un xuna-ro xarid.*
 Ali last-year suddenly that house.DOM bought.3SG
 “Ali bought that house suddenly last year.”

- b. *Vali Ali emsâl yeho un-xuna-ro furuxt.*
 but Ali this-year suddenly that house-DEF sold.3SG
 “But (he) sold (that house) suddenly this year.”

In the additive interpretation, the verb ‘buy’ is identical across the clauses, and the focus particle *ham* ‘also’ follows the contrastive external argument *Araz* in (110b).

(110) Modified from Rasekhi (2018, p. 72) to remove embedding.

- a. *Aydâ âpârtêmân xarid.*
 Ayda apartment bought.3SG
 “Ayda bought an apartment.”
- b. *Araz ham âpârtêmân xarid.*
 Araz also apartment bought.3SG
 “Araz also bought (an apartment).”

In verb-stranding ellipsis cases with a corrective interpretation, verbs can be either contrastive across (111a) and (111b) or identical across (111a) and (111c). If the verbs are lexically identical, however, some other element must be contrastive with its correlate in the antecedent clause.

(111) Modified from Rasekhi (2018, p. 73) to remove embedding.

- a. *Aydâ benz-a-ro xarid.*
 Ayda BENZ-DEF-DEF bought.3SG
 “Ayda bought the Mercedes.”
- b. *na, Aydâ benz-a-ro furuxt.*
 no Ayda BENZ-DEF-DEF sold.3SG
 “No, (Ayda) sold (the Mercedes).”
- c. *na, Araz benz-a-ro xarid.*
 no, Araz BENZ-DEF-DEF bought.3SG
 “no, Araz bought the Mercedes.”

The following pairs in (112) and (113) further reveal that a polarity contrast on the verb (affirmative vs. negative) qualifies as the verbs being in a contrastive relationship.

(112) Contrastive interpretation similar to (109).

- a. *Ali pârsâl yeho ye xune xarid.*
 Ali last-year suddenly a house bought.3SG
 “Ali suddenly bought a house last year.”
- b. *Vali Ali emsâl yeho ye-xune na-xarid.*
 but Ali this-year suddenly that house.DOM sold.3SG
 “But (he) didn’t suddenly buy (a house) this year.”

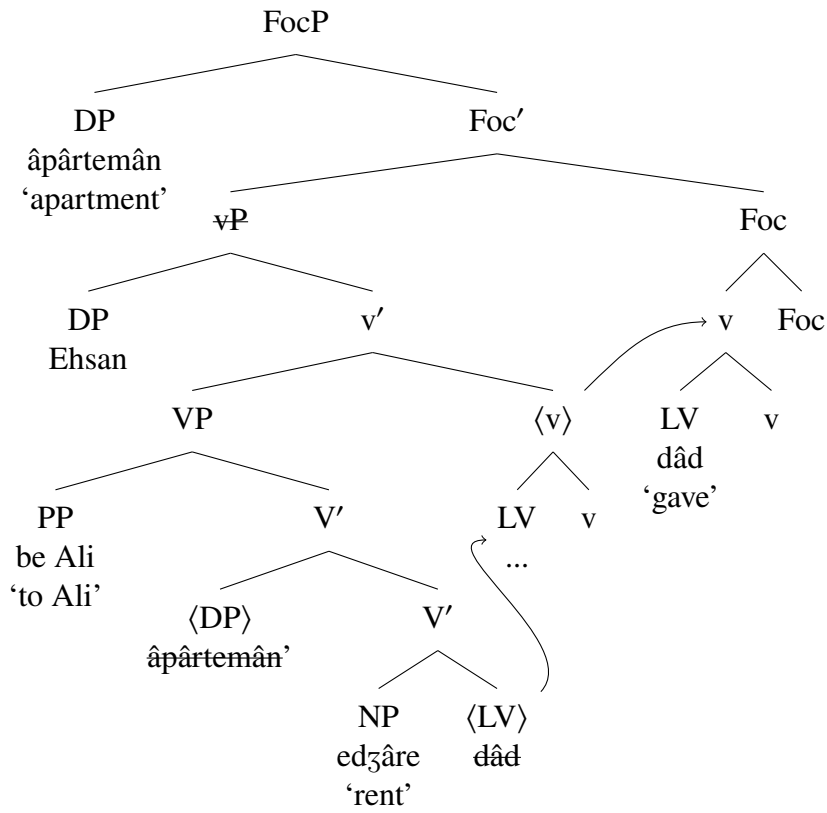
(113) Corrective interpretation similar to (111a) and (111b).

- a. *Aydâ benz-a-ro xarid.*
 Ayda BENZ-DEF-DOM bought.3SG
 “Ayda bought the Mercedes.”
- b. *na, Aydâ benz-a-ro na-xarid.*
 no Ayda BENZ-DEF-DOM NEG-bought.3SG
 “No, (Ayda) didn’t buy (the Mercedes).”

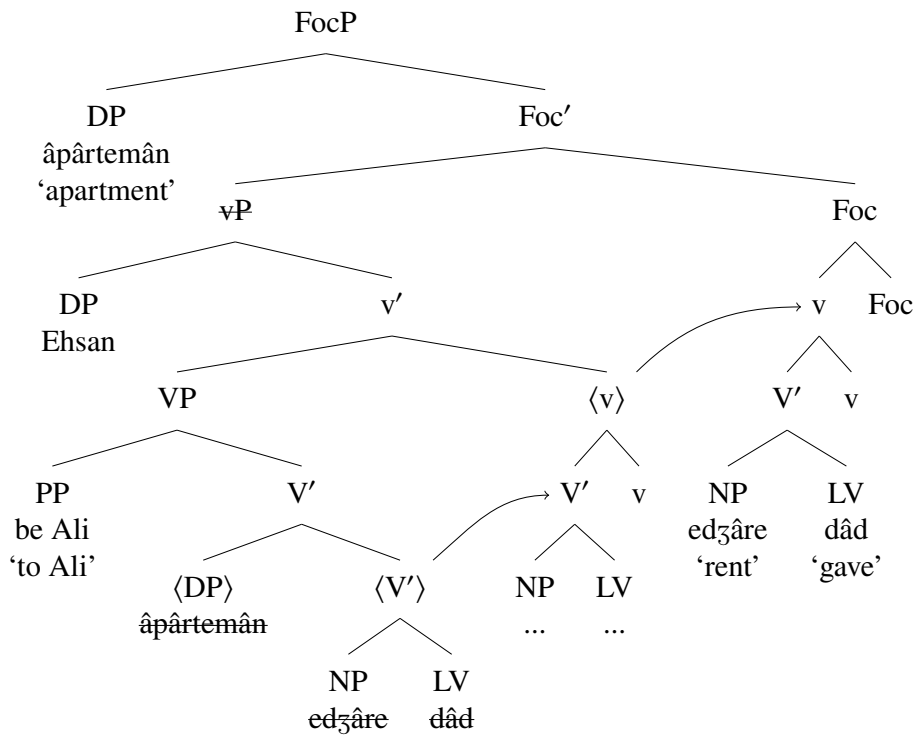
Rasekhi (2018) proposes that there is TP-internal Foc(us) P(hrase) above the vP. This Foc head is bundled with a number of features: (a) an [E] feature that licenses the deletion of the vP complement at PF; (b) an uninterpretable V feature that is satisfied through the movement of the verb to this Foc head; and (c) an uninterpretable Contrastive Focus feature (ConF) that must be valued through the movement of a contrastively focused argument (contrastive with a correlate in the antecedent clause) to its specifier. To revisit how Rasekhi’s analysis works, the ellipsis data in (39) and the derivations in (40) and (41) are repeated in (114), (115), and (116) respectively.

- (114) a. *Ehsân be Ali xune eďâre na-dâd.*
 Ehsan to Ali house rent NEG-gave.3SG
 “Ehsan did not rent house(s) to Ali.”
- b. *Vali Ehsân be Ali âpârtêmân eďâre dâd.*
 but Ehsan to Ali apartment rent gave.3SG
 “But (Ehsan) rented apartment(s) (to Ali).”
- c. *Vali Ehsân be Ali âpârtêmân eďâre dâd.*
 but Ehsan to Ali apartment rent gave.3SG

(115) Raising the LV alone



(116) Raising the NP-LV complex



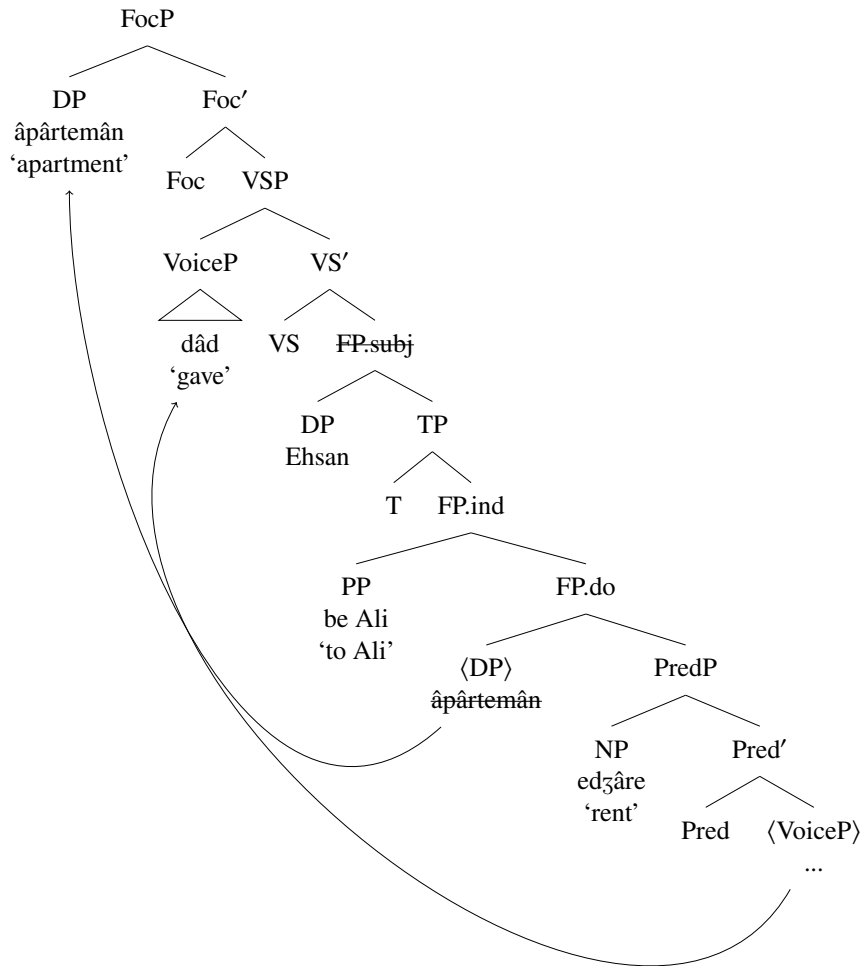
My account of the ellipsis data however builds on the alternative derivation laid out in (106). I propose that a functional projection exists in the left periphery of the clause to which a verbal constituent moves from the ellipsis site. This means that whenever an additional contrastive element escapes the ellipsis site, it moves to the specifier position of a separate functional projection than the one utilized for stranding the verb. It was shown in (109)-(111) that there are three contrastive interpretations available to the verb-stranding ellipsis data. It can be further observed that the contrastive focus requirements are not exactly the same for all three interpretations, as summarized in (117):

- (117) a. **Contrastive focus interpretation** requires (a) **the verbs to be contrastive** across the ellipsis clause and the antecedent clause, and (b) minimally **one other element to be in a contrastive focus relation** with its correlate.
- b. **Additive interpretation** requires (a) **the verbs to be identical** across the clauses, and (b) **a contrastive element followed by the focus particle *ham* ‘also’** to escape ellipsis.
- c. **Corrective interpretation** can be of two forms:
- i. **The verbs are contrastive** across the clauses and **no additional contrastive element** is needed.
 - ii. **The verbs are identical** and **one other element is contrastive** with its correlate in the antecedent clause.

Crucially, while all the ellipsis data show verb stranding, not all constructions have the same identity requirement on the verb and not all constructions require an additional contrastive XP to move to the left periphery (as is the case in (117c-i)). Consequently, I claim that the only projection constant for deriving the different constructions is the one attracting a verbal constituent, which I will label Verb Stranding Phrase (VSP). The VS head of this phrase will carry either a Contrastive Verb (ConV) or Identical Verb (IdentV) feature depending on the particular ellipsis construction. In the context of a contrastive construction such as (114), the VS head will necessarily carry the ConV feature. This feature must be locally satisfied by raising either the VoiceP (containing only the

LV *dâd*) or the PredP (containing both the NV and the LV *edžâre dâd*) to account for (114b) and (114c), respectively. Above this VSP, Farsi requires another projection for raising the contrastive internal argument *âpârtêmân*, which I’ve labeled FocP in line with Rasekhi (2018). Therefore, this construction requires both the verbs and minimally an argument to be contrastive.¹² The two verb cluster stranding options are shown in (118) and (119), respectively.

(118) The VoiceP can move to satisfy the VS head’s ConV feature

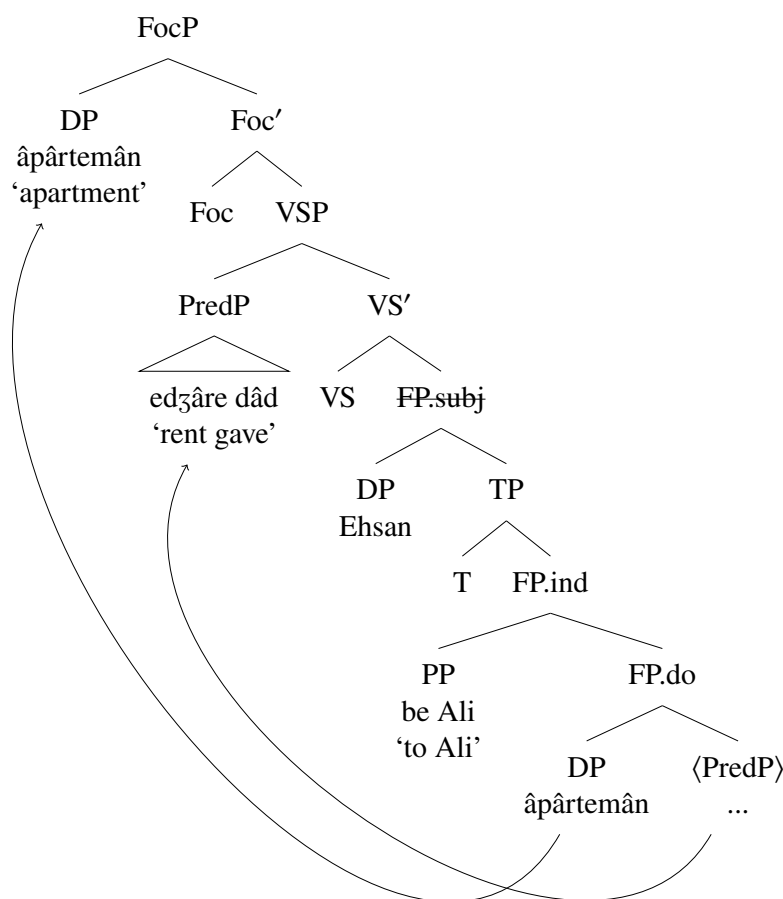


¹²I’m claiming that the predicates in (114) are contrasted based on the observation that the negative-affirmative verb pairs in (112) and (113) behave like lexically contrastive verbs in (109) and (111a-b). Additionally, Rasekhi (2018) suggests that the verbs in the following pair (*na-dâd* ‘didn’t give’ and *dâd* ‘gave’) are contrastive, which means that the ones in (114) must be as well.

- (i) a. *Az in-ke Ayda ketâb-ro be pesar-esh na-dâd ta’ajob na-kard-am.*
 from this-that Ayda book-DOM to son-CL.3SG NEG-gave.3SG surprise NEG-did-1SG
 “The fact that Ayda did not give the book to her son did not surprise me.”
 b. *Vali az in-ke majalla-ro be pesar-esh dâd ta’ajob kardam.*
 but from this-that magazine-DOM to son-CL.3SG gave.3SG surprise did.1SG
 “But the fact that (she) gave the magazine (to her son) surprised me.”

(Rasekhi 2018, p. 83)

(119) The PredP can move to satisfy the VS head's ConV feature



2.5.2 Complex predicates with the Perfect Auxiliary

With the addition of auxiliaries into the mix, I follow Zwart (1996) and Barbiers, Cornips, and Corrigan (2005) in assuming that all auxiliary verbs enter the syntax according to a hierarchical base structure and syntactic operations further follow a feature checking approach. The Perfect auxiliary *budan* in (120) is one such auxiliary and it licenses participles, as evident from the participle marking ‘-e’ on the light verb.

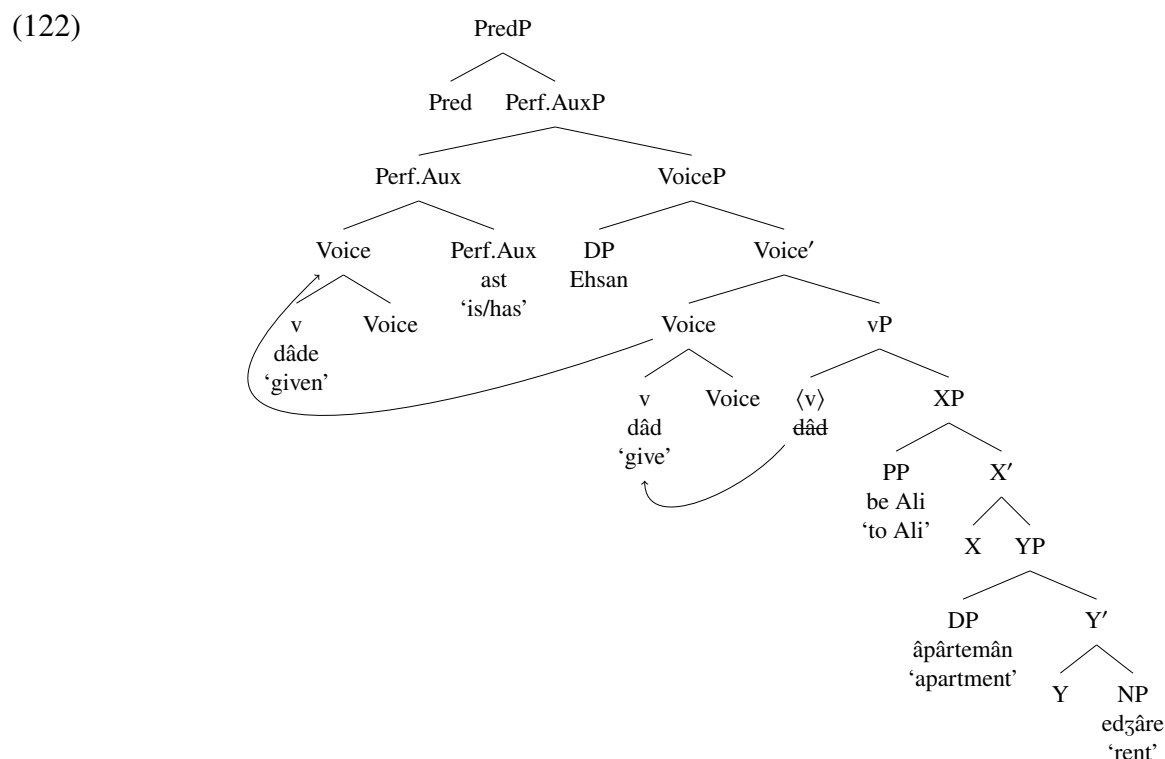
- (120) *Ehsân be Ali apârtêmân edzâre dâd-e ast.*
 Ehsan to Ali apartment rent give-PTCP AUX.PRF.3SG
 ‘Ehsan has rented apartments(s) to Ali.’

I propose that the participle LV adjoins to the Perfect Aux head resulting in a complex head, and this is the only Auxiliary verb that requires the head movement of the participle it selects for. This explains the observation pointed out above regarding the placement of the negative prefix in such

constructions. As (121) shows, the Neg- attaches to the left of the sequence [Participle-Perf.Aux] and never intervenes between the two heads. This is expected if the participle and the Perfect Aux form a complex head, and negation is always realized on the highest verbal head in the structure (in this case the complex Aux head).

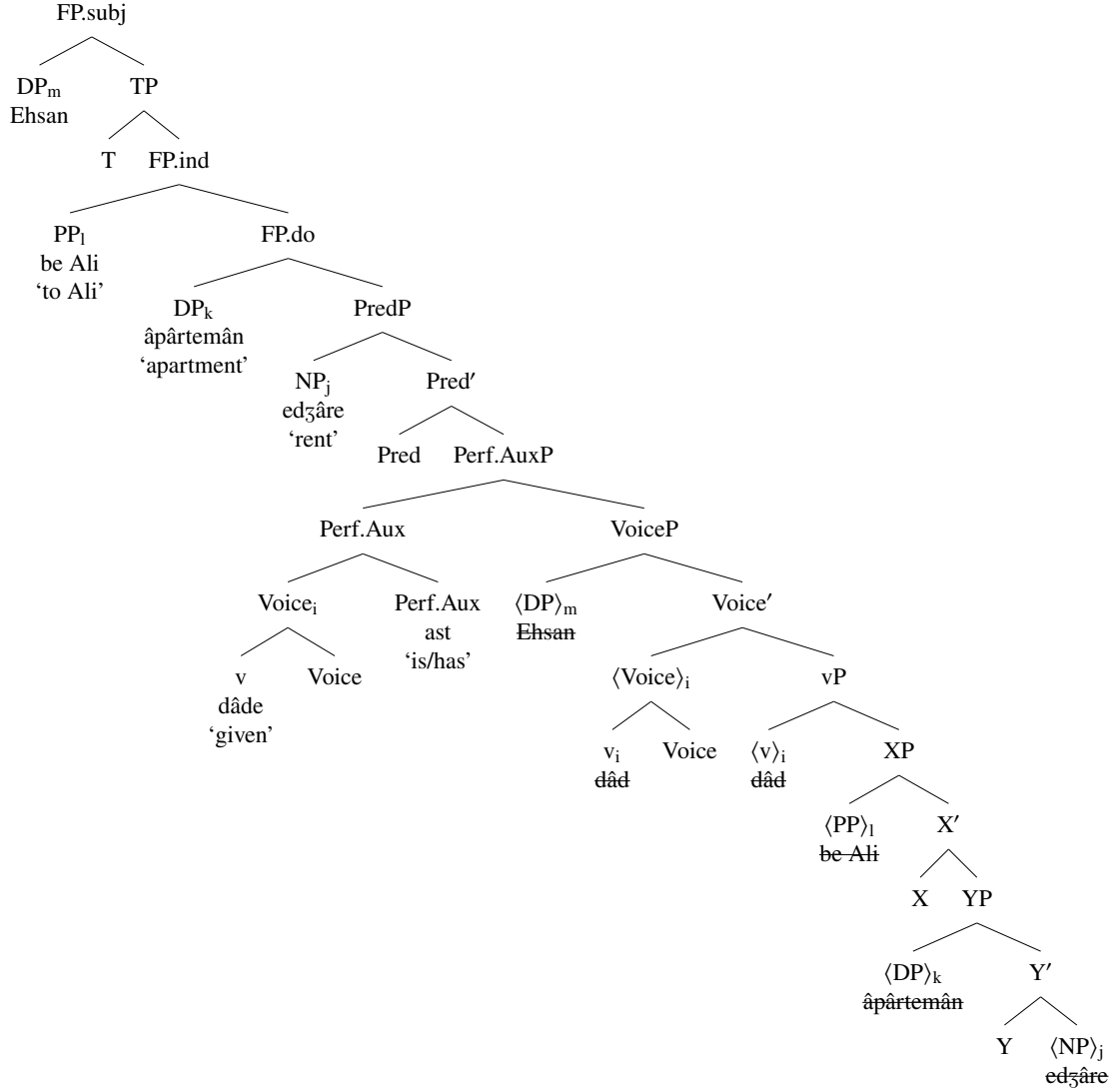
- (121) *Ehsân be Ali âpârtêmân edžâre na-dâd-e ast.*
 Ehsan to Ali apartment rent NEG-give-PTCP AUX.PRF.3SG
 ‘Ehsan has not rented house(s) to Ali.’

Building on the complex predicate derivation, I assume that the Aux hierarchically projects higher and selects for the VoiceP; this accounts for the subject agreement morphology realized on the Aux as the highest verbal head in (120). The agreement would otherwise show up on the Voice head if it were the highest verb head below PredP, as was the case in (107). Therefore, all that is required to account for (120) is to include an AuxP projection above the vP and below the PredP.



The rest of the derivation proceeds in (123) as proposed above; the nominal predicate preposes to [Spec.PredP], the internal arguments raise to FP.do and FP.ind, and lastly the external argument is licensed in FP.subj.

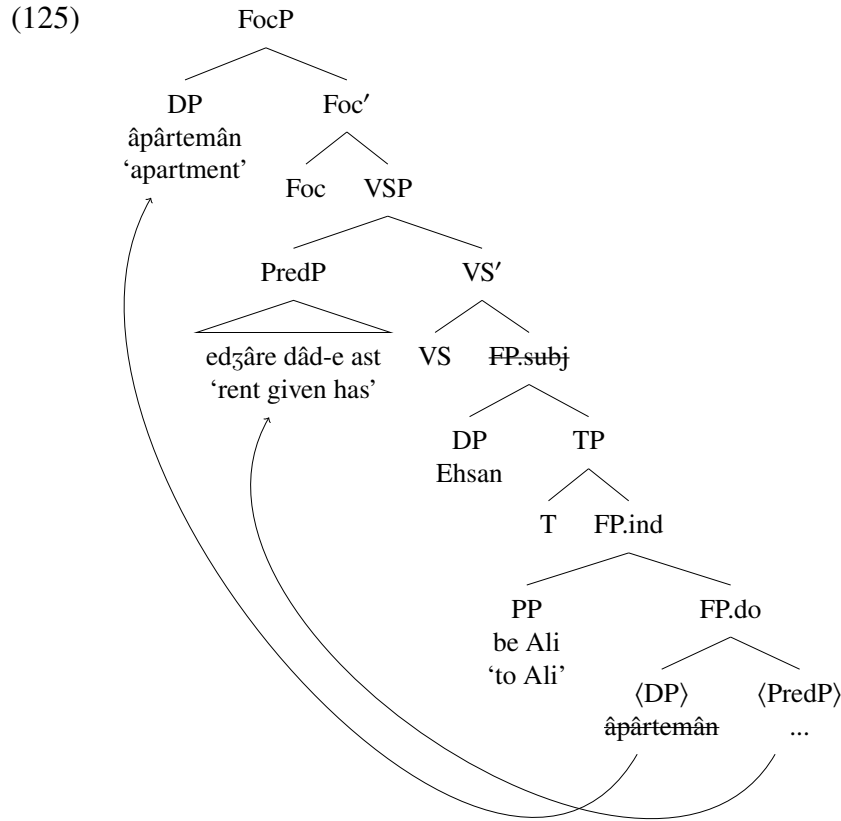
(123)



In addition to straightforwardly deriving the canonical NV-LV-Perf.Aux order through uniform leftward movements and left-adjunction, the derivation in (123) is a desirable outcome for the elipsis cases in (47), repeated below as (124).

- (124) a. *Ehsân be Ali xune ed3âre na-dâd-e ast.*
 Ehsan to Ali house rent NEG-give-PTCP NEG-AUX.PRF.3SG
 “Ehsan has not rented house(s) to Ali.”
- b. *Vali Ehsân be Ali âpârtêmân ed3âre dâd-e ast.*
 but Ehsan to Ali apartment rent give-PTCP AUX.PRF.3SG
 “But (Ehsan) has rented apartment(s) (to Ali).”
- c. *Vali Ehsân be Ali âpârtêmân ed3âre dâd-e ast.*
 but Ehsan to Ali apartment(s) rent give-PTCP AUX.PRF.3SG

In (124), the direct object *âpârtêmân* ‘apartment’ contrasts with *xune* ‘house’ and survives ellipsis. The verbal clusters *edžâre na-dâde ast* ‘has not rented’ (in the antecedent clause) and *edžâre dâde ast* ‘has rented’ (in the ellipsis clause) are also contrastive, with the option of either the Aux+LV in (124b) or the Aux+NV+LV in (124c) escaping ellipsis. The two choices are captured through raising either the Perf.AuxP or the PredP out of the ellipsis site and to the specifier of the Verb Stranding Phrase proposed above. The derivation for the latter option in (124c) is given in (125).



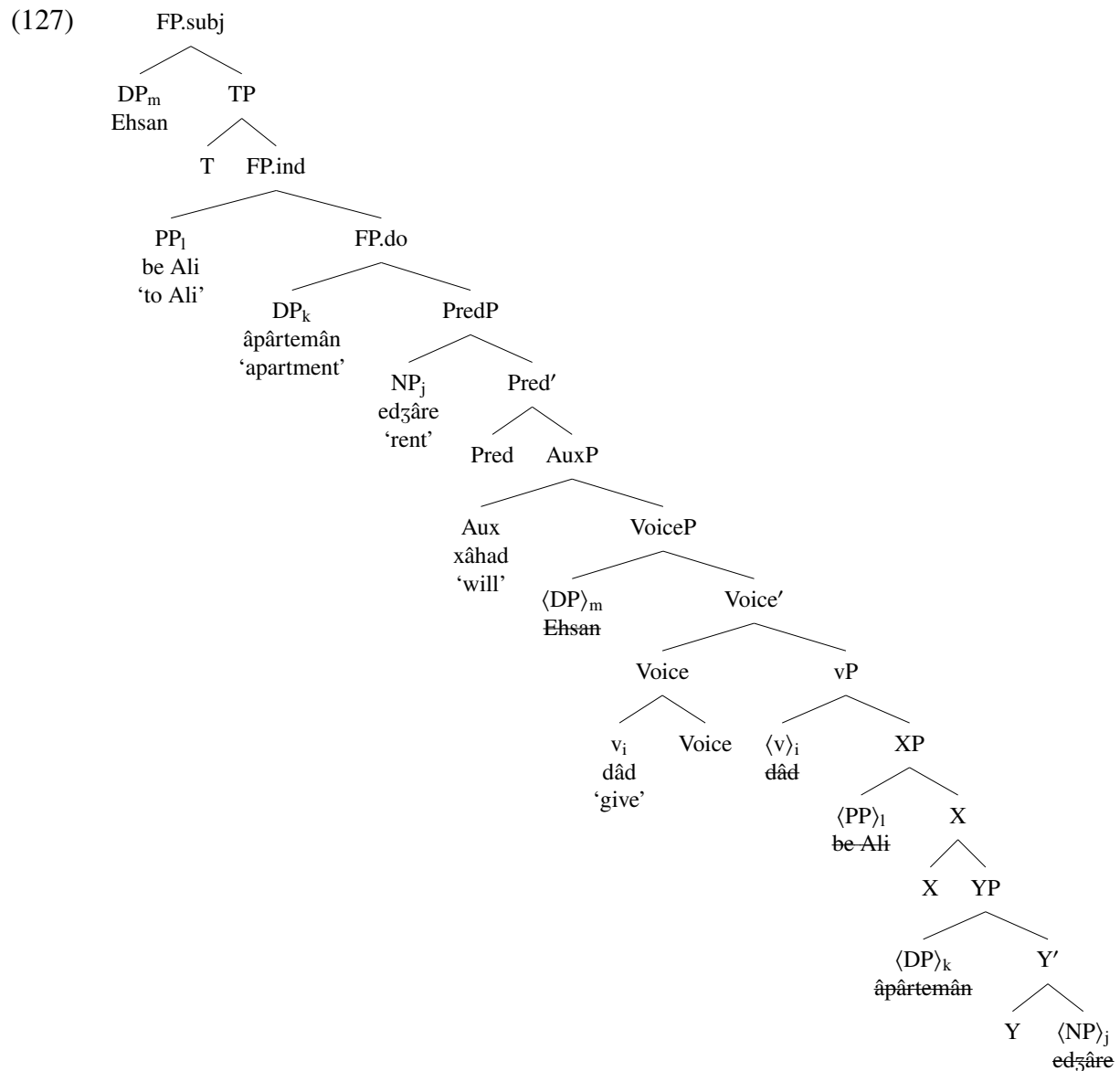
2.5.3 Complex predicates with the Future Auxiliary

This section offers an analysis for deriving the fact that the Future Aux *xâstan* surfaces penultimately and breaks up the components of a complex predicate in data such as (126).

- (126) *Ehsân be Ali âpârtêmân edžâre xâhad dâd.*
 Ehsan to Ali apartment rent AUX.FUT.3SG give
 ‘Ehsan will rent apartment(s) to Ali.’

I propose that this auxiliary also hierarchically projects above and selects for the VoiceP to which the light verb adjoins. This is once again supported by the subject agreement inflection on the

auxiliary as the highest verbal head in the structure. In comparison to the Perfect Aux, however, the feature checking between the Future Aux and the (short)-infinitival verb it selects for takes place remotely and there is no v-to-Aux head adjunction in the narrow syntax. Therefore, the inflected aux *xâhad* linearly precedes the LV *dâd* in (126) as a reflection of the hierarchical base order of the two heads. Following previous assumptions, a PredP in turn merges above the Aux projection and preposes the NV predicate to its specifier. The rest of the derivation follows as expected in (127); the internal arguments raise to FP.do and FP.ind, and the external argument is licensed in FP.subj.

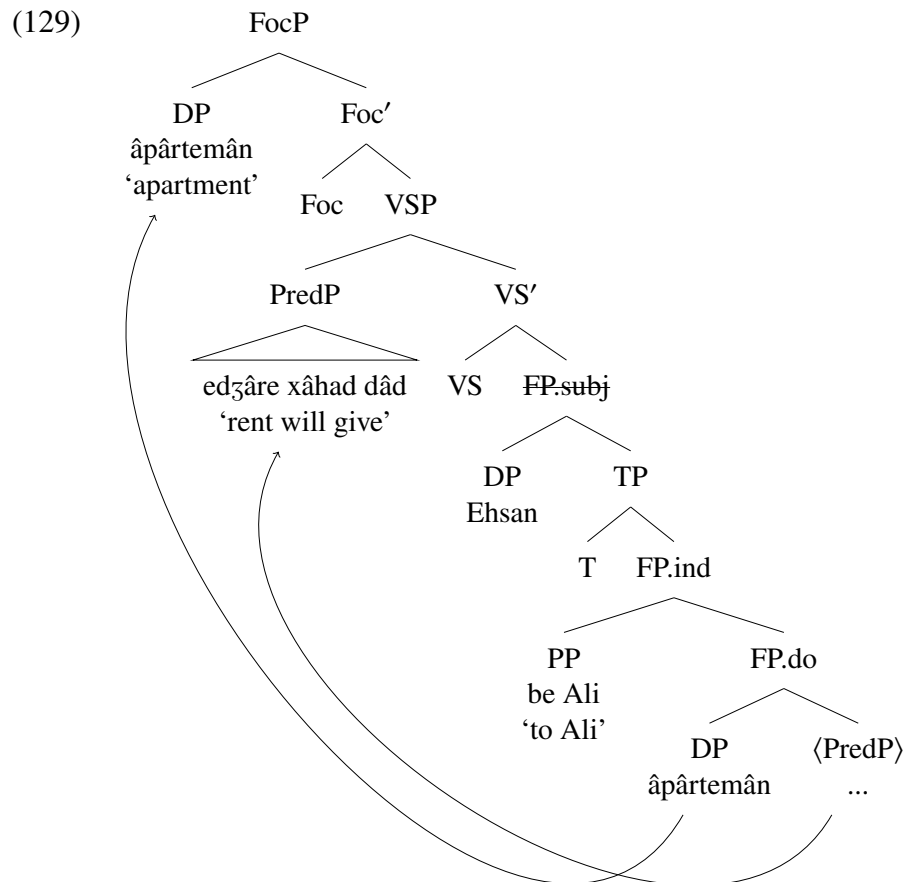


This account has the advantage of eliminating the need to assume rightward movement/right

-adjunction and the need to commit to an excorporation operation for deriving the strict NV-Fut.Aux-LV order. The derivation of the ellipsis constructions in (61), repeated below as (128), naturally proceeds from the structure in (127).

- (128) a. *Ehsân be Ali xune ed̤âre na-xâhad dâd.*
 Ehsan to Ali house rent NEG-AUX.FUT.3SG give
 ‘Ehsan will not rent house(s) to Ali.’
- b. *Vali Ehsân be Ali apârtêmân ed̤âre xâhad dâd.*
 but Ehsan to Ali apartment rent AUX.FUT.3SG give.
 ‘But (Ehsan) will rent apartment(s) (to Ali).’
- c. *Vali Ehsân be Ali apârtêmân ed̤âre xâhad dâd.*
 but Ehsan to Ali apartment rent AUX.FUT.3SG give.

Ultimately, the option for the entire complex cluster *ed̤âre xâhad dâd* ‘rent will give’ to escape ellipsis in (128c) is captured through the movement of PredP out of the ellipsis site, as shown in (129).



2.6 Complex predicate derivation and Topic/Focus fronting

Given the proposed phrasal movement account of verb stranding in Section 2.4 and the already existing independent evidence for leftward scrambling of XPs in Farsi, I further analyze “verb preposing” as an XP movement that targets the highest extended projection of the verb. This approach is in line with a unified treatment of leftward movement in Farsi as broadly XP movement (Ganjavi 2007; Ghomeshi 1996; Kahnemuyipour 2004, 2009; a.o) based primarily on similar data as those discussed in Section 2.3.3.1 of this paper.

Ghomeshi (1996) proposes that while the data documented in Karimi (1989, 1994) suggests for a separation between XP movement as the source of argument scrambling vs. X^0 raising as the source of verb movement, the inseparability of the non-verbal and the verbal components of a complex predicate is not expected under the X^0 movement story. This was observed for *gharz dâdan* ‘rent giving’ in (91), repeated here in (130).

- (130) a. *Kimeâ ketâb-ro be dust-esh gharz dâd.*
 Kimea book-DOM to friend-CL.3SG loan gave.3SG
 “Kimea loaned the book to her friend.”
- b. * *KIMEA_i dâd_j <Kimeâ_i ketâb-ro be dust-esh gharz <dâd>_j*
 Kimea gave.3SG book-DOM to friend-CL.3SG loan
 “It was Kimea who loaned the book to her friend.”
- c. *KIMEA_i gharz dâd_j <Kimeâ_i ketâb-ro be dust-esh <gharz dâd>_j*
 Kimea loan gave.3SG book-DOM to friend-CL.3SG

Based on Ghomeshi’s suggestion that subjects merge external to VP and “definite direct objects and PPs are adjoined to VP” (or alternatively definite/specific DPs raise outside of VP per Karimi (2005)), cases such as (130c) can be analyzed as the leftward movement of the verbal phrase constituent containing just the verb and its non-verbal XP complement. This also explains Karimi’s (1989, 1994) general observation that arguments may only surface in a position following the verb when they are specific/definite and may not do so otherwise. In (131b), the specific direct object *hayât-o* ‘the courtyard’ can surface after the complex verb *tamiz kardan* ‘to clean’, whereas (132b) shows that the non-specific object *hayât* ‘courtyard’ may not do so.

- (131) a. *Sepide emruz hayât-o tamiz kard.*
 Sepide today courtyard-DOM clean did.3SG
 “Sepide cleaned the courtyard today.”
- b. *Sepide emruz tamiz kard hayât-o.*
 Sepide today clean did.3SG courtyard-DOM
 “Sepide CLEANED the courtyard today” (Karimi 1989, p. 213)
- (132) a. *Sepide emruz hayât tamiz kard.*
 Sepide today courtyard clean did.3SG
 “Sepide cleaned courtyards today.”
- b. * *Sepide emruz tamiz kard hayât.*
 Sepide today clean did.3SG courtyard
 (Karimi 1989, p. 213)

Ghameshi (1996) suggests that this difference can be explained if “XP-preposing”, as opposed to object postposing is at play. In (131a), the VP contains only the complex predicate and excludes the specific direct object ‘the courtyard’ because such objects are external to the VP under either the base-generation or the movement analysis. So, VP should be able to move to a (sentence internal) position above it in (131b). Since non-specific objects are VP internal, however, the object ‘yard’ in (132a) must move along with the complex predicate whenever VP undergoes leftward movement; so, a VP movement that leaves a non-specific direct object stranded in (132b) is not predicted to be possible.

Ganjavi (2007) discusses the option of forming yes-no questions in Farsi either with the verb in-situ and with rising intonation at the end of the sentence in (133a), or through the movement of the verb to the front in (133c). Ganjavi also observes that such movement targets both the verb and the direct object whenever the object is non-specific. The same holds in (134b) as the modified direct object *film-e xeyli ahmaghane* ‘very stupid movie’ and the verb ‘see’ must both front.

- (133) a. *Nâder ghazâ pox?*
 Nader food cooked.3SG
 “Did Nader cook food?”

- b. **pox* *Nâder ghazâ* *⟨pox⟩*?
 cooked.3SG Nader food
 “Did Nader cook food?”
- c. *ghazâ pox* *Nâder* *⟨ghazâ pox⟩*?
 food cooked.3SG Nader

(Ganjavi 2007, p. 32)

- (134) a. **mi-bin-e* *Rezâ har shab film-e* *xeyli ahmaghâne* *⟨mi-bin-e⟩*?
 IPFV-see.3SG Reza every night movie-Ezf very stupid
 “Does Reza watch very stupid movies every night?”
- b. *film-e* *xeyli ahmaghâne* *mi-bin-e* *Rezâ har shab*
 movie-Ezf very stupid IPFV-see.3SG Reza every night
⟨film-e xeyli ahmaghâne mi-bin-e⟩?

(Ganjavi 2007, p. 32)

In line with Ghomeshi (1996), Ganjavi (2007) takes (133c) and (134b) to argue for a “VP fronting” analysis. Ganjavi rules out such cases as involving subject postposing based on the following in (135) and (136). In (135a), the adverb *hanuz* ‘still’ follows the subject ‘Nader’ and precedes the objects *bara dustamun* ‘for our friends’ and *ghazâ* ‘food’. As (135c) reveals, the relative ordering of the subject and the adverb remains unchanged when the object and the verb move. As (136) also shows, the preposing of the non-specific direct object *ketab* ‘book’ and the verb *xaridan* ‘to buy’ in (136c) does not alter the order in which the subject ‘Homeyra’ and the adverb *hamishe* ‘always’ appear. This is to show that the subject hasn’t postposed and what’s at play is really XP fronting.

- (135) a. *Nâder hanuz barâ dust-â-mun* *ghazâ* *na-pox*-e.
 Nader still for frined-PL-CL.1PL food NEG-cooked-PTCP
 “Nader still hasn’t cooked food for our friends.”
- b. **na-pox*-e *Nâder hanuz barâ dust-â-tun* *ghazâ* *⟨na-pox-e⟩*?
 NEG-cooked-PTCP Nader still for frined-PL-CL.2PL food
 “Hasn’t Nader cooked for your friends yet?”
- c. *ghazâ na-pox*-e *Nâder hanuz barâ dust-â-tun* *⟨ghazâ na-pox-e⟩*?
 food NEG-cooked-PTCP Nader still for frined-PL-CL.2PL

(Ganjavi 2007, pp. 34–35)

- (136) a. *Homeyrâ hamishe ketâb mi-xa-re.*
Homeyra always book IPFV.buys.3SG
“Homeyra always buys books.”
- b. **mi-xa-re Homeyrâ hamishe ketâb <mi-xa-re>?*
IPFV.buys.3SG Homeyra always book
“Does Homeyra always buy books?”
- c. *ketâb mi-xa-re Homeyrâ hamishe <ketâb mi-xa-re>?*
book IPFV.buys.3SG Homeyra always

(Ganjavi 2007, p. 35)

Lastly, Kahnemuyipour (2004, 2009) proposes that the obligatory movement of manner adverbs along with the verbs, as shown in (90b) and further exemplified in (137) and (138), operates on the maximal projections (vP) that contain the adverbs and cannot strand them.

- (137) a. *Ali ârum ghazâ mi-xor-e.*
Ali slowly food IPFV-eat.3SG
“Ali eats slowly.”
- b. **ghazâ mi-xor-e Ali ârum <ghazâ mi-xor-e>.*
food IPFV-eat.3SG Ali slowly
- c. *ârum ghazâ mi-xor-e Ali <ârum ghazâ mi-xor-e>.*
slowly food IPFV-eat.3SG Ali

(Kahnemuyipour 2009, p. 89)

- (138) a. *Ali xub bâzi kard.*
Ali well play did.3SG
“Ali played well.”
- b. **bâzi kard Ali xub <bâzi kard>.*
play did.3SG Ali well
- c. *xub bâzi kard Ali <xub bâzi kard>.*
well play did.3SG Ali

(Kahnemuyipour 2009, p. 90)

In line with the XP movement view of “verb preposing”, I propose that the same Topic/Focus projections to which arguments scramble can host verbal constituents as well.¹³ Where fronting

¹³This raises the relevant question of whether it is possible to have long distance focus movement of an embedded

seems to diverge from ellipsis to some extent is the strong requirement for Farsi speakers that all verbal elements plus additional material in the extended functional projection of the verb move to the left. This is a restriction that must be encoded in the grammar regardless of whether one assumes the base structure of the VP/vP to be head-final or head-initial.¹⁴ This means that scrambling of the maximal functional projection is what derives the verb preposing facts in cases such as the following in (139). The grammatical case of verb fronting in (139f) shows that the manner adverb *bezur* ‘reluctantly’/‘forcibly’ and the non-specific direct object *ketâb* ‘book’ must front together with the auxiliary containing cluster *gharz xâhad dâd* ‘will loan’.

- (139) a. *Ehsân be Ali be-zur âpârtemân edzâre xâhad dâd.*
 Ehsan to Ali with-force apartment rent AUX.FUT.3SG give
 “Ehsan will reluctantly/forcibly rent apartment(s) to Ali.”

constituent, accompanied by long distance movement of the verb. If so, an argument can be made that what crosses the clause boundary is a maximal projection and not a head. Karimi (2005) argues that verb fronting can only happen within the same clause and cannot be long distance, as shown in (i) below.

- (i) a. [*un film-ro did-an*] *unâ.*
 that movie-râ saw-3PL they
 “They saw that movie.”
 b. * [*un film-ro did-an*] *Kimea goft unâ.*
 that movie-râ saw-3PL Kimea said-3PL they

Karimi analyzes (ia) as the topic movement of the direct object *un film-ro* ‘that movie’ to [Spec, TP] and the movement of the verb *didan* ‘to see’ to T. The ungrammaticality of (ib) is therefore argued to be due to the locality of V movement. However, counter cases of the long-distance movement of the verb can be found, such as in the following from Bonami and Samvelian (2015, ex. 22b).

- (ii) *dâd Maryam fekr mi-kon-am [in tâblo-râ be Omid diruz].*
 gave.3SG Maryam thought IPFV-do.1SG this painting-râ to Omid yesterday
 “I think that Maryam gave this painting to Omid yesterday.”

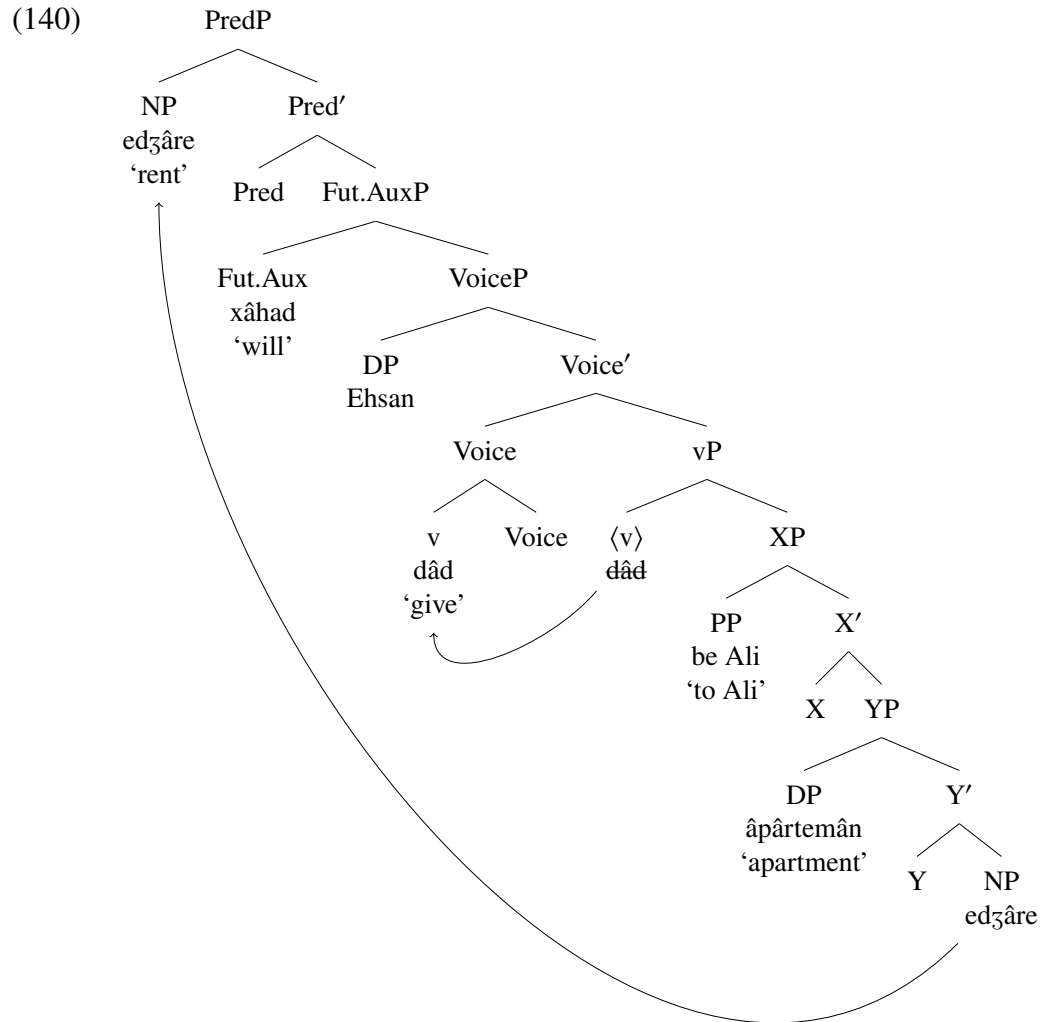
In (ii), the inflected embedded verb *dâd* ‘gave’ and the embedded subject ‘Maryam’ have moved to the matrix clause, and they must have done so separately given that the subject and the verb alone do not form a constituent. The same is possible with the complex predicate *gharz dâd* ‘lent’ in (iii).

- (iii) *gharz dâd Maryam fekr mi-kon-am [in tâblo-râ be Omid diruz].*
 loan gave.3SG Maryam thought IPFV-do.1SG this painting-râ to Omid yesterday
 “I think that Maryam lent this painting to Omid yesterday.”

¹⁴I do not have a conclusive hypothesis as to what this requirement stems from. As far as the Farsi data is concerned, I’ve shown in relevant examples that “verb movement” necessarily involves the movement of the non-verbal predicate, the light verb, and the auxiliary + certain other phrasal elements (non-specific objects, manner adverbs, etc.). That’s why I’ve analyzed these cases as the topicalization/focalization of the functional projection that will contain all the items that have not moved out or have not originated external to this constituent. To this effect, verb preposing as discussed here is to be thought of as FP-fronting.

- b. * *dâd_i Ehsân be Ali be-zur âpârtêmân eďzâre xâhad* *⟨dâd⟩_i*
 give Ehsan to Ali with-force apartment rent AUX.FUT.3SG
- c. * *xâhad dâd_i Ehsân be Ali be-zur âpârtêmân eďzâre* *⟨xâhad dâd⟩_i*
 AUX.FUT.3SG give Ehsan to Ali with-force apartment rent
- d. * *eďzâre xâhad dâd_i Ehsân be Ali be-zur âpârtêmân* *⟨eďzâre xâhad dâd⟩_i*
 rent AUX.FUT.3SG give Ehsan to Ali with-force apartment
- e. * *âpârtêmân eďzâre xâhad dâd_i Ehsân be Ali be-zur*
 apartment rent AUX.FUT.3SG give Ehsan to Ali with-force
⟨âpârtêmân eďzâre xâhad dâd⟩_i
- f. *be-zur âpârtêmân eďzâre xâhad dâd_i Ehsân be Ali*
 with-force apartment rent AUX.FUT.3SG give Ehsan to Ali
⟨be-zur âpârtêmân eďzâre xâhad dâd⟩_i

To account for the gramamtical movement possibility in (139f), I will first derive the neutral case in (139a). I start with the previously laid out assumptions; (i) the NV predicate (*eďzâre* 'rent') and its arguments (*be Ali* 'to Ali' and *âpârtêman* 'apartment') are introduced in the extended projection (XP) of the predicate, which is further selected for by the light verbal v head *dâd* 'give'; (ii) the external argument 'Ehsan' is introduced externally in VoiceP; (iii) the auxiliary (Future Aux *xâstan*) selects for VoiceP; and (iv) the PredP merges above the AuxP and licenses the NV predicate locally in its specifier.

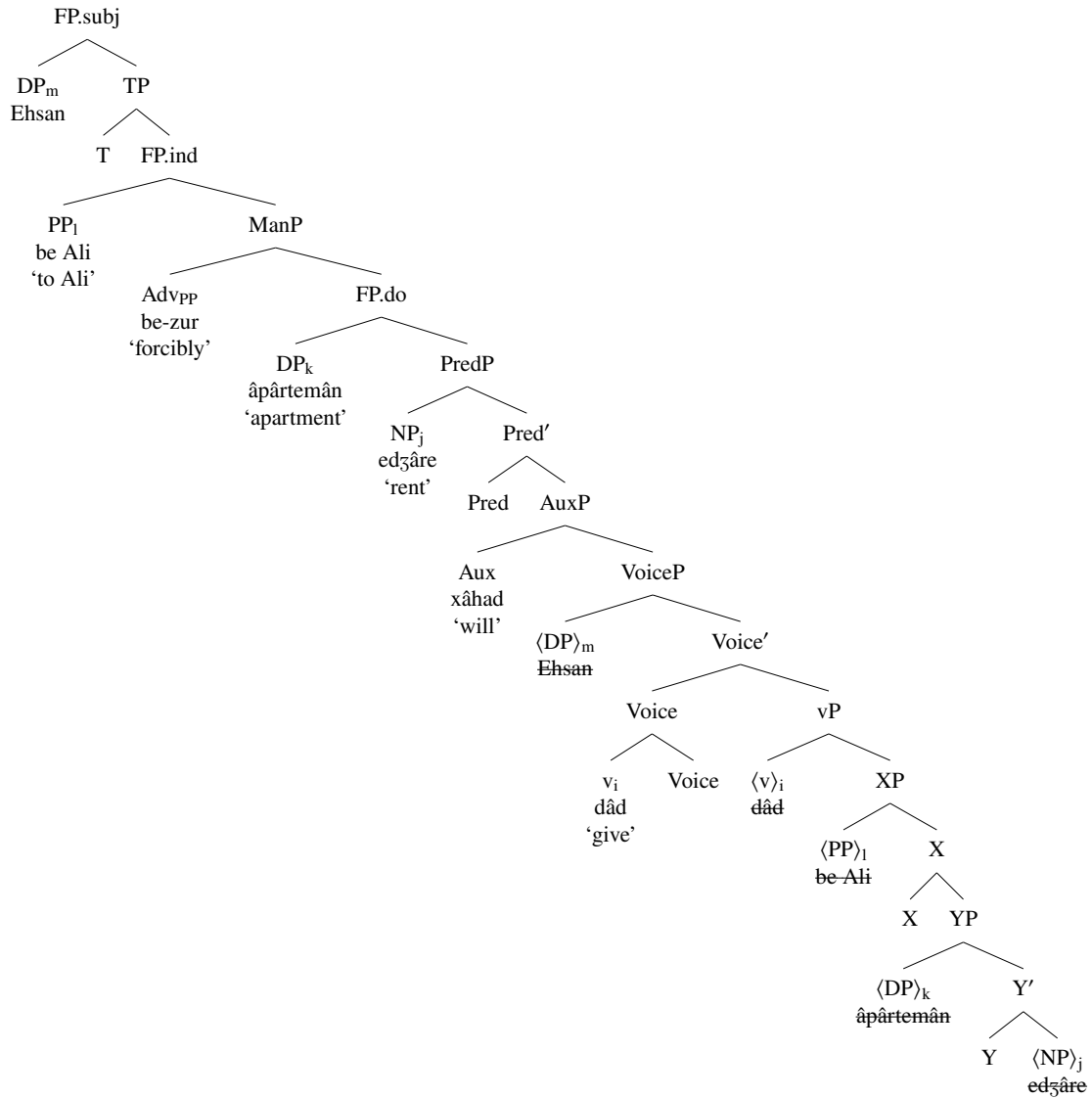


In Kahnemuyipour (2009, p. 89), vP is argued to be the maximal projection that moves in “VP-fronting”. Since manner adverbs must front with the verb and cannot be stranded (as evident in (139e)), Kahnemuyipour proposes that these adverbs originate in a ManP projection below vP.

(141) Persian: . . . [_{VP} t_{subj} v [_{ManP} Adv_{Man} [_{AspP} DP_{obj} Asp [_{VP} V t_{obj}]

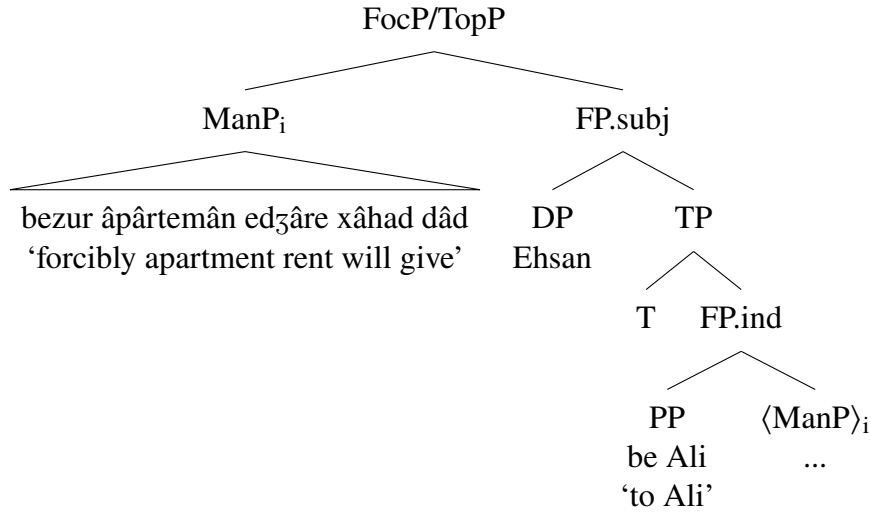
In my analysis however the ManP merges above the Functional Projection to which the direct object always shifts to and below the derived position of the indirect object PP. The rest of the derivation proceeds and the subject is subsequently licensed in the FP.subj.

(142)



The next step of deriving (139f) therefore involves the movement of the ManP to a left-peripheral Focus/Topic position, as suggested in (143):

(143)



2.7 Conclusion

In this chapter, I’ve argued that the widely adopted VP/vP head final analyses for Farsi run into some theoretical inconsistencies, which were highlighted in the discussion of Rasekhi’s (2018) verb-stranding vP ellipsis and Karimi’s (2005) optional verb fronting/scrambling in sentences involving Farsi complex predicates and the Future auxiliary verb in particular. I showed that relying on verb movement alone as the source behind ellipsis and fronting runs into issues, such as the need to (i) adopt both leftward and rightward movement and adjunction to both the left and the right of licensing heads, (ii) resort to head-excorporation of the sort that requires additional stipulations under the head-adjunction theory of excorporation in the first place, and (iii) posit rules for adjoining non-heads to heads. I’ve proposed that all phrases in Farsi start from a base Spec-Head-Comp (S-H-C) (Kayne 1994) configuration and that the mixed word orders are derived from syntactic movements result in configurations conforming S-H-C. By deriving complex predicates through consistent licensing of phrases in designated functional projections, I’ve shown that verbal elements become clusters contained under maximal XP projections through the course of the derivation. As a result, both ellipsis and verb preposing are analyzed as instances of XP movements to Focus and Topic positions.

The proposed antisymmetric framework building on Kayne (1994) has the advantage of elimi-

nating the need to appeal to excorporation, an operation that is at odds with standard assumptions about head movement, violates locality constraints, and when adopted is highly limited/restricted. The current theory further does away with mixed head directionality, given that all movement operations (head and phrasal movement) are proposed to be uniformly leftward. The leftward movements adopted in the current theory are shown to derive complex predicates and verbal clusters of various complexities in a systematic way. In particular, the proposal that data from “verb-stranding” ellipsis and Focus/Topic fronting involve certain XP-movements correctly captures the facts of previously unaccounted for cases with auxiliary verbs such as *shodan* ‘become’, *budan* ‘be’, and *xâstan* ‘want/will’. In comparison, the traditional head-final vP analysis is weakened on theoretical grounds for the following reasons. These analyses assume an underlying head-final structure for the verb phrase domain subject to certain scrambling operations, yet they generally assume a base head-initial structure for projections of other lexical categories. Additionally, the apparent asymmetry in the head-adjunction properties of different auxiliaries (e.g., left-adjunction and consistent head-incorporation for the Perfect Aux *budan*, right-adjunction and head-excorporation for the Future Aux *xâstan*) is highly stipulative and does not follow from a principled theory. Furthermore, such theories leave an explanatory gap as to what properties underlie head-adjunction in the context of lexical vPs in contrast to that involved in “verb movement” to left peripheral Topic/Focus phrases for information structure.

The current theory also has a number of lingering disadvantages. It derives the canonical SOV order in neutral contexts from a series of movements whose trigger seems to lack sufficient motivation. Relatedly, while the current theory is more elegant and systematic in its uniform leftward phrasal and limited head movements, it appears to be less economical given the larger number of movement operations assumed. I claim however that the discussion provided in this chapter points to enough issues for the standard analyses to merit a reconsideration of the assumptions about head-finality in Farsi verb phrase domain and to pursue an alternative account. The following chapters aim to address some of the concerns about the licensing operations by presenting additional evidence in favor of such. In doing so, I turn to post-verbal clausal CP arguments in Chapter 3.

CHAPTER 3

THE ISSUE OF THE POSTVERBAL CLAUSAL COMPLEMENTS

In this chapter, I examine the linear order of Farsi sentential CP complements with respect to the predicate and argue that the post-predicate position of the CP complements significantly challenges the base head-final vP analyses. I propose an alternative account for the distribution of postverbal CP arguments that builds on the antisymmetric analysis proposed in Chapter 2. In particular, I argue that clausal and non-clausal arguments alike originate in specifier positions of higher FPs above the non-verbal predicate and below the light verbs and auxiliaries. However, only non-clausal DP and PP arguments move leftward to licensing positions located in the functional domain above the predicate, and CP arguments remain in their base-generated position. To show how this account gains advantages over a standard head-final verb phrase analysis, I will continue to utilize constructions with complex verbs and bigger verbal clusters to highlight several lingering issues for the previous analyses. In Section 3.1 I review and discuss how the base-generation hypotheses for the merged position of CP arguments (such as Aghaei 2006; Karimi 2005) cannot be maintained in light of constructions where the predicate is larger than a simplex verb. To maintain such hypotheses, some theory of rightward extraposition of the CP must be assumed. Section 3.2 shows that rightward extraposition ultimately runs into the issues raised in favor of the base-generation accounts, such as the transparency of the CP to subextraction (as argued in Aghaei 2006; Darzi 1996; Karimi 2005; a.o) and binding (as argued in Aghaei 2006; Darzi 1996; a.o). In Section 3.2.2, I show that elliptical constructions in which CPs are deleted provide further evidence that CPs are interpreted in a low structural position. Section 3.3 looks at some previous alternative antisymmetric approaches. I argue that these accounts, while in the same spirit as the framework adopted here, are still insufficient in accounting for cases where the verbal clusters contain other elements such as the auxiliary verbs. In Section 3.4, I offer an alternative analysis that systematically derives the linear ordering of clausal arguments with respect to other elements in the structure and accounts for the subextraction and binding facts, utilizing only leftward movements and eliminating the need to resort to rightward extraposition. Lastly, Section 3.5 concludes.

3.1 Base-generation hypotheses

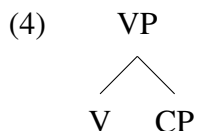
As discussed in Chapter 2, Farsi is an SOV language and has widely been analyzed as head-final in the verb phrase with verbs selecting their complements to the left, and head-initial in all other categories (i.e NPs, PPs, CPs) where heads select their complements to the right. The head finality facts are repeated below in (1) and (2), where the predicate ‘give’ follows the direct object ‘money’ and the indirect object ‘to me’ regardless of whether the direct object is [+/- specific].

- (1) *Ehsân be man pul dâd.*
Ehsan to me money gave.3SG
“Ehsan gave money to me.”
- (2) *Ehsân pul-râ be man dâd.*
Ehsan money-DOM to me gave.3SG
“Ehsan gave the money to me.”

Unlike non-clausal arguments, however, it has widely been observed in the literature that sentential arguments of verbs always surface post-verbally. As seen in (3a) and (3b), the CP *ke Kimea xune nist* ‘that Kimea isn’t home’ must follow the simplex verb *goftan* ‘tell/say’.

- (3) a. *Ehsân be man goft [ke Ali xune n-ist].*
Ehsan to me told/said.3SG that Ali home NEG-is.3SG
“Ehsan told me that Ali isn’t home.”
- b. **Ehsân be man [ke Ali xune n-ist] goft.*
Ehsan to me that Ali home NEG-is.3SG told/said.3SG

One approach taken to account for the post-verbal CP arguments assumes that the CP merges as a right complement to the verb (Aghaei 2006; Darzi 1996; Karimi 2005; among others). Such analyses maintain a head-final verb phrase structure for Farsi where verbs take their non-clausal arguments to the left whereas they select for CPs to the right. Aghaei (2006) argues for the base-generation analysis schematically represented in (4) based on (i) the ability of the complementizer to be covert due to the CP being governed by the V head, and (ii) the possibility to extract out of such CPs, as shown in (5).



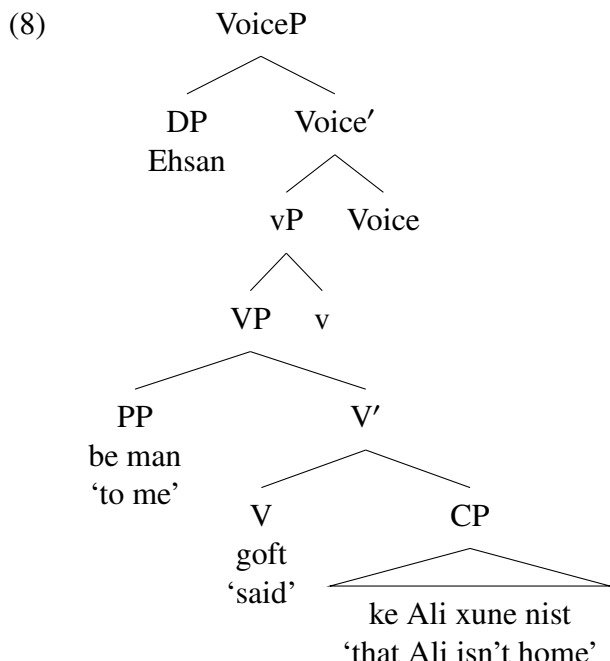
- (5) *ki_i fekr mi-kon-i [(ke) <ki>_i fardâ mosâbeghe-ro be-bar-e]*
 who thought IPFV-do.2SG (that) tomorrow competition-DOM SBJV-win.3SG
 “Who do you think will win the game tomorrow?” (Aghaei 2006, p. 128)

Aghaei further discusses cases in which there is a c-command relationship between a DP and CP when they are co-arguments of the verb, suggesting that the CP stays low. The variable binding data in (6) shows that the QP *har bazikoni* ‘every player’ takes scope and c-commands into the CP and therefore binds the pronoun *un* ‘he/she’. Additionally, (7) reveals that the pronoun *u* ‘him’ cannot c-command and be correferential with the R-expression ‘Ali’ due to a violation of the Condition C of the binding theory.

- (6) *dâvar-â [be har bâzikon-i]_i ghol dâd-an [ke faghat [‘un]_i bâzi-ro be-bar-e].*
 referee-PL to every player-INDEF promise gave.3PL that only he/she game-DOM
 SBJV-win.3SG
 “The referees promised to every player that he/she only will win the game.”
 (Aghaei 2006, p. 157)

- (7) **dâvar [be u]_i goft [ke [Ali]_i bâzi-ro mi-bar-e].*
 referee to him said.3SG that Ali game-DOM IPFV-win.3SG
 “The referee told him that Ali will win the game.”
 (Aghaei 2006, p. 157)

The transparency of the clausal argument to extraction in data such as (5) is also raised in Darzi (1996), Karimi (2005), and others as an argument in favor of the post-verbal base-generation hypothesis and against a rightward extraposition operation. Karimi further argues that the CP does not merge pre-verbally and its post-verbal position cannot be the result of leftward V-to-T movement, because such movement does not obligatorily take place in other constructions like the ones in (1) and (2) where the internal arguments are non-clausal. The application of Karimi and Smith’s (2020) phrase structure and Karimi’s (2005) proposal to the CP data in (3a) therefore derives the following structure in (8).



While Karimi's (2005) base-generation approach for the post-verbal CPs seems to account for cases such as (8), it cannot be straightforwardly extended to sentences with Complex Predicates and clusters of even larger sizes. We have already seen that Farsi has a wide range of complex verbs formed by a Non-Verbal (NV) predicate and a Light Verb (LV), and it was shown previously that such predicates follow their non-clausal arguments as well (as evident from (9) and (10)).

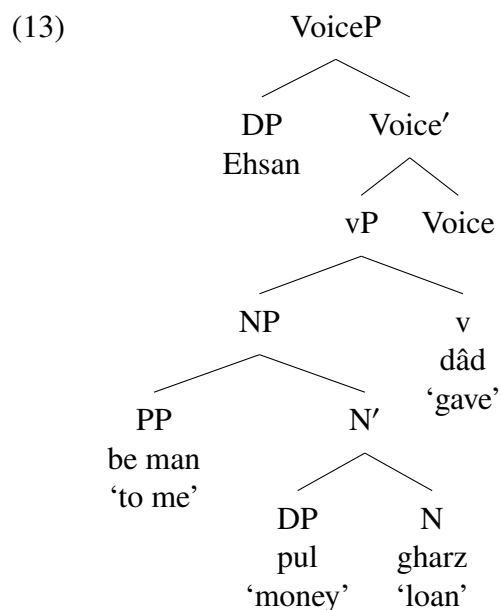
- (9) *Ehsân be man pul gharz dâd.*
 Ehsan to me money loan gave.3SG
 "Ehsan lent Money to me."

- (10) *Ehsân pul-o be man gharz dâd.*
 Ehsan money-DOM to me loan gave.3SG
 "Ehsan lent the money to me."

Farsi is consistent with respect to the post-verbal position of CPs whether the predicate is of the simplex or the complex type. In (11) and (12), the sentential complements *ke ketâb-e Ali-ro be-xune* 'to read Ali's book' and *ke ketâbe Ali jâleb-e* 'that Ali's book is interesting' must follow the full complex predicates *ghol dâdan* 'to promise' and *ghâne' kardan* 'to convince', respectively, and they can neither precede the predicate nor intervene between its NV and LV components.

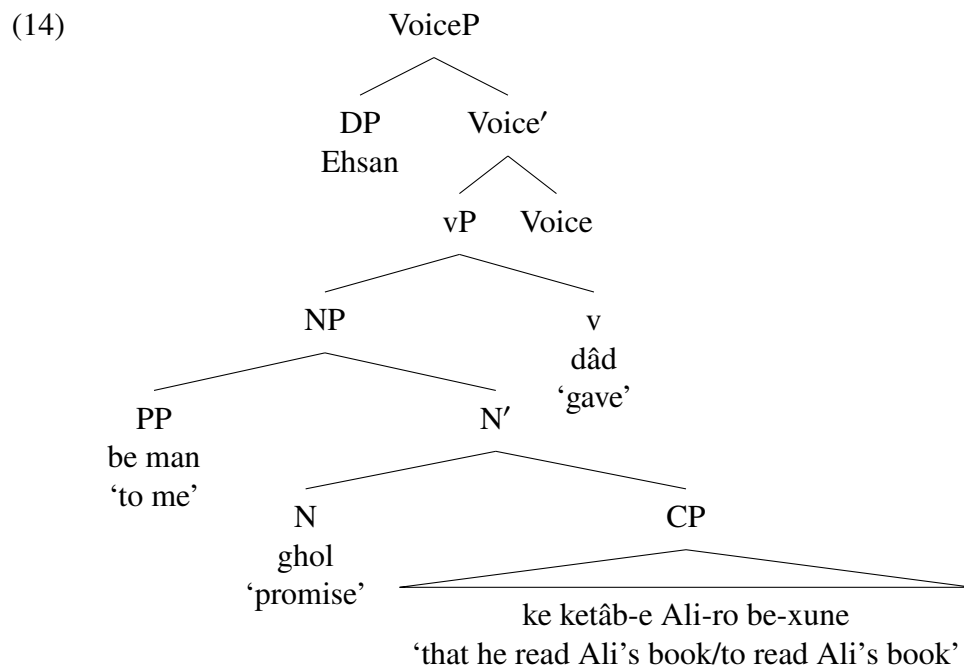
- (11) a. *Ehsân be man ghol dad* [ke ketâb-e Ali-ro *be-xune*].
 Ehsan to me promise gave.3SG that book-Ezf Ali-DOM SBJV-read.3SG
 “Ehsan promised me to read Ali’s book.”
- b. **Ehsân be man* [ke ketâb-e Ali-ro *be-xune*] *ghol dâd*.
 Ehsan to me that book-Ezf Ali-DOM SBJV-read.3SG promise gave.3SG
- c. **Ehsân be man ghol* [ke ketâb-e Ali-ro *be-xune*] *dâd*.
 Ehsan to me promise that book-Ezf Ali-DOM SBJV-read.3SG gave.3SG
- (12) a. *Ehsan man-o ghane’ kard* [ke ketab-e Ali *jaleb-e*].
 Ehsan me-DOM convinced did.3SG that book-Ezf Ali interesting-is.3SG
 “Ehsan convinced me that Ali’s book is interesting.”
- b. **Ehsan man-o* [ke ketab-e Ali *jaleb-e*] *ghane’ kard*.
 Ehsan me-DOM that book-Ezf Ali interesting-is.3SG convinced did.3SG
- c. **Ehsan man-o ghane’* [ke ketab-e Ali *jaleb-e*] *kard*.
 Ehsan me-DOM convinced that book-Ezf Ali interesting-is.3SG did.3SG

As was discussed in Chapter 2, Folli et al.’s (2005) and Karimi’s (2005) head-final vP account for complex predicates argues for the treatment the NV predicate as the head of its own small clause projection that introduces the internal arguments, while the LV heads a higher vP and selects for the NV. Applying this to (9) yields the structure in (13).



Given the proposal that Non-Verbal predicates select their arguments like simplex Vs do, and adopt-

ing the CP base-generation hypothesis, clausal complements are therefore wrongly predicted to surface to the right of the NV predicate and intervening between the NV and the LV, yielding the structure in (14) for the data in (11). There is, however, no plausible verb movement in this structure that would derive the only permitted order NV - LV- CP.



The distribution of the CP with respect to complex verbs is a puzzle for Darzi (1996) and Aghaei (2006) as well. Both Darzi and Aghaei include constructions containing complex predicates, but these predicates are either represented as single V nodes in Darzi (1996) or they are not given a structural account in Aghaei (2006). Given the evidence for the separability of the components of the complex predicate and their XP status, it remains a question as to how the linear order facts with CPs are accounted for under their analyses. The base-generation analysis runs into further complications when bigger verbal clusters are considered. In sentences that consist of a complex verb in addition to an auxiliary verb, CP arguments of verbs consistently surface to the right of all the elements in the verbal cluster. This is observed for the Perfect Aux *budan* ‘to be’ in (15a), where the clausal argument *ke ketâb-e Ali-ro bexune* follows the NV-LV-Aux sequence *ghol dâde ast* ‘has promised’. It is otherwise infelicitous for the CP to surface to the left of the Aux in (15b), to the left of the LV in (15c), or the left of the entire verbal cluster in (15d).

- (15) a. *Ehsân be man ghol dad-e ast [ke ketâb-e Ali-ro be-xune].*
 Ehsan to me promise give-PTCP AUX.PRF.3SG that book-Ezf Ali-DOM
be-xune.
 SBJV-read.3SG
 “Ehsan has promised me to read Ali’s book.”
- b. **Ehsân be man ghol dâd-e [ke ketâb-e Ali-ro be-xune]*
 Ehsan to me promise give-PTCP that book-Ezf Ali-DOM SBJV-read.3SG
ast.
 AUX.PRF.3SG
- c. **Ehsân be man ghol [ke ketâb-e Ali-ro be-xune] dâd-e*
 Ehsan to me promise that book-Ezf Ali-DOM SBJV-read.3SG give-PTCP
ast.
 AUX.PRF.3SG
- d. **Ehsan be man [ke ketâb-e Ali-ro be-xune] ghol dad-e*
 Ehsân to me that book-Ezf Ali-DOM SBJV-read.3SG promise give-PTCP
ast.
 AUX.PRF.3SG

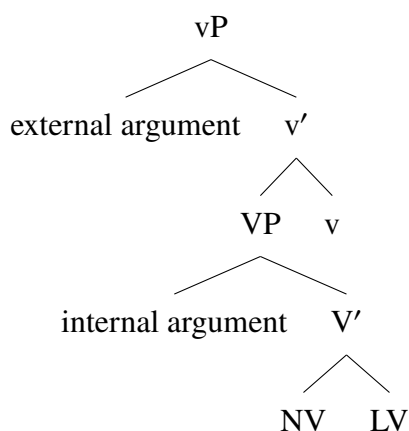
The same observation holds for the Future Aux constructions in (16). As (16a) shows, the clause surfaces to the right of the NV-Aux-LV cluster *ghol xâhad dâd* ‘will promise’; it may not precede the LV in (16b), precede the Aux in (16c), or precede the entire cluster in (16d).

- (16) a. *Ehsân be man ghol xâhad dâd [ke ketâb-e Ali-ro be-xune].*
 Ehsan to me promise AUX.FUT.3SG give that book-Ezf Ali-DOM SBJV-read.3SG
 “Ehsan will promise me to read Ali’s book.”
- b. **Ehsân be man ghol xâhad [ke ketâb-e Ali-ro be-xune] dâd.*
 Ehsan to me promise AUX.FUT.3SG that book-Ezf Ali-DOM SBJV-read.3SG give
- c. **Ehsân be man ghol [ke ketâb-e Ali-ro be-xune] xâhad dâd.*
 Ehsan to me promise that book-Ezf Ali-DOM SBJV-read.3SG AUX.FUT.3SG give
- d. **Ehsân be man [ke ketâb-e Ali-ro be-xune] ghol xâhad dâd.*
 Ehsan to me that book-Ezf Ali-DOM SBJV-read.3SG promise AUX.FUT.3SG give

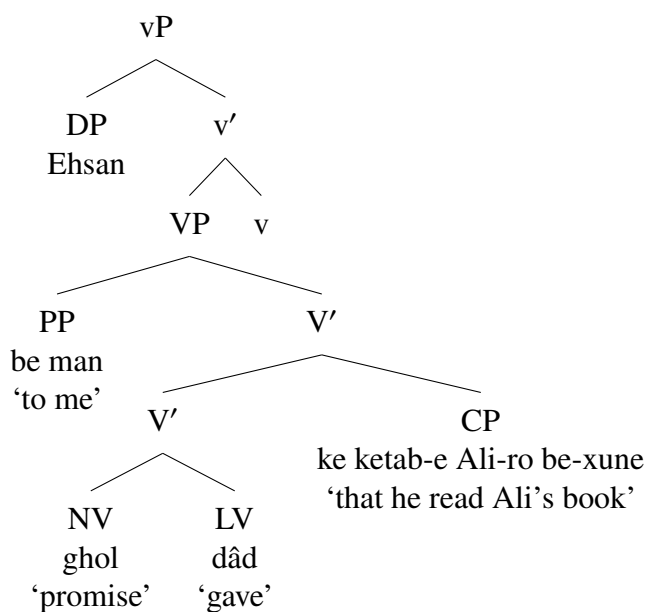
Again if we assume that clausal complements originate and remain in-situ as the right hand sister of their selecting predicates (either a simplex V or a NV predicate), then the CP should be able to surface inside a verb cluster, contrary to fact.

It should be clear that the treatment of complex predicates as a constituent, as proposed by Rasekhi (2018) and discussed in Chapter 2, similarly falls short in adequately explaining the puzzle of the consistent final position of CP arguments. Under Rasekhi's (2018) complex predicate structure in (17), the complex verb constituent can be proposed to select for a CP argument to the right and therefore accounting for the data in (12), as suggested in (18).

(17) (Rasekhi 2018, p. 86)



(18)



However, this structure fails to account for the auxiliary verb constructions in (15) and (16) as it wrongly predicts that the auxiliaries follow the CP, if the CP stays in its suggested base-generated position.

(19) * [AuxP [vP Ehsân [VP [PP be man] ghol dâd-e [CP ke ketâb-e Ali-ro be-xune]]] ast]

- (20) * [_{AuxP} [_{VP} Ehsân [_{VP} [_{PP} be man] ghol dâd [_{CP} ke ketâb-e Ali-ro be-xune]]] xâhad]

This shows that a base-generation hypothesis operating under a head-final vP fails to hold consistently unless some theory of rightward extraposition of the clausal complement is assumed. Whenever Aux heads are involved, the CP must right-adjoin to a position outside of just the smallest vP that contains the component of a complex verb because the auxiliaries are expected to merge hierarchically higher. In the following section, I argue that rightward extraposition as an operation is similarly inadequate as it fails to explain the same transparency to extraction and scope facts discussed in this section, making the issues for the base-generation and the extraposition approaches circular. However, additional empirical evidence from ellipsis will be provided to suggest that CP complements must be low relative to the predicate and the nominal arguments of the predicate.

3.2 Against rightward extraposition

3.2.1 More on transparency to movement and binding

It was shown in Section 3.1 that the proponents of the base-generation analyses argue against a rightward extraposition account partly based on empirical evidence for the possibility to extract elements out of sentential complements. Additional examples highlighting the CP's transparency are given in (21-23), to show that this fact holds in auxiliary containing complex clusters in (22) and (23) as well.

- (21) *un ketâb-o_i man be Ehsân pishnahâd kardam* [*ke* *⟨un ketâb-o⟩_i be-tadrij*
that book I to Ehsan suggestion did.1SG that gradually
be-xune].
SBJV-read.3SG

“As for that book, I recommended that Ehsan reads it gradually.”

- (22) *un ketâb-o_i man be Ehsân pishnahâd kard-e am* [*ke* *⟨un ketâb-o⟩_i be-tadrij*
that book I to Ehsan suggestion do-PTCP AUX.PRF.1SG that gradually
be-xune].
SBJV-read.3SG

“As for that book, I have recommended that Ehsan reads it gradually.”

- (23) *un ketâb-o_i man be Ehsân pishnahâd xâham kard ke* [*⟨un ketâb-o⟩_i be-tadrij*
that book I to Ehsan suggestion AUX.FUT.1SG do that gradually

be-xune].

SBJV-read.3SG

“As for that book, I will recommend that Ehsan reads it gradually.”

Scrambling of this kind can target not only arguments, but adjunct XPs as well. This is shown in (24)-(26), where the focalized embedded adjunct *be-tadrij* ‘gradually’ has moved out of the CP.

- (24) *BE-TADRIJ_i man be Ehsân pishnahâd kardam [ke un ketâb-o <be-tadrij>_i*

GRADUALLY I to Ehsan suggestion did.1SG that that book-DOM

be-xune].

SBJV-read.3SG

“I recommended Ehsan to read that book GRADUALLY”

- (25) *BE-TADRIJ_i man be Ehsân pishnahâd karde am [ke un ketâb-o*

GRADUALLY I to Ehsan suggestion do-PTCP AUX.PRF.1SG that that book-DOM

<be-tadrij>_i be-xune].

SBJV-read.3SG

“I have recommended Ehsan to read that book GRADUALLY”

- (26) *BE-TADRIJ_i man be Ehsân pishnahâd xâham kard [ke un ketâb-o*

GRADUALLY I to Ehsan suggestion AUX.FUT.1SG do that that book-DOM

<be-tadrij>_i be-xune].

SBJV-read.3SG

“I will recommend Ehsan to read that book GRADUALLY”

Rightward extraposition accounts therefore still have to deal with CP subextraction, considering the extended discussion in the literature that extraposed clauses should become islands to extraction and Zwart’s (1993) further argument that movement to the right of the verb isn’t motivated for feature checking as no such position is known to exist. Farsi patterns with other SOV languages with respect to the transparency of postverbal clausal complements, such as German (Koster 1987), Dutch (Koster 1987; Zwart 1993), Hindi (Mahajan 1997), among others. This reinforces the larger drawback of a head-final verb phrase analysis needing to stipulate that DP/PP versus CP arguments differ in their direction of selection, while also needing to posit additional movement operations to capture the linear order facts beyond the simplex verb constructions.

We have already looked at evidence from binding (Aghaei 2006) that sentential arguments are in the scope and c-command domain of non-clausal arguments, as highlighted by variable binding

in (27) in and Condition C effects in (28). In (27), the Quantifier Phrase *har dânesâmuz* ‘every student’ binds *ketâb-ash* ‘his book’.¹⁵ The data in (28) shows that the embedded R-expression ‘Ali’ inside the possessive DP *ketâb-e Ali* ‘Ali’s book’ cannot be co-referential with the main clause indirect object pronoun ‘him’ due to a Principle C violation. This is expected if the embedded R-expression is in the c-command domain of the main clause indirect object. Crucially, (27b-27c) and (28b-28c) show that such binding relations also hold when the CP follows larger verbal clusters.

- (27) a. *Parviz be har dânesâmuz_i ghol dâd [ke ketâb-e u_i râ be-xânad].*
 Parviz to every student promise gave-3SG that book-Ezf he DOM SBJV-read.3SG
 “Parviz promised every student to read his (every student’s) book.”
- b. *Parviz be har dânesâmuz_i ghol dâd-e ast [ke ketâb-e u_i râ be-xânad].*
 Parviz to every student promise give-PTCP AUX.PRF.3SG that book-Ezf he DOM SBJV-read.3SG
 “Parviz has promised every student to read his (every student’s) book.”
- c. *Parviz be har dânesâmuz_i ghol xâhad dâd [ke ketâb-e u_i râ be-xânad].*
 Parviz to every student promise AUX.FUT.3SG give that book-Ezf he DOM SBJV-read.3SG
 “Parviz will promise every student to read his (every student’s) book.”
- (28) a. **Parviz be u_i ghol dâd [ke ketâb-e Ali_{*i} râ be-xânad].*
 Parviz to him promise gave-3SG that book-Ezf Ali DOM SBJV-read.3SG
 “Parviz promised him to read Ali’s book.”
- b. **Parviz be u_i ghol dâd-e ast [ke ketâb-e Ali_{*i} râ be-xânad].*
 Parviz to him promise give-PTCP AUX.PRF.3SG that book-Ezf Ali DOM SBJV-read.3SG
 “Parviz has promised him to read Ali’s book.”
- c. **Parviz be u_i ghol xâhad dâd [ke ketâb-e Ali_{*i} râ be-xânad].*
 Parviz to him promise AUX.FUT.3SG give that book-Ezf him DOM SBJV-read.3SG
 “Parviz will promise him to read Ali’s book.”

¹⁵Because ‘every student’ appears to be inside a PP, I assume that the QP can bind out of the PP; this is perhaps due to the P being a case marker or something of the sort, or due to the QP being in [Spec,PP] and the specifier of a specifier being able to c-command and bind into the sister of the PP (Kayne 1994).

Since PP arguments are generally analyzed as VP in-situ in VP/vP head-final analyses, rightward extraposition of CP in sentences like the ones in (27) and (28) would position the CP outside of the c-command domain of the PP argument. In the auxiliary constructions in particular, CPs cannot be argued to adjoin as low as VP or vP, and they must extrapose to a position to the right of the clusters *ghol dâde ast* ‘has promised’ and *ghol xâhad dâd* ‘will promise’ and outside of the c-command domain of the (in)direct object. Movement to a position as high as the auxiliaries would place the CP outside of the c-command domain of the subject as well, which is problematic for the cases in (29a) and (29b) where the same variable binding and Principle C violation facts hold with a subject DP antecedent.

- (29) a. *har dâneseshâmu_i be moallem goft-e (ast) [ke u_i bâyard emtahân-o*
every student to teacher tell-PTCP (AUX.PRF.3SG) that he must test-DOM
pass kone].
pass SBJV-do.3SG
“Every student_i has told the teacher that he_i must pass the test.”
- b. * *u_i be moallem goft-e (ast) [ke Ehsân_i bâyard emtahân-o pass*
he to teacher tell-PTCP (AUX.PRF.3SG) that Ehsan must test-DOM pass
kone].
SBJV-do.3SG
“He_i has told the teacher that Ehsan_i must pass the test.”

Additionally, Darzi (1996) argues that the proponents of the extraposition analyses cannot argue that the binding relations (at least with respect to Principle C) hold prior to CP extraposition, based on the following data in (30).

- (30) a. * *man u-râ_i [be ostâd-e Rezâ_i] moarrefi kardam.*
I him-DOM to professor-Ezf Reza introduction did.1sg
‘To Reza’s professor, I introduced him.
- b. *[be ostâd-e Rezâ_i] man u-râ_i <be ostâd-e Rezâ_i> moarrefi kardam.*
to professor-Ezf Reza I him-DOM introduction did.1sg
‘To Reza’s professor, I introduced him. (Darzi 1996, p. 69)

In (30), the R-expression ‘Reza’ inside the indirect object PP *be ostâd-e Reza* ‘to Reza’s professor’ is originally c-commanded by of the direct object *u* ‘him’. Therefore, the binding of ‘Reza’ by

the pronoun ‘him’ would constitute a Principle C violation before any movement has taken place. Such violation is however obviated in (30) following the raising of the indirect object PP to a position outside of the c-command domain of the direct object pronoun, which is unexpected if the relevant binding conditions were in fact satisfied at an earlier stage in the derivation. At the same time, evidence from binding data of this kind suggesting that movement does not force Principle C reconstruction is not conclusive. Karimi (2005) provides the following pair in (31) to argue that scrambling does not in fact “bleed” a Principle C violation. In (31a), the embedded R-expression *Kimea* is bound by the embedded subject pronoun *u* ‘she’, therefore violating Principle C. The movement of *Kimea* to the matrix clause and out of the c-command domain of ‘she’ in (31b) does not obviate this violation, which contradicts what’s observed in (30b).

- (31) a. * *Sepide fekr mi-kone* [_{CP} *ke* [_{TP} [_{vP} *un_i Kimea-ro_i* [_{PredP} *dust dâre*]]]]
 Sepide thought IPFV-does.3SG that she Kimea-DOM friend has.3SG
 “Sepide thinks that she_i likes Kimea_i.”
- b. * [_{TP} [*Kimea-ro_i*]_j *Sepide fekr mi-kone* [_{CP} *ke* [_{TP} [_{vP} *un_i <Kimea-ro_i>_j* [_{PredP} *dust dâre*]]]]]
 Kimea-DOM Sepide thought IPFV-does.3SG that she
 friend has.3SG
 “As for Kimea, Sepide thinks that she_i likes her_i.”

(Adapted from Karimi 2005, p. 180)

I have no explanation as to what the contrast between (30b) and (31b) stems from. However, given that (i) an obligatory CP extraposition movement is a mystery with respect to its A/A’ status and its consequences on binding relations, and that (ii) the patterns in (27), (28), and (29) uniformly suggest a lower interpretation site for the CP than the verb’s non-clausal arguments, I maintain that binding stands as an argument against the rightward (upward) movement analyses.

3.2.2 CP deletion

Lastly, I revisit the verb-stranding vP ellipsis phenomena discussed in Chapter 2 and extend it to constructions where an entire clause is elided to argue that the structural position of sentential complements must be lower than the predicate, and therefore not in a rightward extraposed position. It was shown by Shafiei (2015, 2016) and Rasekhi (2018) that in data such as (32), arguments that

stand in contrastive relationship with their correlates in the antecedent clause survive ellipsis while everything else is deleted. In (32b), the indirect object PP *be pesaresh* ‘to her son’ (in contrast with *be doxtaresh* ‘to her daughter’) and the complex predicate *hedye dâdan* ‘gift giving’ survive ellipsis, while the subject DP ‘Ayda’ and the direct object DP *ye ketâb* ‘a book’ are deleted.

- (32) a. *Âydâ be dokhtar-esh ye ketâb hedye na-dâd.*
 Ayda to daughter-CL.3SG a book gift NEG.give.3SG
 “Ayda didn’t gift a book to her daughter.”
- b. *Vali Âydâ be pesar-esh ~~ye ketâb~~ hedye dâd.*
 but Ayda to son-CL.3SG ~~a book~~ gift gave.3SG
 “But (Ayda) gifted (a book) to her son.”

Considering now cases in which one of the internal arguments of the verb is a CP clause, the pair in (33) reveals that except for the contrastive external argument *Maryam* and the simplex predicate *goftan* ‘tell’, both the PP *be pesaresh* ‘to her son’ and the CP *ke emruz madâres tati -and* ‘that schools are closed today’ are deleted in (33b).

- (33) a. *Zahrâ be pesar-esh na-goft [ke emruz madâres tatil -and].*
 Zahra to son-CL.3SG NEG-told.3SG that today school closed are.3PL
 “Zahra didn’t tell her son that schools are closed today.”
- b. *vali Maryam ~~be pesar-esh~~ goft [ke ~~emruz madâres tatil~~ -and].*
 but Maryam ~~to son-CL.3SG~~ told.3SG ~~that today school closed are.3PL~~
 “But Maryam told (her son) (that schools are closed today).”

Since Rasekhi analyzes verb phrase ellipsis as an instance of the movement of the contrastive elements and the predicate out of the vP (the elided constituent) to a TP-internal Focus phrase, the elision of the CP arguments must mean that they are trapped inside vP and therefore in a low position. Sentential arguments can also be elided following a complex predicate. The sentence (34b) shows that the CP *ke present konam* ‘that I present’ and the subject ‘Ali’ both get deleted, with the direct object ‘me’ and the predicate *majbur kard* ‘obliged’ as the remnants. So, the same assumption for the low vP-internal position of the clause would need to hold for such cases as well.

- (34) a. *Ali hama-ro majbur kard [ke present konan].*
 Ali everyone-DOM obliged did.3SG that present SBJV-do.3PL

“Ali forced everyone to present/that they present.”

- b. *vali Ali man-o majbur na-kard [ke-present konam].*
 but Ali me-DOM obliged NEG-did.3SG that-present SBJV-do.1SG
 “But (Ali) didn’t force me (to present/that I present).”

Lastly, clausal arguments can be trapped inside the ellipsis site following an even bigger cluster: complex verb + auxiliary. This is true for both the Perfect Aux (35b) and the Future Aux (36b) constructions, where the CP *ke bâham berand mosâferat* ‘that they go on a trip together’ gets elided alongside the subject ‘Hamid’.

- (35) a. *Hamid dust-esh-o hanuz ghâne’ na-kard-e ast [ke bâham*
 Hamid friend-CL.3SG-DOM yet convinced NEG-do-PTCP AUX.PRF.3SG that together
berand mosâferat].
 SBJV-go.3PL trip/vacation
 “Hamid hasn’t yet convinced his friend to go on a trip together.”

- b. *vali Hamid xânom-esh-o ghâne’ kard-e ast*
 but Hamid wife-CL.3SG-DOM convinced do-PTCP AUX.PRF.3SG
[ke bâham berand mosâferat].
 that together SBJV-go.3PL trip/vacation
 “But (he) has convinced his wife (to go on a trip together).”

- (36) a. *Hamid dust-esh-o ghâne’ na-xâhad kard [ke bâham*
 Hamid friend-CL.3SG-DOM convinced NEG-AUX.FUT.3SG do that together
berand mosâferat].
 SBJV-go.3PL trip/vacation
 “Hamid will not convince his friend to go on a trip together.”

- b. *vali Hamid xânom-esh-o ghâne’ xâhad kard*
 but Hamid wife-CL.3SG-DOM convinced AUX.FUT.3SG do
[ke bâham berand mosâferat].
 that together SBJV-go.3PL trip/vacation
 “But (he) will convince his wife (to go on a trip together).”

CPs that follow a verbal cluster involving an Aux head must be assumed to have adjoined higher than the vP, potentially to AuxP or TP. Since Rasekhi analyzes such ellipsis cases as vP deletion, however, the CP cannot have been raised above the vP in (35) and (36). Based on the data discussed here, I argue that the CP’s derived position must be lower than that of the predicate and all the nominal

argument(s) in the structure. This, however, cannot be sufficiently achieved through a rightward extraposition approach.

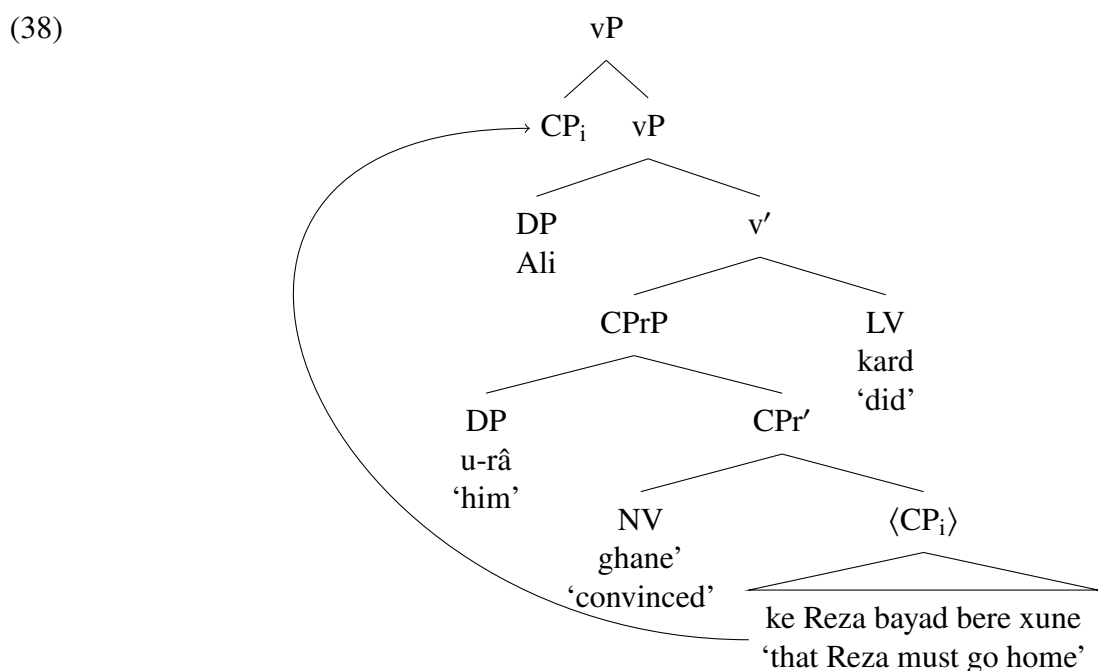
3.3 Some shortcomings of alternative proposals

3.3.1 Remnant movement

Taleghani (2006) considers Karimi's (2001) vP remnant movement analysis for Persian Complex DPs as a potential extension to Farsi sentential complements. By this hypothesis, in complex predicate constructions such as (37), the CP originates as the righthand sister of the NV element and must move to a vP phase edge due to a [-focus] feature as suggested in (38). Taleghani takes this movement to be an instance of object shift, following Holmberg (1999).

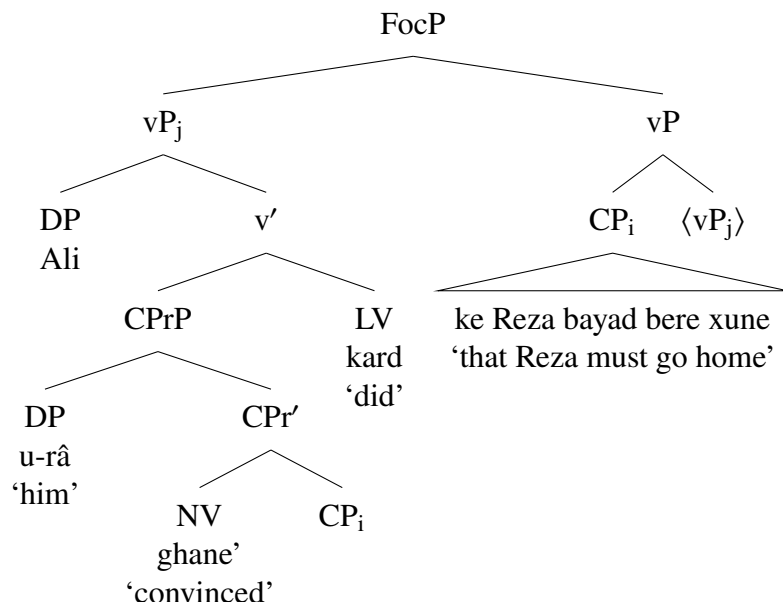
- (37) * *Ali u-râ_i ghâne' kard ke [Rezâ_i bâyard be-re xune].*
 Ali him-DOM convinced did.3SG that Reza should/must SBJV-go.3SG home
 "Ali convinced him that Reza should go home."

(Adapted from Darzi 1996, p. 68)



This movement achieves the NV LV adjacency inside the vP, and it is followed by the subsequent remnant movement of the vP to the specifier of a Focus phrase, yielding the NV-LV-CP order. This is illustrated in (39).

(39)



Despite the observation that the trace of the CP is no longer bound by its antecedent, Taleghani considers antecedent binding having taken place prior to the movement of the CP to the phase edge. However, the vP-remnant movement analysis is questionable for familiar reasons. For one, it lacks sufficient motivation considering the fact that it would only ever take place when the verb takes a CP object. Additionally, it's problematic for the same binding data discussed previously, show again here in (40) and (41). Since the application of vP to FocP raising as suggested in (39) would place the embedded clause arguments out of the c-command domain of the matrix indirect objects (40a) and (41a) and the matrix subjects (40b) and (41b), we would expect the quantifier binding relations in (40) and the Condition C violations in (41) to disappear, contrary to fact.

- (40) a. *Parviz be har dânesâmu_z_i ghol dâd-e ast [ke ketâb-e u_i râ be-xânad].*
 Parviz to every student promise give-PTCP AUX.PRF.3SG that book-Ezf he
 DOM SBJV-read.3SG
 “Parviz has promised every student to read his (every student’s) book.”
- b. *har dânesâmu_z_i be moallem goft-e (ast) [ke u_i bâyard emtahân-o pass kone].*
 every student to teacher tell-PTCP (AUX.PRF.3SG) that he must test-DOM
 pass SBJV-do.3SG
 “Every student_i has told the teacher that he_i must pass the test.”

- (41) a. * *Parviz be u_i ghol dâd-e ast [ke ketâb-e Ali*_i râ*
 Parviz to him promise give-PTCP AUX.PRF.3SG that book-Ezf Ali DOM
be-xânad].
 SBJV-read.3SG
 “Parviz has promised him to read Ali’s book.”
- b. * *u_i be moallem goft-e (ast) [ke Ehsân_i bâyard emtahân-o pass*
 he to teacher tell-PTCP (AUX.PRF.3SG) that Ehsan must test-DOM pass
kone].
 SBJV-do.3SG
 “He_i has told the teacher that Ehsan_i must pass the test.”

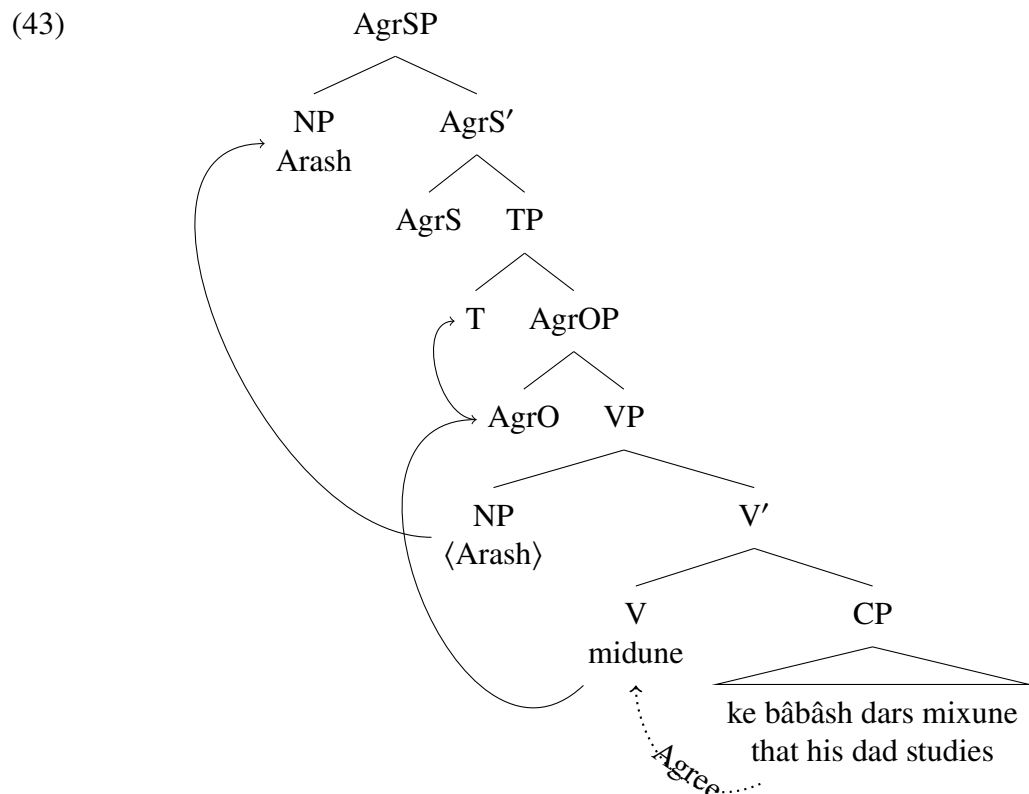
Taleghani’s (2006) vP-remnant movement analysis therefore faces the same challenge as the rightward extraposition accounts when it comes to accounting for the binding facts.

3.3.2 Antisymmetric alternatives

Moinzadeh (2001) analyzes Farsi VPs as head-initial in-line with Kayne’s (1994) LCA and takes the post-verbal position of CP complements as an evidence in favor of such hypothesis. In Moinzadeh’s (2001) analysis, both clausal and non-clausal complements merge to the right of the verb in a consistent fashion; so like many other Kanyean analyses, the OV order with non-clausal arguments is attributed to their leftward movement for case checking while the VO order with CPs is due to them remaining in their base position.¹⁶ On this basis, Moinzadeh derives the data in (42) as shown in (43).

- (42) *Ârash mi-dune [ke bâbâ-sh dars mi-xune]*.
 Arash IPFV-knows.3SG that dad-CL.3SG lesson IPFV-reads.3SG
 “Arash knows that his dad studies”.

¹⁶Moinzadeh (2001) assumes that NP arguments check their features via *Move* in specifier positions of AgrPs (AgrOP for the DO, AgrIOP for the IO, and AgrSP for the Subj) while feature checking with the CP occurs via *Agree* (Chomsky 1998, 2000) in-situ.



Moinzadeh's (2001) derivations represent complex predicates as single V nodes in the structure, and he does not address auxiliary verb containing constructions. Therefore, it is not clear how the proposed analysis deals with the CP linearization puzzles revisited in this chapter so far.

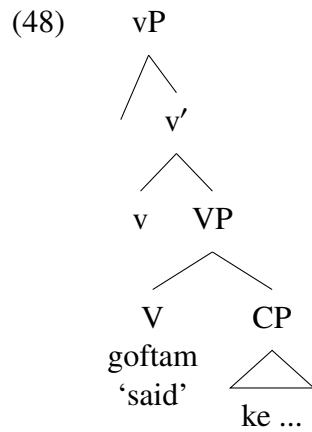
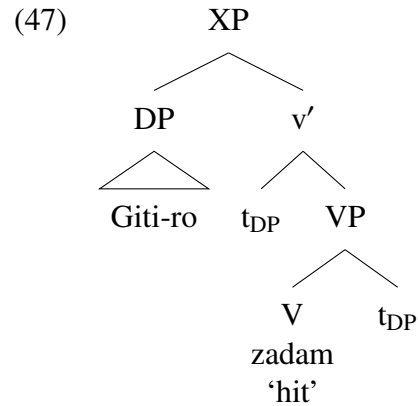
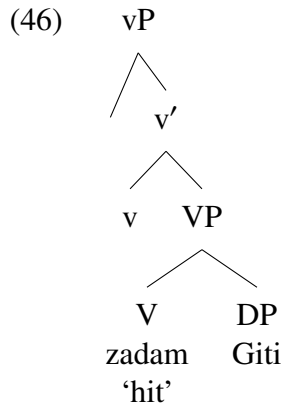
Farudi (2007) also proposes an analysis within the antisymmetric framework of Kayne (1994). In the case of DPs/CPs following a simplex predicate in (44) and (45), respectively, Farudi proposes the structures in (46) and (48). She argues that while the specific *râ* marked DP argument 'Giti' moves for case and even higher to some XP in (47), the CP argument remains to the right of the verb and is interpreted in its base position in (48).

- (44) *man* [_{DP} *Giti-ro*] *zadam*.
 I Giti-DOM hit.1SG
 "I hit Giti."

(Farudi 2007, ex. 51)

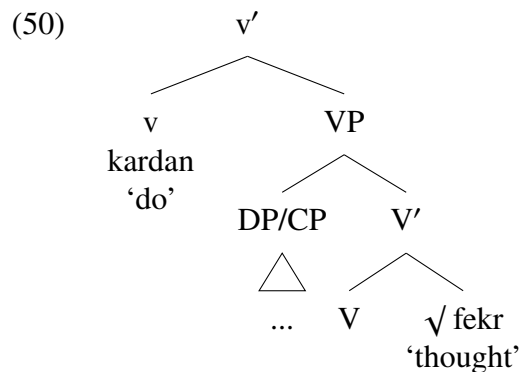
- (45) *man goftam* [_{CP} *ke Giti-ro da'vat kardam*].
 I said.1SG that Giti-DOM invitation did.1SG
 "I said that I invited Giti."

(Farudi 2007, ex. 53)

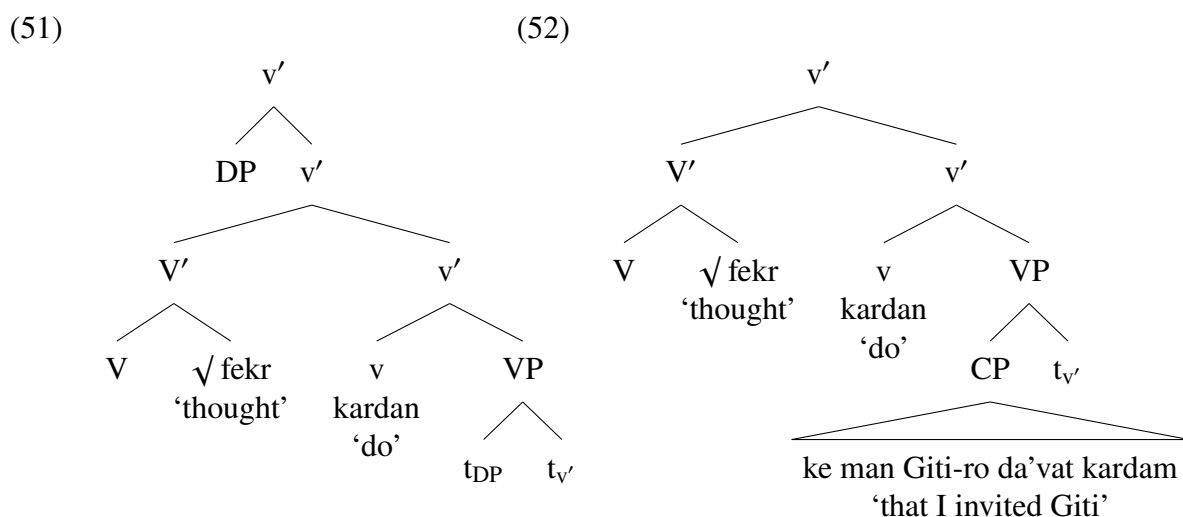


With respect to the CP arguments of complex predicates, as in (49), Farudi (2007) proposes the underlying complex predicate structure in (50). In this structure, it is assumed that the Non-Verbal element (in this case the nominal *fekr* ‘thought’) is a root and combines with a verbalizing head V, resulting in the projection of the VP that introduces the DP/CP argument in is specifier. This VP is in turn selected for by the LV *kardan* ‘to do’ that occupies the v position.

- (49) *Sârâ fekr mi-kone* [_{CP} *ke man Giti-ro da’vat kardam*].
 Sara thought IPFV-does.3SG that I Giti-DOM invitation did.1SG
 “Sara thinks that I invited Giti.”



Farudi further proposes that the V' constituent in (51) containing the verbalizing head and the NV element is probed and moves up to the left of the light verb 'do', independent of DP argument that moves from the specifier VP position to get case. In comparison, the CP remains in the specifier VP position and only the V' projection raises in (52).



Although this analysis derives the post-verbal behavior of CPs with simplex and complex verbs, it similarly fails to address the auxiliary verb constructions discussed in the previous sections. In fact, Farudi (2013) argues that the Aux merges outside of the vP, and as a result, she concludes that the account pursued in Farudi (2007) falls short in accounting for such constructions.

3.4 Restating the Puzzle and the Solution

The discussion of the post-verbal position of Farsi sentential complements, in the context of the widely adopted head-final VP/vP analyses, revisits a puzzle that can be restated as follows: CP arguments show properties of verbal complements, yet it cannot be maintained that they remain

VP-internal under a head-final verb phrase approach. Imposing rightward extraposition complicates things further as it poses its own problems and comes short in accounting for the data discussed in Section 3.2. In this section, I propose an alternative account for the distribution of postverbal CP arguments that builds on the antisymmetric analysis proposed in Chapter 2. In particular, I assume that clausal and non-clausal arguments alike originate in specifier positions of higher FPs above the non-verbal predicate and below the light verbs and auxiliaries. However, only non-clausal DP and PP arguments move leftward to licensing positions located in the functional domain above the predicate, and CP arguments remain in their base-generated position. While this analysis maintains the basic intuition behind base-generation hypothesis criticised above, it rejects the stipulation that the post-verbal behavior of CP arguments as opposed to the preverbal status of DP/PP arguments is governed by a parametric difference in their direction of selection.

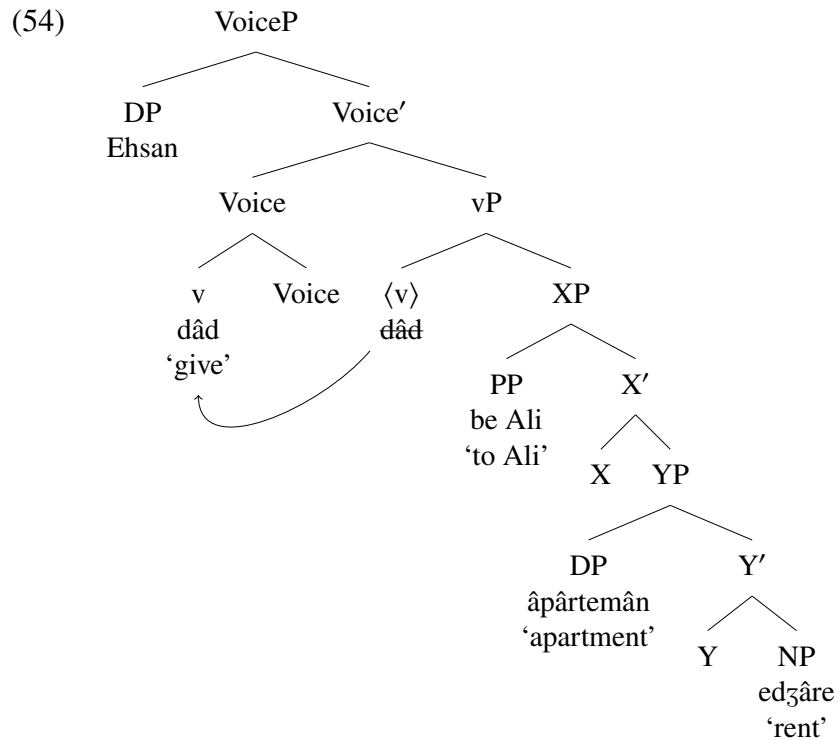
3.4.1 The alternative antisymmetric analysis

I will begin with a brief recap of the analysis proposed in Chapter 2 for deriving complex predicate constructions containing only non-clausal DP and PP arguments, as in (53).

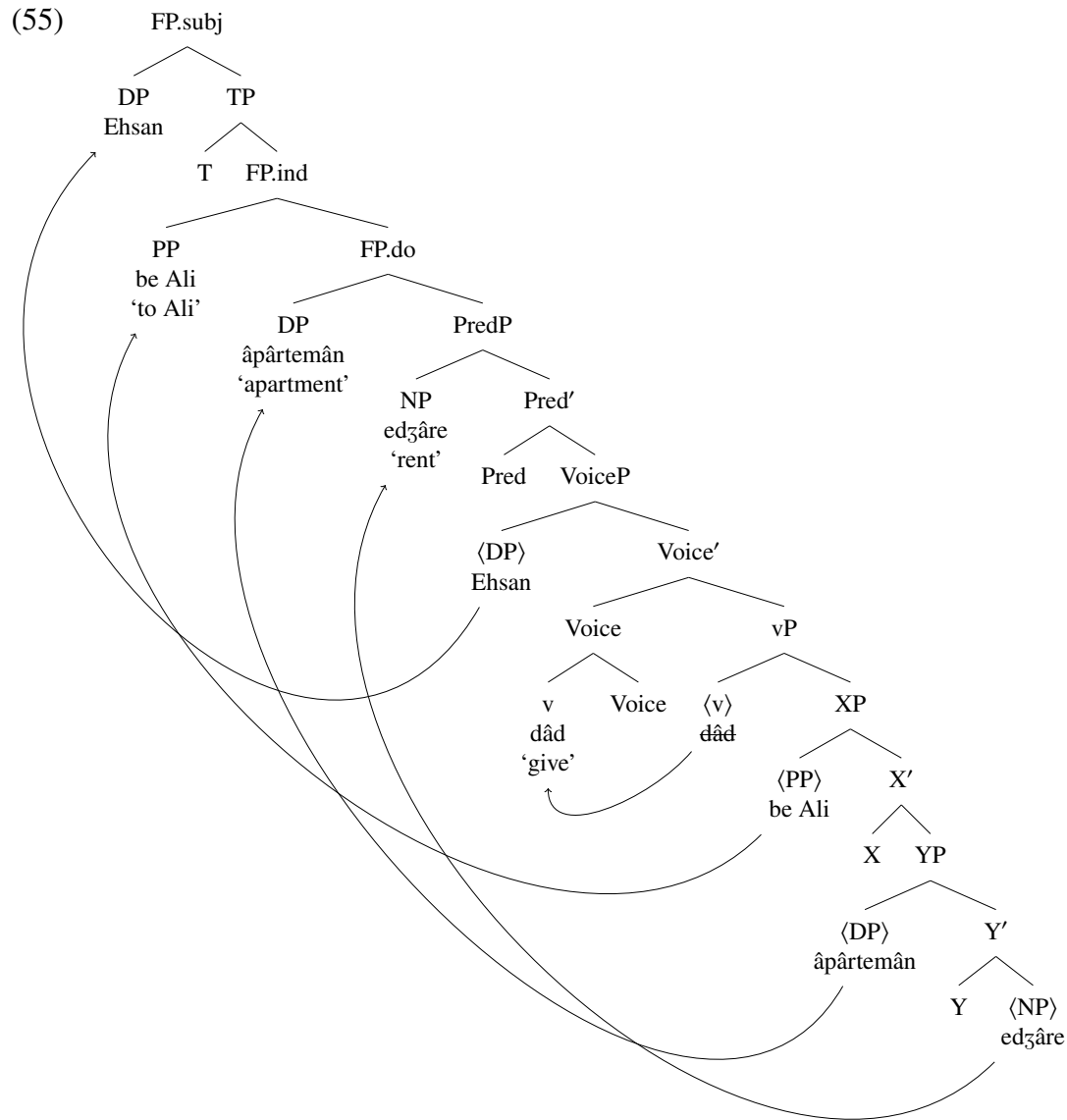
- (53) *Ehsân be Ali âpârtemân ed₃âre dâd.*
 Ehsan to Ali apartment rent gave.3sg
 ‘Ehsan rented houses to Ali.’

Several assumptions were laid out regarding the phrase structure, starting with the proposal that all phrases are head initial (Kayne 1994), and all word orders are derivable from a base hierarchical structure and movement operations that conform to a Specifier-Head-Complement (S-H-C) configuration. This translated to analyzing the derivation of complex predicates and complex verbal clusters in general as involving consistent leftward movements to various licensing Functional Projections. In line with this framework, I proposed that in the base structure of constructions such as (53), the relation between the NV element *ed₃âre* ‘rent’ and its internal arguments *âpârtemân* ‘apartment’ and *be Ali* ‘to Ali’ is captured via merging the arguments in the specifier positions of Functional Projections (labeled YP and XP) above the predicate, as suggested in (54). The X and Y heads are treated as equivalent to Pr⁰ heads (Bowers 1993) or Relator heads (Den Dikken 2006)

mediating the selection and theta role assignment between the predicate and its subject (in this case, the objects). Next, the light verb *dâd* ‘give’ selects for the extended projection of the NV predicate (labeled XP) and heads a vP that is further dominated by the external argument-introducing VoiceP.



I further proposed that the Non-Verbal predicates obligatorily front to a licensing functional projection PredP (Koster 1999, 2000; Zwart 1993, 1996, 1997) located above all other verbal items (light verbs and auxiliaries) and below the derived position of the object(s). Applied to (54), this step raises the maximal NP projection headed by the NV element *edžâre* to [Spec,PredP], as suggested in (67). Lastly, the non-clausal (nominal) internal arguments and external argument are locally licensed in the specifiers of their designated Functional Projections above the PredP.

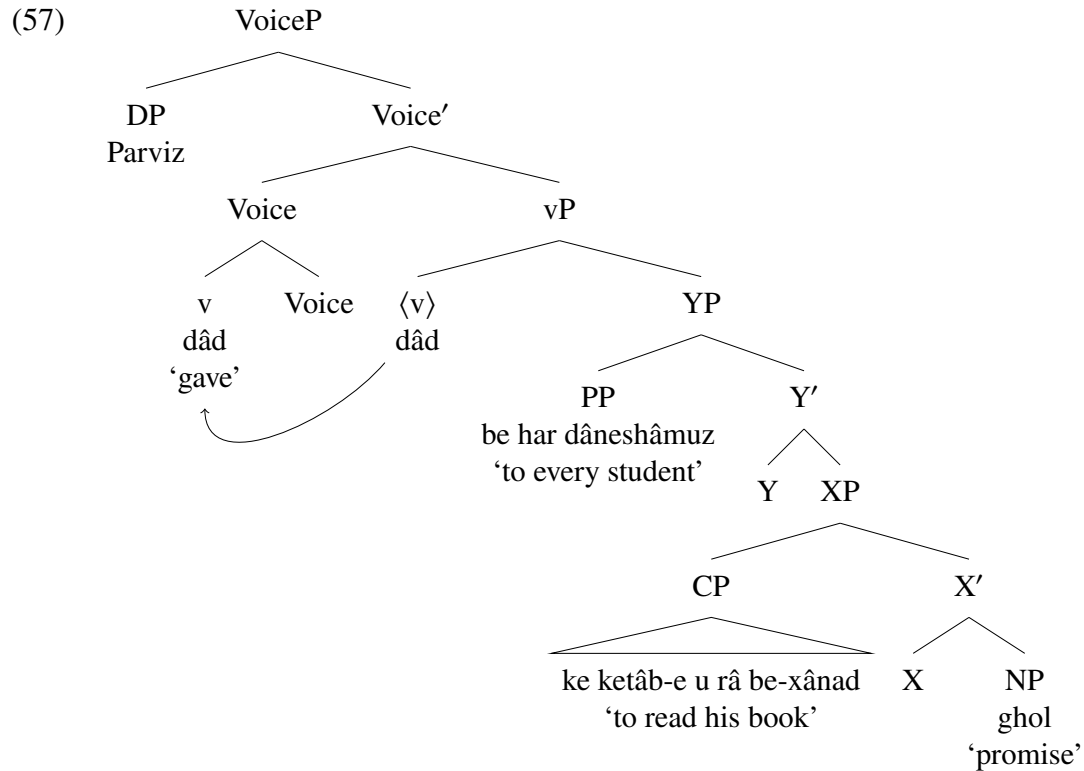


With this analysis in place, I now turn to complex predicate constructions involving sentential arguments, as in (56), and discuss how the postverbal distribution of CPs can be accounted for under the current theory.

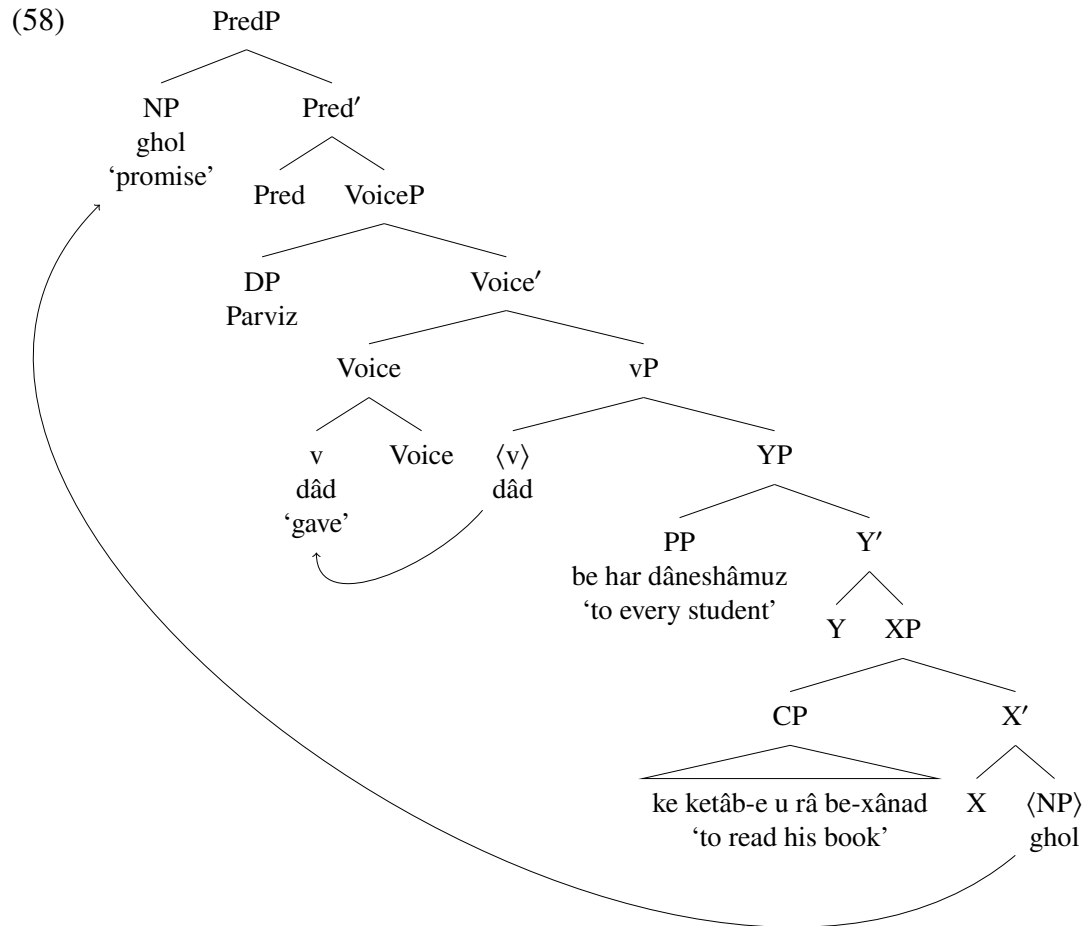
- (56) *Parviz be har dânesâmooz ghol dâd* [ke ketâb-e u râ *be-xânad*].
 Parviz to every student promise gave.3SG that book-Ezf he DOM SBJV-read.3SG
 ‘Parviz promised every student to read his (every student’s) book.’

Consistent with the hypothesis that internal arguments are introduced in the Spec of FPs above the NV predicate, I assume that both the PP *be har dêneshâmuz* ‘to every student’ and the CP *ke ketâb-e u râ be-xânad* ‘to read his book’ in (56) are merged above the NV *ghol* ‘promise’. This eliminates

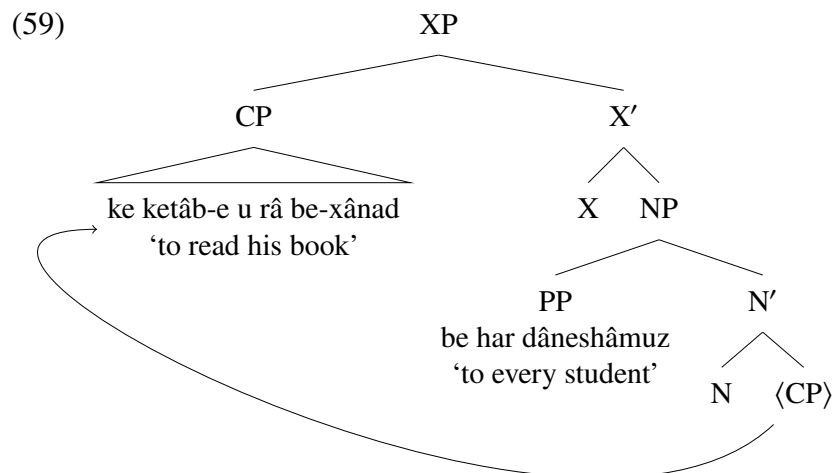
the need to stipulate that the base merge position of the arguments varies depending on their clausal vs. non-clausal status. As before, the light verb merges above the NV element's extended projection and is further selected for by the VoiceP, as suggested in (57).



There is a theoretical advantage to separating the internal arguments from the predicate in this way, as opposed to including them as complements and specifiers of the nominal inside the NP. Given the theory that NV predicates in Farsi must raise to the Spec of the PredP licensing projection above the VoiceP (or AuxPs), this operation as applied to (56) directly moves the NP to this position, as shown in (58).



If we were to instead treat the CP as the complement of the nominal ‘rent’, we would have to stipulate that at least the CP initially evacuates the NP, as suggested in (59), prior to the leftward movement of NP. This would be a necessary step, considering we never find a bare CP argument of the verb in any position to the left of (members of) the predicate, and raising the NP without first extracting out the CP would derive exactly the unattested order. Therefore, some mechanism for stranding the CP in the postverbal region would be required.



Given the observation that such CPs behave like verbal complements with respect to their transparency to subextraction, the initial movement of the CP to XP in (59) poses the question as to why this operation wouldn't turn the clause into an island in the same way that rightward extraposition is argued to do so in Section 3.2. The alternative approach taken in (58) is therefore more conceptually desirable as it avoids this problem altogether.

There is additional empirical evidence in favor of introducing the NV predicates and their internal arguments as I have in (57) and (58). The relevant data concerns complex verb constructions in which the NV element is modified. A few pairs are provided here to illustrate this. Starting with the construction in (56), repeated below as (60a), we observe that the NV predicate *ghol* can be modified by the NP *sharaf* 'honor' in (60b), in which case the modifier follows the NV predicate and the precedes the Light Verb.

- (60) a. *Parviz be har dâreshâmooz ghol dâd* [*ke ketâb-e u râ be-xânad*].
 Parviz to every student promise gave.3SG that book-Ezf he DOM SBJV-read.3SG
 "Parviz promised every student to read his (every student's) book."
- b. *Parviz be har dâreshâmooz ghol-e sharaf dâd* [*ke ketâb-e u râ be-xânad*].
 Parviz to every student promise-Ezf honor gave.3SG that book-Ezf he DOM
be-xânad.
 SBJV-read.3SG
 "Parviz made an honorable promise to every student to read his (every student's) book."

The same ordering facts are observed with the complex predicates *tasmim gereftan* 'decide' (61) and *talâsh kardan* 'make an effort' (62). As (61b) and (62b) show, when the NV elements *tasmim*

‘decision’ and *talâsh* ‘effort’ are modified by *ghatei* ‘certain’ and *ziyâd* ‘much’, the modifiers are directly to the right of the predicates.

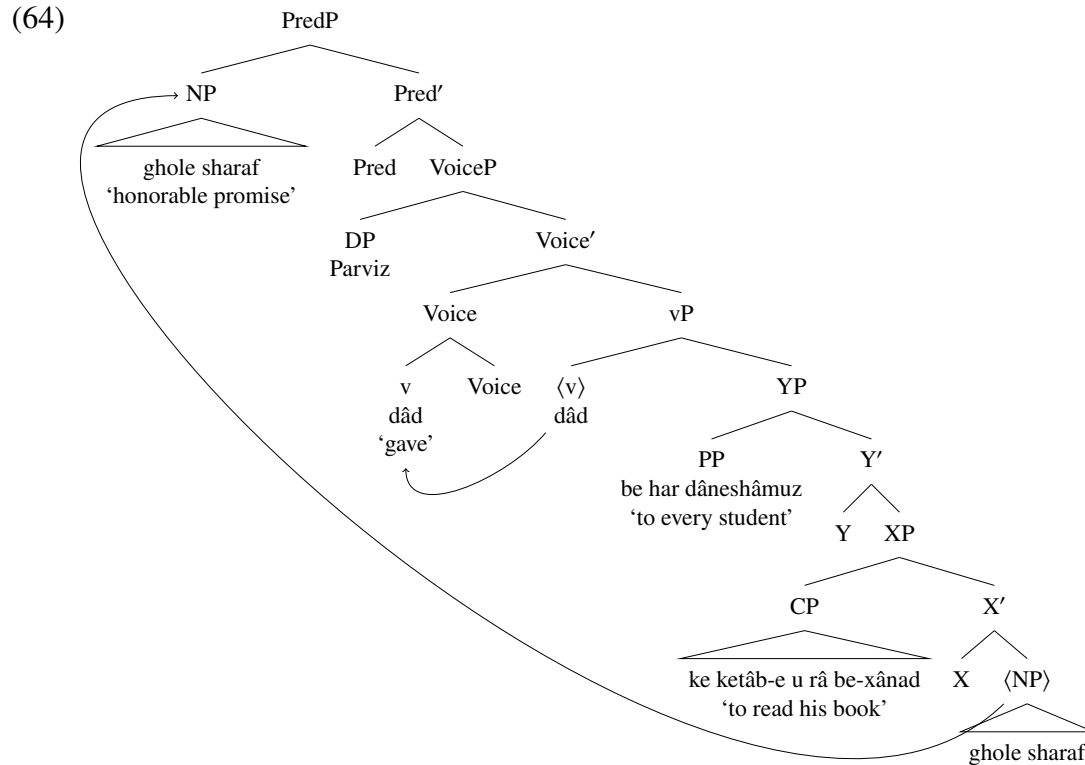
- (61) a. *man tasmin gereftam [ke mashin-o be-forusham].*
 I decision took.1SG that car-DOM SBJV-sell.1SG
 “I decided to sell the car.”
- b. *man tasmin-e ghereftam [ke mashin-o be-forusham].*
 I decision-EZF certain took.1SG that car-DOM SBJV-sell.1SG
 “I made the certain decision to sell the car.”
- (62) a. *dâneshâmuz-an talâsh kardan [ke emtehân-ro pass konan].*
 student-PL effort did/made.3PL that test-DOM pass do.3PL
 “The student made an effort/attempted to pass the test.”
- b. *dâneshâmuz-an talâsh-e ziyâdi kardan [ke emtehân-ro pass konan].*
 student-PL effort-EZF much did/made.3PL that test-DOM pass do.3PL
 “The student made a strong effort to pass the test.”

This is not restricted to subjunctive clause taking predicates. As shown in (63), when the non-verb *shahâdat* ‘testimony’, which subcategorizes for a factive CP, is modified in (63b), the same Predicate-Modifier ordering is observed.

- (63) a. *Ali dar dâdgah shahâdat dâd [ke hamsâye mâshine-e u ro dozdid-e (ast)].*
 Ali in court testimony gave.3SG that neighbor car-EZF he DOM steal-PTCP
 AUX.PRF.3SG
 Ali testified in court that the neighbor has stolen his car.”
- b. *Ali dar dâdgah shahâdat-e dorugh dâd [ke hamsâye mâshine-e u ro dozdid-e (ast)].*
 Ali in court testimony-EZF lie gave.3SG that neighbor car-EZF he DOM
 steal-PTCP AUX.PRF.3SG
 Ali gave a false testimony in court that the neighbor has stolen his car.”

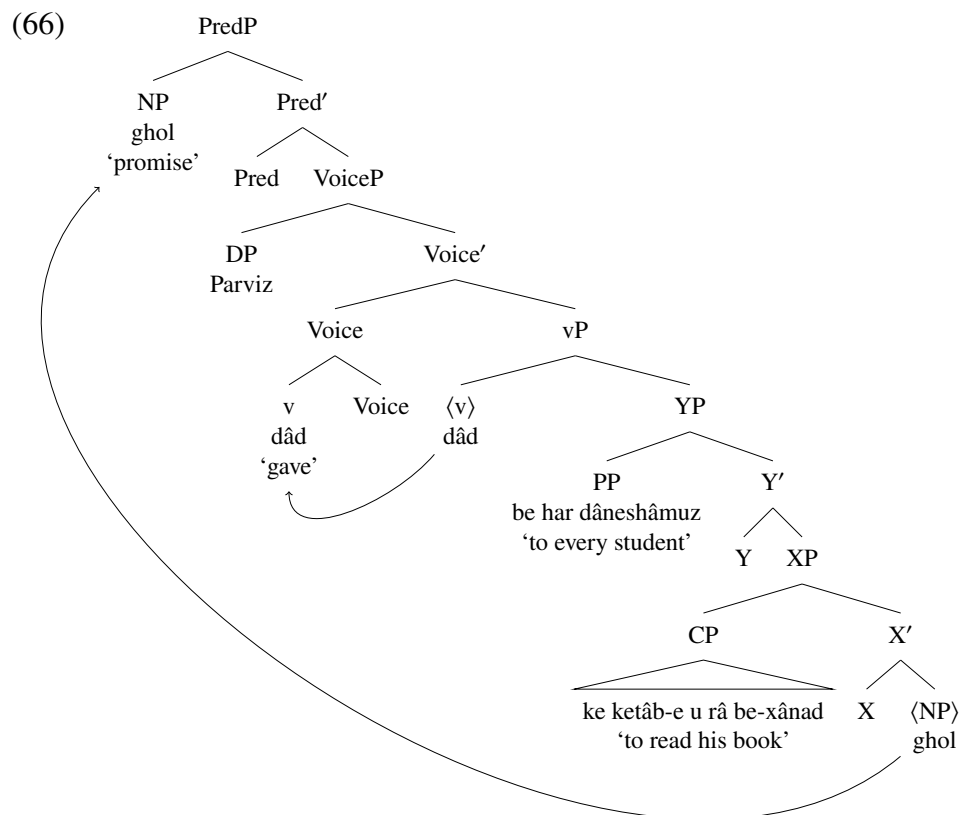
I leave the details of how the non-verbal predicate modification is derived to Chapter 4, where I discuss the puzzle of post-predicate constituents to a larger extent. For the moment, I argue that (60b), (61b), (62b), and (63b) provide sufficient evidence for the phrasal status of the NV predicates, and I make the claim that in such constructions the non-verbal XPs are internally modified prior to

predicating of their CP and PP arguments. Extending this to (60b) yields the structure in (64), which has the observed consequence that the operation licensing the predicate in [Spec, PredP] raises the complex NP constituent containing both the NV element and its modifier. This straightforwardly derives the the correct [NV Modifier LV] order *ghol-e sharaf dâd* .



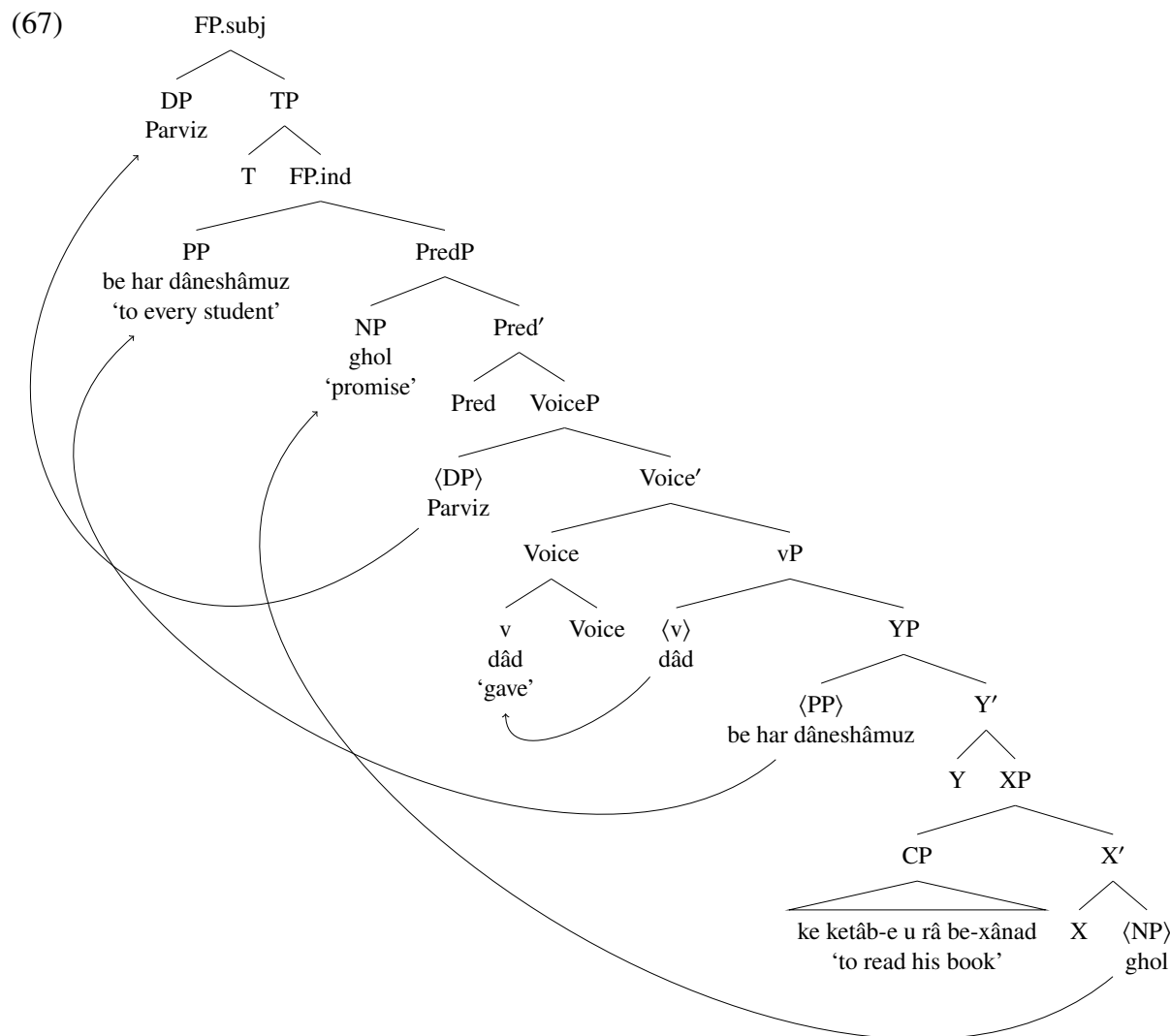
Returning to our original sentence under discussion in (65) and the partial derivation in (66), I will now address how the derivation proceeds to account for the difference in the distribution of the PP and the CP relative to the predicate, considering they are both in a post-predicate position at this stage in the derivation.

- (65) *Parviz be har dânesshâmooz **ghol** dâd [ke ketâb-e u râ **be-xânad**].*
 Parviz to every student promise gave.3SG that book-Ezf he DOM SBJV-read.3SG
 “Parviz has promised every student to read his (every student’s) book.”



I propose that the linear order contrast is driven by the requirement that non-clausal arguments (DPs and PPs) move to be licensed in FPs above the predicate, whereas CP arguments are exempt from this requirement and remain in their base position.¹⁷ This is presumably due to the nominal status of non-CP arguments, which is uncontroversially absent on finite CPs, or perhaps some other factor such as a case requirement or lack thereof for XPs of distinct categories. Assuming this hypothesis is correct, the facts in (65) are captured systematically as follows. The indirect object ‘to every student’ and subject *Parviz* pattern together and are subject to movements to the specifier position of their respective licensing FPs (FP.ind and FP.subj). The CP, on the other hand, remains in its base position and linearizes to the right of the complex predicate. These subsequent steps are shown in (67).

¹⁷The literature generally takes the view consistent with the Case Resistance Principle (Stowell 1981) by which DPs need to be case licensed (and move), while CPs do not. This is the hypothesis pursued in Farudi (2007), where it is argued that in Farsi DPs move to the left of the predicate to be case licensed, and CPs lack case and remain in-situ. While I follow Farudi (2007) in assuming a distinction in the licensing requirements of arguments of different categories, I remain agnostic about whether it is case or something else (such as categorial features themselves) that underlies this distinction.



As I have already alluded to, this analysis departs from previous work that propose sentential arguments (and not just nominal ones) move out of the projection of the predicate and do not remain in-situ (Smith 2024a,b for Mende; Holmberg 2000 for Finnish; Hinterhölzl 2000 for German; Aboh 2004 for Gbe; among others).¹⁸ This is the view previously adopted for Farsi in Ahari (2024),

¹⁸For instance, Holmberg (2000) and Hinterhölzl (2000) assume a short leftward movement of the CP to a functional projection that is lower than the surface position of the verbs and their non-clausal arguments. Holmberg (2000) accounts for the Finnish data in (i) based on the structure in (ib). Here, the goal object *sinulle* and the CP *että hän on lähdössä Pariisiin* originate inside an XP, and the CP has moved to the specifier of an abstract functional head above XP, prior to the selection of the XP by the verb.

- (i) Finnish (Holmberg 2000, ex. 62 & ex. 63)
- a. *Milloin Jussi sinulle kertoi että hän on lähdössä Pariisiin?*
 when Jussi to.you told that he is on.his.way to.Paris
 “When did Jussi tell you that he is on his way to Paris?”
- b. [_{VP} V [_{FP} CP [_{F'} F [_{XP} PP X *t_{CP}*]]]]

where I proposed that CPs originate as the righthand sister to the NV predicate and must raise to a licensing projection that is hierarchically lower than the derived position of predicate and its non-clausal arguments. Within the updated analysis pursued here however, this assumption is no longer necessary as the CP remains its base position and is not subject to the same licensing requirements that is satisfied locally for non-clausal arguments to the left of the verb.

3.4.2 Implications of the alternative analysis for auxiliaries, subextraction and binding

In Section 3.1, I showed that previous base-generation accounts that maintain a head-final verb phrase structure fall short in explaining the behavior of CPs in auxiliary verb constructions, as in (68) and (69). To reiterate, such analyses assume that verbs select for non-clausal complements to the left, while CP complements are selected to the right and remain in this position. This predicts that CPs immediately follow their selecting predicates and precede any additional aspectual and auxiliary heads merging higher in the structure. This is not the case however, based the observation that clauses consistently surface after the entire Verb-Aux sequences *ghol dâde ast* ‘has promised’ in (68) and *ghol xâhad dâd* ‘will promise’ in (69).

- (68) a. *Parviz be har dâneseshâmooz ghol dâd-e ast* [ke ketâb-e u
 Parviz to every student promise give-PTCP AUX.PRF.3SG that book-Ezf he
râ be-xânad].
 DOM SBJV-read.3SG
 “Parviz has promised every student to read his book.”
- b. * *Parviz be har dâneseshâmooz ghol* [ke ketâb-e u *râ be-xânad*]
 Parviz to every student promise that book-Ezf he DOM SBJV-read.3SG
dâd-e ast.
 give-PTCP AUX.PRF.3SG

As mentioned in Chapter 1, Hinterhölzl (2000) argues that the postverbal position of the CP argument in (iia) is derived via raising the CP to the specifier of a Functional Projection (SpecF3P) right above the VP, with the verb also having moved to the head of a F2 above the CP. This is shown in (iib) and (iic).

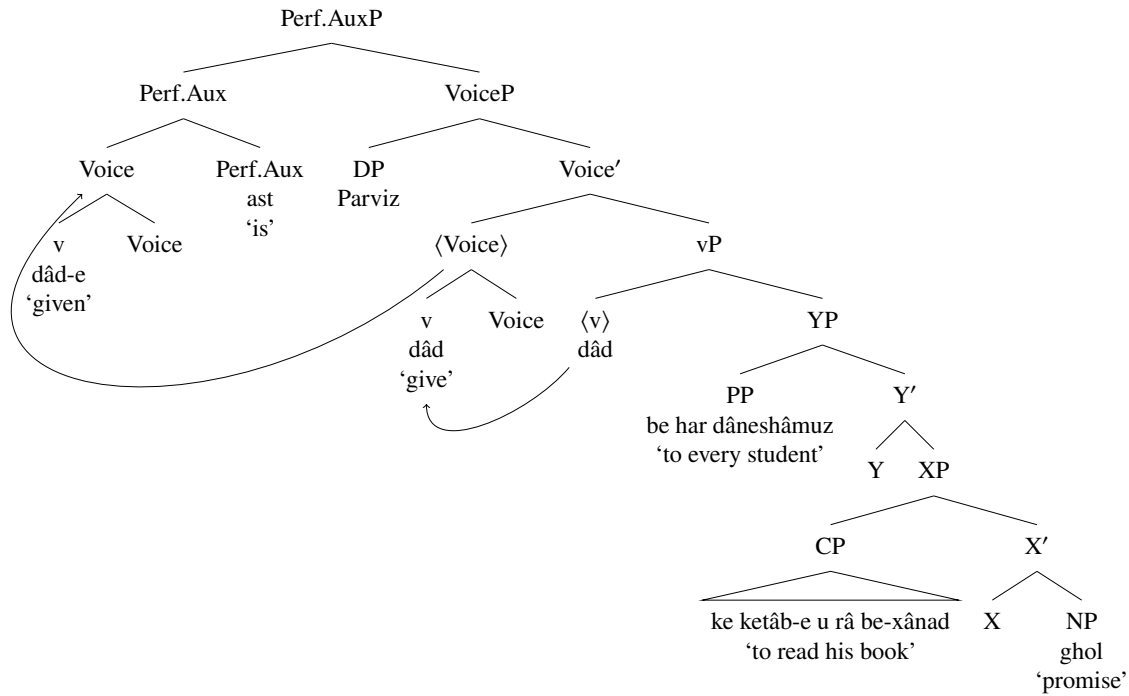
- (ii) German (Hinterhölzl 2000, ex. 38b-d)
- a. *ohne der Maria zu sagen daß Peter krank ist*.
 without the Maria.DAT to say that Peter sick is
 “without telling Maria that Peter is sick”
- b. *ohne der Maria* [F1 *zu* [VP *sagen daß Peter krank ist*]]
- c. *ohne der Maria* [F1 *zu* [F2 *sagen* [F3 CP [VP *tV tCP*]]]]

- (69) a. *Parviz be har dânesâmooz ghol xâhad dâd* [*ke ketâb-e u râ be-xânad*].
 Parviz to every student promise AUX.FUT.3SG give that book-Ezf he DOM
 SBJV-read.3SG
 “Parviz will promise every student to read his book.”
- b. * *Parviz be har dânesâmooz ghol* [*ke ketâb-e u râ be-xânad*]
 Parviz to every student promise that book-Ezf he DOM SBJV-read.3SG
xâhad dâd.
 AUX.FUT.3SG give

This again highlights the fact that a parametric difference in the directionality of argument selection does not successfully rule out the cases in (68b) and (69b), unless some theory of rightward extraposition is stipulated in order to move to the CP to a position high enough to follow all the elements in the verbal string. The alternative analysis offered in the previous section, on the other hand, can straightforwardly derive the grammatical constructions in (68a) and (69a) without any additional mechanism or modification to the assumptions already laid out. All that’s needed is to include the auxiliary verbs in the structure in line with the the account advanced in Chapter 2.

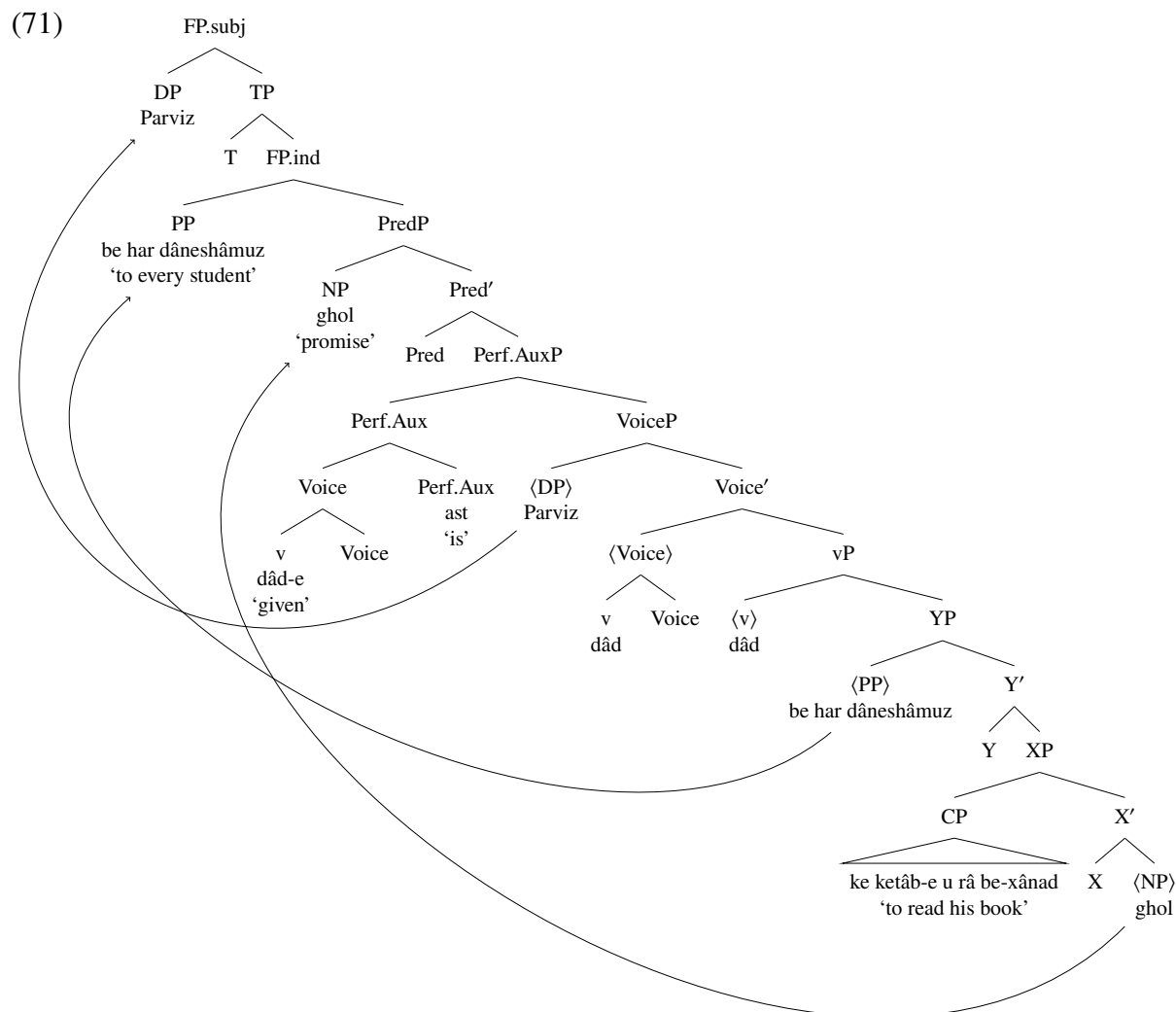
Starting with the derivation of (68a) involving the Perfect Aux *budan*, I assume the underlying structure in (70). Everything up to the VoiceP reflects the already stated assumptions about the base positions of the NV predicate, the internal arguments, the light verb, and the external argument. The only addition is the Perfect Aux *ast* ‘be’ which hierarchically projects higher than the VoiceP and attracts the light verb via the suggested v-Voice-Aux movement, as previously motivated in Chapter 2.

(70)



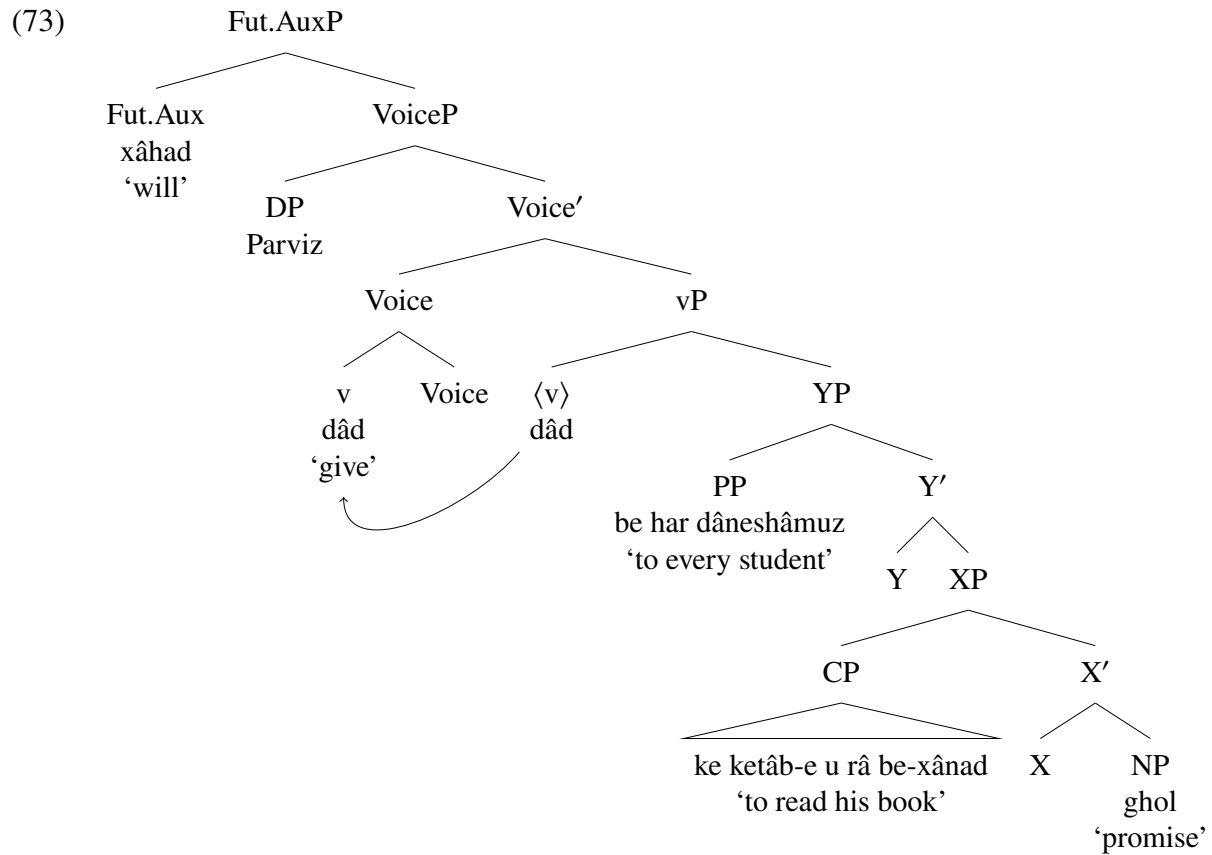
The derivation proceeds with the by-now-familiar licensing operations that move the Non-Verbal predicate as well as the non-clausal arguments to functional projections to the left. Consequently, in the full derivation in (71), *ghol* 'promise' raises to the Spec of PredP above the AuxP,¹⁹ the indirect object *be har dânesshâmuz* raises to the Spec of FP.ind, and the subject *Parviz* moves to the Spec of FP.subj. The finite CP argument *ke ketâb-ash râ be-xânad* however simply stays low and naturally surfaces to right of the verbal complex.

¹⁹This is in line with the generalization that PredP merges above all the *v* heads (including auxiliaries), accounting for the leftmost position of the NV predicates in any verbal cluster.

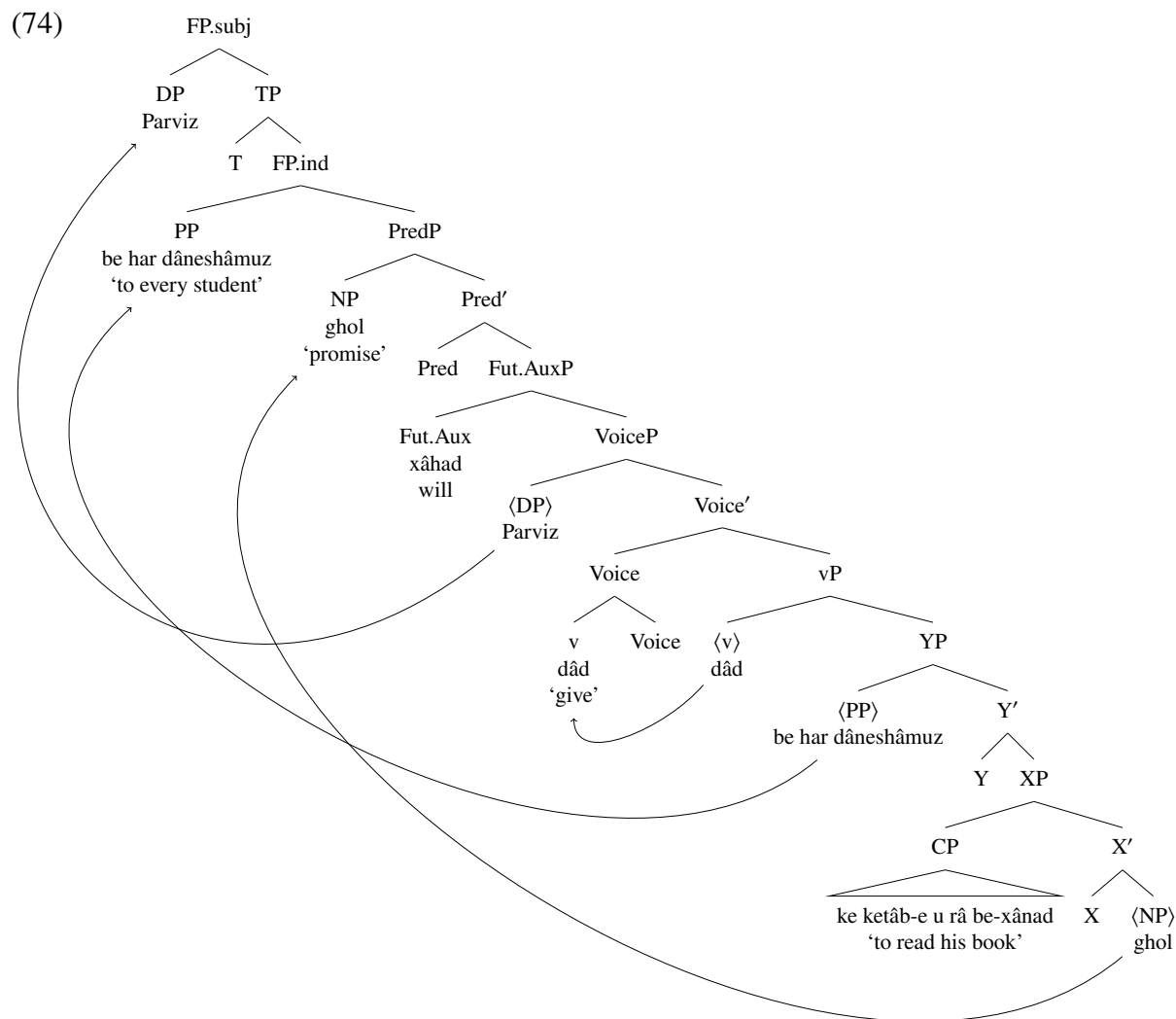


The facts in (72) can be similarly derived from the same set of assumptions. As with the Perfect Aux, the Future Aux hierarchically projects above and selects for the VoiceP in (73). However, no adjunction of the complex Voice head to the Future Aux takes place in the narrow syntax. Therefore, inflected aux *xāhad* linearly precedes the LV *dād* as a reflection of the hierarchical base order of the two heads.

- (72) *Parviz be har dāneshāmooz ghol xāhad dād* [*ke ketāb-e u rā*
 Parviz to every student promise AUX.FUT.3SG give that book-Ezf he DOM
be-xānad].
 SBJV-read.3SG
 “Parviz will promise every student to read his book.”



The rest of the derivation proceeds as expected. The NV element, the indirect object PP, and the subject DP move to their respective licensing projections to the left of the AuxP, while the CP remains in-situ.



In all constructions in which the predicates takes a clausal CP argument, I have accounted for the sentence-final position of the CP through uniform and predictable leftward movements and without appealing to rightward CP extraposition. Considering the general tendency for movements to target positions to the left and the lack of a clear feature checking trigger attracting the CP to a right edge position, the proposed analysis is a welcoming result and one that is clearly more favorable on theoretical grounds.

Additionally, the current analysis provides a straightforward explanation for the subextraction and binding facts shown earlier, repeated below in (75) and (76). Since CP arguments do not move from their original argument position in (75a) and (75b), the fact that they remain transparent to extraction is as expected. The non-movement of CPs further means that they remain in the c-

command domain of the non-clausal arguments at every stage of the derivation. Therefore, the variable binding relation in (76a) and the Principle C violation in (76b) are accounted for without any additional assumptions.

- (75) a. *un ketâb-o_i man be Ehsan pishnahâd xâham kard ke* [*un ketâb-o_i*
that book I to Ehsan suggestion AUX.FUT.1SG do that
be-tadrij be-xune].
gradually SBJV-read.3SG
“As for that book, I will recommend that Ehsan reads it gradually.”
- b. *BE-TADRIJ_i man be Ehsân pishnahâd xâham kard* [*ke un ketâb-o*
GRADUALLY I to Ehsan suggestion AUX.FUT.1SG do that that book-DOM
be-tadrij_i be-xune].
SBJV-read.3SG
“I will recommend Ehsan to read that book GRADUALLY”
- (76) a. *Parviz be har dânesâmuzi_i ghol dâd-e ast* [*ke ketâb-e u_i*
Parviz to every student promise give-PTCP AUX.PRF.3SG that book-Ezf he
râ be-xânad].
DOM SBJV-read.3SG
“Parviz has promised every student to read his (every student’s) book.”
- b. **Parviz be u_i ghol dâd-e ast* [*ke ketâb-e Ali_{*i} râ*
Parviz to him promise give-PTCP AUX.PRF.3SG that book-Ezf Ali DOM
be-xânad].
SBJV-read.3SG
“Parviz has promised him to read Ali’s book.”

3.4.3 The alternative analysis and ellipsis

In this section, I return to the verb-stranding ellipsis constructions presented in Section 3.2.2, such as (77), and offer a preliminary account of how the proposed analysis handles the deletion of the CP argument *ke bâham berand mosâferat* as well as the DP subject *Hamid* in (77b).

- (77) a. *Hamid dust-esh-o ghâne’ na-xâhad kard* [*ke bâham berand*
Hamid friend-her-DOM convinced NEG-AUX.FUT.3SG do that together SBJV-go.3PL
mosâferat].
trip/vacation
“Hamid will not convince his friend to go on a trip together.”

- b. *vali Hamid xânom-esh-o ghâne' xâhad kard*
 but Hamid wife-her-DOM convinced AUX.FUT.3SG do
 [*ke bâham berand mosâferat*].
 that together SBJV-go.3PL trip/vacation
 “But (he) will convince his wife (to go on a trip together).”

The previous chapter provided an overview of data representing what the literature typically refers to verb-stranding verb phrase ellipsis. There, I argued in detail that it is inadequate to analyze such constructions as involving the movement of the verb to a vP-external Foc head and the movement of the contrastive argument to the Spec of that FocP, followed by vP deletion (Rasekhi 2018). I discussed empirical evidence suggesting that what appears to be head movement of the verb out of the ellipsis site is actually XP movement of a maximal verbal projection, and ellipsis should instead be analyzed as the fronting of the contrastive XPs (the remnants) followed by the deletion of the whole *FP.subj* clause. Before discussing how this proposal extends to ellipsis cases involving CP-deletion as in (77b), I will review the derivation of a sample ellipsis pair in (78), consisting of the non-clausal arguments *xune* ‘house’, *be Ali* ‘to Ali’ and *Ehsân*, the complex verb *edzâre dâdan* ‘rent’, and the Future Aux *xâstan* ‘want’.

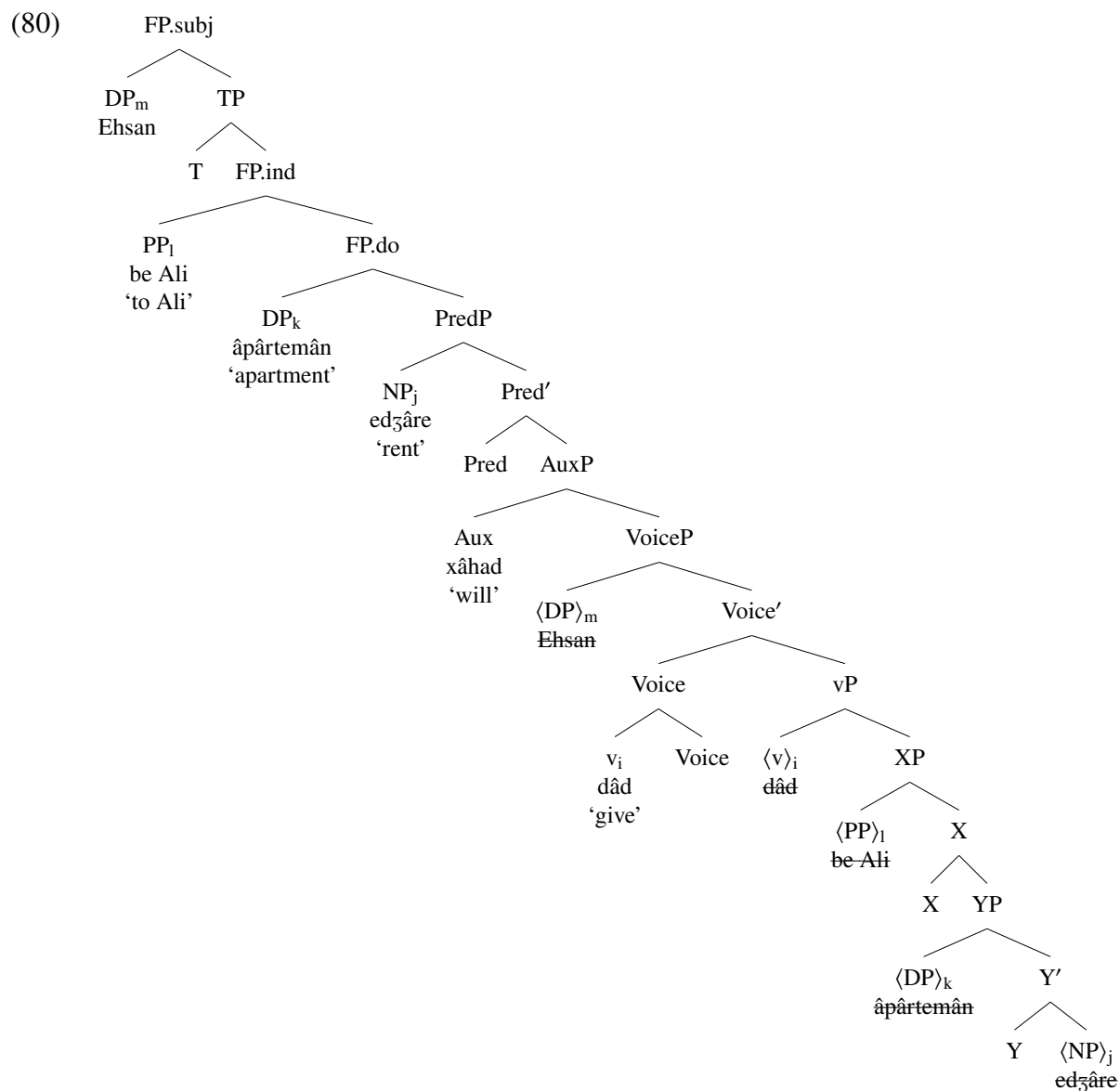
- (78) a. *Ehsân be Ali xune edzâre na-xâhad dâd.*
 Ehsan to Ali house rent NEG-AUX.FUT.3SG give
 “Ehsan will not rent house(s) to Ali.”
- b. *vali Ehsân be Ali apârtêmân edzâre xâhad dâd.*
 but Ehsan to Ali apartment rent AUX.FUT.3SG give.
 “But (Ehsan) will rent apartments (to Ali).”

Starting from the non-elliptical counterpart to (78b), the derivation follows a set of previously stated assumptions summarized in (79).

- (79) a. The relation between the NV predicate ‘rent’ and its internal arguments is captured via the extended projection labeled XP, mediated by abstract Functional heads above the predicate.
- b. The light verb ‘give’ selects for the XP and projects a vP.
- c. The external argument ‘Ehsân’ is introduced in a VoiceP above the vP, and v further adjoins to Voice.

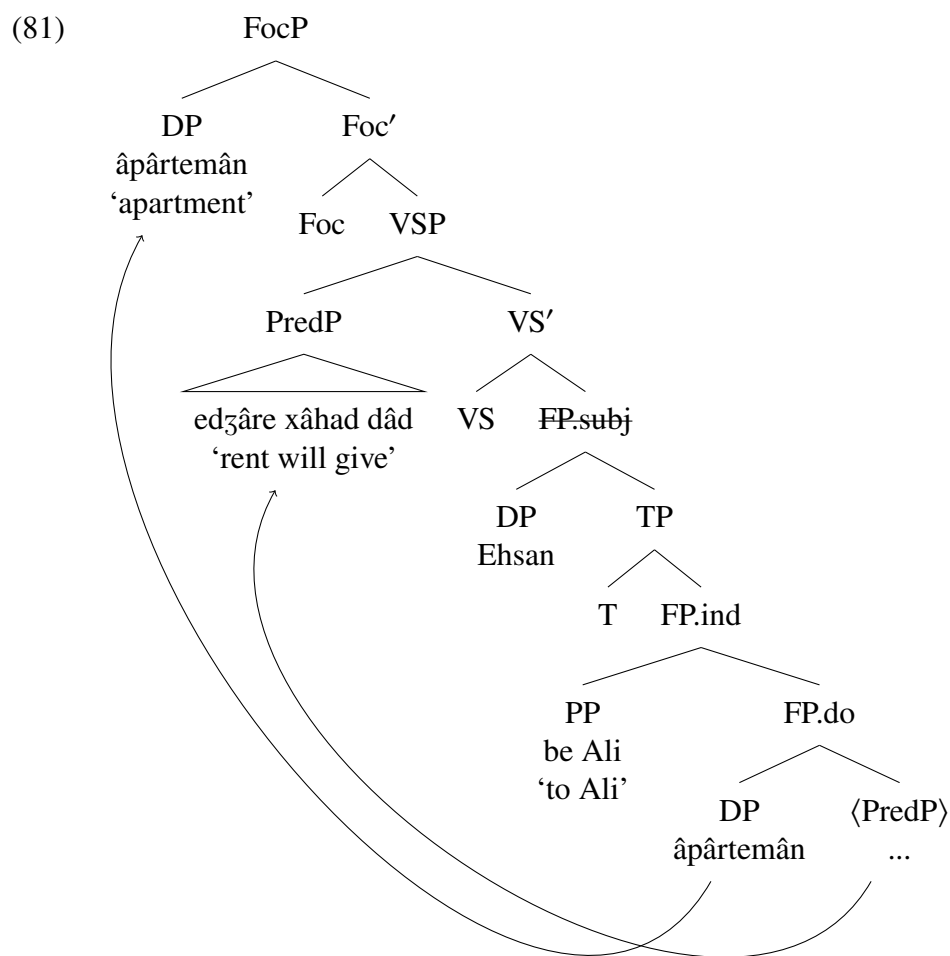
- d. The Aux ‘want’ selects for the VoiceP and the Fut.Aux-LV order reflects the hierarchical base order of the two heads.
- e. PredP merges about AuxP and preposes the NP (headed by the NV element) to its specifier.
- f. The internal arguments raise to FP.do and FP.ind, and the external argument is licensed in FP.subj.

These steps systematically derive the preverbal position of the arguments as well as the strict [NV Aux LV] order *edžâre xâhad dâd* in the verbal cluster, as reflected in the structure in (80)



As (78b) shows, both the contrastive DP argument *âpartemân* ‘apartment’ and the entire cluster

edžâre xâhad dâd ‘will rent’ can escape ellipsis. I argue that this cluster movement cannot be an instance of verb to Focus movement in line with Rasekhi (2018) as it’s not clear how the complex predicate and the intervening Fut.Aux ever form a complex head that can further be probed by another head. In the structure in (80), the preposing of the NV element ‘rent’ to the [Spec,PredP] alternatively derives the [NV Aux LV] cluster as a maximal projection PredP. As a result, the raising of the verbal cluster out of the ellipsis site is analyzed as the movement of PredP to the relevant FP (labeled VSP) followed by *FP.subj* deletion, as shown in (81).



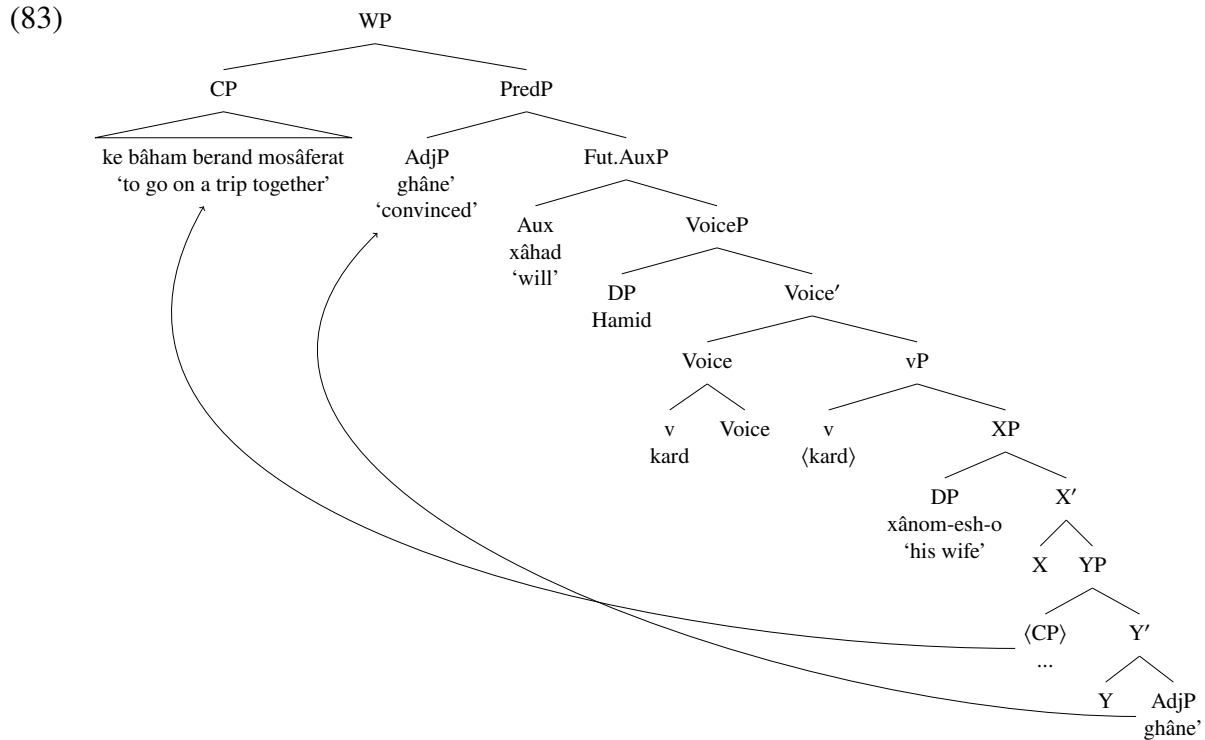
Extending this idea to (82), one additional assumption needs to be made to account for (82b).

- (82) a. *Hamid dust-esh-o ghâne' na-xâhad kard* [*ke bâham berand mosâferat*].
 Hamid friend-CL.3SG-DOM convinced NEG-AUX.FUT.3SG do that together
 SBJV-go.3PL trip/vacation

“Hamid will not convince his friend to go on a trip together.”

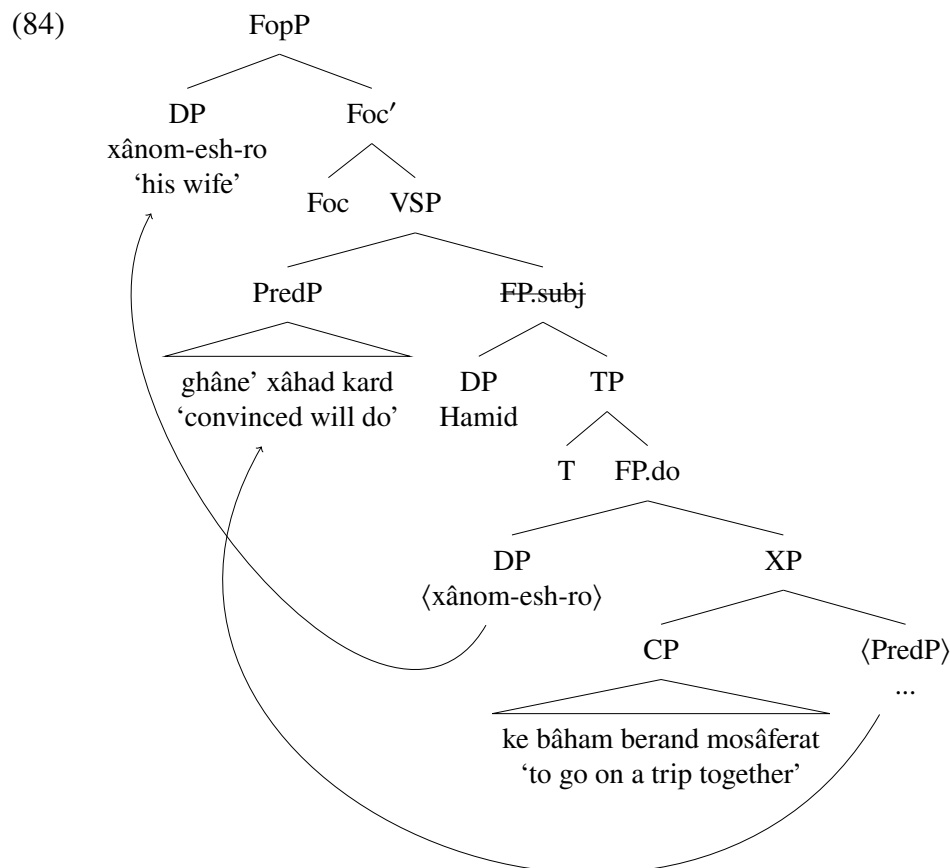
- b. *vali Hamid xānom-esh-o ghāne' xāhad kard*
 but Hamid wife-CL.3SG-DOM convinced AUX.FUT.3SG do
 [*ke bâham berand mosâferat*].
 that together SUBJ-V-go.3PL trip/vacation
 “But (he) will convince his wife (to go on a trip together).”

Since the CP argument is also deleted in (82b) and is therefore not escaping ellipsis alongside the verbal cluster *ghāne' xāhad kard* ‘will convince’, it must raise out of PredP first to some projection (labeled WP), prior to the movement of PredP out of *FP.subj*. This is shown in (83) below:



I have proposed the evacuating movement of the CP out the PredP in (83) as a tentative solution to the problem of CPs simultaneously (i) behaving as arguments low in the structure and (ii) not being able to stay in the position when the complex verb raises out of the clause in such ellipsis cases. As it stands, it’s not really obvious what is the feature motivating this movement and why it should proceed as soon as the PredP constituent is derived. I claim the motivation behind this seemingly vacuous movement is the general requirement of ellipsis that the remnants are in a contrastive relationship with the constituents in the antecedent clause. Since the CPs are identical across the

antecedent clause and the ellipsis clause in (82) and therefore do not meet this requirement, the one in (82b) must remain in the ellipsis site in the course of the derivation. For now, I capture this intuition by positing the movement step in (83) and leave a further developed analysis for future work. As the structure in (84) shows, the rest of the derivation is as expected; the PredP raises out of the ellipsis site alongside the contrastive DP ‘his wife’ and *FP.subj* gets deleted.



Importantly, it is possible for a CP argument to escape ellipsis alongside the verbal cluster if it stands in contrast with the CP in the antecedent clause. An example is given in (85) below. In this case, the PredP that raises out of *FP.subj* in (85b) contains the CP as well.

- (85) a. *Ali [be bache-hâ-sh] ghol dâd [ke dir biyâd].*
 Ali to child-PL-CL.3SG promise gave.3SG that late SBJV-come.3SG
 “Ali promised his children to come late.”
- b. *vali Zahrâ [be bache-hâ-sh] ghol dâd [ke zood biyâd].*
 but Zahra to-child-PL-CL.3SG promise gave.3SG that early SBJV-come.3SG
 “But Zahra promised (her children) to come early.”

3.5 Conclusion

In this chapter, I've revisited the puzzle of post-verbal sentential arguments in Farsi. I have argued that base-generation accounts assuming an underlying head-final verb phrase configuration fall short in accounting for the surface position of clausal CP arguments in sentences involving complex verbs and verbal clusters, unless some CP-movement operation is assumed. I have pointed out how a movement account poses problems for empirical data, especially with respect to CP-deletion in verb phrase ellipsis data. Alternatively, I have proposed an antisymmetric (Kayne 1994) analysis that only relies on leftward movements and assumes that the surface position of sentential arguments is due to such arguments remaining in their base position in the course of the derivation. Against the traditional head-final verb phrase accounts, the proposed antisymmetric analysis fares better with a broader set of empirical patterns based on the following detailed observations.

Amongst the traditional head-final vP analyses, the base-generation accounts generally assume that the verbs exceptionally select for their CP complements directly to the right. This assumption explains the pattern with simplex verbs, but it fails to account for more complex verbal clusters involving complex predicates and auxiliaries. Since CPs uniformly surface sentence-finally and never intervene between the components of the verbal complex, such accounts have limited empirical coverage as it stands. Therefore, some additional mechanism for deriving the correct linear order, such as rightward extraposition or remnant movement, is required.

Extraposition and remnant movement are in turn problematic on both theoretical and empirical grounds. In particular, CP arguments of verbs are transparent to subextraction, and their internal structure is accessible to binding relations which rely on CPs being in the c-command domain of the preverbal arguments. This is incompatible with either extraposition or remnant movement, considering such operations should turn CPs into islands and would place them outside of the scope of preverbal arguments. Therefore, additional stipulations are necessary in order to reconcile the movement accounts with the observed subextraction and binding facts.

In contrast, the analysis developed in this chapter avoids the stipulative asymmetry in directionality of argument selection required in head-final accounts, considering arguments (clausal and

non-clausal alike) are uniformly introduced in the extended projection of the predicate. This analysis derives for the postverbal distribution of CP arguments in a systematic way; DP and PP arguments undergo further movements to higher licensing functional projections, while CPs remain in their base-generated positions due to their distinct licensing requirements or categorial properties. Consequently, CPs surface to the right of all the elements in the verbal complex because they are hierchically lower than all the v heads, including the auxiliary verbs. The extraction and binding patterns are also straightforwardly accounted for since CPs do not move to a position from which they are expected to become an island or fall out of the scope of preverbal arguments.

CHAPTER 4

THE NON-VERBAL DOMAIN: THE PUZZLE OF POST-PREDICATE CONSTITUENTS

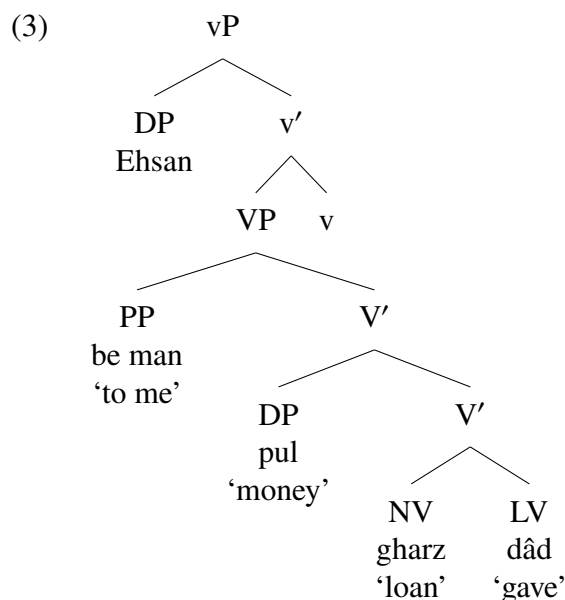
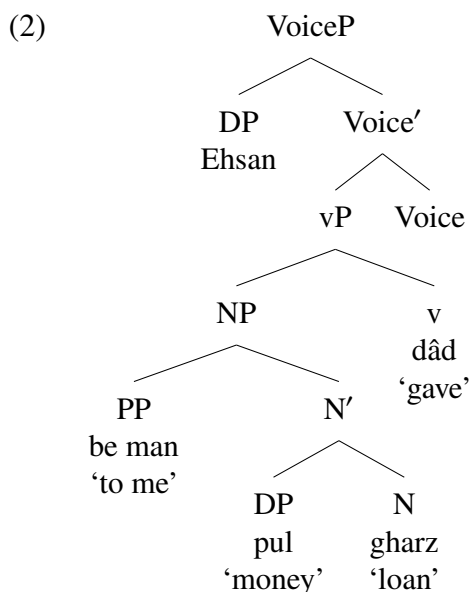
In this chapter, I investigate data involving post-predicate constituents in copular and complex predicate sentences, and show that such constructions provide further support for the theory that Non-Verbal predicates and their (non-clausal) arguments must undergo movement operations in order to be licensed. In particular, I argue that sentences in which predicate-internal arguments and modifiers follow the non-verbal predicate challenge the standard head-final verb phrase analyses that assume the constituent ordering within the predicate domain reflects the base-generated order. These constructions frequently feature the *ezâfe* marker (-e), which is generally understood as an element linking nouns (and lexical categories of nominal status) with their modifiers and complements. I discuss two primary syntactic accounts of *ezâfe* and adopt the movement-based analysis of Kahnemuyipour (2014) to deal with both the predicate+modifier and the predicate+argument contexts. I argue that the operation realizing *ezâfe* in these contexts ultimately reflects a licensing mechanism in line with the general licensing requirements for the arguments in the clause. The chapter is broken down as follows. In section 4.1, I introduce the primary data in which arguments and modifiers follow non-verbal predicates and lay out the implications of such constructions for some previous analyses. Section 4.2 focuses on two previous syntactic accounts of *ezâfe*: (i) the case licensing analysis, and (ii) the phrasal (roll up) movement. I adopt the latter account since it deals better with a range of challenging data for the case licensing analyses, and can be naturally extended to predicate-modifier/argument orderings in non-verbal predicate constructions. Section 4.4 provides detailed derivations of various copular and complex predicate constructions, highlighting the licensing possibilities available to the internal arguments (DPs and PPs) and modifiers. Section 4.5 concludes.

4.1 Post-predicate constituents

Given the consensus in the standard head-final VP/vP analyses that in Farsi complex predicates the non-verbal elements select their arguments to the left like a simplex V does, we should perhaps expect all nominal predicates to follow their arguments in such constructions. This is so whether

one adopts a base complex predicate structure in line with Karimi (2005) and Karimi and Smith (2020) in (2) or Rasekhi (2018) and Shafiei (2015, 2016) in (3) for the sentence in (1).

- (1) *Ehsân be man pul gharz dâd.*
 Ehsan to me money loan gave.3SG
 “Ehsan lent money to me”



Despite this assumption, we encounter constructions involving non-verbal predicates preceding an argument or modifier.

4.1.1 Post-nominal arguments/complements

One such case is predicational copular clauses in which Adjectival elements take PP arguments and predicate of subject experiencer DPs. In (4a) and (5a), the adjectival predicates *alâghemand* ‘interested’ and *râzi* ‘satisfied’ follow their respective PP arguments *be riâzi* ‘in maths’ and *az nomarâtash* ‘from/of her grades’, whereas the order of the predicates and the arguments is reversed in (4b) and (5b).

- (4) a. *Ali [PP be riâzi] alâghemand ast.*
 Ali to math interested is.3SG
 “Ali is interested in mathematics.”

- b. Ali *alâghemand* (-e) [_{PP} *be riâz*] *ast*.
 Ali interested -Ezf to math is.3SG
- (5) a. Ali [_{PP} *az nomarât-ash*] *râzi* *n-ist*.
 Ali from/of grades-CL.3SG satisfied NEG-is.3SG
 “Ali isn’t satisfied with his grades.”
- b. Ali *râzi* (-e) [_{PP} *az nomarât-ash*] *n-ist*.
 Ali satisfied -Ezf from/of grades-CL.3SG NEG-is.3SG

Additionally, there are copular constructions in which the argument of the predicate typically only surfaces after the predicate. This is observed whenever the predicate subcategorizes for DP objects. In (6) and (7), the DPs ‘Mitra’/‘linguistics’ and ‘Ali’/‘ice cream’ must follow the adjectival predicates *âshegh* ‘enamored/in love’ and *divâne* ‘crazy’, respectively.

- (6) a. Hasan *âshegh-e* [_{DP} *Mitrâ / zabânshenâsi*] *ast*.
 Hasan in-love-Ezf Mitra / linguistics is.3SG
 “Hasan is in love with Mitra / linguistics.”
- b. *Hasan [_{DP} *Mitrâ-râ / zabânshenâsi*] *âshegh* *ast*.
 Hasan Mitra-DOM / linguistics in-love is.3SG
- (7) a. Minâ *divâne-ye* [_{DP} *Ali / bastani*] *ast*.
 Mina crazy-Ezf Ali / ice-cream is.3SG
 “Mina is crazy for Ali / ice cream.”
- b. *Minâ [_{DP} *Ali-râ / bastani*] *divâne* *ast*.
 Mina Ali-DOM / ice-cream crazy is.3SG

Data of this sort aren’t exclusive to copular constructions. We also find complex verb sentences in which the non-verbal predicates appear to both precede and follow their complements. In (8a), the non-finite clause embedding PP *be dars xundan* ‘to study/studying’ precedes the full complex predicate *vâdâr kardan* ‘force’, while in (8b), it intervenes between the NV predicate *vâdâr* and the LV *kardan*. In (9a), the PP *be lebâs shostan* ‘to wash clothes’ precedes the entire predicate *mashghul shodan* ‘get busy’, while it follows the NV element *mashghul* ‘busy’ in (9b).

- (8) a. *moallem dâneshamuz-o* [_{PP} *be dars xundan*] *vâdâr* *kard*.
 teacher student-DOM to lesson read-INF forced did.3SG
 “The teacher forced the student to study.”

- b. *moallem dâneshamuz-o vâdâr* (-e) [*PP be dars xundan*] *kard*.
 teacher student-DOM forced -Ezf to lesson read-INF did.3SG

- (9) a. *Ali* [*PP be lebâs shostan*] ***mashghul*** *shod*.
 Ali to clothes wash-INF busy became.3SG
 “Ali got busy washing clothes.”

- b. *Ali mashghul* (-e) [*PP be lebâs shostan*] *shod*.
 Ali busy -Ezf to clothes wash-INF became.3SG

In some complex predicate constructions, the complement of the NV predicate seems to only surface after it. This is observed with predicates like *ghol* ‘promise’ in (10), where the nominal propositional argument *mosâferat raftan* ‘to travel/to go on a trip’ intervenes between the nominal predicate *ghol* ‘promise’ and the light verb ‘*dâdan* ‘give’. The reduced clause in such cases is ultimately a DP given that it patterns with smaller DP constituents in (11) such as *mâshin* ‘car’, *mosâferat* ‘trip’, etc.

- (10) *Ehsân be bache-hâ-sh ghol-e* [*DP mosâferat raft-an*] *dâd*.
 Ehsan to child-PL-CL.3SG promise-Ezf trip go-INF gave.3SG
 “Ehsan promised his children to travel/go on a trip.”

- (11) *Ehsan be bache-hâ-sh ghol-e* [*DP mosâferat / mâshin*] *dâd*.
 Ehsan to child-PL-CL.3SG promise-Ezf trip / car gave.3SG
 “Ehsan promised his children a trip / cars.”

Additionally, examples of double object constructions can be constructed in which the predicates *tajviz* ‘prescription’ (12) and *darxâst* ‘request’ (13) can surface to either side of their direct object DP internal arguments.

- (12) a. *man be Ali* [*DP dâru*] ***tajviz*** *kardam*.
 I to Ali medicine prescription did.1SG
 “I prescribed Ali medicines.”

- b. *man be Ali tajviz-e* [*DP dâru*] *kardam*.
 I to Ali prescription-Ezf medicine did.1SG

- (13) a. *kârmand-ân az ra’is-e edâre* [*DP moraxasi*] ***darxâst*** *kardand*.
 employee-PL from boss-Ezf office time-off request did.3PL
 “The employees requested some time off from the office boss.”

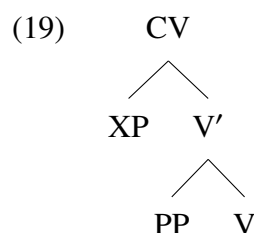
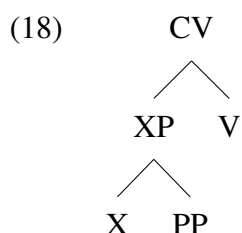
- b. *kârmand-ân az ra'is-e edâre darxâst-e* [_{DP} *moraxasi*] *kardand*.
 employee-PL from boss-Ezf office request-Ezf time-off did.3PL

The initial observation that the internal arguments of the non-verbal elements may appear post-nominally in certain constructions, and must even do so in other cases, poses a real challenge for the standard head-final verb phrase analyses that take these arguments to be interpreted in their base position to the left of the predicate. The separability of the non-verbal predicate and the light verb by a PP in (4b), (5b), (8b), and (9b) or by a DP in (6a), (7a), (10), and (11) would require that such configurations are either base-generated as such or derived due to some movement operation. Another explanation would be to assume that different complex predicate constructions are captured via different base structures. Such approach was taken in earlier work by Karimi (1997) with respect to the argument structure of (the components of) complex predicates, where Karimi assumes a different subcategorization specification for the pairs in (14) and (15), on the one hand, and the pairs in (16) and (17), on the other.

- (14) a. *Kimeâ unâ-ro* [_{PP} *be mehmâni*] *da'vat* *kard*.
 Kimea they-DOM to party invitation did.3SG
 “Kimea invited them to the party.” (Karimi 1997, ex. 80a)
- b. *Kimeâ unâ-ro* *da'vat* [_{PP} *be mehmâni*] *kard*.
 Kimea they-DOM invitation to party did.3SG
 “Kimea invited them to the party.” (Karimi 1997, ex. 10)
- (15) a. *Kimeâ in otâgh-ro* [_{PP} *be mehmân*] *extesâs* *dâd*.
 Kimea this room-DOM to guest allocation gave.3SG
 “Kimea allocated this room to guests.” (Karimi 1997, ex. 80b)
- b. *Kimeâ in otâgh-ro* *extesâs* [_{PP} *be mehmân*] *dâd*.
 Kimea this room-DOM allocation to guest gave.3SG
 “Kimea allocated this room to guests.” (Karimi 1997, ex. 76)
- (16) a. *Sepide âtash* [_{PP} *be zendegi-sh*] *zad*.
 Sepide fire to life-CL.3SG hit.3SG
 “Sepide destroyed her life.” (Karimi 1997, ex. 81a)
- b. *Sepide* [_{PP} *be zendegi-sh*] *âtash* *zad*.
 Sepide to life-CL.3SG fire hit.3SG
 “Sepide destroyed her life.” (Karimi 1997, ex. 83a)

- (17) a. *Sepide aks [PP az bache-hâ] gereft.*
 Sepide picture of child-PL caught.3SG
 “Sepide took pictures of children.” (Karimi 1997, ex. 81b)
- b. *Sepide [PP az bache-hâ] aks gereft.*
 Sepide of child-PL picture caught.3SG
 “Sepide took pictures of children.” (Karimi 1997, ex. 83b)

Karimi (1997) proposed that in (14) and (15), the NV element selects for the complement to the right as given in the configuration in (18). In the case of (16) and (17), however, Karimi (1997) assumed that it is the light verb that selects for the PP argument as in (19).



Therefore, sentences (14a) and (15a) reflect the base-generated position of PP in (18), and (16a) and (17a) reflect the structure in (19). As for the (b) sentences, in which the PP arguments precede the predicate, Karimi (1997) proposes that these arguments have scrambled to a topic or focus position. This is however at odds with the later development of complex predicate structures in Folli, Harley, and Karimi (2005) and Karimi (2005) according to which non-verbal predicates are treated as similar to V heads in selecting their arguments to the left. Furthermore, the more recent analyses assume that non-specific direct object DPs as well as indirect object PPs typically remain in their base position to the left of the predicate. However, such an assumption cannot be maintained in light of the post-predicate complement data in (6a) and (11), repeated here as (20) and (21), where the direct objects (non-specific or specific) or some other element must have obligatorily moved even in neutral contexts.

- (20) *Hasan âshegh-e [DP Mitrâ / zabânshenâsi] ast.*
 Hasan in-love-Ezf Mitra / linguistics is.3SG
 “Hasan is in love with Mitra / linguistics.”

- (21) *Ehsan be bache-hâ-sh ghol-e* [_{DP} *mosâferat / mâshin*] *dâd*.
 Ehsan to child-PL-CL.3SG promise-Ezf trip / car gave.3SG
 “Ehsan promised his children a trip / cars.”

4.1.2 Post-nominal internal modification in complex predicates

In complex predicate sentences, modifiers follow the non-verbal predicate they modify. In cases where the predicate is a noun, it is possible to attributively modify the noun if it is even-itive (as observed by Fleischhauer (2021), Fleischhauer and Neisani (2020), Karimi (1997), and Karimi-Doostan (2011)). As the data in (22) shows, the modifier *ziyâd* ‘much’ precedes the entire complex predicate *komak kardan* ‘help doing’ when used adverbially in (22b), and it surfaces internal to the complex verb and to the right of the non-verbal element *komak* ‘help’ when used attributively in (22c). The same modification options are observed with respect to the adjective ‘bad’ and the predicate *latme* ‘damage’ in (23).

- (22) a. *Ehsân dust-esh-ro komak kard*.
 Ehsan friend-CL.3SG-DOM help did.3SG
 “Ehsan helped his friend.”
- b. *Ehsân dust-esh-ro [ziyâd] komak kard*.
 Ehsan friend-CL.3SG-DOM much help did.3SG
 “Ehsan helped his friend often.”
- c. *Ehsân dust-esh-ro komak-e [ziyâdi] kard*.
 Ehsan friend-CL.3SG-DOM help-Ezf much did.3SG
 “Ehsan helped his friend a lot.”
- (23) a. *tagarg be bâgh-e man latme zad*.
 hail to garden-Ezf I damage hit3SG
 “The hail damaged my garden.”
- b. *tagarg be bâgh-e man [bad] latme zad*.
 hail to garden-Ezf I bad damage hit3SG
 “The hail damaged my garden often/a lot.”
- c. *tagarg be bâgh-e man latme-ye [badi] zad*.
 hail to garden-Ezf I damage-Ezf bad hit3SG
 “The hail caused a bad damage to my garden.”

The internal modification of complex predicates is possible with concrete nominals as well (as observed by Fleischhauer 2021; Fleischhauer and Neisani 2020). In (24b) and (25b), the modifiers *xub* ‘good’ and *garm* ‘warm’ surface post-nominally with respect to the non-verbal elements *xune* ‘comb/brush’ and *dast* ‘hand’, respectively.

- (24) a. *Mohammad moo-sh-o shune kard.*
 Mohammad hair-CL.3SG-DOM comb did.3SG
 “Mohammad combed his hair.”
- b. *Mohammad moo-sh-o shune-ye [xubi] kard.*
 Mohammad hair-CL.3SG-DOM comb-Ezf good did.3SG
 “Mohammad did a good job combing his hair.”
- (25) a. *Ehsân ba mehmân dast dâd.*
 Ehsan with guest hand gave.3SG
 “Ehsan shook hands with the guest.”
- b. *Ehsân ba mehmân dast-e [garmi] dâd.*
 Ehsan with guest hand-Ezf warm gave.3SG
 “Ehsan gave the guest a warm handshake.”

In the majority of the data discussed in this section, the examples that show the [Predicate Argument/Modifier] order contain the *ezâfe* element *-e* between the two constituents. As such, it may be argued that since the post-predicate constituent data only concern the nominal domain, they should have no bearing on the head final analysis of verb phrases. I propose however that the possibility of linking non-verbal predicates with their complements and modifiers via *ezâfe* in certain cases is evidence for the more principled requirement that internal arguments (as well as modifiers) are licensed throughout the derivation, and that this may take place in various syntactic positions. Taking the pairs in (12) and (13) for instance, repeated below as (26) and (27), the DP complements are either licensed in a position to the right of the non-verbal predicate and accompanied by *ezâfe*, or they raise to some licensing projection above the entire complex verb. When the arguments of the predicates are of the finite clausal kind, such as in (28) and (29), they can only follow the entire complex verb in line with the hypothesis that clausal arguments do not move to the same licensing

positions for non-clausal arguments.²⁰

- (26) a. *man be Ali* [_{DP} *dâru*] ***tajviz*** *kardam*.
 I to Ali medicine prescription did.1SG
 “I prescribed Ali medicines.”
- b. *man be Ali* ***tajviz-e*** [_{DP} *dâru*] *kardam*.
 I to Ali prescription-Ezf medicine did.1SG
- (27) a. *kârmand-ân az ra’is-e edâre* [_{DP} *moraxasi*] ***darxâst*** *kardand*.
 employee-PL from boss-Ezf office time-off request did.3PL
 “The employees requested some time off from the office boss.”
- b. *kârmand-ân az ra’is-e edâre* ***darxâst-e*** [_{DP} *moraxasi*] *kardand*.
 employee-PL from boss-Ezf office request-Ezf time-off did.3PL
- (28) *moallem dâneshamuz-o* ***vâdâr*** *kard* [_{CP} *ke dars be-xune*].
 teacher student-DOM forced did.3SG that lesson SBJV-read.3SG
 “The teacher forced the student to study.”
- (29) *Ehsân be bache-hâ-sh* ***ghol*** *dâd* [_{CP} *ke be-ran mosâferat*].
 Ehsan to child-PL-CL.3SG promise-Ezf gave.3SG that SBJV-go.3PL trip
 “Ehsan promised his children to travel/go on a trip (together).”

Laying out the details of how the licensing options are derived therefore necessitates providing a discussion of the *ezâfe* analyses and how they fit within the current framework. The following section focuses on two primary syntactic accounts of this element and their implications for the [Predicate -Ezf Arg/Mod] ordering in copular and complex predicate clauses.

4.2 Previous syntactic analyses of the *ezâfe* construction

The *ezâfe* element has been researched extensively in the literature and has received various analyses. It has been analyzed as a case marker in Larson and Yamakido (2008), Larson and Samiian (2020, 2021), and Samiian (1983, 1994), as a PF-inserted vowel in Ghaniabadi (2010) and Ghomeshi (1996, 1997), as an affix marking a syntactic dependent in Samvelian (2007, 2008), and as the spell-out of a roll-up phrasal movement operation inverting nouns around their modifiers via intermediate Agr projections in Kahnemuyipour (2014, 2016). As documented by these authors

²⁰See Chapter 3 for the detailed analysis.

and others, *ezâfe* is observed with postnominal XPs of various categories (NP, AdjPs, PPs, possessor DPs, etc.), as shown in (30a-30d). Multiple modifiers can occur post-nominally in which case *ezâfe* is repeated, with the requirement that the possessor comes last after all the modifiers in (30e).

- (30) a. *ostâd* -e [_{NP} *dânesgh*]
 professor -Ezf university
 “university professor”
- b. *pirhan* -e [_{AdjP} *âbi*]
 shirt -Ezf blue
 “(the) blue shirt”
- c. *gol* -e [_{PP} *posht* -e *panjere*]
 flower -Ezf behind -Ezf window
 “(the) flower behind the window”
- d. *mâshin* -e [_{DP} *Ehsân*]
 car -Ezf Ehsan
 “Ehsan’s car”
- e. *mâshin* -e [_{AdjP} *âbi*] -e [_{PP} *tu-ye garaj*] -e [_{DP} *Ehsân*]
 car -Ezf blue -Ezf in-Ezf garage -Ezf Ehsan
 “Ehsan’s blue car in the garage”

As it relates to this chapter and the data concerning post-predicate modifiers/arguments, I only discuss two primary syntactic accounts of *ezâfe*; (i) the case analysis of Larson and Samiian (2020, 2021) and Samiian (1983, 1994) and (ii) the roll-up phrasal movement analysis of Kahnemuyipour (2014). I argue in favor of adopting the phrasal movement analysis and extending it to the Predicate+Modifier/Argument cases highlighted in section 4.1.

4.2.1 *Ezâfe* as a case marker

With respect to the case marking analyses, I will mainly focus on the developments made in Larson and Samiian (2020, 2021) and Samiian (1994). Based on the earlier proposal in Samiian (1994), Larson and Samiian (2020, 2021) treat *ezâfe* (and the preposition *az* ‘of’) as the equivalent of the English *of*. In this sense, the function of *ezâfe* is to assign case to [+N] elements (modifiers and complements) that are base-generated postnominally, because the head noun can’t assign case

(31) a. [_{NP} *shâm*] *(-e) [_{PP} *ba Hasan*] Class 1 P
dinner (-Ezf) with Hasan
“dinner with Hasan”

b. [_{NP} *divâr*] *(-e) [_{PP} *jelo Ali*] Class 2 P
wall (-Ezf) in-front-of Ali
“wall in front of Ali”

c. [_{NP} *divâr*] *(-e) [_{PP} *jelo -ye Ali*] Class 2 P
wall (-Ezf) in-front-of -Ezf Ali
“wall in front of Ali”

d. [_{NP} *miz*] *(-e) [_{PP} *baghal -e Hasan*] Class 3 P
table (-Ezf) near -Ezf Hasan
“table near Hassan”

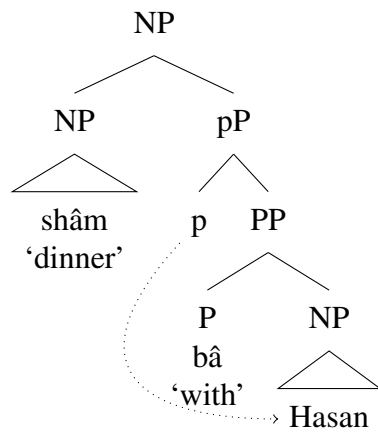
Larson and Samiian (2020, 2021) view the distribution of *ezâfe* with PPs in (31a)-(31d) as parallel to that of the English *of* in the following constructions involving verbs (32a), gerunds (32b-32c), and derived nominals (32d).

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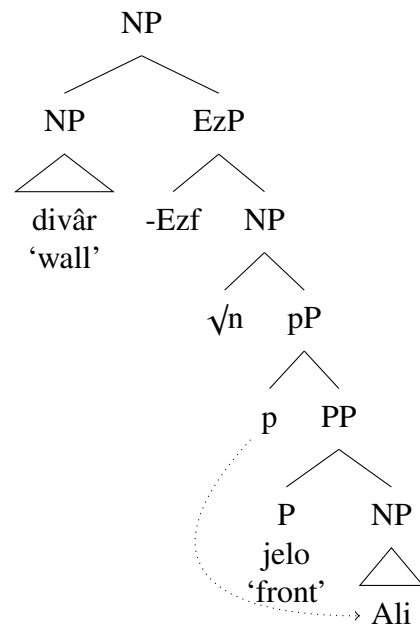
In (32a), the VP/vP *destroy the evidence* is both “internally” and “externally verbal”, in that it has an accusative object and combines with the modal *will*. In parallel, the PP/pP *bâ Hasan* in (31a) is both “internally” and “externally prepositional”, and therefore appears with an accusative object and is not preceded by *ezâfe*. The verbal gerund *destroying the evidence* in (32b) is “internally verbal” as it has an accusative object, but “externally nominal” since it’s preceded by a possessor. Similarly, *jelo Ali* in (31b) is “internally prepositional” and includes an accusative object, but “externally nominal” as it follows *ezâfe*. Like *destroying of the evidence* in (32c), which is “internally” and “externally nominal”, in needing *of* and in combining with a possessor, *jelo-ye Ali* in (31c) is also “internally” and “externally nominal” and therefore requires *ezâfe* on either side of the preposition. Lastly, the nominals *destruction of the evidence* (32d) and *baghal-e Hasan* (31d) are “internally” and “externally nominal”.

To capture these assumptions, Larson and Samiian (2020, ex. 76) propose the following structures for (31a)-(31d) (in line with Jackendoff’s (1977) analysis of nominalization in English). (33) corresponds to the “true” preposition contexts as in (31a), where the p head (like v) assigns case to the object. With respect to Class 2 prepositions that are not followed by *ezâfe* but are preceded by it, Larson and Samiian assume the p head still assigns case, but the pP becomes an NP via the nominalizer $\sqrt{n}$ morpheme. When Class 2 prepositions are followed by *ezâfe* as in (31c), the $\sqrt{n}$ morpheme attaches to the P head directly and turns it into an N (35). As a result, both the object *Ali* and the NP *jelo-ye Ali* need to get case from the preceding -Ezf heads. Lastly, (36) reflects the assumptions for true nominal prepositions, such as *baghal* in (31d).

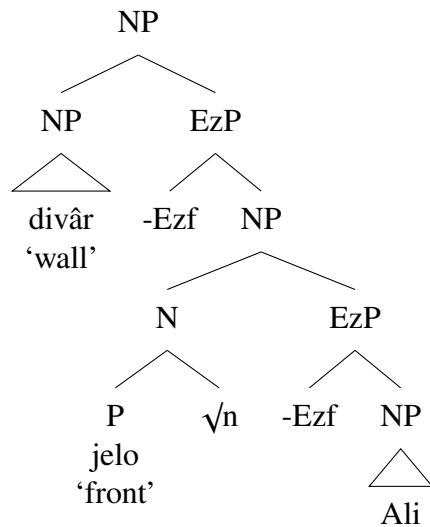
(33) Class 1 (pure pP)



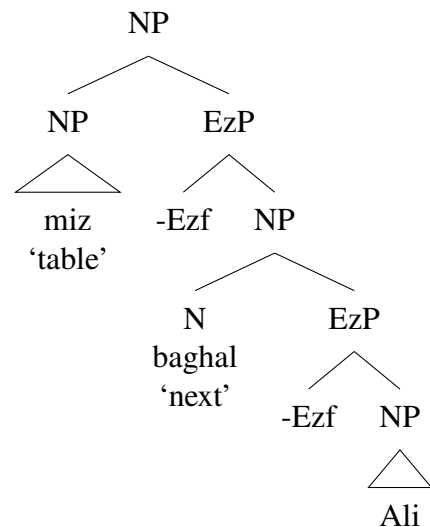
(34) Class 2 P (nominalized pP)



(35) Class 2 P (nominalized P)



(36) Class 3 (de-prepositional N)



Larson and Samiian (2020) further show that a case-marking analysis predicts that *ezâfe* should be present with nominal clauses, but not with finite clauses. They argue that such a prediction holds based on data involving relative and complement clauses. In (37), the noun *ghol* 'promise' takes the non-finite propositional complement *amadan(-e) be Pâris* 'coming to Paris', and in (38), the noun *aks* 'photo' is modified by the non-finite relative clause *châp shod-e dar ruznâme* 'printed in the

newspaper’. In both situations, *ezâfe* is required on either side of the non-finite modifier/argument.

- (37) a. **ghol** *(-e) [*âmadan(-e) be Paris*]
 promise -Ezf come.INF-Ezf to Paris
 “the promise of coming/to come to Paris”
- b. **ghol** -e [*âmadan(-e) be Paris*] -e [*Ali*]
 promise -Ezf come.INF-Ezf to Paris -Ezf Ali
 “Ali’s promise of coming/to come to Paris”
- (38) a. **aks** *(-e) [*châp shod-e dar ruznâme*]
 photo -Ezf publish become-PTCP in newspaper
 “the photo published in the newspaper”
- b. **aks** -e [*châp shod-e dar ruznâme*]-ye [*Ali*]
 photo -Ezf publish become-PTCP in newspaper Ali
 “Ali’s photo published in the newspaper”

However, when ‘promise’ takes a finite CP complement in (39), and ‘photo’ is modified by a full relative clause CP in (40), *ezâfe* seems to be absent before the modifier/argument.

- (39) a. *in ghol* (*-e) [*ke miyâd Pâris*]
 this promise -Ezf that come-to.3SG Paris
 “the promise that he’ll come to Paris”
- b. *in ghol* -e [*Ali*] [*ke miyâd Pâris*]
 this promise -Ezf Ali that come-to.3SG Paris
 “Ali’s promise that he’ll come to Paris”
- (40) a. **aks** (*-e) [*ke chap shod-e bud dar ruzname*]
 photo -Ezf that publish become-PTCP was.3SG in newspaper
 “the photo that had been published in the newspaper”
- b. **aks** -e [*Ali*] [*ke chap shod-e bud dar ruzname*]
 photo -Ezf Ali that publish become-PTCP was.3SG in newspaper
 “Ali’s photo that had been published in the newspaper”

In short, the case-licensing account reviewed here assumes that [+N] modifiers and arguments of nouns originate post-nominally and need to be case licensed there by the *ezâfe* element whenever the selecting heads themselves are unable to assign case.

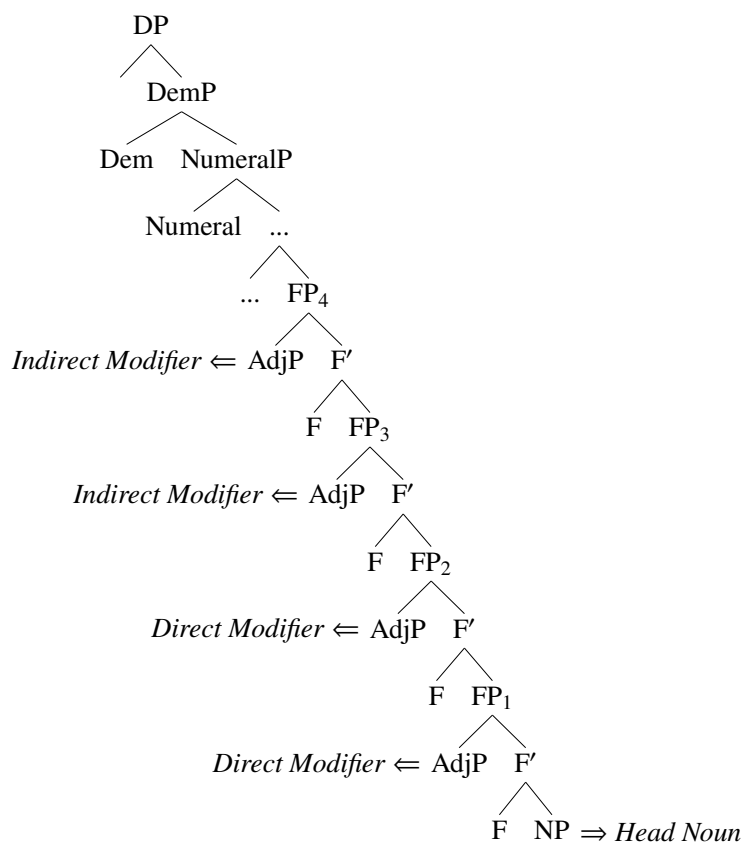
4.2.2 *Ezâfe* as the realization of phrasal movement and Agreement

Kahnemuyipour (2014) proposes that the modifiers following the head noun are base-generated to the left of the noun, and they surface postnominally due to syntactic movements inverting the noun around the modifiers. In this analysis, *ezâfe* is analyzed as the morphological spellout of such inversion operations, and the absence of *ezâfe* with elements preceding the nominal reflects the lack of movement. This correlation is illustrated in the following constructions, where the post-nominal modifiers *ghatur* ‘thick’ (41a), *jâleb* ‘interesting’ (41a), and *bi-ma’ni* ‘meaningless’ (41b) are all accompanied by *ezâfe*. In contrast, the pre-nominal demonstrative *in* ‘this’ (41a-41b), the numerals *do* ‘two’ (41a) and *se* ‘three’ (41c), the adjective *tanhâ* ‘sole/only’ (41b), and the quantifier *har* ‘every’ (41c) lack any *ezâfe* marking.

- (41) a. *in do ketâb-e [ghatur]-e [jâleb]*
 this two book-Ezf think-Ezf interesting
 “these two interesting thick books.” (Kahnemuyipour 2014, ex. 3a)
- b. *in tanhâ so‘âl-e [bi-ma’ni]*
 this only question-Ezf meaningless
 “this only meaningless question.” (Kahnemuyipour 2014, ex. 3b)
- c. *har se ketâb-e [âbi]*
 every three book-Ezf blue
 “all three blue books.” (Kahnemuyipour 2014, ex. 3d)

Kahnemuyipour (2014) follows Cinque (2010) in assuming a base universal structure for DPs in line with (42), where the adjectival modifiers merge above the noun (NP) in a hierarchical order. This configuration reflects the noun final order observed in languages such as English, where no movements have taken place. It further captures the crosslinguistic observation that direct modification adjectives appear closer to the noun and in a more restricted order compared to the indirect modification ones.

(42) Structure of DP based on Kahnemuyipour (2014, ex. 30))

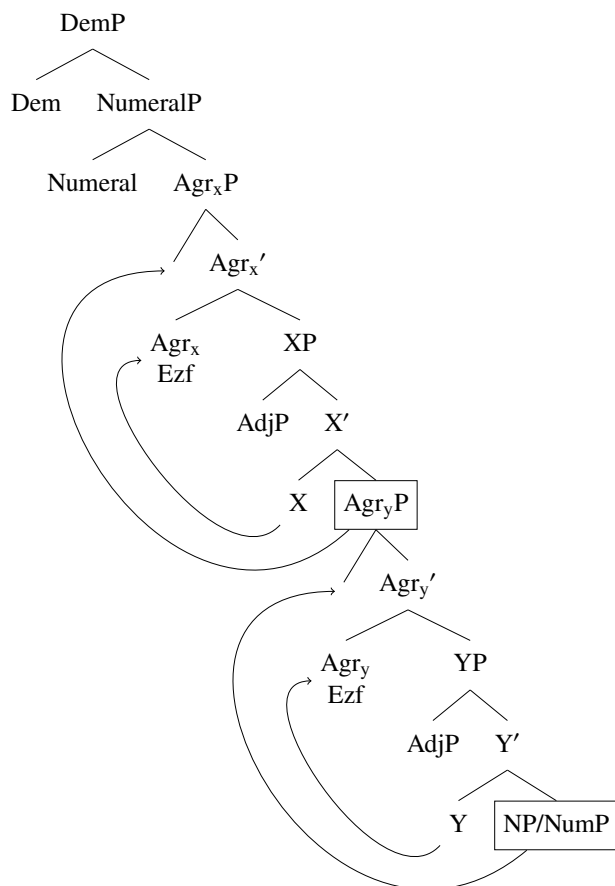


Crucially, any word order diverging from the *Mod-Noun* order in (42), such as the [Noun -Ezf- Mod] order observed in Farsi, is assumed to be the outcome of Cinque-style roll-up movement operations that raise the noun and constituents containing the noun around the modifiers. Kahnemuyipour extends this analysis to Farsi DPs and provides the schematic derivation in (43). Initially, the head noun is merged as (part of) an NP/NumP lowest in the structure, and the modifiers are uniformly introduced as phrases in specifier positions of functional projections (YP, XP, etc.). Additionally, there are intermediate Agr projections (Agr_yP, Agr_xP, etc.) that enable the (roll-up) movements as shown. Each inversion process is accompanied by the overt realization of *ezâfe* on the Agr heads. Fundamentally then, *ezafe* represents Agreement relations established between the nominals and their dependents via movement.²¹ The lack of *ezâfe* with the elements preceding the nominal (such

²¹Kahnemuyipour (2014) highlights data from other Iranian languages, such as the Kurmanji dialect of Kurdish, where *ezâfe* displays person, number, and gender agreement. This is taken as evidence that the intermediate Agr heads resulting in the roll-up movements do in fact involve Agreement relations.

as Demonstratives, Numerals, etc.) is attributed to them merging as heads (as opposed to phrases) high up in the structure and therefore not being subject to the movements linking the head NP to its dependents.

(43) Deriving the *ezâfe* construction via roll-up movement (Kahnemuyipour 2014, ex. 31)



To derive the sample construction in (41a), repeated below as (44), the noun *ketâb* ‘book’ is base-generated at the bottom of the tree in (45), with the functional YP projection introducing the modifier *ghatur* ‘thick’ just above it. An AgrYP projection immediately merges next. The movement of the NP to the Spec of AgrYP and the raising of Y to the AgrY head realizes *ezâfe*. Subsequently, in (46) the modifier *jâleb* ‘interesting’ is introduced in an XP above the AgrYP. The derivation

- (i) *mâl -â mâzin- â Narmîn-e*
house -Ezf.FEM.SG big -Ezf.FEM.SG Narmin-OBL.FEM
“Narmin’s big house.”

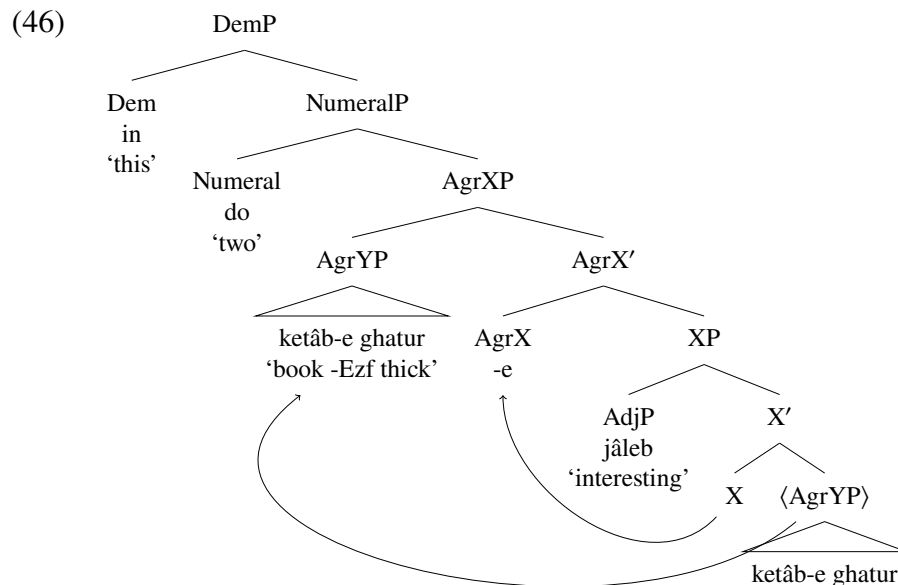
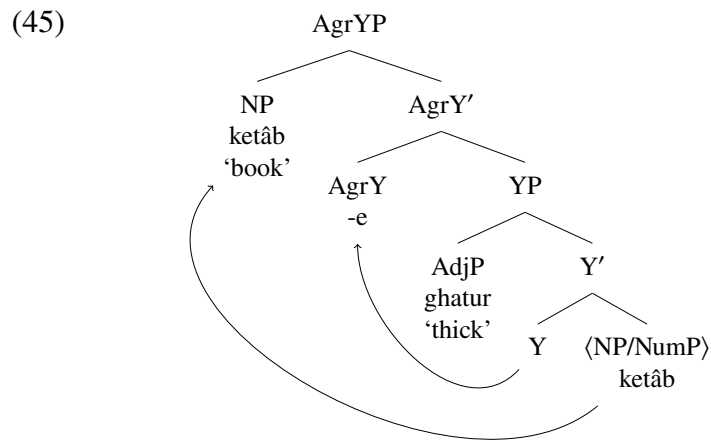
(Samvelian 2008, ex. 10)

- (ii) *mirov -ê ku min dît-î*
man -Ezf.MASC.SG that I.OBL saw-PAST
“the man whom I saw”

(Samvelian 2008, ex. 34)

proceeds with another intermediate projection AgrXP, which in turn raises the immediately lower AgrP to its specifier. This results in *ketâb-e ghatur* ‘book -Ezf thick’ surfacing to left of *jâleb* and another spellout of *ezâfe* between the two constituents. Any additional modifiers and inversion processes are assumed to follow from further roll-up movements in the same fashion, as triggered by additional AgrPs.

- (44) *in do ketâb-e [ghatur]-e [jâleb]*
 this two book-Ezf thick-Ezf interesting
 “these two interesting thick books.”



As a consequence of this derivation, whenever there are multiple modifiers as in (44), their relative order with respect to the noun matches the order in which they are initially merged in the structure. Relatedly, this derivation predicts that Farsi should display the mirror-image of English, based on the assumption that in English the head noun stays low, whereas in Farsi roll-up movements are involved. Kahnemuyipour argues that this is indeed the case based on the data in (47a)-(49a). In these examples, the modifiers preceding or following the nouns are “unambiguously direct modification adjectives” and are therefore assumed to appear in a rigid order (for the intended meanings).

- (47) a. *fizikdân* -e [*hasteyi*] -ye [*javân*]
physicist -Ezf nuclear -Ezf young
“young nuclear physicist”
- b. **fizikdân* -e [*javân*] -e [*hasteyi*]
- (48) a. *mashrubxor* -e [*sangin*] -e [*ghabli*]
drinker -Ezf heavy -Ezf former/previous
“the former/previous heavy drinker”
- b. **mashrubxor* -e [*ghabli*] -e [*sangin*]
- (49) a. *zabânshenâs* -e [*nazari*] -ye [*bâ-este’dâd*]
linguist -Ezf theoretical -Ezf with-talented
“talented theoretical linguist”
- b. ??*zabânshenâs* -e [*bâ-este’dâd*] -e [*nazari*]

(Kahnemuyipour 2014, ex. 32a-c)

4.3 Assessing the two analyses - In favor of the phrasal movement account

Larson and Samiian (2020) argue against Kahnemuyipour (2014) based on several observations. One such case concerns the distribution of *ezâfe* with respect to relative clauses, as observed in (38a) and (40a) and repeated below as (50) and (51). Larson and Samiian argue that since Kahnemuyipour’s (2014) movement analysis does not make a distinction with respect to the [-N] or [+N] status of postnominal modifiers, it should make no prediction about the presence of *ezâfe* with the non-finite relative clause in (50) and its absence with the finite relative clause in (51).

- (50) *aks* *(-e) [*châp shod-e dar ruznâme*]
photo -Ezf publish become-PTCP in newspaper
“the photo published in the newspaper”
- (51) *aks* *(-e) [*ke chap shod-e bud dar ruzname*]
photo -Ezf that publish become-PTCP was.3SG in newspaper
“the photo that had been published in the newspaper”

Relatedly, Larson and Samiian argue that since Kahnemuyipour’s (2014) analysis primarily involves modifiers, it should have no prediction to make about the distribution of *ezâfe* with respect to arguments, such as complement clauses in (37a) and (39a), repeated below as (52) and (53).

- (52) *ghol* *(-e) [*âmadan(-e) be Paris*]
promise -Ezf come.INF-Ezf to Paris
“the promise of coming/to come to Paris”
- (53) *in ghol* *(-e) [*ke miyâd Pâris*]
this promise -Ezf that come-to.3SG Paris
“the promise that he’ll come to Paris”

Addressing these criticisms, Gündoğdu, Kahnemuyipour, and Den Dikken (2024) propose that *ezâfe* is in fact found in the context of finite clause CPs, which they take as an argument that *ezâfe* is not dependent on the [+N] or [-N] feature of the postnominal element. They assume that (restrictive) relative clauses, as in (54), are preceded by an allomorph of *ezâfe*, the *-i* marker. They provide further support for this assumption by showing that two other Iranian languages, Kurmanji and Zazaki, utilize their regular form of *ezâfe* before both finite and non-finite restrictive relative clauses.²²

- (54) *zan* -i [_{CP} *ke az Tehrân umad-e*] *xeyli bahush-e*.
woman -i that from Tehran come-PRF.3SG very smart-is3SG
“The woman who has come from Tehrân is very smart.”

(Gündoğdu, Kahnemuyipour, and Den Dikken 2024, ex. 8a)

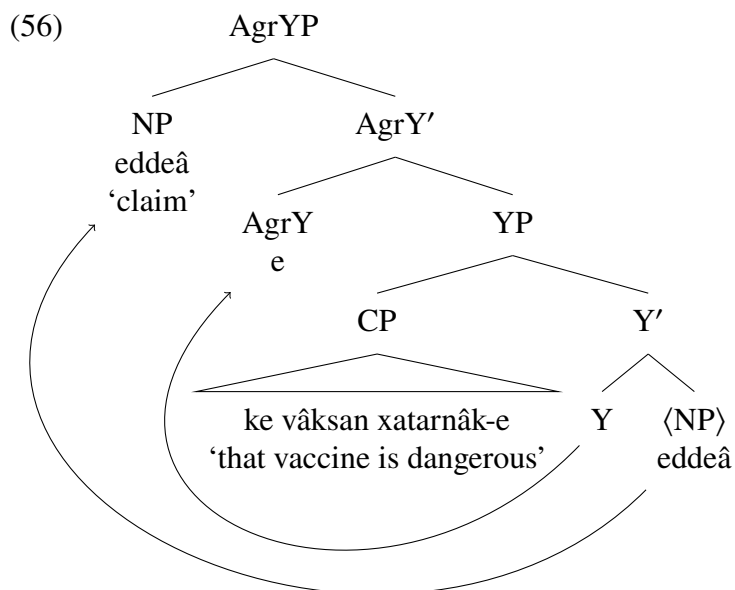
²²See Gündoğdu, Kahnemuyipour, and Den Dikken (2024) for the data and the full discussion.

Additionally, Gündoğdu, Kahnemuyipour, and Den Dikken (2024) document constructions in which the same allomorph of *ezêfe* (-i) is optionally used between nouns and their finite CP complements. An example is given in (55).

- (55) [in/un **eddeâ** / **eddeâ** -yi][_{CP} *ke vâksan xatarnâk-e*] -ro *man matrah na-kardam*.
 that/that claim / claim -i that vaccine dangerous-is -DOM I mention NEG-did.3SG
 “I didn’t bring up the claim that the vaccine is dangerous.”

(Gündoğdu, Kahnemuyipour, and Den Dikken 2024, ex. 26)

They analyze the -i bearing case as the reflex of a structure in which the CP argument originates above the head noun, and the noun inverts around the CP, as suggested in (56). The consistent occurrence of *ezêfe* with CP complements in Kurmanji and Zazaki is taken as further evidence in support of their argument.



In the remainder of this section, I focus on the following data in (57)-(60) further raised by Larson and Samiiian (2020) as incompatible with Kahnemuyipour’s phrasal movement. I illustrate that not only such constructions are unproblematic for the movement analysis but they are also better dealt with and receive a more straightforward explanation under this analysis. These constructions involve cases where the postnominal dependents are arguments (rather than modifiers) of their preceding head, and should therefore be treated as complements. This is observed with re-

spect to heads of various categories, such as nouns (57), adjectives (58), certain prepositions (59), and certain quantifiers (60).

(57) *Noun -Ezf Complement*

- a. **taxrib** -e [DP shahr]
destruction -Ezf city
“destruction of the city”
- b. **xordan** -e [DP âb]
drinking -Ezf water
“drinking of water”

(58) *Adjective -Ezf Complement*

- a. **âshegh** -e [DP Hasan]
enamored -Ezf Hasan
“in love with Hasan”
- b. **negarân** -e [DP bache-ha]
worried -Ezf child-PL
”worried for the children”

(59) *Preposition -Ezf Complement* (with some prepositions)

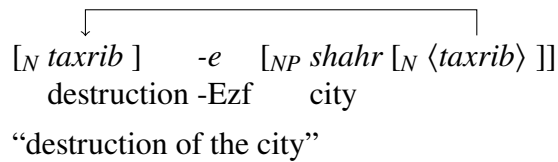
- a. **zir** -e [DP miz]
under -Ezf table
“under the table”
- b. **jelo** -ye [DP xune]
front -Ezf house
“(in) front of the house”

(60) *Quantifier -Ezf Complement* (with some partitives)

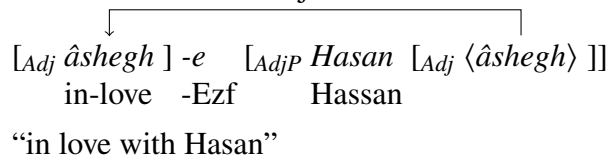
- a. **tamâm** -e [DP sherkat-hâ]
all -Ezf company-PL
“all/the-totality-of companies”
- b. **bishtar** -e [DP ketâb-hâ]
most -Ezf book-PL
“most of the books”

Larson and Samiian (2020) argue that Kahnemuyipour’s (2014) analysis does not deal with such cases because the assumed roll-up movements are consistently phrasal, and primarily involve movements to the left of modifiers. Since the nouns, adjectives, prepositions, and quantifiers in (57)-(60) are all X^0 heads, and not XPs, they cannot have undergone phrasal movements to the left of their complements. Instead, some movement operation raising the heads to specifier positions, as schematically suggested in (61), would have to be posited. Therefore, Larson and Samiian (2020) conclude that ultimately there is no clear way to extend the phrasal movement account to explain the *ezâfe* data concerning head-complement relations.

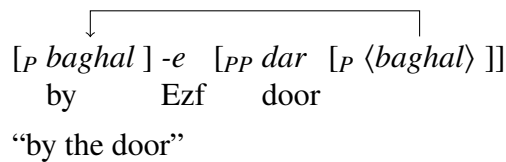
(61) a. Head movement of N



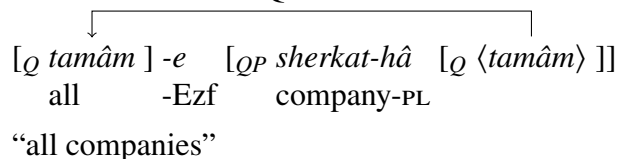
b. Head movement of Adj



c. Head movement of P

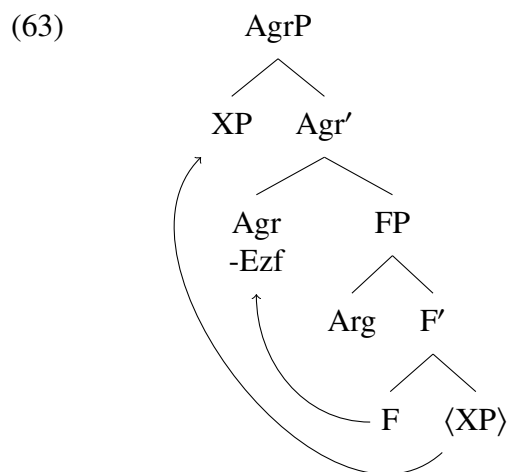
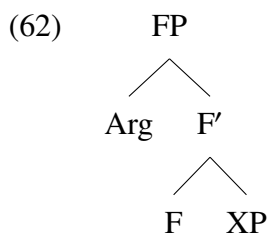


d. Head movement of Q



(Adapted from Larson and Samiian 2020, ex. 22)

I argue however that this data may only be an issue for Kahnemuyipour (2014) if we commit to the assumption that in all such cases, the subcategorized argument is merged as the sister to the head. If we consider the alternative base structure in (62), where the (Arg)uments of the heads originate in specifiers of separate FPs above the head XP, then the same phrasal movements assumed for the [Noun -Ezf Mod] cases should be able to derive the [Head -Ezf Arg] order as well, as suggested in (63).



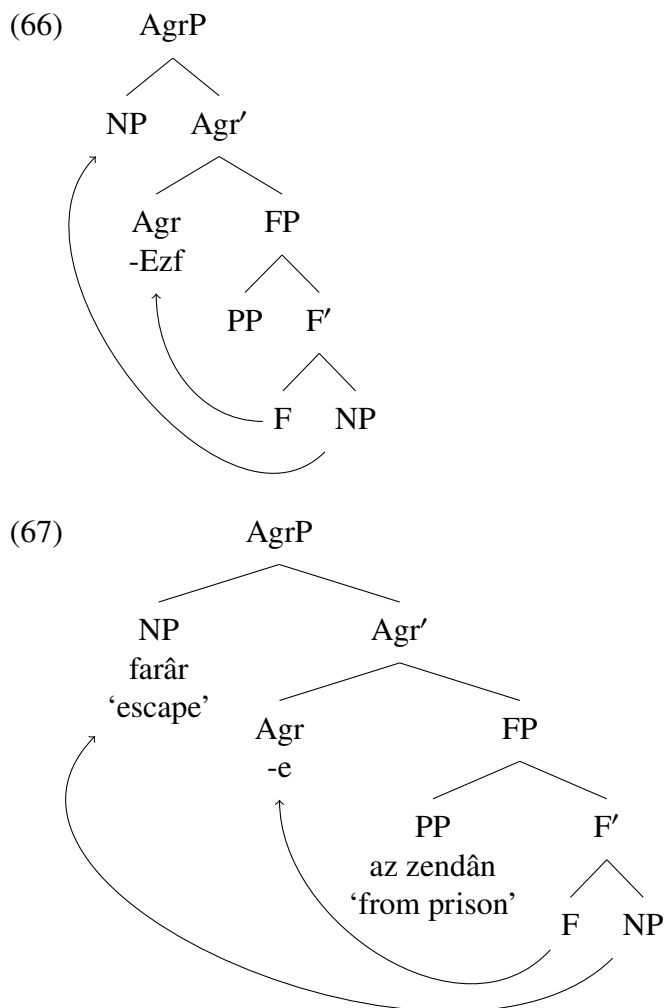
This idea has previously been explored in Kahnemuyipour and Taghipour (2024) as it relates to PP arguments. In particular, they discuss the optionality of *ezâfe* with PP arguments of eventive nominals, as shown in (64) and (65), and argue that this optionality stems from the availability of two possible derivations; one that involves movement and results in the spellout of *ezâfe*, and one that doesn't.

- (64) *farâr* (-e) [_{PP} az zendân]
 escape -Ezf from prison
 “escape from prison”

- (65) *tamarkoz* (-e) [_{PP} ru(-ye) entexâbât]
 focus -Ezf on-Ezf election
 “focus on the election”

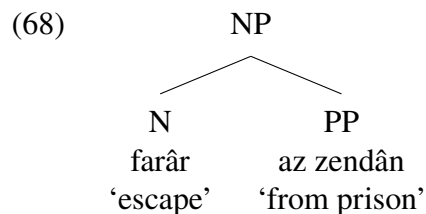
In the former case, Kahnemuyipour and Taghipour (2024) assume the structure in (66) in line with the inversion analysis of Kahnemuyipour (2014). Crucially, according to this structure, PP argu-

ments (like modifiers) are introduced in specifier positions above the noun, and their postnominal distribution is derived from the suggested inversion process (NP to [Spec,AgrP]). As expected, when this derivation is involved, *ezâfe* obligatorily surfaces between the noun and the PP. Applying this analysis to the sample construction in (64) yields the following structure in (67).

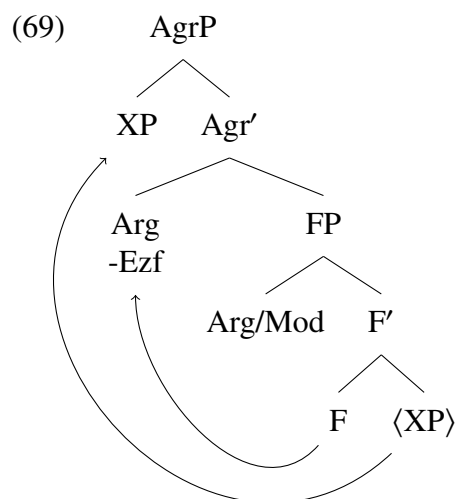


Kahnemuyipour and Taghipour assume that PPs can also merge low in the structure as the complement of the noun, as suggested in (68). In this case, no inversion process is involved, and therefore no *ezâfe* is observed.²³

²³Kahnemuyipour and Taghipour (2024) take the preliminary stance that there is a difference in interpretation of such constructions depending on the presence or absence of *ezâfe*. Based to their initial judgments, *farâr -e az zendân* ‘escape -Ezf from prison’ seems to refer to a specific incident (“the escape from prison”), whereas *farâr az zendân* ‘escape from prison’ seems to convey a more general event (“escaping from prison”). I agree with their judgments.



Building on the analysis in (66) that both arguments and modifiers can be base-generated in specifiers of FPs above the noun (as also argued in Atlamaz 2016), I propose that the broader range of *ezâfe* data concerning head-complement relations can be derived from the structure in (69), where the XP stands for heads of various categories.

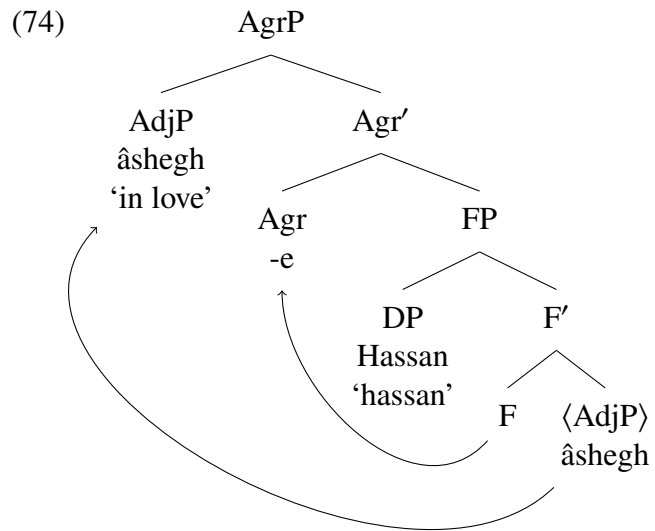
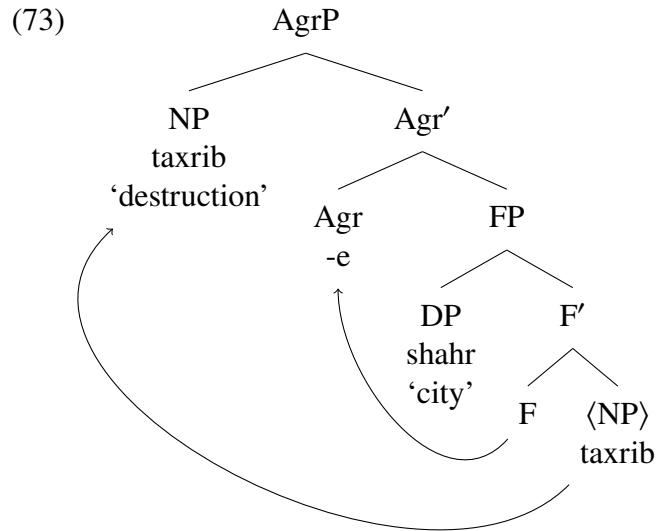


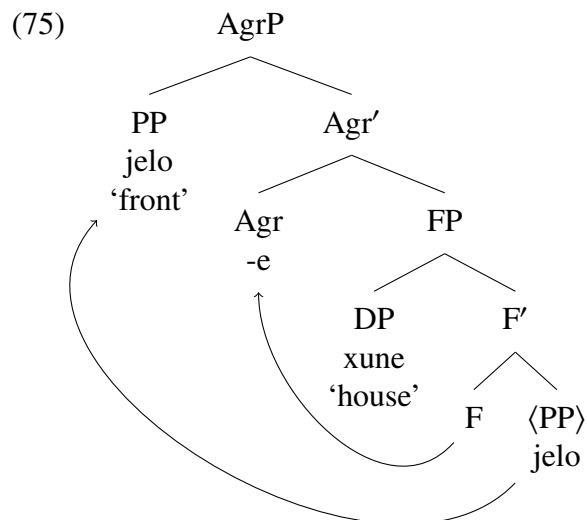
As a consequence, the crucial cases viewed by Larson and Samiian (2020) as challenging for Kahnemuyipour's (2014) phrasal movement analysis, such as *taxrib-e shahr* 'destruction-Ezf city' (70), *âshegh-e Hassan* 'in love-Ezf Hassan' (71), and *jelo-ye xune* 'front-Ezf house' (72), receive the straightforward derivations in (73), (74), and (75), respectively.

- (70) ***taxrib*** *-e* [*shahr*]
 destruction -Ezf city
 "destruction of the city"

- (71) ***âshegh*** *-e* [*Hasan*]
 in-love -Ezf Hasan
 "in love with Hasan"

- (72) *jelo* -ye [*xune*]
 front -Ezf house
 “(in) front of the house”





A closer investigation of the [Head -Ezf Arg] data provides further evidence that arguments appearing with *ezâfe* are not introduced as the sister to the head. The relevant cases involve constructions in which the heads behave like phrases and can be modified by adjectives. Crucially, in such constructions, the modifier appears adjacent to the head and precedes the head's apparent complement. One such example is documented in Samiian (1994) and given in (76), where the modifier *nâgahâni* 'sudden' intervenes between the nominal head *taxrib* 'destruction' and its argument *shahr* 'city'.

- (76) *taxrib-e* [nâgahani]-ye [shahr] *tavasot-e doshman*
 destruction-Ezf sudden-Ezf city by-Ezf enemy
 "the sudden destruction of the city by the enemy"

Another instance is provided by Larson and Samiian (2020, 2021) in their discussion of "nominalized" prepositions (Class 2 Ps), such as *zir* 'under' and *jelo* 'front'. Larson and Samiian show that such prepositions can follow demonstratives (77), can be pluralized (78), and crucially can be modified by an adjective (79).

- (77) a. *in/un zir-e* [miz]
 this/that under-Ezf table
 "that/that underspace of the table"
- b. *un jelo-ye* [xune]
 that front-Ezf house

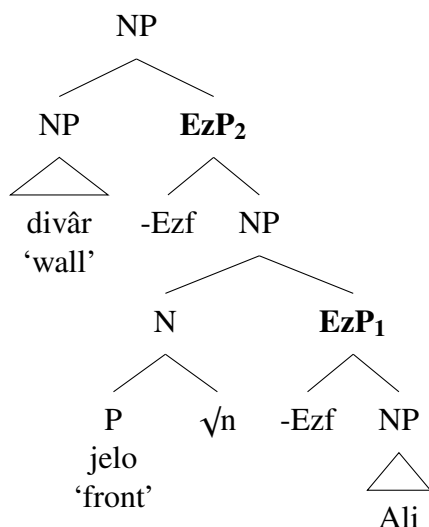
“that front of the house”

- (78) a. *un zir-â-ye* [miz]
that under-PL-Ezf table
“those underspaces of the table”
- b. *un jelo-â-ye* [xune]
that front-PL-Ezf house
“those front areas of the house.”
- (79) a. *zir-e* [kasif]-e [miz]
under-Ezf dirty-Ezf table
“that dirty underspace of the table.”
- b. *in jelo-ye* [ghashang]-e [xune]
this front-Ezf beautiful-Ezf house
“this beautiful front area of the house”

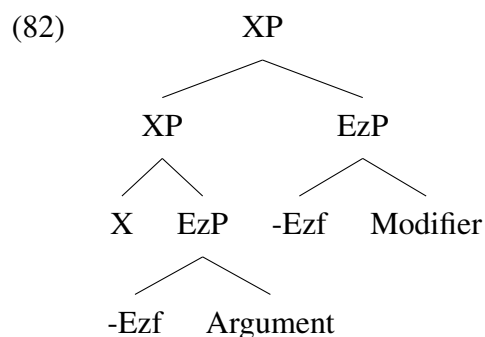
The possibility to modify head nouns (76) and head prepositions (79) in this way directly challenges Larson and Samiian’s structural analysis of the *ezâfe* constructions. As we saw earlier, Larson and Samiian (2020) assume that the construction in (80) involving the nominalized preposition *jelo* ‘front’ has the following derivation in (81). In this structure, the object of the nominalized P, as well as the (NP) projection of the nominalized P must be licensed by -Ezf heads.

- (80) *divâr -e* [jelo] -ye [Ali]
wall -Ezf front -Ezf Ali
“the wall in front of Ali”

(81) Nominalized Class 2 Preposition (Larson and Samiian 2020, ex. 76a)



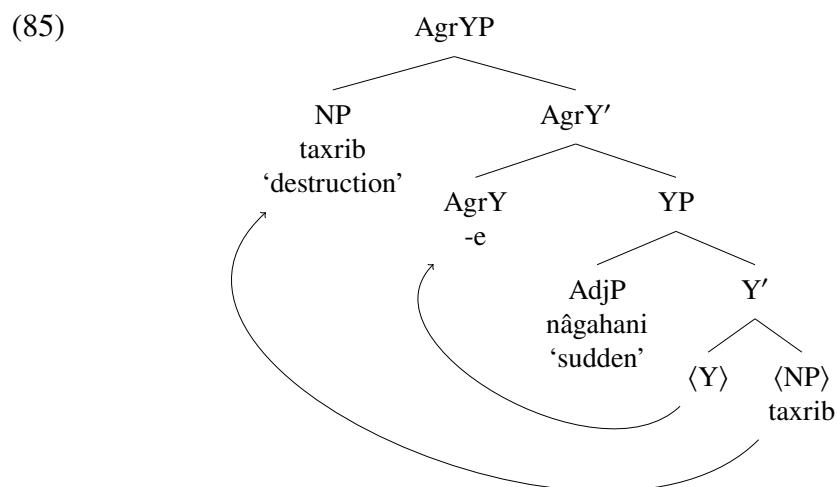
This structure suggests that an Ezâfe Phrase (EzP) can be merged either as the complement to an X head, to case-license an argument (**EzP₁** in the tree), or as an adjunct to an XP phrase, to case-license a modifier (**EzP₂** in the tree). This makes the prediction that whenever a head is followed by both an argument and a modifier, the argument should surface closer to the head than the modifier, as suggested in (82).



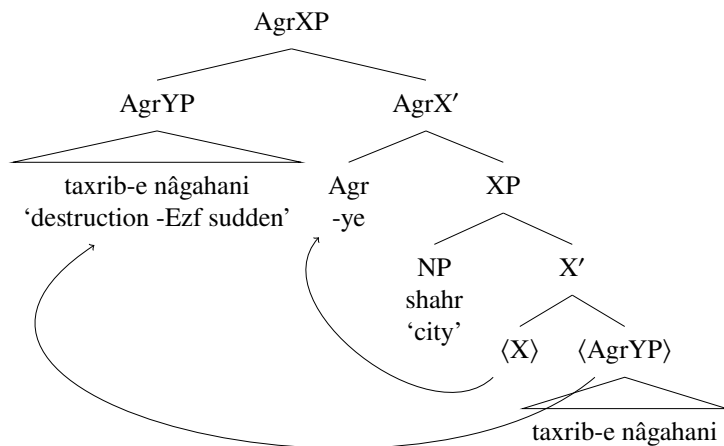
However, constructions such as (76) and (79b), repeated below as (83) and (84), show that this prediction is not always borne out. Contrary to (82), the [N -Ezf Mod -Ezf Arg] order in (83) suggests that the adjunct *nâgahan* ‘sudden’ appears closer to the head than the complement *shahr* ‘city’. Similarly, (84) shows the same word order facts for the prepositional head *jelo* ‘front’ and its dependents. Therefore, it’s not clear how Larson and Samiian (2020) would derive the problematic cases in (83) and (84) without additional assumptions.

- (83) *taxrib-e* [nâgahani]-ye [shahr] tavasot-e doshman
 destruction-Ezf sudden-Ezf city by-Ezf enemy
 “the sudden destruction of the city by the enemy”
- (84) *in jelo-ye* [ghashang]-e [xune]
 this front-Ezf beautiful-Ezf house
 “this beautiful front area of the house”

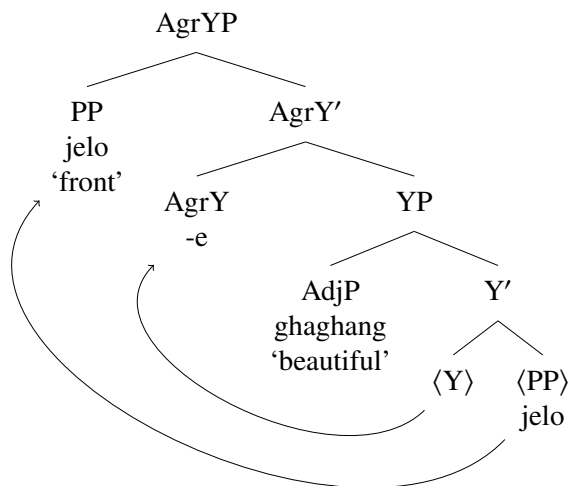
Alternatively, I propose that the version of Kahnemuyipour’s (2014) analysis where both modifiers and arguments are introduced in specifiers of functional projections above the head can systematically derive (83) and (84) via the application of (roll-up) phrasal movements to the base order [Arg Mod Head]. In both constructions, the modifiers are introduced in the Spec of YPs right above the heads, and the heads invert around the modifiers as mediated by intermediate AgrYPs. The arguments are merged next in the Spec of XPs above the AgrYPs, and intermediate AgrXPs enable the subsequent movement of the AgrYPs around the arguments. The relevant steps deriving (83) are given in (85) and (86), and those responsible for (84) are shown in (87) and (88).



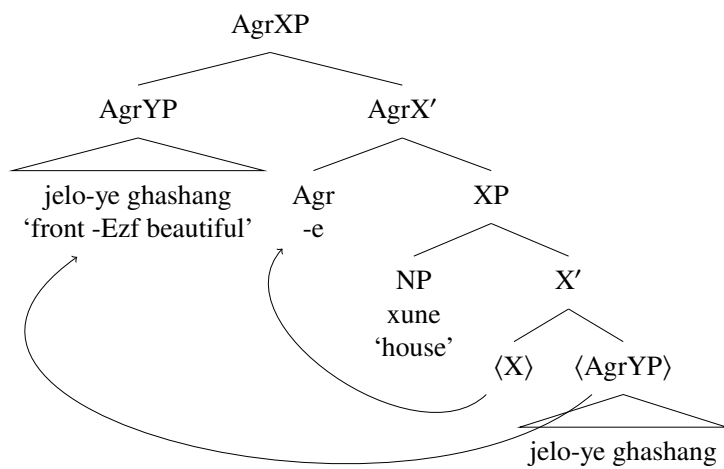
(86)



(87)



(88)



Based on this discussion, I argue that the phrasal movement analysis (Kahnemuyipour 2014 and subsequent work) offers broader empirical coverage of the postnominal Arg/Mod data than the case-marking account reviewed in this section.

4.4 The unified analysis of *ezâfe* in the Predicate - Modifier/Argument contexts

Having argued in the previous section that post-nominal modifiers and arguments can be accounted for via a unified overt phrasal movement analysis, I will now turn to the data concerning post-predicate constituents introduced in section 4.1. I begin with discussing the derivation of the [Predicate -Ezf Modifier] order and move on to the [Predicate -Ezf Argument] cases in section 4.4.2.

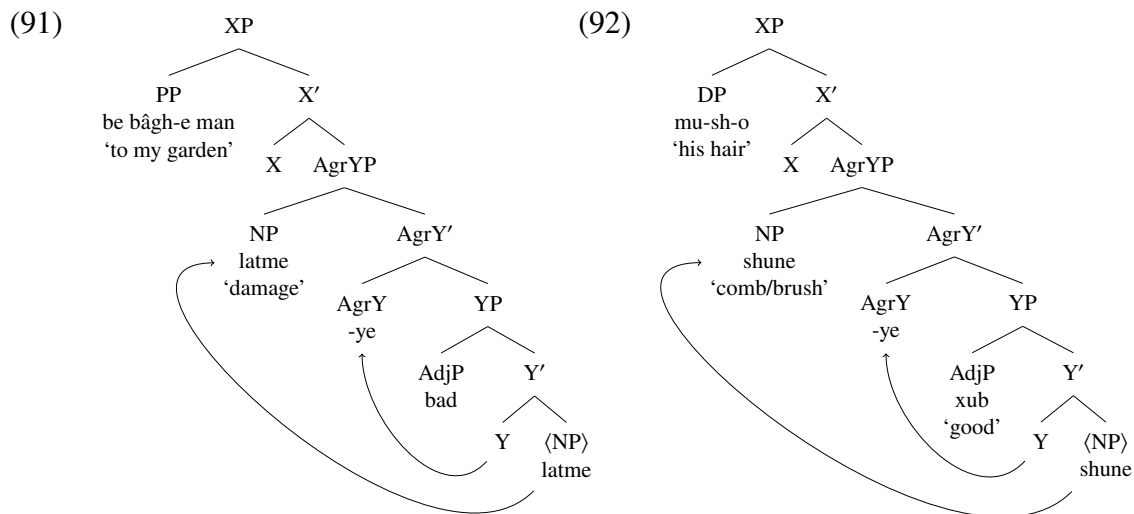
4.4.1 An account of the Predicate -Ezf Modifier order

As introduced in Section 4.1.2, Farsi allows for the internal modification of complex predicates via attributively modifying the non-verbal component of these predicates. Some examples are repeated here in (89) and (90) to highlight that when this happens, the modifiers *ziyâd* ‘much’ and *xub* ‘good’ follow the non-verbal predicates and are preceded by *ezâfe*.

- (89) a. *tagarg be bâgh-e man latme zad.*
 hail to garden-Ezf I damage hit3sg
 “The hail damaged my garden.”
- b. *tagarg be bâgh-e man latme-ye [badi] zad.*
 hail to garden-Ezf I damage-Ezf bad hit3sg
 “The hail caused a bad damage to my garden.”
- (90) a. *Mohammad mu-sh-o shune kard.*
 Mohammad hair-CL.3SG-DOM comb did.3sg
 “Mohammad combed his hair.”
- b. *Mohammad mu-sh-o shune-ye [xubi] kard.*
 Mohammad hair-CL.3SG-DOM comb-Ezf good did.3sg
 “Mohammad did a good job combing his hair.”

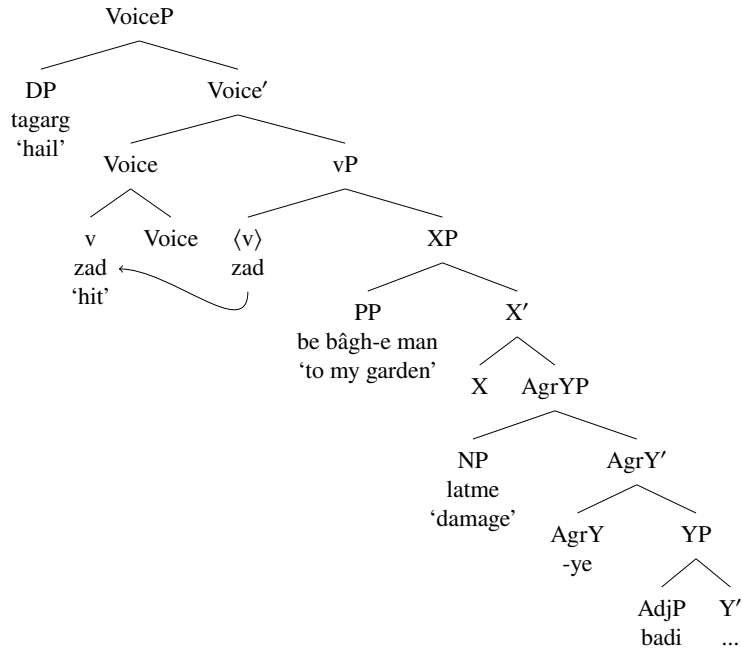
In line with Kahnemuyipour’s analysis that in the NP/DP domain, the head noun starts low in the structure and its modifiers/arguments are base-generated in specifiers of higher functional heads, I propose that non-verbal predicates and their arguments are introduced in the same fashion.

With respect to the [Predicate -Ezf Mod] data under discussion, I assume that the modifiers merge first above the non-verbal predicate. This is immediately followed by the movement of the non-verbal predicate NP around the modifier to an intermediate AgrP projection, therefore licensing the modifier and realizing *ezâfe* on the Agr head. I assume that this step takes place prior to the introduction of the DP/PP argument. This yields the sub-structures (91) and (92) for (89b) and (90b), respectively.

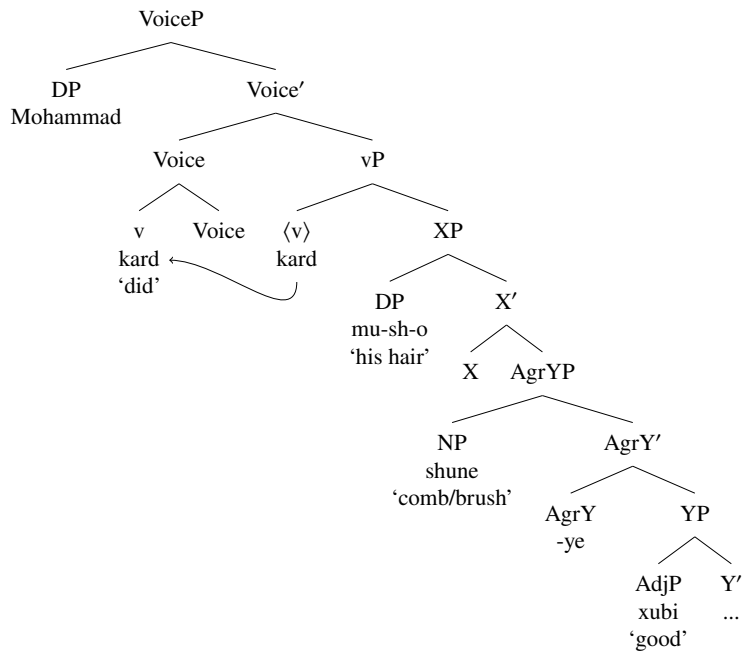


The maximal projections labeled XP represent the internal structures of the projection of the NV predicate, where the modifiers and the arguments have already been introduced. Subsequently, the light verbs select for these XPs and project to vPs. The vPs are in turn dominated by the VoiceP projections that host the external arguments. These steps are shown in (93) and (94).

(93)

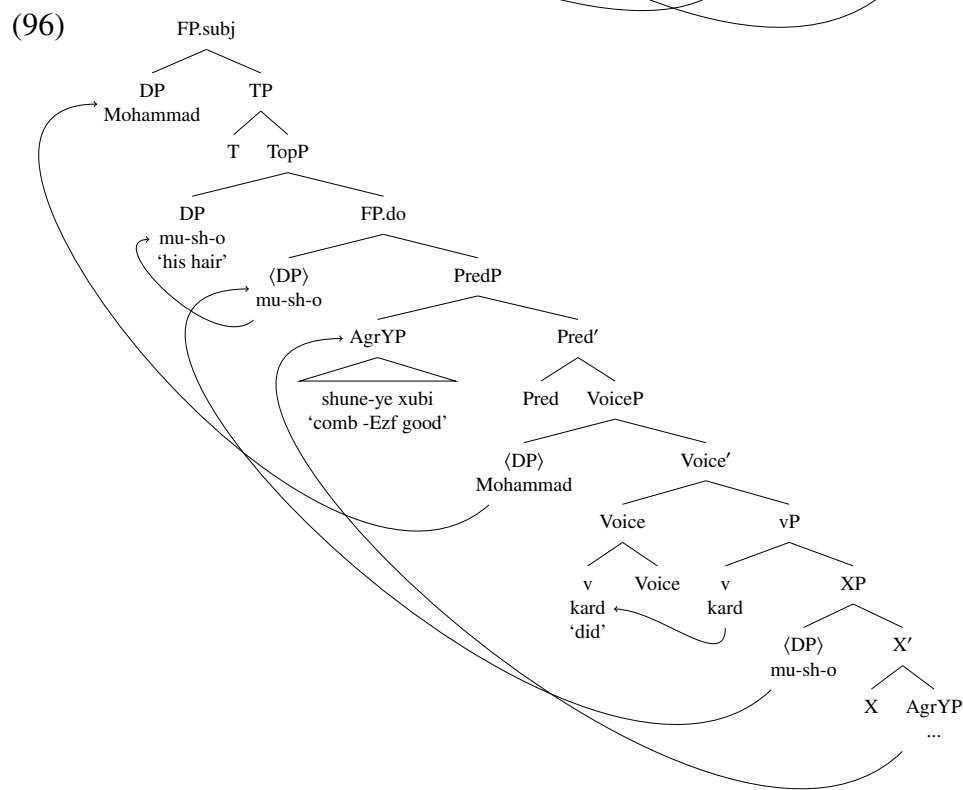
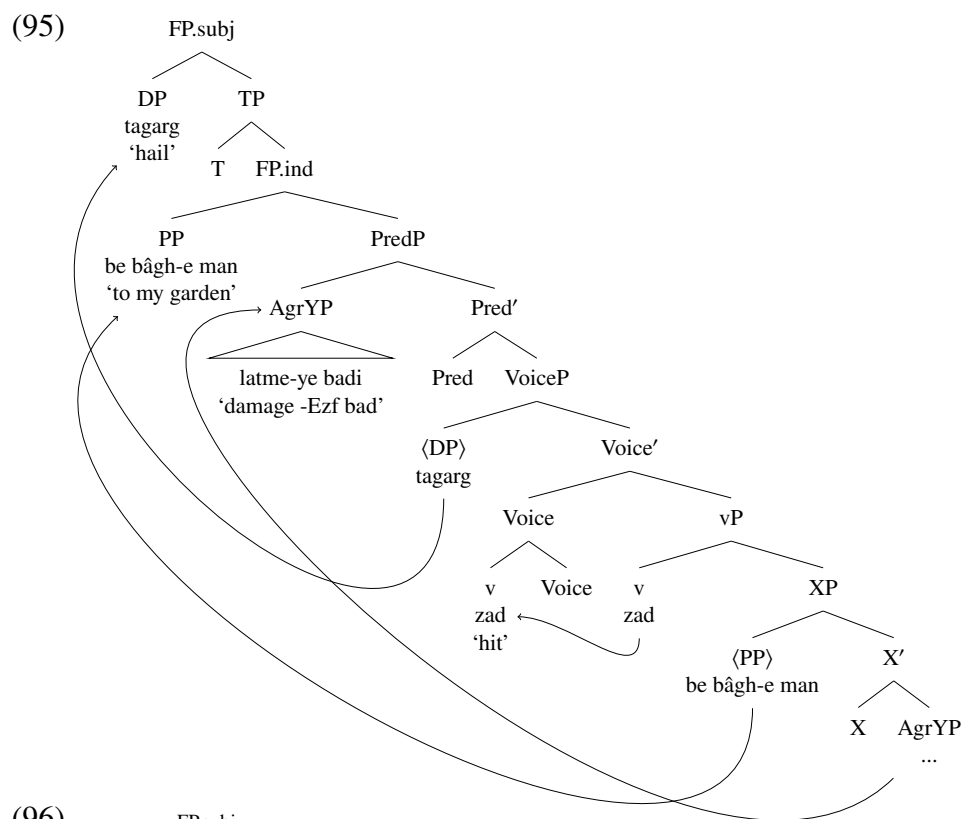


(94)



In line with my proposal in Chapters 2 and 3 that the NV predicate must be licensed in the functional domain in [Spec, PredP], I assume that in the case of (93) and (94) where the predicates have moved to the specifiers of AgrYP, it is the AgrP that raises to PredP, as represented in (95) and (96). The rest of the derivation follows as expected, with the internal DP and PP arguments as well as the

external argument moving to their licensing functional projections to be licensed.²⁴

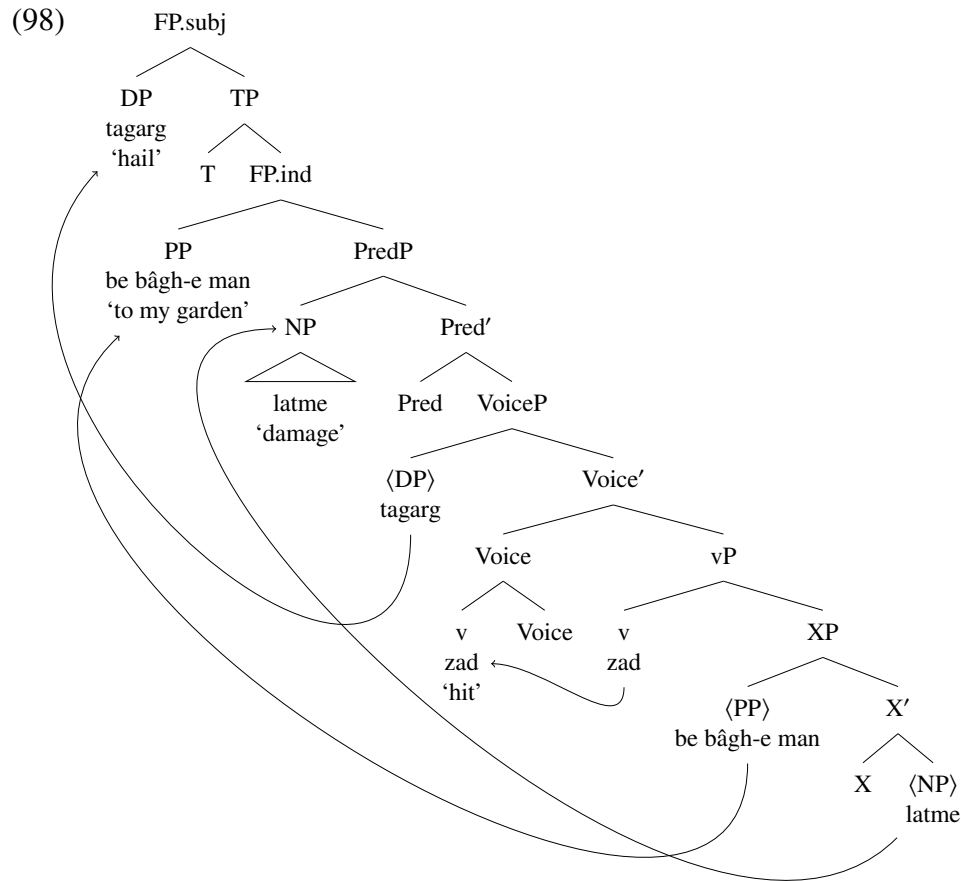


²⁴In (95), (96), and subsequent trees, only the specifiers of the argument licensing FPs are represented.

As a consequence of embedding the modifiers inside the AgrPs that ultimately end up in [Spec, PredP], I claim that the predicate and the modifier form a syntactic unit and that the modifier is interpreted as part of the full complex predicate. A piece of evidence in favor of this claim comes from the interaction between such constructions and the verb-stranding ellipsis phenomena investigated in Chapter 2. To illustrate this point, I will first discuss the sample data in (97), in which the NV predicate *latme* ‘damage’ appears without a modifier. The crucial observation here is that in the ellipsis clauses in (97b) and (97c), not only the contrastive PP argument *be bâgh-e man*, but also either the sole LV component *zad* ‘hit’ (97b) or the NV+LV *latme zad* ‘damage hit’ together (97c) must escape ellipsis.

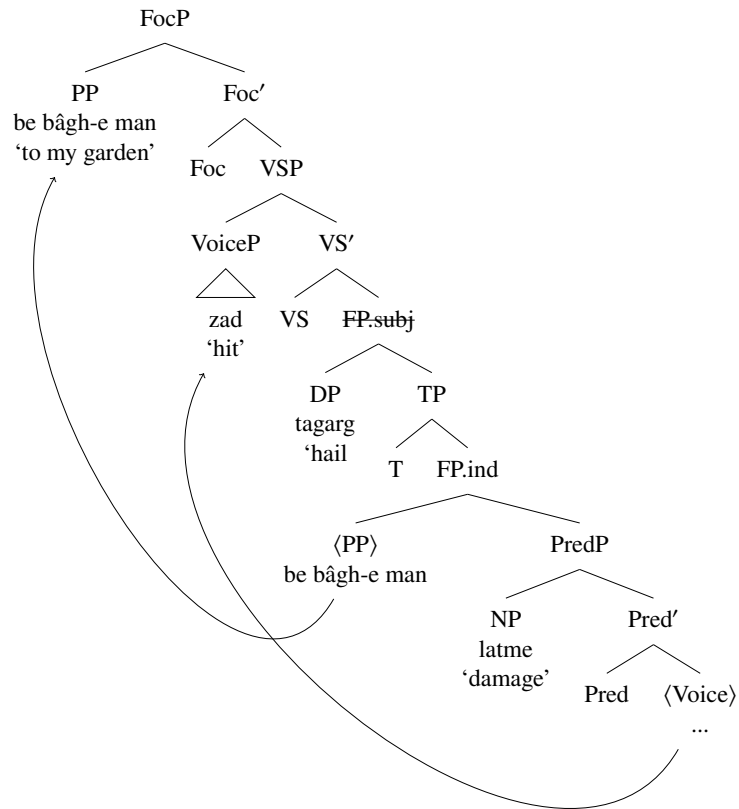
- (97) a. *tagarg be bâgh-e Ali latme na-zad.*
 hail to garden-Ezf Ali damage NEG-hit3SG
 “The hail didn’t damage Ali’s garden.”
- b. *vali tagarg be bâgh-e man latme zad.*
 but hail to garden-Ezf I damage hit.3SG
 “But (the hail) did (damage) my garden.”
- c. *vali tagarg be bâgh-e man latme zad.*
 but hail to garden-Ezf I damage hit.3SG
 “But (the hail) did damage my garden.”

Under the account of complex predicate constructions developed thus far, the non-elliptical counterpart to (97b) and (97c) receives the straightforward derivation in (98), where all the licensing requirements (movement of the NV predicate to [Spec, PredP], the PP argument to [Spec, FP.ind], and the subject DP in [Spec, FP.subj]) have already taken place.

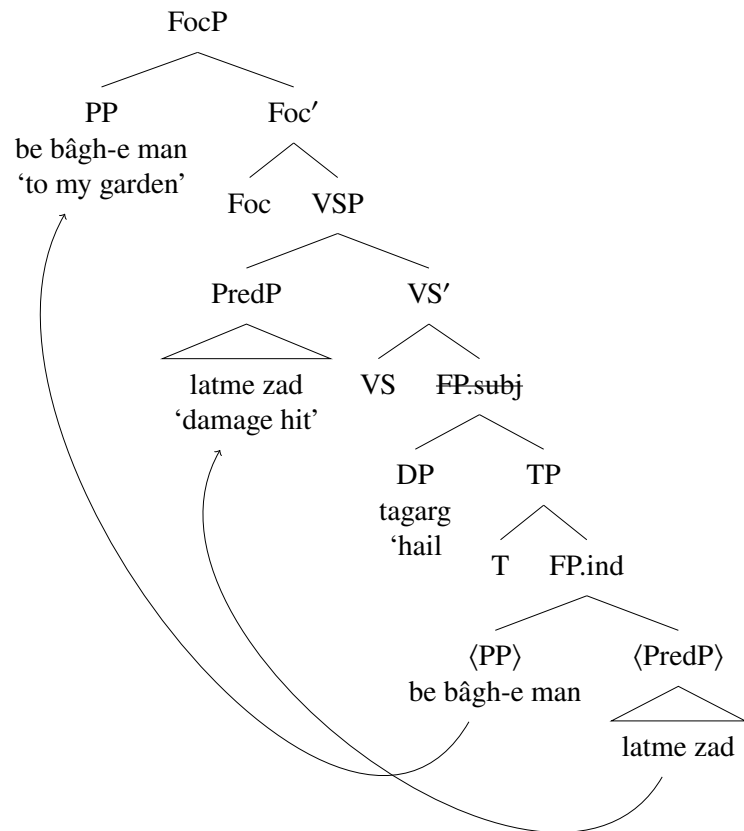


Operating on the full derivation in (98), I've proposed that the verb-stranding options in (97b) and (97c) can be derived from two possible phrasal movements. In the case of (97b), the verb-stranding requirement is satisfied by raising the verbal projection immediately below PredP (the VoiceP spelling just the Light Verb) to a functional projection (VSP) outside of the ellipsis site, which I take to be *FP.subj*. With respect to (97c), this requirement is met through the movement of the PredP constituent (spelling out the full predicate) to the relevant functional head outside of *FP.subj*. The two options are captured in (99) and (100), respectively.

(99)

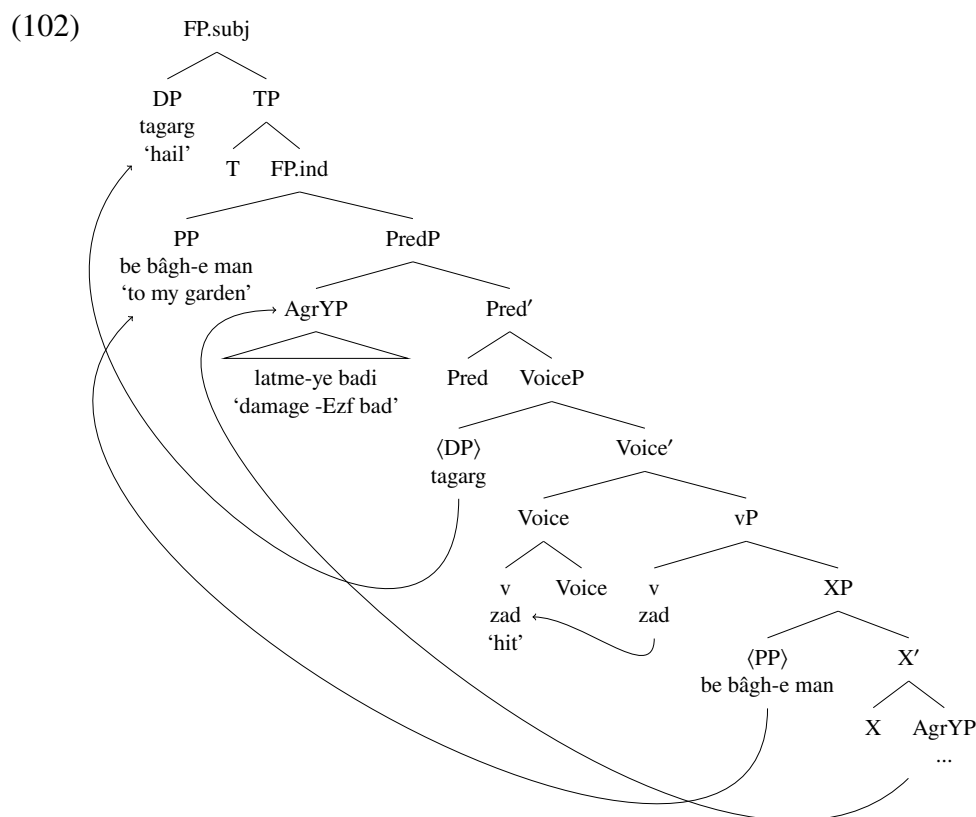


(100)



I now return to constructions in which the NV predicate is modified, such as in (89b), repeated below as (101). The derivation of this sentence is given in (102).

- (101) *tagarg be bâgh-e man latme-ye [badi] zad.*
 hail to garden-Ezf I damage-Ezf bad hit3sg
 “The hail caused a bad damage to my garden.”



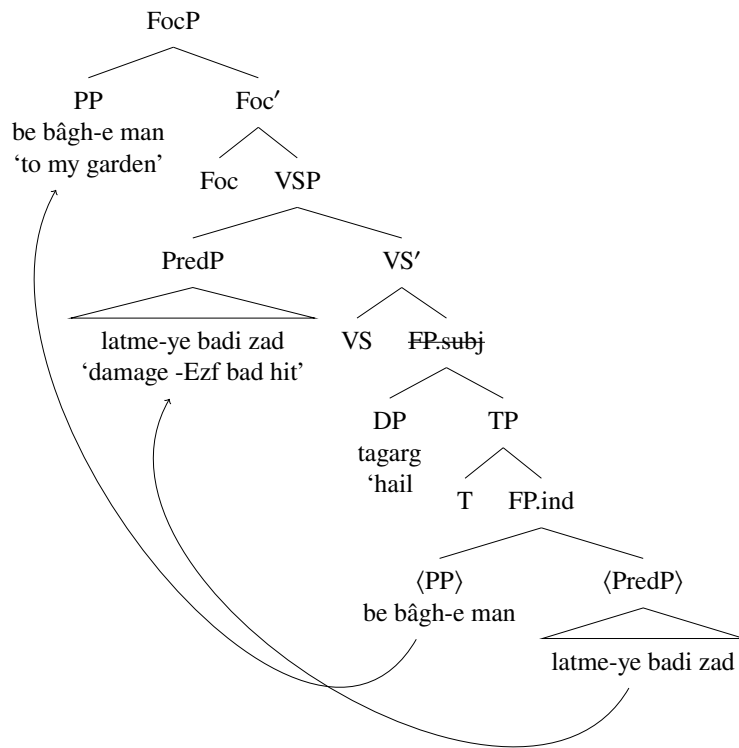
The current analysis predicts that in ellipsis data involving (101) and the derivation in (102), two verb-stranding possibilities should be observed. One in which the VoiceP below PredP moves out of *FP.subj* and therefore, only the the LV *zad* ‘hit’ is stranded. Another one in which the PredP raises out of the ellipsis site, and given that fact that the modifier is embedded in the AgrYP occupying the Spec of PredP, the entire sequence *latme-ye badi zad* ‘caused a bad damage’ including the modifier should be stranded. The data in (103) reveals that this predication is indeed borne out. Crucially, (103c) represents the latter option in which PredP has moved out of the ellipsis site alongside the contrastive PP ‘to my garden’, as suggested in the structure in (104).

- (103) a. *tagarg be bâgh-e Ali latme-ye badi na-zad.*
 hail to garden-Ezf Ali damage-Ezf bad NEG-hit3sg

“The hail didn’t cause a bad damage to Ali’s garden.”

- b. *vali tagarg be bâgh-e man latme-ye badi zad.*
 but hail to garden-Ezf I damage-Ezf bad hit3SG
 “But (the hail) did (cause a bad damage) to my garden.”
- c. *vali tagarg be bâgh-e man latme-ye badi zad.*
 but hail to garden-Ezf I damage-Ezf bad hit3SG
 “But (the hail) did cause a bad damage to my garden.”
- d. **vali tagarg be bâgh-e man latme-ye badi na-zad.*
 but hail to garden-Ezf I damage-Ezf bad hit3SG
 “But (the hail) did cause a (bad) damage to my garden.”

(104)



I take the facts in (103c), and the observation that in (103d) the modifier cannot be deleted by itself, as confirmation that the NV predicate and its modifier move to [Spec, PredP] as a unit. The following example further shows that this is the general pattern observed whenever verb-stranding ellipsis interacts with Predicate+Modifier constructions. Essentially, the stranding of the LV *kard* ‘did’ in (105b) is derived from the movement of the projection below PredP out of the ellipsis site, whereas in (105c), it is the PredP (containing the entire verbal complex *xune-ye xubi kard*) that’s

stranded.

- (105) a. *Mohammad mu-sh-o shune-ye xubi na-kard.*
 Mohammad hair-CL.3SG-DOM comb-Ezf good NEG-did.3SG
 “Mohammad didn’t do a good job combing his hair.”
- b. *vali Zahrâ mu-sh-o shune-ye xubi kard.*
 but Zahra hair-CL.3SG-DOM comb-Ezf good did.3SG
 “But Zahra did (a good job combing) (her hair).”
- c. *vali Zahrâ mu-sh-o shune-ye xubi kard.*
 but Zahra hair-CL.3SG-DOM comb-Ezf good did.3SG
 “But Zahra did a good job combing (her hair).”

So far, I have shown how postnominal modifiers of non-verbal predicates can be accounted for and licensed by *ezâfe* via the application of Kahnemuyipour’s (2014) phrasal movement analysis, and without imposing any additional mechanisms to the ones already proposed for licensing XPs in the verbal functional domain. I now move on to discussing how the derivation of [Predicate -Ezf Complement] constructions follow from the implementation of the same framework.

4.4.2 An account of the Predicate -Ezf Argument data

4.4.2.1 Copular constructions

I will start with revisiting the copula constructions presented in section 4.1.2, in which the adjectival non-verbal predicate subcategorizes for either a PP or DP argument and the subject can be understood as assuming an experiencer role assigned by the predicate. It was observed in (4a) and (5a), repeated here as (106) and (107), that predicates ‘interested’ and ‘satisfied’ take indirect object PP arguments that can surface either to the left of the predicate in the (a) sentences or optionally with *ezâfe* to the right of the predicates in the (b) sentences.

- (106) a. *Ali [PP be riâzi] alâghemand ast.*
 Ali to math interested is.3SG
 “Ali is interested in mathematics.”
- b. *Ali alâghemand-(e) [PP be riâzi] ast.*
 Ali interested-Ezf to math is.3SG

- (107) a. *Ali* [_{PP} *az nomarât-ash*] ***râzi*** *n-ist*.
 Ali from/of grades-CL.3SG satisfied NEG-is.3SG
 “Ali isn’t satisfied with his grades.”
- b. *Ali* ***râzi-(e)*** [_{PP} *az nomarât-ash*] *n-ist*.
 Ali satisfied-Ezf from/of grades-CL.3SG NEG-is.3SG

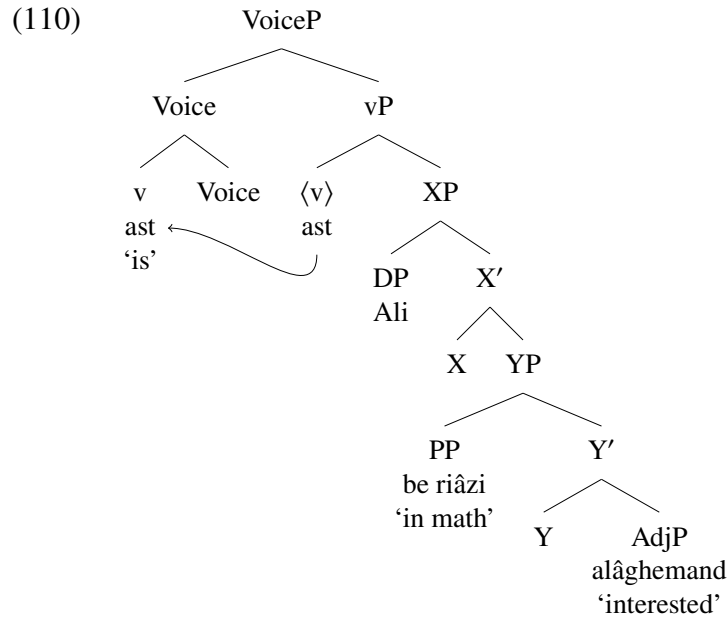
In the case of DP argument taking predicates such as ‘in-love’ and ‘crazy’ in (108) and (109), however, the two options are not readily available. The DPs must occur with the *Ezâfe* to the right of the predicate .

- (108) a. *Hasan* ***âshegh-e*** [_{DP} *Mitrâ / zabânshenâsi*] *ast*.
 Hasan in-love-Ezf Mitra / linguistics is.3SG
 “Hasan is in love with with Mitra / linguistics.”
- b. * *Hasan* [_{DP} *Mitrâ-râ / zabânshenâsi*] ***âshegh*** *ast*.
 Hasan Mitra-DOM / linguistics in-love is.3SG
- (109) a. *Minâ* ***divâne-ye*** [_{DP} *Ali / bastani*] *ast*.
 Mina crazy-Ezf Ali / ice-cream is.3SG
 “Mina is crazy for Ali / ice cream.”
- b. * *Minâ* [_{DP} *Ali-râ / bastani*] ***divâne*** *ast*.
 Mina Ali-DOM / ice-cream crazy is.3SG

Consistent with the requirement that arguments must be functionally licensed, I propose that cases such as (106) and (107) ultimately reveal that the PP arguments in these constructions can be licensed in one of the two ways; either (i) via the verbal functional system or (ii) in the nominal functional domain of the non-verbal predicate with *ezâfe*. The internal DP arguments of the predicates in (108) and (109), on the other hand, only have the latter licensing option available to them. I will begin with the PP cases and lay out how the sample pair in (106) is accounted for.

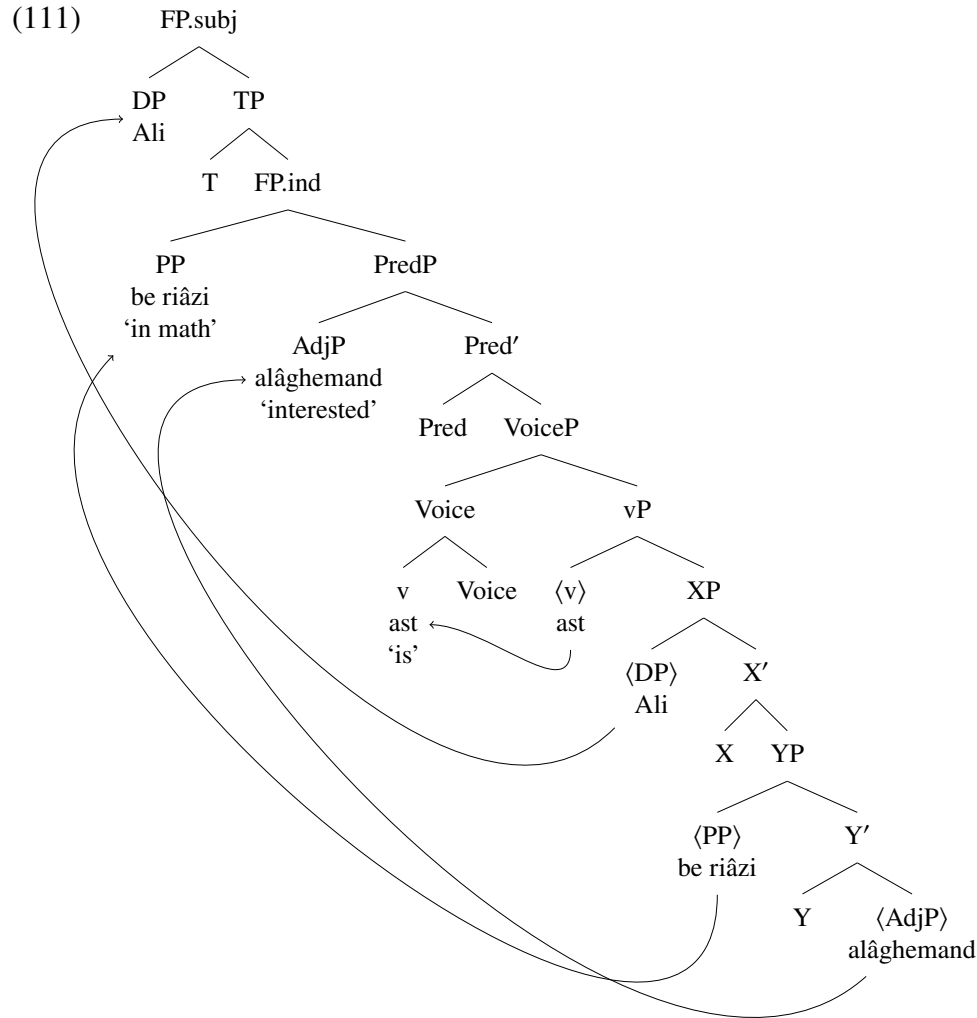
Starting with (106a), I assume the base structure in (110) for the NV phrase, where the adjectival predicate is merged low in the tree, and the PP argument *be riâzi* ‘in math’ occupies the specifier position of a YP immediately above it. The DP *Ali* is in turn base generated as the subject of a small clause in the spec of the higher XP. I assume that the copula *ast* ‘is’ originates as the head the vP that selects for the lower XP and further moves to the Voice head. However, VoiceP does not

project a specifier in this case.²⁵



The next derivational step is the same licensing operation proposed so far for non-verbal predicates in complex predicate constructions. That is, I assume that the NV elements similarly raise to [Spec, PredP] in copular constructions, accounting for their pre-copula distribution. In this respect, the movement operation targeting the NV predicate is systematic and occurs in all constructions that involve such predicates. Subsequent to the movement of ‘interested’ to the specifier of PredP in (111), the PP ‘in math’ is licensed in FP.ind, and lastly the subject raises to the FP.subj projection to complete the derivation. This captures the option of licensing the PP in the verbal functional domain.

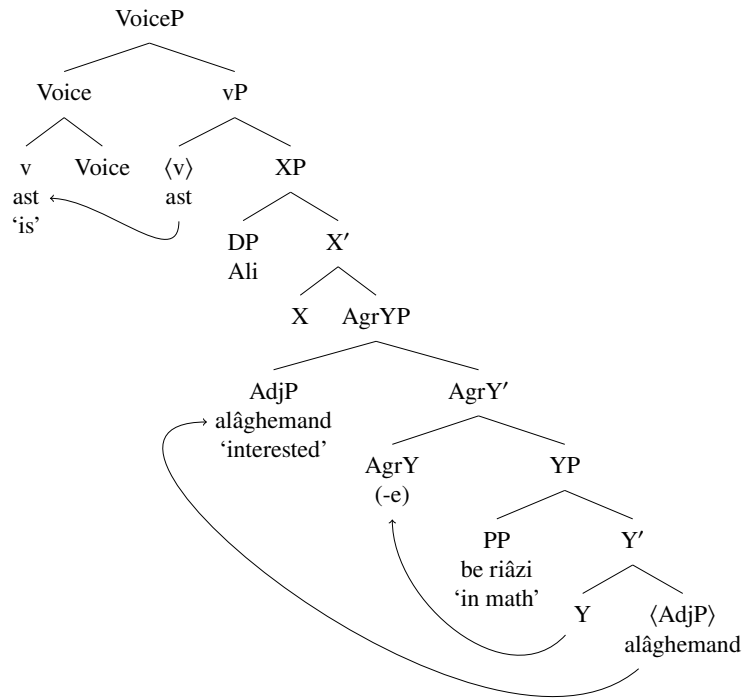
²⁵Consistent with all previous derivations, I maintain that Voice is always present. It could alternatively be the case that such constructions simply lack a VoiceP projection in their intransitive form and in the absence of an external argument.



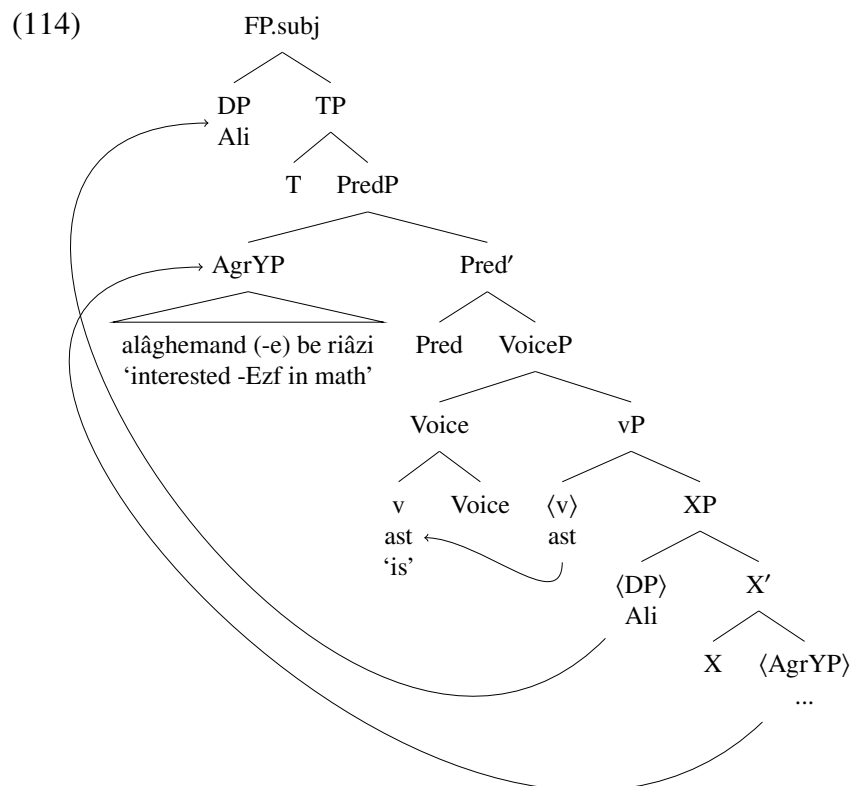
When it comes to the [Predicate -Ezf PP] case in (106b), repeated below as (112), I assume the same base structure for the projection of the non-verbal predicate and its arguments. However, in the case of (112), the PP argument *be riâzi* is licensed internal to the projection of the NV phrase and via the same inversion process linking the predicate and its modifier in section 4.4.1. This is implemented as shown in (113), where an intermediate AgrYP projects directly above YP and raises the NV predicate to its specifier.

- (112) *Ali alâghemand-(e) [PP be riâzi] ast.*
 Ali interested-Ezf to math is.3sg
 “Ali is interested in mathematics.”

(113)



As a result of linking the NV and the PP argument in this way, the constituent that raises to PredP at a later stage in the derivation is the full AgrYP. Since the PP argument has already been licensed via *Agr/ezâfe*, it follows that no subsequent licensing in FP.ind takes place in (114). The only argument that remains in need of licensing is the subject DP *Ali*, and this is achieved through movement to FP.subj as expected.



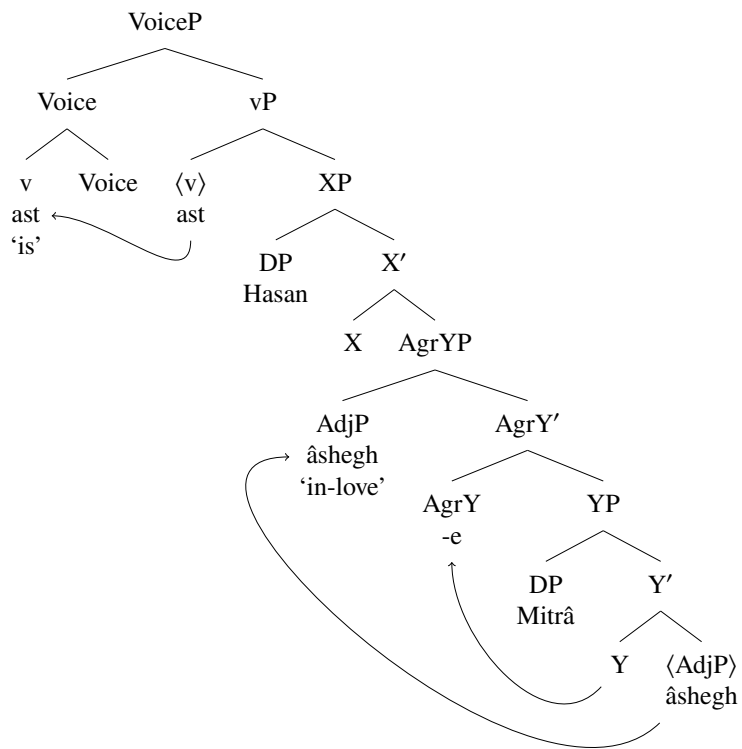
I now proceed to discussing the derivation of the [Predicate -Ezf DP] order shown in (108), repeated below as (115a). The crucial observation is that the DP argument *Mitrâ* cannot surface to the left of the predicate as in (115b) whether it's in its bare form or morphologically marked by the differential object marker *-râ*.

- (115) a. *Hasan âshegh-e* [_{DP} *Mitrâ*] *ast*.
 Hasan in-love-Ezf Mitra is.3sg
 “Hasan is in love with/enamored with Mitra.”
- b. **Hasan* [_{DP} *Mitrâ* / *Mitrâ-râ*] *âshegh ast*.
 Hasan Mitra / Mitra-DOM in-love is.3sg
 “Hasan is in love with/enamored with Mitra.”

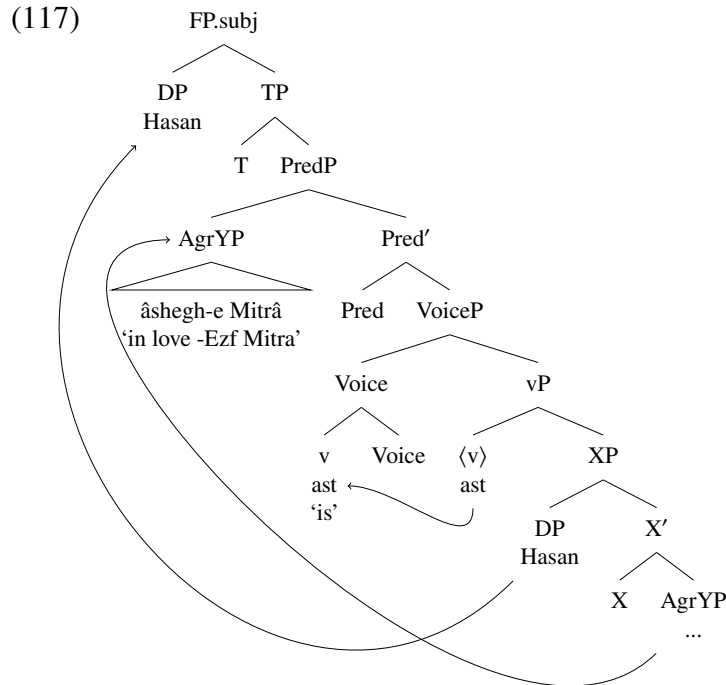
The adjectival predicate *âshegh* ‘in love’ and its arguments are introduced in the same fashion as is proposed in (113). That is the predicate is merged low and the arguments are base-generated in the specifier positions of higher functional projections (YP, XP). Crucially, the DP internal argument must be licensed in the lower nominal functional domain, enabled by the movement of the predicate to the intermediate AgrYP. This takes place prior to the introduction of the small clause subject DP

and the subsequent vP and VoiceP, as suggested in (116).

(116)



The rest of the derivation parallels that in (114); the AgrYP constituent *âshegh-e Mitrâ* ‘in love -Ezf Mitrâ’ raises to be licensed in [Spec,PredP], and the subject *Hasan* moves to [Spec,FP.subj], as illustrated in (117).



This raises the question of why the option of argument licensing to the left of the predicate is not available to DP internal arguments in these clauses, whereas it is available to PP arguments. I propose that this is due to the syntactic properties of predicational copular clauses and their licensing properties in general. Crucial to these constructions is the fact that subjects are not introduced as external arguments in what I have labeled VoiceP but instead as specifiers of small clause XPs and as arguments of the predicate. Relatedly, these constructions do not have the capabilities to license and assign an objective case to DPs; the only case assignment observed is the Nominative Case on the referential subject DPs. In the current theory, this reflects the lack of a licensing FP in the verbal functional domain for canonical direct object DPs. Given that copular constructions are fundamentally intransitive, it follows that FP.do does not project and no licensing operation to the left of the predicate is accessible to the DP argument *Mitrâ* in (115b). Consequently, in such constructions the DP complements are always licensed internal to the functional domain of the adjectival predicate, as suggested in (116). In the case of PP complements as in (106) and (107), however, I hypothesize that the FP licensing projection for PPs is always available and is not contingent on whether the clause is transitive. Therefore, PPs can be licensed in one of two ways. The causative counterparts to (115) and (106), involving the light verb *kardan* ‘made’ and

the external argument *Ehsân*, further confirm that the word order facts for the arguments of the predicate remain constant. In (118), the DP *Mitrâ* must occur with *ezâfe* and follow the predicate ‘in love’. In (119), the PP *be riâzi* ‘in math’ can either precede the predicate ‘interested’ or follow it. In both cases, the subjects of the predicate now carry the *-râ* marking typically observed on specific/referential direct objects, which means that they have undergone movement to the relevant DO licensing positions to the left of the predicate.

- (118) *Ehsân Hasan-râ âshegh-e* [_{DP} *Mitrâ*] *kard*.
 Ehsan Hasan-DOM in-love-Ezf Mitra made.3SG
 “Ehsan made Hasan fall in love with/enamored with Mitra.”

- (119) a. *Ehsân Ali-râ* [_{PP} *be riâzi*] *alâghemand kard*.
 Ehsan Ali-DOM to math interested made.3SG
 “Ehsan caused Ali to be interested in mathematics.”
 b. *Ehsân Ali-râ alâghemand (-e)* [_{PP} *be riâzi*] *kard*.
 Ehsan Ali-DOM interested -Ezf to math made.3SG

4.4.2.2 Complex verb constructions

I will now turn our attention to the Predicate+Complement cases observed in the context of complex verbs constructions. One environment in which the PP argument of the NV predicate can surface on either side of it is the Direct Object Obligatory Control constructions (Karimi, Samiian, and Stilo 2008) in (8) and (9), repeated below as (120) and (121). This data shows that the non-finite clause embedding PP arguments *be dars xundan* ‘to study’ (120) and *be lebâs shostan* ‘to wash clothes’ (121) may either precede the adjectival predicates ‘forced’ and ‘busy’ in the (a) examples or follow them as linked via the *ezâfe* marker in the (b) examples.

- (120) a. *moallem dâneshamuz-o* [_{PP} *be dars xundan*] *vâdâr kard*.
 teacher student-DOM to lesson read-INF forced did.3SG
 “The teacher forced the student to study.”
 b. *moallem dâneshamuz-o vâdâr(-e)* [_{PP} *be dars xundan*] *kard*.
 teacher student-DOM forced-(Ezf) to lesson read-INF did.3SG
 (121) a. *Ali* [_{PP} *be lebâs shostan*] *mashghul shod*.
 Ali to clothes wash-INF busy became.3SG

“Ali got busy washing clothes.”

- b. *Ali mashghul-(e) [PP be lebâs shostan] shod.*
 Ali busy-(Ezf) to clothes wash-INF became.3SG

Fundamental to such control clauses is the requirement that the direct object DP is always licensed in the verbal functional domain. Since these DPs are referential and carry the differential object marker *-râ*, it is not surprising that they undergo movement up to the projection where they receive the marking. This reflects both the selectional property of the non-verbal predicates that participate in such direct object control constructions as well as the property of the clauses themselves. It also follows that if the predicate subcategorizes for another DP argument in such constructions, this argument can only be licensed in a position to the right of the predicate. The sentence (122a) (documented in Ghomeshi (1996)) reveals that this is indeed the case, given that the DP argument *ye safar-e tulâni* ‘a long trip’ follows the predicate ‘ready’ and occurs with *ezâfe*. As the data shows, this DP cannot precede the predicate (122b) unless it’s embedded under a PP (122c).

- (122) a. *Ali man-o âmade-ye [DP ye safar-e tulâni] kard.*
 Ali I-DOM ready-Ezf one trip-Ezf long made.3SG
 “Ali made me ready for a long trip.” (Ghomeshi 1996, p. 269)
- b. **Ali man-o [DP ye safar-e tulâni] âmade kard.*
 Ali I-DOM one trip-Ezf long ready-Ezf made.3SG
- c. *Ali man-o [PP barâ-ye ye safar-e tulâni] âmade kard.*
 Ali I-DOM for-Ezf one trip-Ezf long ready made.ft3sg

The optionality of licensing the PP arguments on either side of the predicate in (120b) and (121b) is further consistent with the behavior of the PPs observed in copular constructions such as in (106-109), repeated here as (123) and (124).

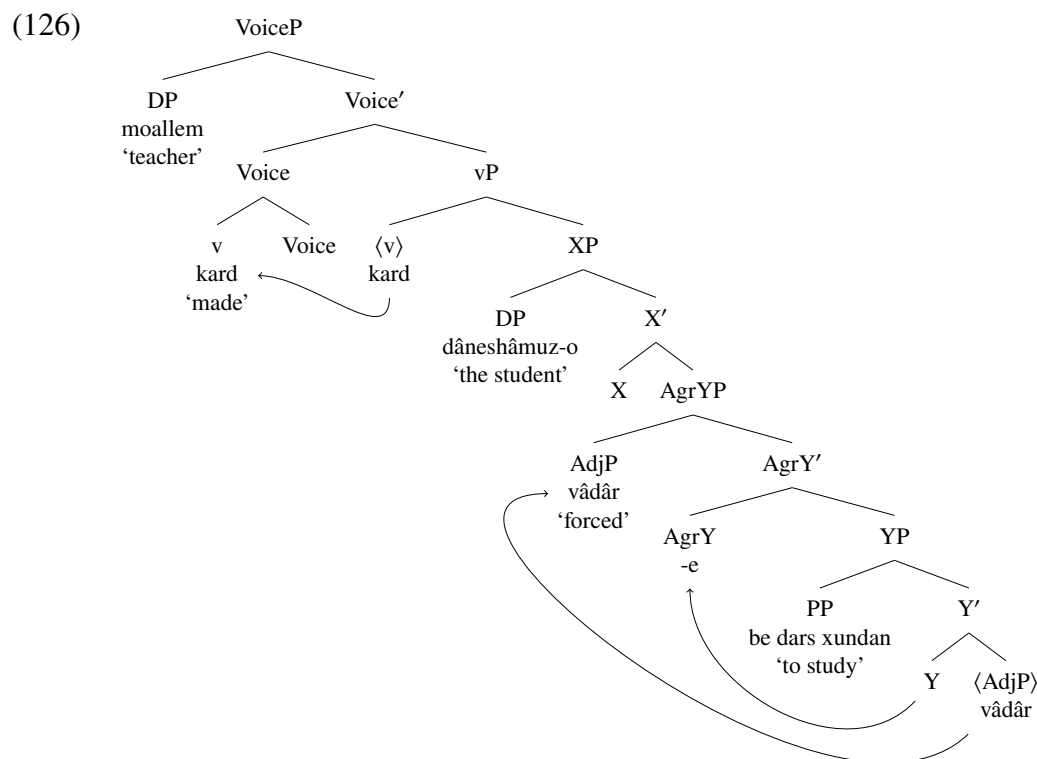
- (123) a. *Ali [PP be riâzi] alâghemand ast.*
 Ali to math interested is.3SG
 “Ali is interested in mathematics.”
- b. *Ali alâghemand-(e) [PP be riâzi] ast.*
 Ali interested-Ezf to math is.3SG
- (124) a. *Ali [PP az nomarât-ash] râzi n-ist.*
 Ali from/of grades-CL.3SG satisfied NEG-is.3SG

“Ali isn’t satisfied with his grades.”

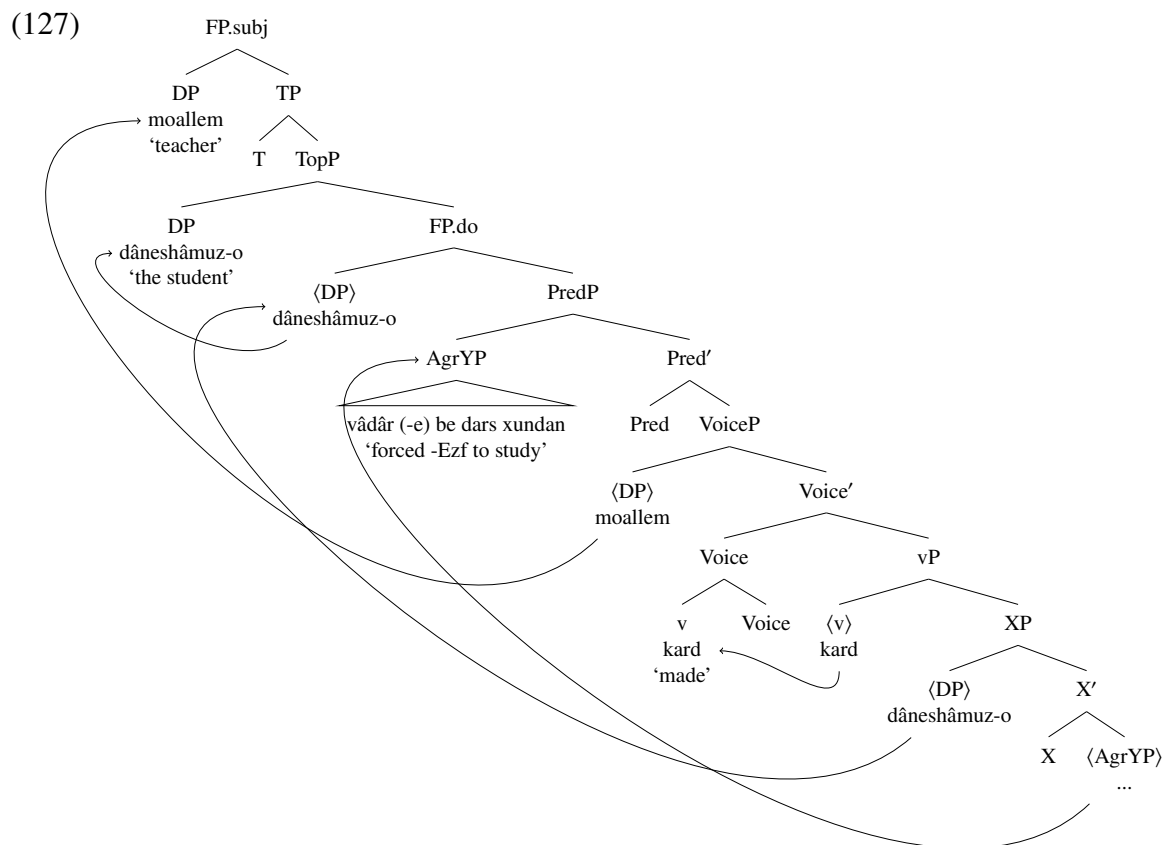
- b. *Ali râzi-(e) [PP az nomarât-ash] n-ist.*
 Ali satisfied-Ezf from/of grades-CL.3SG NEG-is.3SG

All that is required to derive the post-predicate position of the PP is to license this argument via an AgrP in the extended domain of the predicate. Applied to the control construction in (125), this step yields the partial derivation in (126).

- (125) *moallem dâneshamuz-o vâdâr-(e) [PP be dars xundan] kard.*
 teacher student-DOM forced-(Ezf) to lesson read-INF did.3SG
 “The teacher forced the student to study.”



Following the current assumptions, the inversion of the predicate around the PP has the consequence that the entire AgrYP *vâdâr (-e) be dars xundan* ‘forced -Ezf to study’ raises to PredP in (127). The PP, having been licensed, undergoes no further movement. Therefore, only the DO and the Subject move to designated FPs above the predicate.



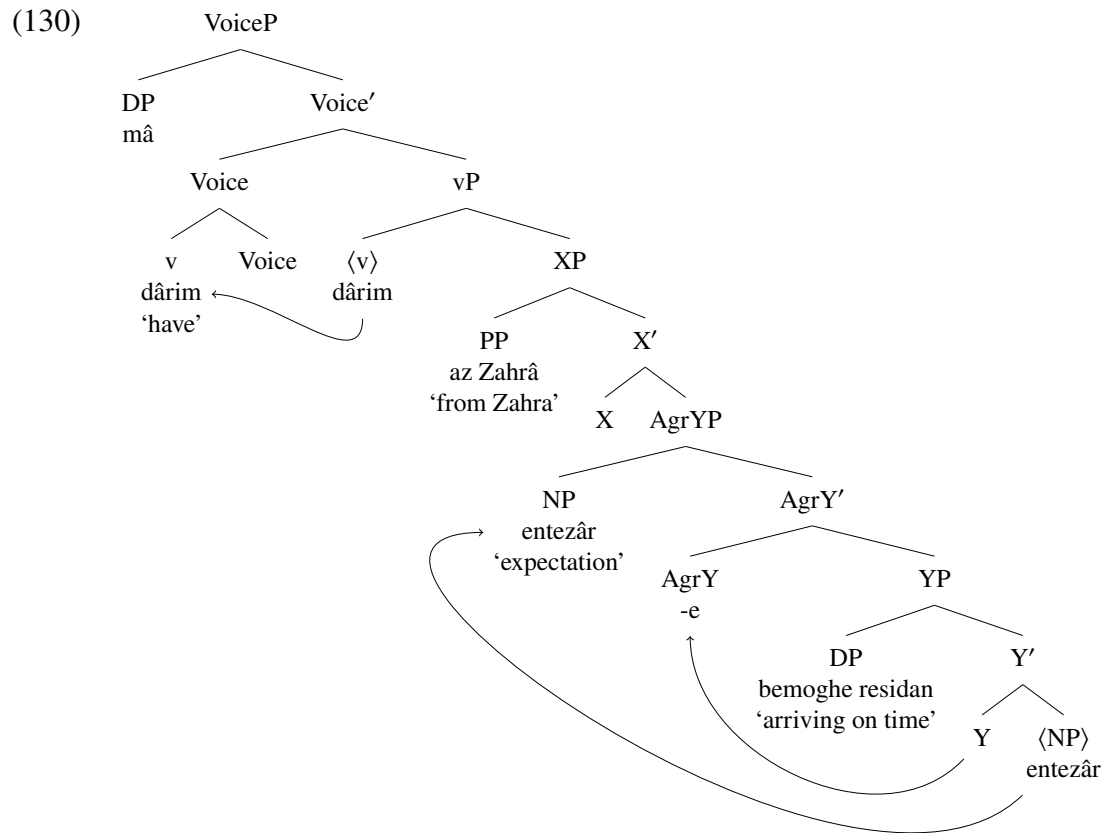
In Indirect Object Control constructions, the indirect object PP arguments can only precede the non-verbal predicate, as shown in (128) and (129). Additionally, DP complements in such constructions typically follow the predicates in (128a) and (129a) and are linked by *ezâfe*. Sentences in which DPs precede the predicate as in (128b) and (129b) are only marginally acceptable.

- (128) a. *doctor* [_{PP} *be bimâr-ân*] *tosiye*-ye [_{DP} *râh raft-an*] *kard*.
 doctor to patient-PL recommendation-Ezf way/path go-INF did.3SG
 “The doctor recommended the patients to walk/go on walks.”
- b. ? *doctor* [_{PP} *be bimâr-ân*] [_{DP} *râh raft-an*] *tosiye kard*.
 doctor to patient-PL way/path go-INF recommendation did.3SG
- (129) a. *mâ* [_{PP} *az Zahrâ*] *entezâr-e* [_{DP} *bemoghe resid-an*] *dârim*.
 we from Zahra expectation-e on-time arrive-INF have.1PL
 “We expect Zahra to arrive on time.”
- b. ?? *mâ* [_{PP} *az Zahrâ*] [_{DP} *bemoghe resid-an*] *entezâr dârim*.
 we from Zahra on-time arrive-INF expectation have.1PL

In the current theory, the fact that indirect objects do not appear with *ezâfe* to the right of the

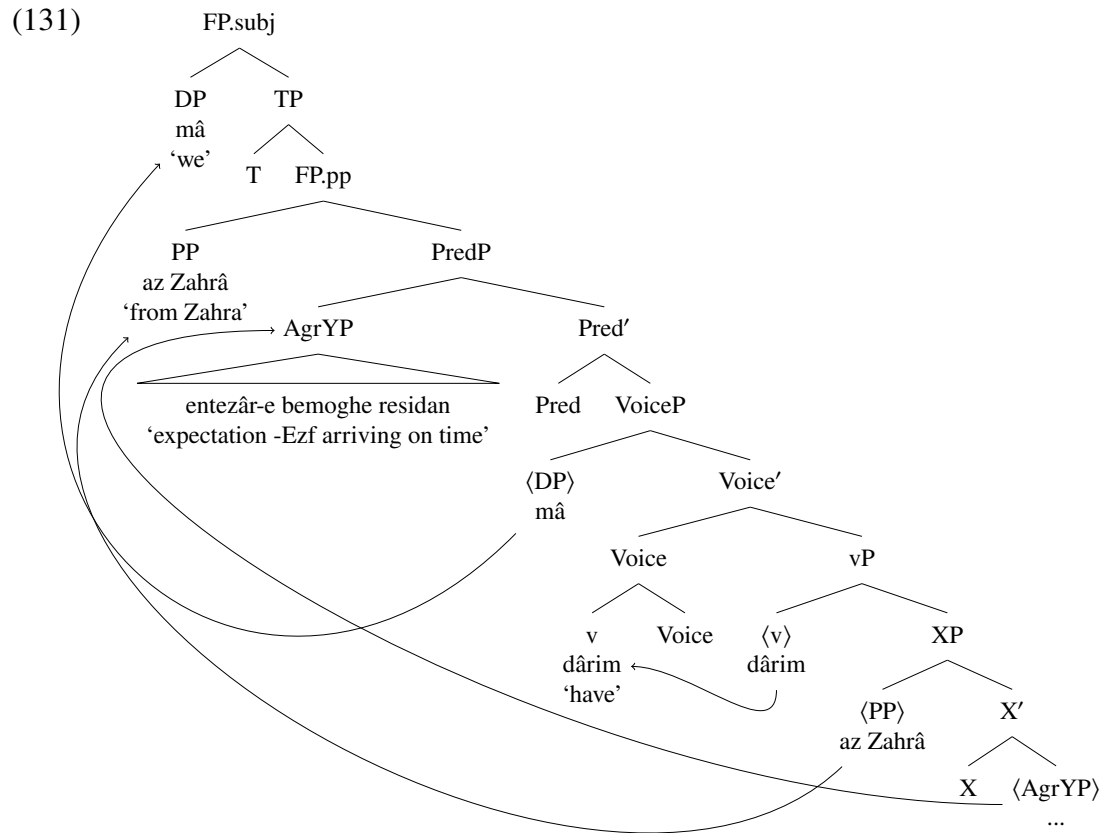
predicate can be captured by the generalization that in such constructions, the inversion process licensing an internal argument in the extended NV predicate domain is simply not available to the PPs. This is not surprising given that the locus of such control constructions is the indirect object, and as such, I hypothesize that this argument must be licensed in the verbal functional domain. Additionally, I assume that whatever factor drives the strong dispreference towards the direct object DPs preceding the predicates, as observed in (128b) and (129b), it must be an inherent property of these clauses or of the NV predicates themselves. Crucially, any alternative theory of phrase structure, including the head-final verb phrase analyses, must address and offer an explanation for these patterns.

Focusing on the preferred sentences in (128a) and (129a), the [Predicate -Ezf DP] order reflects the by-now-familiar syntactic operation that links the predicate and its DP argument via the intermediate AgrP early on in the derivation.²⁶ The structure in (130) illustrates this for (129a).



²⁶For the marginal cases in (128b) and (129b), I simply assume that this inversion process does not apply, and as a result, the direct object is licensed in FP.do to the left of the predicate at a later stage in the derivation.

Once again, this has the consequence that the predicate and the DO form a unit and move together to PredP in (131). The obligatory licensing of the indirect object PP and the subject DP follow as predicted.



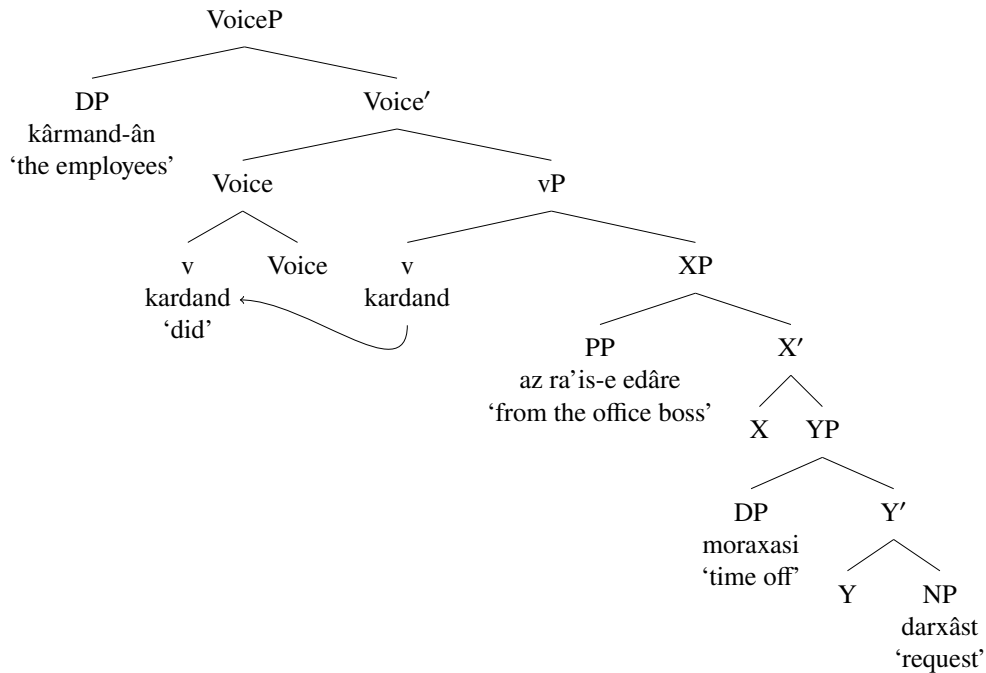
Lastly, I will discuss regular double object constructions, in which certain nominal predicates such as *tajviz* ‘prescription’ in (132) and *darxâst* ‘request’ in (133) can both follow or precede their direct object DP internal arguments.

- (132) a. *man* [_{PP} *be Ali*] [_{DP} *dâru*] ***tajviz*** *kardam*.
 I to Ali medicine prescription did.1SG
 “I prescribed Ali medicines.”
- b. *man* [_{PP} *be Ali*] ***tajviz-e*** [_{DP} *dâru*] *kardam*.
 I to Ali prescription-Ezf medicine did.1SG
- (133) a. *kârmand-ân* [_{PP} *az ra’is-e edâre*] [_{DP} *moraxasi*] ***darxâst*** *kardand*.
 employee-PL from boss-Ezf office time-off request did.3PL
 “The employees requested some time off from the office boss.”

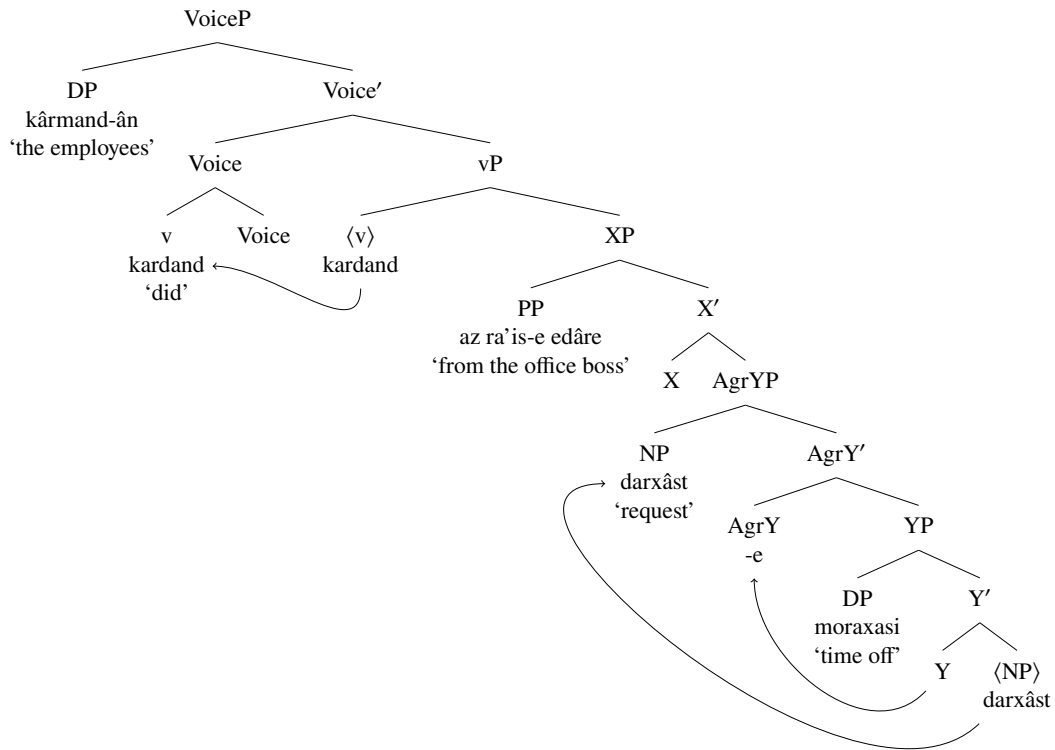
- b. *kârmand-ân* [_{PP} *az ra'is-e edâre*] *darxâst-e* [_{PP} *moraxasi*] *kardand*.
 employee-PL from boss-Ezf office request-Ezf time-off did.3PL

This again confirms that the predicate's arguments must be licensed, and with the class of predicates represented in (132) and (133), the licensing of direct objects can take place in the FP.do to the left of the predicate or alternatively via predicate inversion and Agr/*ezâfe* linking. Deriving the sample pair in (133), I first assume the partial structures in (134) and (135) up to the Voice projection. (135) represents the case in (133b) where the direct object DP *moraxasi* has already been licensed.

(134)

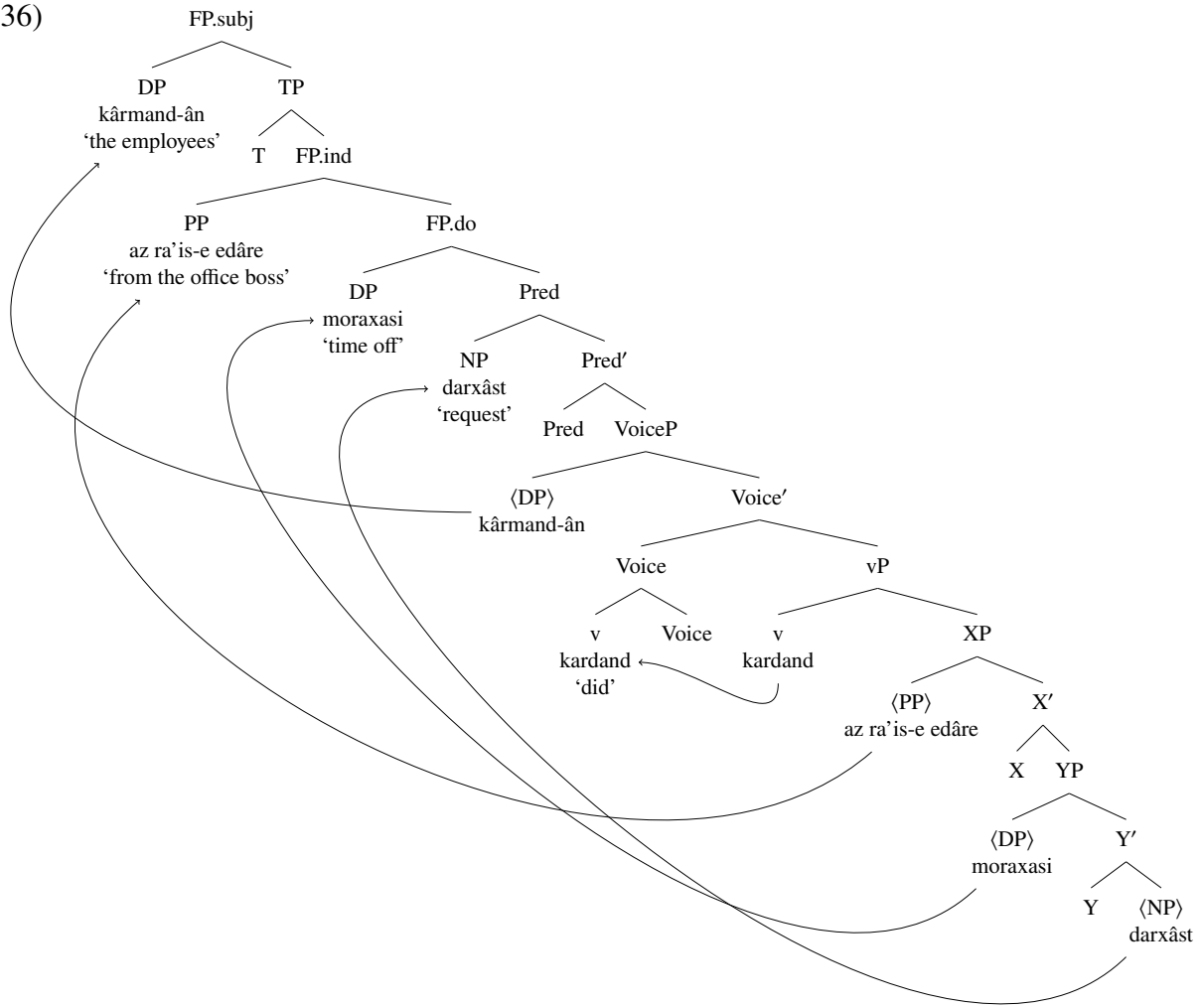


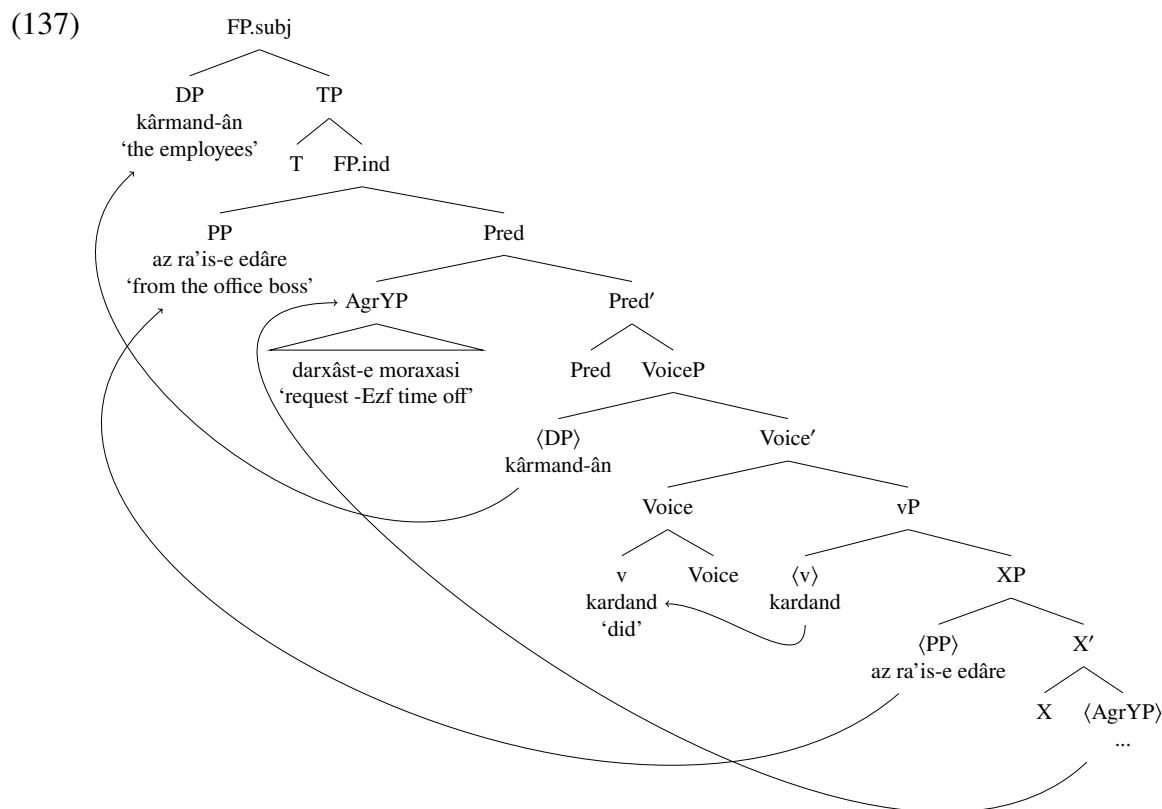
(135)



From here, both derivations proceed with the movement of the NV predicate to the specifier of PredP, and in the case of (135), this operation targets the AgrYP that contains the already licensed direct object. As a consequence, the full derivation of (133a) requires the movement of both the direct object DP and the indirect object PP (in addition to the subject DP) to their designated FPs, as shown in (136). However, in (133b) only the PP internal argument remains in need of licensing and moves to FP.ind, as suggested in (137).

(136)





Verb-stranding ellipsis phenomena once again supports the claim that in constructions in which the argument follows the non-verbal predicate, the two elements form a constituent and that the argument forms part of the complex predicate. Recall that in such elliptical construction, a verbal element (in addition to the contrastive argument(s)) must escape deletion. In complex predicate contexts, this verb-stranding requirement is typically met via stranding either the LV component of the predicate alone or the entire complex predicate (including both the NV and the LV). In the data in (138)-(140), the (b) examples reflect the former possibility, where only the light verbs *kardan* ‘do’ and *dâshtan* ‘have’ have avoided deletion. The examples in (138c), (139c), and (140c) correspond to the latter option, where entire complex predicates have escaped deletion. Crucially, these cases reveal that not only the NV and the LV elements but also the post-predicate arguments *be dars xundan* ‘to study’ (138c), *bemoghe resid-an* ‘arriving on time’ (139c), and *moraxasi* ‘time off’ (140c) must be included as part of the stranded complex verb constituent.

- (138) a. *moallem Ali-ro vâdâr-(e) be dars xundan kard.*
 teacher Ali-DOM forced-(Ezf) to lesson read-INF did.3SG
 “The teacher forced Ali to study.”

- b. *vali moallem Rezâ-ro vâdâr-(e) be dars xundan na-kard.*
 but teacher Reza-DOM forced-(Ezf) to lesson read-INF NEG-did.3SG
 “But (the teacher) didn’t (force) Reza (to study).”
- c. *vali moallem Rezâ-ro vâdâr-(e) be dars xundan na-kard.*
 but teacher Reza-DOM forced-(Ezf) to lesson read-INF NEG-did.3SG
 “But (the teacher) didn’t force Reza to study.”
- (139) a. *mâ az Ehsân entezâr-e bemoghe resid-an na-dârim.*
 we from Ehsân expectation-Ezf on-time arrive-INF NEG-have.1PL
 “We don’t expect Ehsan to arrive on time.”
- b. *Vali mâ az Zahrâ entezâr-e bemoghe resid-an dârim.*
 but we from Zahra expectation-Ezf on-time arrive-INF have.1PL
 “But (we) do (expect) Zahra (to arrive on time).”
- c. *Vali mâ az Zahrâ entezâr-e bemoghe resid-an dârim.*
 but we from Zahra expectation-Ezf on-time arrive-INF have.1PL
 “But (we) do expect Zahra to arrive on time.”
- (140) a. *Ali az ra’is-e edâr-ash darxâst-e moraxasi kard.*
 Ali from boss-Ezf office-CL.3SG request-Ezf time-off did.3SG
 “Ali requested some time off from his office boss.”
- b. *vali man az ra’is-e edâr-am darxâst-e moraxasi na-kardam.*
 but I from boss-Ezf office-CL.1SG request-Ezf time-off NEG-did.1SG
 “But I didn’t (request time off) (from my office boss).”
- c. *vali man az ra’is-e edâr-am darxâst-e moraxasi na-kardam.*
 but I from boss-Ezf office-CL.1SG request-Ezf time-off NEG-did.1SG
 “But I didn’t request time off (from my office boss).”

4.5 Conclusion

In this chapter, I have investigated constructions in which modifiers and arguments surface to the right of the predicate in both copular and complex predicate contexts, in comparison to their counterparts in which arguments precede the predicate as typically observed. I have argued that this data is problematic for the traditional head-final VP/vP analyses that assume the linear order to reflect the underlying structure of the predicate domain. I have provided an analysis for deriving the post-predicate constituent ordering and have proposed that such constructions represent a licens-

ing operation available to arguments and modifiers. Maintaining the view that elements must be functionally licensed within the antisymmetric framework pursued in the dissertation, I have argued that the distribution of arguments to either the left or the right of the predicate narrows down to two licensing options, one that is mediated by Agr at the nominal level and one that raises the licensee to a FP projection in the verbal system. In this way, the linear orders are derived in a principled way from syntactic operations that target various elements in the structure.

CHAPTER 5

CONCLUSION

This dissertation set out to closely examine the structure of the predicate domain in Farsi and evaluate whether its verb-finality is best explained by the widely adopted head-final VP/vP analyses or by an alternative antisymmetric framework that derives all linear orders from the same base structure. Through detailed investigation of complex predicates and their interaction with auxiliary ordering, ellipsis, and verb fronting, I have shown that head-final verb phrase accounts face significant empirical and theoretical problems, including inconsistent directionality in head-movement and reliance on problematic excorporation. Additionally, I have shown that accounting for the postverbal distribution of CPs under a head-final verb phrase analysis requires additional assumptions. Alternatively, an antisymmetric analysis captures the full range of ordering facts with the single, consistent derivational mechanism of leftward movement, eliminates the need for ad hoc operations, and unifies the treatment of predicate-argument and predicate-modifier configurations. In doing so, the dissertation aims to not only resolve long-standing issues in the analysis of Farsi syntax but also contribute to broader debates on head directionality and the derivation of mixed word order in natural language.

Chapter 2 provided the alternative antisymmetric analysis of phrase structure that derives the surface orders through the movement of elements from their base position to various dedicated licensing functional projections. In the context of complex predicate constructions interacting with auxiliary verbs, such as the Perfect Aux (1) and the Future Aux (2), I proposed that not only the internal and external arguments, but also the Non-Verbal (NV) predicate of complex verbs undergo leftward movements.

- (1) *Ehsân be Ali xune edžâre dâd-e ast.*
Ehsan to Ali house rent give-PTCP AUX.PRF.3SG
“Ehsan has rented house(s) to Ali.”
- (2) *Ehsân be Ali xune edžâre xâhad dâd.*
Ehsan to Ali house rent AUX.FUT.3SG give
“Ehsan will rent houses to Ali.”

More specifically, taking inspiration from Zwart (1993, 1997), I argued that the NV predicates consistently raise to a PredP licensing projection above all other verbal heads in the cluster (including auxiliary verbs when present), and this has the effect that they consistently surface as the leftmost element in any verbal sequence. The internal arguments (direct object DPs and indirect object PPs) are licensed in the functional projections FP.do and FP.ind, respectively, located above PredP. Lastly, the external argument is also taken to undergo movement to its designated FP.subj licensing projection highest in the TP domain. Another important assumption of this theory is that all auxiliary heads hierarchically project above the V heads, but they differ with respect to their selectional properties and whether they require the head they select for to adjoin to them. In the case of the Perfect Aux which selects for a participle verb, the participle adjoins to the auxiliary in the narrow syntax, resulting in the LV-Aux order in (1). With respect to the Future Aux which selects for a (short)-infinitival verb however, no v-to-Aux head adjunction takes place. Therefore, the Aux-LV observed in (2) is simply a reflection of the hierarchical base order of the two heads.

In Chapter 3, I revisited the puzzle of postverbal clausal complements in Farsi, which can be recast as the question of why in constructions in which verbs select for a CP argument, the CP must invariably surface to the right of the verb in comparison to the DP and PP arguments, which precede the verb.

- (3) *Parviz be har dânesâmooz **ghol** dâd* [ke ketâb-e u râ **be-xânad**].
 Parviz to every student promise gave.3SG that book-Ezf he DOM SBJV-read.3SG
 “Parviz promised every student to read his book.”
- (4) *Parviz be har dânesâmooz **ghol** dâd-e ast* [ke ketâb-e u râ
 Parviz to every student promise give-PTCP AUX.PRF.3SG that book-Ezf he DOM
be-xânad].
 SBJV-read.3SG
 “Parviz has promised every student to read his book.”
- (5) *Parviz be har dânesâmooz **ghol** xâhad dâd* [ke ketâb-e u râ
 Parviz to every student promise AUX.FUT.3SG give that book-Ezf he DOM
be-xânad].
 SBJV-read.3SG
 “Parviz will promise every student to read his book.”

Building on the developed antisymmetric analysis, I proposed that in such constructions the same licensing operations moving the Non-Verbal predicate as well the non-clausal arguments (DPs, and PPs) to positions above the light verb and the auxiliaries still apply. However, the key assumption is that CP arguments remain in their base position and are not subject to the same licensing requirements above the predicate. Since the base-generated position of CP arguments is always hierarchically lower than the projections of the light verb and the auxiliary head, it follows that they are always in a final position in the clause independent of whether the verb is simplex, complex, or part of a larger verbal cluster. Crucially, this proposal correctly accounts for the subextraction, binding, and ellipsis facts that are indicative of the low interpretation site of the CP.

Chapter 4 investigated the puzzle of post-predicate constituents in the non-verbal domain of Farsi, focusing on copular and complex predicate constructions where internal arguments or modifiers appear to the right of the predicate, often linked by *ezâfe*. Based on a close investigation of the relevant patterns, I have argued that this data is problematic for the traditional head-final VP/vP analyses that assume the linear order to reflect the underlying structure of the predicate domain. Instead, I have adopted and extended Kahnemuyipour's (2014) roll-up phrasal movement analysis, originally proposed for postnominal dependents in the NP/DP domain, to also cover predicate–modifier and predicate–argument orders in the verbal domain. Under this proposal, both modifiers and arguments are base-generated in specifiers of functional projections above the predicate head (whether nominal, adjectival, or prepositional). Roll-up movement of the predicate (or a projection containing it) across these specifiers derives the observed [Predicate –Ezf– XP] orders, while the *ezâfe* morpheme spells out the agreement relation established in this process. Crucially, I have claimed that this derivation represents a licensing operation available to arguments and modifiers in certain constructions and is further evidence that elements must be functionally licensed within the antisymmetric framework pursued in the dissertation. This is achieved via two licensing options; one that is mediated by Agr at the nominal level (deriving the post-predicate order) and one that raises the licensee to a FP projection in the verbal system (deriving the pre-predicate order). In this way, the linear orders are derived in a principled way from syntactic operations that target various

elements in the structure.

Taken together, the chapters of this dissertation have provided grounds for re-evaluating the syntactic analysis of Farsi word order, demonstrating that its apparent combination of head-final and head-initial properties can be derived from the interaction between a universally head-initial base structure and independently motivated leftward movement operations. In dispensing with the Head-Directionality Parameter, the framework developed here further eliminates the need for exceptional, within-category stipulations, such as positing special mechanisms for predicates selecting clausal arguments (in contrast to non-clausal ones) or for the occurrence of post-predicate constituents. Rather, it shows that the same fundamental principles, applied consistently, suffice to capture the full range of patterns observed across the grammar.

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